HOCHSCHULE RHEIN-WAAL
Rhine-Waal University of Applied Sciences

SunnyPro

Group No. 3

Faculty of Technology and Bionics

Submitted by :

Ahmad Azzeh,
(Degree : Mechanical Engineering)

Ahmed Yehia Mounir Moawad Mohamed,
(Degree : Electrical and Electronics Engineering)

Evelyn Kowal,
(Degree : Industrial Engineering)

Farhan Sirajudeen,
(Degree : Industrial Engineering)

Omar Neazy Ahmed Abdelsalam Ibrahim,
(Degree : Mechatronics Systems Engineering)

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Statement of Originality

We hereby certify that, to the best of our knowledge, all text passages and other content of this report are the original work of the listed members of our student group, unless explicitly referenced otherwise. All external sources, including datasheets, application notes, textbooks, scientific publications, and online resources, are cited at the point of use and listed in the bibliography.

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Ahmad Azzeh

Ahmed Yehia Mounir Moawad Mohamed

Evelyn Kowal

Farhan Sirajudeen

Omar Neazy Ahmed Abdelsalam Ibrahim

Project Supervisors:

Christian Berendonk

Thorsten Brandt

Jure Cubelic

Özgür Egici

Martin Hellwig

Angelika Michel

Martin Schlösser

Dirk Untiedt

Executive Summary

In today's competitive hospitality landscape, delivering an exceptional breakfast experience is a defining standard of quality that guests remember long after they check out. Labour costs continue to climb, skilled staff are harder to retain, and demand for fresh, made-to-order breakfast dishes peaks precisely when operational stress is at its highest. The SunnyPro is engineered to meet this challenge head-on — bringing together culinary precision, industrial reliability, and elegant design in one compact, counter-ready unit. It is not merely a piece of kitchen equipment; it is a strategic investment in the consistency and quality your guests deserve.

What sets SunnyPro apart is its complete end-to-end automation. There is no pre-cracking, no manual preparation, and no need for a cook to be present at the station. The machine stores a generous supply of raw eggs internally, automatically cracks each one with precision, cooks it to a flawless sunny-side-up finish, and dispenses it ready to serve — all without staff intervention. The integrated eggshell collection system captures and safely contains all shell waste within the unit, maintaining a clean working environment while freeing your team to focus on higher-value tasks elsewhere in the kitchen.

At its heart, SunnyPro is a guest-facing solution. A clean, intuitive user interface allows guests or service staff to select their preferred quantity with ease, making it perfectly suited for buffet environments, à la carte breakfast stations, or self-service setups. Whether a guest is in a hurry before an early morning meeting or enjoying a leisurely weekend brunch, SunnyPro delivers with the speed and consistency that leaves a lasting impression. Egg quality is tightly controlled throughout — the yolk and egg white structure are preserved with every cook cycle to a standard that rivals a trained chef, and the system remains fully refillable during live operation, ensuring service is never interrupted.

In professional food service, hygiene is a legal obligation and a cornerstone of brand trust. SunnyPro has been designed from the ground up to comply fully with Regulation (EC) No. 852/2004 of the European Parliament and Council on the hygiene of foodstuffs. Every surface and component in contact with food meets the rigorous requirements of European food safety law. The eggshell containment system prevents cross-contamination, and the machine's internal architecture supports straightforward cleaning routines — making compliance a matter of routine rather than effort.

SunnyPro's purposefully compact footprint fits comfortably on a standard kitchen counter without requiring structural changes to your existing setup. Power supply flexibility ensures compatibility with the full range of professional kitchen electrical configurations. Labour shortages are not a temporary disruption — they are a structural

shift in the industry. By automating one of the most consistently demanded breakfast items, your operation gains reliability, speed, and quality assurance around the clock, without a dedicated member of staff at the egg station.

Keywords:

Automation

Catering

Hygiene

Efficiency

Hospitality

Precision

Compliance

Reliability

Innovation

Consistency

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Milestone 1

1. Project Plan

1.1. Work Packages showing Responsibilities

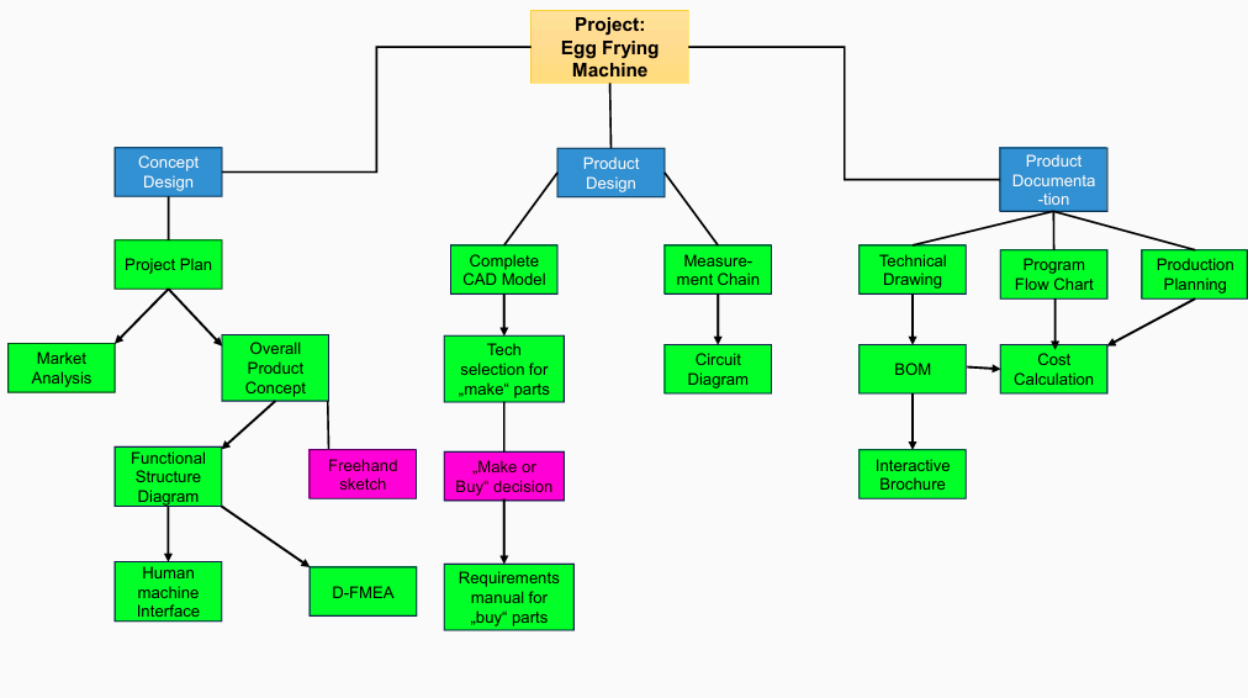
Work Packages	Responsibilities to	Sub-Tasks
1.Milestone: Concept Design		
1. Project Plan	IE	- Work Breakdown Structure - Gantt Chart
2. Market Analysis	IE	- Market Overview - Target Market Analysis - 4P Marketing Mix - Marketing Strategy - Planned Sales & Revenue Forecast
3. Overall Product Concept	ME, MSE, EE	- Bill of Requirements - Morphological Box - Design Sketch - Written Description - BOM
4. Functional Structure Diagram	MSE, EE	-
5. Human Machine Interface	ME, MSE, EE	- Description of Interface Design - List of In- & Outputs
6. D-FMEA	All	-
2.Milestone: Product Design		
7. Complete CAD Model	ME:MSE	-
8. Measurement Chain	MSE, EE	- Selection of Signal Conditioning Circuits - Calculation of required values
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		- BOM of Electrical Components
10. Tech selection for „make“ parts	All	- “Make or Buy” decision - Material Selection - List of Assembly Technologies
11. Requirements manual for “buy” parts	IE	-
3.Milestone: Project Documentation		
12. Technical Drawing	ME,MSE	-
13. Program Flow Chart	EE	-
14. BOM	ME,MSE,EE	- Mechanical BOM - Electrical BOM
15. Production Planning	IE	
16. Cost Calculation	IE	
17. Interactive Brochure	All	- Design of User Manual - Video - Design of Brochure

Table 1:Work Packages and Responsibilities

ME- Mechanical Engineering.
MSE- Mechatronics Engineering.
EE- Electric Engineering.
IE- Industrial Engineering.

1.2. Work Breakdown Structure showing Dependencies



Legend:

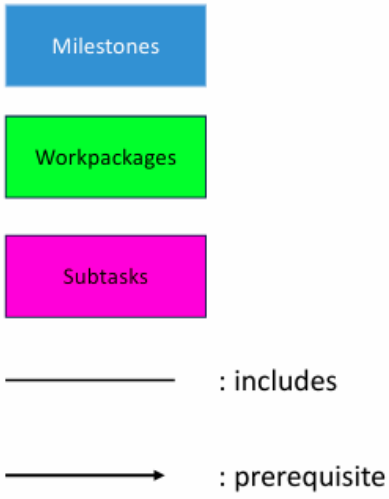


Figure 1:Work breakdown structure with dependencies.

1.3. Gantt Chart

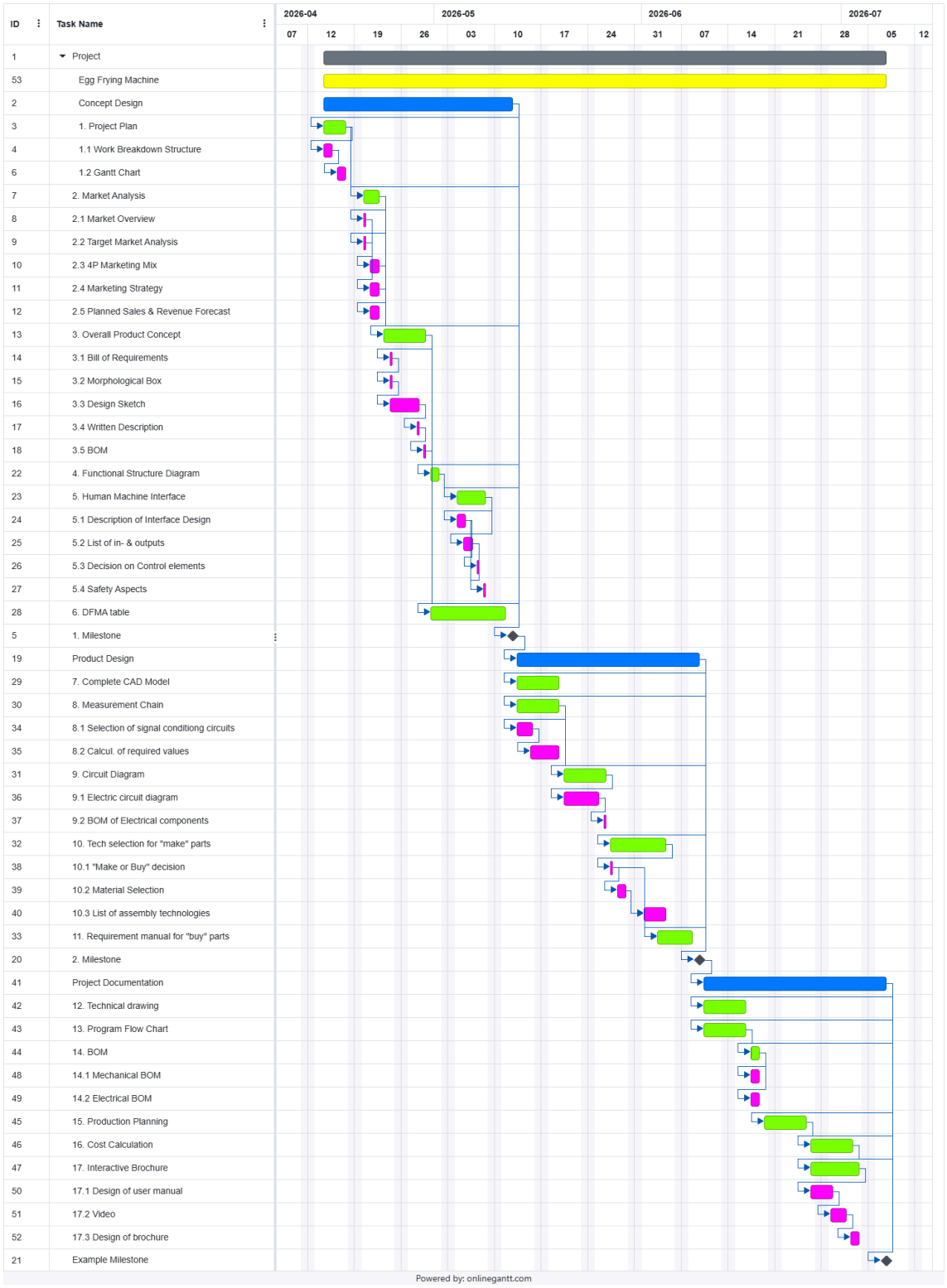


Figure 2:Gantt Chart demonstrating dependencies and milestones.

2. Marketing Decisions

2.1. Market Overview

2.1.1. Current market situation

The European market for automatic egg frying machines is currently niche but expanding steadily in 2026, fueled by labor shortages and high breakfast volumes in cafes and restaurants. Adoption remains low outside major chains, with most sales via imports, but interest is rising amid automation trends.

The data on this specific situation is limited, so we use data on machines like our product to analyze the market situation.

2.1.2. Current Demand

- Global market size: **$1.2 billion** in 2024.[1]
- Projected CAGR: **6.8%** through 2030.[1]
- Key regions: China (largest producer), USA, Germany, Japan.
- Rising demand for automation in breakfast service lines.[2]
- Growth is fueled by Chain Restaurants and Catering Services.

2.1.3. Supply Landscape

Asian manufacturers (e.g., XuZhong, SMK Cooking[3]) dominate via Alibaba and AliExpress, offering CE-certified 220V units at 2000€-4500€ that fry 120 eggs/hour with minimal oil. Local distributors like ChefsDeal provide limited stock, but supply chains face EU tariff pressures and delays.

Retailers like Amazon also sell generic automatic fried egg machines which cost around the range of 4000€-5000€.

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[1] *Market Data Forecast*. Europe egg processing market. Published April, 2026. Accessed May 8, 2026. URL: https://www.marketdataforecast.com/market-reports/europe-egg-processing-market

[2] Automatic Egg Processing Machines Market Type Application. *LinkedIn*. Accessed May 8, 2026. https://www.linkedin.com/pulse/automatic-egg-processing-machines-market-type-application-zxdbc

[3] Foshan Semikron Electronic Technology Co., Ltd. *Automatic Egg Frying Machine Hamburger Fried Egg Hotel Breakfast Snack Shop Batch Fried Eggs*. Accessed May 9, 2026. https://www.smkcooking.com/product-automatic-egg-frying-machine-hamburger-fried-egg.html

On analysis there is no local retailer selling this product in Germany or Europe as most of the machines are manufactured and sold in Asia.

2.1.4. Growth Indicators

Specific market size data for automatic egg frying machines in Europe is scarce, as they form a niche subset of the broader egg processing equipment sector without dedicated reports. However, growth proxies from related categories indicate solid potential at 3-7% CAGR through 2032, driven by automation in hospitality.

Major Market Drivers & Challenges

- Drivers: Labor shortage in foodservice[4]; rising consumer demand for consistent, high-quality breakfast items.
- Challenges: High upfront costs for premium models.

[4] HOTREC. *Hospitality Europe: Skills and Labour Shortages—A Roadmap for Action*. Published January 2026. Accessed May 9, 2026. https://www.hotrec.eu/media/static/files/import/all_news_2026_2026_5/hotrec-hospitality-europe_skills-and-labour-shortages_a-roadmap-for-action_january-2026.pdf

2.2. Target Market Analysis

2.2.1. Buyer Characteristics

1. Restaurants and Hotel Breakfast Operations

This buyer includes midscale and upscale hotels, breakfast restaurants, buffet operators, and café concepts serving eggs as a core breakfast item. The operational pattern is concentrated demand over a short breakfast period, where delays and inconsistent cooking directly affect guest satisfaction and queue times.

Daily egg demand ranges from roughly 80 to 300 eggs for independent restaurants and smaller hotel breakfast outlets, and from 300 to 800 eggs for larger buffet-oriented properties, based on the machine throughout available from vendors and the breakfast-service examples seen in hospitality and institutional sources. Budget tolerance is typically in the low-to-mid commercial equipment band, with buyers favoring a single-purpose unit if labor savings, output speed, and footprint clearly justify the investment.

Pain points include difficulty staffing breakfast shifts, inconsistent cooking quality between team members, greasy and repetitive manual work, and service delays during morning peaks. These buyers prefer one-button operation, easy cleaning, compact size, configurable doneness, and reliable temperature control because breakfast service leaves little room for training-heavy workflows.[5]

2. Fast-Food Chains and Quick-Service Operators

This segment includes chain breakfast concepts, convenience-led food service, travel hubs, and quick-service brands that need standardized output across locations. Their buying logic is driven by speed, repeatability, labor substitution, and ease of rollout across multiple units.

Estimated daily egg volume is often 300 to 1200 eggs per site for breakfast-focused or all-day breakfast formats, especially where eggs appear in sandwiches, combo meals, or grab-and-go offers. These operators are typically more willing than independents to pay for automation when it reduces training time, improves speed of service, and protects product consistency across stores.

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[5] Arka & Co. *Transforming the European Restaurant Sector: Key Trends Shaping the Industry in 2025*. Published March 8, 2025. Accessed May 9, 2026. https://arkapandco.com/insights/industries/retail/transforming-the-european-restaurant-sector-key-trends-shaping-the-industry-in-2025/

The main pain points are staff turnover, inconsistent execution across shifts, and the need to simplify back-of-house processes for semi-skilled crews. Preferred features include programmable settings, precise and repeatable heat control, throughput at or near 120 eggs per hour, easy integration into compact kitchen lines, and lower energy consumption because these sites are highly sensitive to unit economics.

3. Catering Operators and Contract Foodservice

This group includes event caterers, corporate dining providers, airport and rail caterers, and outsourced foodservice operators handling breakfast on a larger scale. Their demand pattern is less continuous than that of chain restaurants, but they value output predictability and labor flexibility because staffing varies by contract, venue, and event schedule.

A reasonable daily egg demand range is 200 to 1000 eggs depending on venue size, breakfast program design, and whether eggs are plated to order or served as buffet items. Budget decisions are usually tied to total cost of ownership rather than sticker price alone, so buyers pay more attention to reliability, cleaning time, portability within kitchen layouts, and energy draw.

The largest pain points are labor scheduling, service-time spikes, food waste from batch overproduction, and the need to produce consistent eggs in temporary or mixed-skill environments.[6] Decision criteria often emphasize compact dimensions, simple controls, flexibility of output, and dependable service support because downtime can disrupt contracted meal delivery windows.

[6] Health Care Without Harm Europe. *Fresh, Healthy and Sustainable Food: Best Practices in European Healthcare*. Published December 2016. Accessed May 9, 2026. https://europe.noharm.org/sites/default/files/documents-files/4680/HCWHEurope_Food_Report_Dec2016.pdf

Buyer Segment	Typical Daily Egg Volume	Budget Mindset	Main Pain Points	Preferred Features	Key Decision Criteria
Restaurants and hotel breakfast operations	80-800 eggs/day	Moderate CapEx, Clear payback required.	Breakfast rush, staffing gaps, inconsistent doneness, slow service.	Compact size, one-button use, multiple doneness settings, easy cleaning.	ROI, footprint, reliability, ease of use.
Fast-food chains and QSR	300-1,200 eggs/day	Will pay more for standardization across sites.	High turnover, training burden, speed-of-service pressure	Programmable presets, repeatable heat control, 120 eggs/hour throughput, low energy use.	Multi-site consistency, throughput, Labor substitution, Service support.
Catering and contract food service	200-1,000 eggs/day	Total cost of ownership focus.	Variable staffing, event spikes, waste from overbatching.	Flexible output, compact form, easy cleaning, portable integration into workflow.	Reliability, cleaning time, Adaptability, Maintenance support.

Table 2:Buyer segments and key purchasing criteria for commercial egg-processing equipment.

2.3. 4P Marketing Mix

Product

SunnyPro's product will be positioned as a professional breakfast automation machine, not a consumer appliance. Based on the project brief, the machine must fry raw eggs automatically, produce both individual eggs and batches, and let users choose 1 to 4 eggs on demand.

The machine must also meet several performance targets:

- 4 eggs ready in 30 seconds.
- Total egg order completed within 5 minutes.
- Continuous output of at least 120 fried eggs per hour.
- Storage for at least 60 raw eggs at once.
- Refilling during operation must be possible.
- Compact depth of no more than 600 mm.
- 230V or 400V power supply compatibility.

For buyers, these specs matter because they directly compare against manual breakfast service. For example, 120 eggs per hour means 2 eggs per minute on average, while the 4-egg on-demand mode delivers a small guest order in 30 seconds. That is much faster and more consistent than a manual station staffed by one cook during breakfast rush.

SunnyPro also emphasizes hygiene and safety. The machine must collect and store eggshells safely, avoid contamination of other food, and comply with professional hygiene standards such as Regulation (EC) No. 852/2004. In commercial kitchens, this is not just a technical detail; it is a buying criterion.

Price

The target selling price is around 3000€. That places SunnyPro in the premium-mid commercial equipment segment rather than the low-cost import segment.

This price makes sense when compared with our estimated components costs of approximately 1490€, which are presented later in the document under 3. Overall Product Concept. This figure only represents the direct costs of the components. On top of this, additional costs such as assembly, overhead, freight, service, sales commissions and other indirect expenses must also be included.

Taking these additional costs into account, we expect the total production cost to be around 2400€. At our target selling price of 3000€, this would leave an estimated gross margin of approximately 600€ per unit, or around 20%, depending on the final production cost.

The price is explained through ROI rather than hardware cost. If a hotel breakfast station saves even a few staff-hours per day, the machine can justify itself over time. That is especially relevant in Germany and Europe, where hospitality staffing gaps remain a real issue.

A useful way to frame the price:

- At 3000€, SunnyPro is about 2 to 3 times more expensive than a basic commercial countertop food appliance, but it is also far more specialized.
- For chain buyers, one machine may be cheaper than recurring labor spent on manual egg service.
- For hotels, the machine supports guest satisfaction during peak breakfast periods without adding a full-time breakfast cook.

Place

Germany is the launch market, then SunnyPro will expand into nearby European countries. Germany is a strong base because it has high egg consumption, a dense hospitality sector, and well-developed distributor channels.

The strongest buyer channels are:

- Direct sales to large hotel groups and institutional buyers.
- Distributors for independent restaurants and cafés.
- Kitchen equipment integrators for caterers and ghost kitchens.

Hotel-chain data make the German opportunity especially clear. Accor has 362 hotels and 50 147 rooms in Germany, BWH Hotel Group has 184 hotels and 20 302 rooms, Marriott has 78 hotels and 18 215 rooms, and IHG has 88 hotels and 17 382 rooms. That means a single chain contract could potentially involve dozens or even hundreds of units across a portfolio.[7]

B&B Hotels also has a strong German presence with 151 hotels in the German ranking data. GCH Hotel Group manages around 120 hotels in Europe, which makes it a useful target for multi-site rollout. These numbers matter because large hotel groups are much more attractive than independent buyers for a single standardized machine.

For Europe, SunnyPro should first target markets with similar breakfast culture and high hotel density:

- Germany.

[7] Horwath HTL. *Germany Hotel Chains Report 2022*. Published September 2024. Accessed May 9, 2026. https://horwathhtl.com/wp-content/uploads/2024/09/Germany_HOTELS-CHAINS_2022.pdf

- Netherlands
- Belgium
- Austria and Switzerland
- France.
- The UK.

Promotion

SunnyPro will be marketed around one central promise: **Breakfast Without Bottlenecks**.The machine solves the problem of making fried eggs consistently when labor is short and service needs to stay fast.

The strongest promotion points are numerical:

- 1 to 4 eggs per guest order.
- 30 seconds for a 4-egg order.
- 120 eggs per hour continuous production.
- 60-egg raw capacity.
- 600 mm max depth for space-saving installation.

These numbers are useful because hotel and restaurant buyers think in throughput and kitchen space, not in product features alone. For example, if a breakfast outlet serves 100 guests and 25% order eggs, SunnyPro can handle that demand comfortably within a short service window.

Promotion should also use market data:

- Germany: 252 eggs per person in 2025 vs 236 in 2023.[8]
- EU: projected to reach 14 kg per person by 2035.[9]
- EU market value for eggs continues growing, with Germany among the largest consumers.

The best promotional channels are:

- Trade fairs and hospitality exhibitions.
- Hotel and restaurant demo kitchens.
- Direct outreach to chain operators.
- ROI calculators for procurement teams.
- Distributor brochures and sales training.

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[8] Statista. *Per capita consumption of eggs in Germany from 2006 to 2023*. Accessed May 9, 2026. https://www.statista.com/statistics/533508/eggs-per-capita-consumption-germany/

[9] Zootecnica International. *EU Poultry & Egg Outlook 2025–2035*. Accessed May 9, 2026. https://zootecnicainternational.com/featured/eu-poultry-egg-outlook-2025-2035/

2.4. Market Strategy

2.4.1. Distribution Strategy

As technology continues to evolve and demand for automated, user-friendly kitchen solutions grows, we expect that others will introduce similar concepts in the future. Nevertheless, we are confident that Sunny-Pro will establish a strong position in the market, as it is designed to be the first machine to specifically fry sunny-side-up eggs. With this we are striving to be a key competitor as the market grows.

We are focusing on establishing a presence with SunnyPro in Germany at first, with the intention to expand to nearby countries such as the Netherlands, France, Luxembourg, UK, Belgium and Switzerland.

We want to reach our customers through:

- Online-Retail Partnerships: Collaborating with known websites for gastronomical purposes like Gastro Hero[10], Gastprodo[11], and XXLgastro
- E-commerce platforms: such as Amazon, eBay and others.
- Digital Marketing: Google Ads, Social media sites, own Website
- Trade Shows: Showcasing our product in Internorga[12] (one of Europe's biggest hospitality and foodservice trade fairs), which attracts around 85 000 visitors of the gastronomy branch. Other trade shows include Intergastra and GastroTage West.

[10] GastroHero GmbH. *Über Uns*. Accessed May 9, 2026. https://www.gastro-hero.de/%C3%9Cber-uns/
[11] Gastprodo GmbH. *Gastprodo*. Accessed May 9, 2026. https://www.gastprodo.de/
[12] Hamburg Messe und Congress GmbH. *About INTERNORGA*. Accessed May 9, 2026. https://www.internorga.com/messe/about-internorga

2.4.2. Customer Care and Support

- Instruction and Video tutorials: Will provide clear instructions e.g. how to clean the machine and where to pour in the oil.
- Customer Support: Will enable quick and efficient online customer service through our website to manage inquiries and technical issues.
- Warranty and Maintenance: Which includes a 3-year warranty, with repair or replacement services if a defect or malfunction happens.
- Sustainability initiatives: Implement a plan for recycling components to reduce environmental impact and our company will encourage the reuse of eggshell waste as a natural fertilizer.[13]

2.4.3. Phased Market Expansion Strategy

Phase 1: Pilot testing in Düsseldorf.[14]

Phase 2: Showcase product at trade shows and focus on Digital Marketing.[15]

Phase 3: Expand in major cities in Germany: Cologne, Bonn, Hamburg, Berlin.

Phase 4: Use Düsseldorf's logistics advantage to expand to nearby countries: Brussels, Amsterdam, Luxembourg, Paris.

Phase 4: Growing our product range and image using customer feedback

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[13] MasterClass. *How to Use Eggshells as Fertilizer*. Accessed May 9, 2026. https://www.masterclass.com/articles/egg-shell-fertilizer-guide

[14] Fuel Cycle. *Pilot Testing: How It Can Improve Research*. Published October 24, 2025. Accessed May 9, 2026. https://fuelcycle.com/blog/pilot-testing-how-it-can-improve-research/

[15] International Dairy Foods Association. *7 Reasons Why Exhibiting at Trade Shows Can Help Grow Your Business*. Published May 21, 2025. Accessed May 9, 2026. https://ice-cream.org/7-reasons-why-exhibiting-at-trade-shows-can-help-grow-your-business/

2.5. Planned Sales and Revenue Forecast

Temporarily we estimate production costs of 2000€ - 2500€ per unit, which allows for a sustainable margin at a selling price of 3500€. The costs and prices may be adjusted throughout the project.

Our sales forecast for the first five years reflects strong initial growth, followed by gradually decreasing growth rates over time. We expect performance to develop as follows:

Year	Units sold*	Price per Unit €)	Growth (%)*	Revenue (€)*
2027	500	3000	-	1500000
2028	900	3000	80	2700000
2029	1440	3000	60	4320000
2030	2016	3000	40	6048000
2031	2419	3000	20	7257000

Table 3:Projected Sales and Revenue for initial 5 Years.

We assume:

- In Year 1: 2027 we will focus on Pilot testing to gain validity and reduce uncertainty about our product. We will focus on Düsseldorf and surrounding big cities e.g. Köln, Bonn, Aachen etc. and have only a limited number of customers, as our product is new.
- Through our effort in trade shows and digital marketing, we expect a higher initial growth rate of 80% due to growing brand awareness in 2028.
- From Year 3-4 we expect to have entered major German cities and to start expanding to neighboring countries, where we are also targeting major cities for example, in Belgium: Brussels, Antwerpen, Bruges and Netherlands: Amsterdam, The Hague, Rotterdam etc. We hope to establish a partnership with Gastro Hero and/or XXLGastro.
- From 2030 on we are anticipating the market to become more saturated. Sales continue to grow through repeat customers, larger clients and from our side we will be working on new and improved product versions.
- We are using a decreasing growth rate as we believe it is more realistic for our product. As our business SunnyPro expands, the market will mature and more competitors will arise. Our growth will gradually decline, but result in a sustainable and long-term growth trajectory, driven by our then already established brand awareness.

* We calculated our units with:

*New Units = Old Units * (1+ Growth Rate)*

* Basic Growth Rate formula:

$$\textit{Growth Rate (\%)} = \left(\frac{((Year\ (N)\ Projected\ Sales) - (Year\ (N-1)\ Projected\ Sales)}{(Year\ (N-1)\ Projected\ Sales)}\right) * 100$$

* Revenue formula:

*Revenue = Units sold * Price*

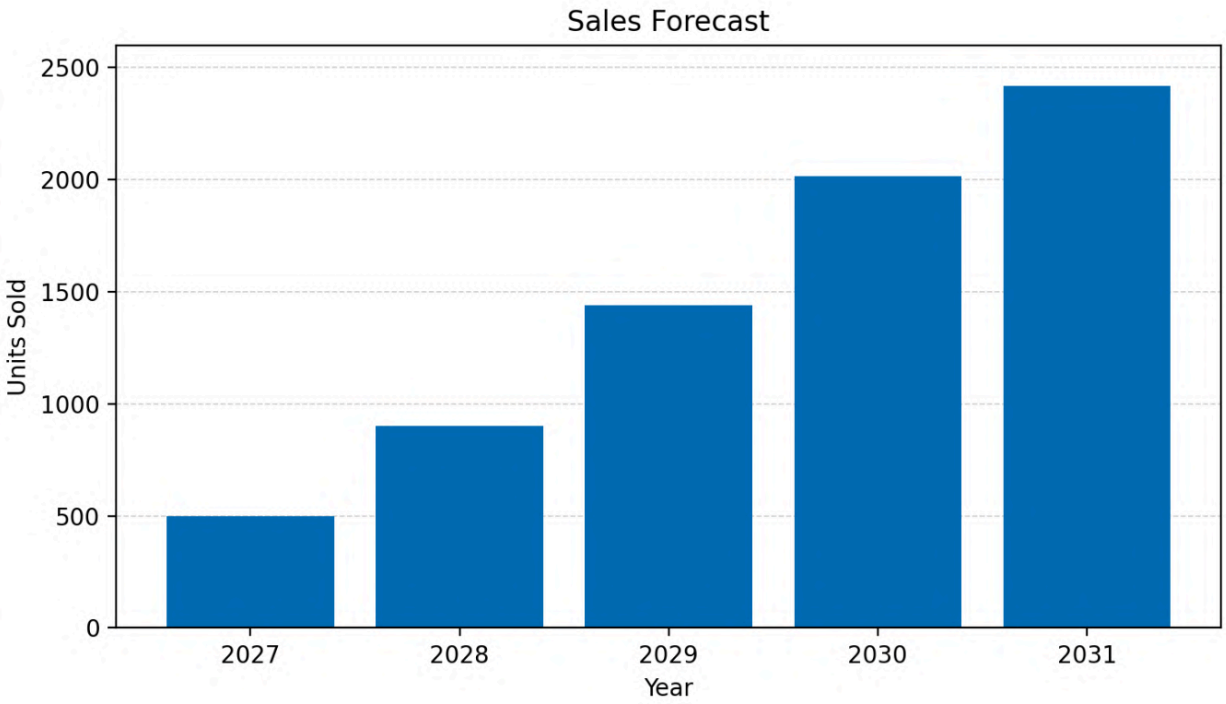


Figure 3:Sales forecast demonstrating units to be sold for each year.

3. Overall Product concept

3.1. List of requirements

- Guests should be able to choose between 1 and a maximum of 4 fried eggs.
- Each egg should be ready within 30 seconds and the total time for the order should be within 5 minutes.
- Continuous production of at least 120 fried eggs per hour must be possible.
- At least 60 raw eggs should be stored at a time.
- Eggs have to be started from scratch and opened and cooked in the machine.
- Sunny side egg and the yolk must not be destroyed , 5% max failure rate.
- User Interface should allow the user to select between 1 to 4 eggs .
- The machine should be refillable during operation.
- Egg shells should be collected and stored safely.
- Design has to be designed according to professional legal hygiene standards (Regulation (EC) No.852/2004 of the European Parliament and the Council on the hygiene of the foodstuffs).[16]
- Maximum depth of the machine on the counter should be 600mm and placeable on a normal counter.
- 230V or 400V power supply possible and operational at all times.
- Aesthetically pleasing design.

[16] Regulation (EC) No 852/2004. Regulation (EC) No 852/2004 of the European Parliament and of the Council of 29 April 2004 on the hygiene of foodstuffs. *Official Journal of the European Union*. 2004;L139:1-54. Accessed May 10, 2026. EUR-Lex

3.2. List of Components

3.2.1. Power Supply System

Electrical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	IEC C20 Power Inlet	Panel-mounted AC inlet used for machine mains power entry	1	8.00
2	16A MCB Circuit Breaker	Protects the system against overload and short-circuit conditions	1	12.00
3	RCCB / RCD (30mA)	Residual current protection device for electrical safety	1	22.00
4	Emergency Stop Switch	Mushroom-type latching switch for emergency shutdown	1	10.00
5	DIN Terminal Blocks (L/N/PE)	AC distribution terminals for live, neutral, and earth wiring	1 set	10.00
6	Protective Earth Terminal Block	Central grounding point for protective earth connections	1	5.00
7	2.5 mm² AC Wire	Heavy-duty AC wiring for PSU and heater connections	5 m	12.00
8	24V Industrial SMPS	Converts AC mains power into regulated 24V DC output	1	32.00
9	24V → 12V Buck Converter	Steps down 24V DC supply to 12V DC	1	8.00
10	12V → 5V Buck Converter	Provides regulated 5V supply for logic and sensors	1	6.00
11	12V → 6V Buck Converter	Dedicated regulated supply for servo operation	1	4.00
12	5V → 3.3V Regulator	Voltage regulator for STM32 logic circuitry	1	2.00

13	DC Terminal Blocks	Distribution terminals for DC voltage rails	1 set	10.00
14	Inline Blade Fuse Holders	Provides replaceable fuse protection for DC circuits	4	6.00
15	Blade Fuse Assortment	Assorted DC protection fuses from 1A–10A	1 set	5.00
16	Shielded Signal Cable	Shielded cable for low-noise sensor and communication signals	5 m	10.00
17	Ferrules and Crimp Terminals	Secure wire termination accessories for electrical connections	1 set	8.00
18	PCB Screw Terminal Connectors	Screw-based connectors for PCB wiring interfaces	1 set	6.00
19	Electrical Cabinet Cooling Fan	Ventilation fan for cooling electrical enclosure components	1	8.00
20	Main Microcontroller STM	Central control MCU for coordinating sensors, motor drivers, solenoid drivers, heating-control signals, touchscreen communication, and system logic.	1	15.00

Table 4.1: Power Supply System Electrical Components List.

Mechanical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	DIN Rail	Standard mounting rail for electrical and control components	1	6.00
2	Cable Glands / Strain Reliefs	Mechanical cable entry protection and strain relief hardware	4	6.00
3	Cable Duct / Wire Raceway	Enclosure ducting used for organized cable routing	2 m	12.00

4	Cable Ties + Mounts	Cable fastening accessories for harness management	1 set	4.00
5	~~Fan Guard / Dust Filter~~	Protective cover and filter for enclosure cooling fan	1	3.00

Table 4.2: Power Supply System Mechanical Components List.

3.2.2. Storage Unit and Breaking Mechanism System

Electrical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	Gate Actuators	Solenoid actuators used to control egg flow gates	2	10.00
2	Flow Sensors	IR/optical sensors for detecting individual egg movement	3	4.00
3	Flip Actuator	MG996R metal gear servo motor for controlled flipping motion	1	3.50
4	Platform Solenoid	12V linear solenoid used to drive spears into the egg	1	10.00
5	Wedge Solenoid	12V linear solenoid used to actuate wedge opening mechanism	1	6.00
6	Axial Fan	80mm 12V DC fan for enclosure ventilation and airflow	2	12.00
7	Solenoid Driver	4-channel MOSFET driver board for controlling solenoids	1	4.00
8	Wire Harness	Sleeved mixed-gauge wiring assembly for power and signals	1	10.90

Table 5.1: Storage Unit and Breaking Mechanism System Electrical Components List.

Mechanical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	Egg Storage Unit	Polypropylene rectangular injection-molded storage component for holding eggs	1	4.26
2	Egg Transportation Unit	Polypropylene rectangular channel guiding eggs through the mechanism	1	3.19
3	Axle Shaft	Aluminum 6061 rod used as rotational support shaft	1	0.18
4	Egg Holder + Disc	Die-cast and CNC-finished aluminum component for egg support and rotation	1	13.75
5	Spear	Stainless steel spear assembly for egg puncturing mechanism	2	16.56
6	Return Spring	Stainless steel compression spring for restoring mechanism position	2	0.30
7	Outer Shell (ABS)	Thermoformed ABS enclosure panels forming machine housing	1	37.69
8	Thermal Insulation	XPS foam insulation layer for thermal retention	1	8.38
9	Inner Lining	Aluminum sheet lining providing internal thermal surface	1	34.51
10	Ventilation Grille	ABS grille protecting ventilation fan openings	2	3.00
11	Vent Duct Channel	Internal polypropylene duct directing airflow across electronics	2	3.00
12	Fasteners Kit	Stainless steel fasteners, standoffs, and inserts for assembly	1	9.38

Table 5.2: Storage Unit and Breaking Mechanism System Mechanical Components List.

3.2.3. Oil System

Electrical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	Peristaltic Pump	Electrically driven pump used to transfer oil at controlled flow rate	1	3.00
2	MOSFET Driver Module	Electronic driver module used to switch and control pump operation	1	2.50

Table 6.1: Oil System Electrical Components List.

Mechanical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	Oil Tank	Stainless steel reservoir used for storing cooking oil	1	12.00
2	Inline Check Valve	Prevents oil backflow and dripping when the pump is off	1	1.00
3	Silicone Tubing	Food-grade flexible tubing for oil transfer between components	1 m	2.00
4	Flat Fan Spray Nozzle	Spray nozzle used for uniform oil distribution across conveyor width	1	2.50
5	Hose Adapter	Mechanical fitting connecting tubing to spray nozzle inlet	1	1.50
6	Nozzle Mounting Bracket	Adjustable bracket used to position spray nozzle above conveyor	1	4.00
7	Pump Vibration Mounts	Rubber isolators used to reduce vibration from pump operation	4 pcs	2.00

Table 6.2: Oil System Mechanical Components List.

3.2.4. Screen System

Electrical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	Microcontroller	Main processing unit for display control and touch interface management	1	18.00
2	TFT LCD Display	Graphical display module for user interface and system monitoring	1	32.00
3	Resistive Touch Controller	Interface IC used to detect touch input from the display panel	1	5.00
4	LED Backlight Driver	Controls power delivery to the display backlight	1	3.50
5	Power Supply	Converts AC mains power into regulated DC output	1	15.00
6	FPC Connector	Connector interface for attaching display ribbon cable to PCB	1	2.50
7	FFC Cable	Flexible cable used for signal transmission between modules	1	3.00
8	Sense Resistor	Sets current level for the backlight driver circuit	1	0.40
9	Pull-up Resistors	Stabilize communication and logic signal lines	2	0.20
10	Decoupling Capacitors	Reduce electrical noise and stabilize supply voltage	3	0.30
11	Bulk Capacitor	Maintains stable voltage during transient load conditions	1	0.60
12	Input Capacitor	Smooths voltage input to the driver circuit	1	0.50

Electrical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	Microcontroller	Main processing unit for display control and touch interface management	1	18.00
2	TFT LCD Display	Graphical display module for user interface and system monitoring	1	32.00
3	Resistive Touch Controller	Interface IC used to detect touch input from the display panel	1	5.00
4	LED Backlight Driver	Controls power delivery to the display backlight	1	3.50
5	Power Supply	Converts AC mains power into regulated DC output	1	15.00
6	FPC Connector	Connector interface for attaching display ribbon cable to PCB	1	2.50
13	Output Capacitor	Stabilizes output from the driver circuit	1	0.40
14	Fuse	Protects the circuit from overcurrent conditions	1	0.70
15	Terminal Block	Provides secure electrical connection for power wiring	1	0.80

Table 7.1: Screen System Electrical Components List.

Mechanical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**

1	Mounting Hardware	Mechanical fasteners used for securing electronic components inside enclosure	1 set	2.00

Table 7.2: Screen System Mechanical Components List.

3.2.5. Transportation System

Electrical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	NEMA 17 Stepper Motor	Drives the conveyor shaft for controlled conveyor movement	1	11.20
2	Stepper Motor Driver	Controls stepper motor speed and motion operation	1	15.00
3	Limit / Home Sensor	Detects conveyor reference and home position	1	3.75
4	Control Wiring and Connectors	Wiring and connection accessories for motor and sensor integration	1 set	7.50

Table 8.1: Transportation System Electrical Components List.

Mechanical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	Roller Chain	Conveyor chain used for continuous plate movement	2 lengths	22.50
2	Drive Sprockets	Transfers motor torque to the conveyor chain	2	9.00

3	Idler Sprockets	Supports and guides the conveyor chain path	2	9.00
4	Drive Shaft	Main rotating shaft supporting conveyor drive sprockets	1	6.00
5	Idler Shaft	Free rotating shaft supporting idler sprockets	1	6.00
6	Support Shafts	Structural shafts used to maintain rolling chain alignment	8	48.00
7	Bearing Blocks	Supports and stabilizes rotating shafts	4	18.00
8	Motor Coupling	Connects the motor shaft to the conveyor drive shaft	1	4.50
9	Chain Tensioning Parts	Mechanical hardware used to adjust conveyor chain tension	2	7.50
10	Side Rails / Guides	Guides conveyor movement and maintains alignment	2	11.25
11	Plate Mounting Pins	Pins used for removable plate attachment to conveyor	~20	9.00
12	Side Support Plates	Structural plates supporting shafts, bearings, and guides	2	22.50
13	Motor Mounting Bracket	Holds and positions the stepper motor on the frame	1	6.00
14	Chain Guards / Covers	Protective covers for conveyor chains and sprockets	2	15.00
15	Fasteners and Spacers	Assembly hardware including screws, nuts, and spacers	1 set	7.50
16	Access Hardware	Hardware components used for maintenance and service access	1 set	7.50

Table 8.2: Transportation System Mechanical Components List.

3.2.6. Containment Subsystem

Mechanical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	Removable Egg Pocket Plates	Food-safe pocket plates used for holding and transporting eggs	4	36.00
2	Non-stick Surface	Food-safe coating or liner used to reduce sticking during cooking	4	22.00
3	Plate Mounting Tabs	Brackets used for attaching pocket plates to conveyor pins	8	8.00
4	Quick-release Pins	Removable pins allowing fast disassembly for cleaning	~20	9.00
5	Side Access Panel	Removable panel providing access for maintenance and cleaning	1	15.00
6	Hinges and Latches	Hardware used for securing and operating access panels	1 set	10.00
7	Food-safe Pin Fasteners	Stainless assembly hardware for food-contact areas	1 set	7.50
8	Food-grade Seals / Gap Strips	Sealing components used to reduce food residue accumulation	1 set	8.00
9	Cleaning / Drip Catch Part	Removable tray or feature used for collecting oil and residue	1	12.00

Table 9.1: Containment System Mechanical Components List.

3.2.7. Egg-Shell Storage System

Mechanical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	Main Waste Drawer Tray	Removable stainless steel drawer used for shell waste collection	1	14.00
2	Internal Inclined Floor Plate	Sloped internal plate directing waste toward collection area	1	5.00
3	Non-disposable Inner Liner	Food-grade liner used to protect the drawer interior and simplify cleaning	1	5.00
4	Liner Retaining Strips / Clips	Retains removable liner securely inside the drawer assembly	1 set	3.00
5	Side Inlet Chute / Inlet Box	Guides eggshell fragments into the waste drawer	1	6.00
6	Inclined Pusher Plate	Compresses and moves waste material along the inclined surface	1	8.00
7	Pusher Plate Stiffening Rib	Reinforcement feature reducing pusher plate deformation	1	3.00
8	Guide Rods	Provides guided linear movement for the pusher assembly	2	6.00
9	Linear Ball Bearings / Bushings	Supports smooth sliding motion of guide rods	2	8.00
10	Return Springs	Returns pusher mechanism to default position after operation	2	3.00
11	Rod End Collars / Retaining Stops	Prevents guide rods from over-travelling during operation	4	4.00
12	Bearing / Rod Support Brackets	Supports and aligns guide rods and bearings within assembly	2	5.00

13	Manual Pusher Handle	Handle used for manual operation of the waste compression mechanism	1	5.00
14	Drawer Slide Rails	Sliding rails enabling drawer movement and removal	2	8.00
15	Drawer Front Pull Handle	Handle used for removing and emptying the waste drawer	1	4.00
16	Drawer Latch / Stop	Keeps waste drawer securely closed during operation	1	4.00
17	Front Sealing Strip / Gap Cover	Reduces leakage of shell dust, residue, and splashing	1 set	4.00
18	Stainless Fasteners Kit	Stainless steel assembly hardware for drawer and support components	1 set	5.50

Table 10.1: Egg-Shell Storage System Mechanical Components List.

3.2.8. Bottom Heating Subsystem

Electrical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	Cartridge Heaters	Electrical heating elements used for heating the cooking surface	8	60.00
2	Temperature Sensor	Measures heating block temperature for control and monitoring	1–2	12.00
3	Temperature Controller	Regulates heating temperature using feedback control	1	18.00
4	Solid-state Relay	Switches electrical power to the heaters electronically	1	12.00
5	Thermal Fuse / Safety Cutoff	Provides over-temperature protection for the heating system	1–2	5.00

6	High-temperature Wiring	Heat-resistant wiring used for heater power connections	1 set	12.00
7	Grounding Hardware	Provides protective earth bonding for electrical safety	1 set	4.00

Table 11.1: Bottom Heating Subsystem Electrical Components List.

Mechanical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	Aluminum Heating Block	Main heated block used for cooking surface heat distribution	1	75.00
2	Thermal Insulation Layer	Insulation layer reducing heat loss below the heating block	1	15.00
3	Mounting Brackets	Supports and secures heating block within machine frame	1 set	9.00
4	Thermal Compound	Improves thermal contact between mating surfaces	1	3.00
5	Lower Heat Shield	Reduces heat transfer to lower structural and electrical sections	1	8.00
6	Fasteners and Spacers	Assembly hardware for heating system installation	1 set	6.00

Table 11.2: Bottom Heating Subsystem Mechanical Components List.

3.2.9. Top IR Heating Subsystem

Electrical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	IR Heater Element	Infrared heating element used for top-side cooking and browning	1	30.00

2	IR Power Control / Relay	Controls electrical power supplied to the IR heater	1	10.00
3	High-temperature Wiring	Heat-resistant wiring used for IR heater connections	1 set	8.00
4	Thermal Cutoff / Safety Sensor	Provides temperature monitoring and overheat protection	1	5.00
5	Small Ventilation Fan	Provides airflow and cooling around the IR heating assembly	1	—

Table 12.1: Top IR Heating Subsystem Electrical Components List.

Mechanical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	Reflector Shield	Reflects infrared heat toward the conveyor cooking area	1	18.00
2	Ventilated Guard Cover	Protective cover allowing ventilation around the IR heater	1	18.00
3	IR Support Brackets	Supports and positions the upper IR heating assembly	1 set	12.00
4	Fixed / Adjustable Mount	Maintains heater positioning above the conveyor system	1 set	8.00
5	User Protection Guard	Additional protective barrier around hot upper heating zone	1	10.00
6	Fasteners and Spacers	Assembly hardware for heater and guard installation	1 set	5.00
7	Cable Strain Relief / Grommets	Protects heater wiring at enclosure entry points	1 set	4.00

Table 12.2: Top IR Heating Subsystem Mechanical Components List.

3.2.10. Rotating Plate System

Electrical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	Stepper Motor	Drives rotational movement of the serving plate	1	1.25
2	Stepper Driver Module	Controls stepping operation of the motor	1	1.25
3	Microswitch	Detects rotational home/reference position	1	3.00
4	Load Cell	Detects plate presence through weight sensing	1	2.00
5	HX711 Amplifier Module	Amplifies load cell output signal for controller processing	1	1.00
6	Wiring	Electrical wiring and connectors for sensors and motor integration	1 set	5.00

Table 13.1: Rotating Plate System Electrical Components List.

Mechanical Components				
S.No	**Component**	**Description**	**Quantity**	**Approx. Cost (EUR)**
1	Stainless Steel Plate	Rotating surface used for customer dish placement	1	20.00
2	Centering Ring	Keeps serving dish aligned at the center of the plate	1	8.00
3	Lazy Susan Bearing	Provides smooth rotational support for the plate assembly	1	12.00

4	Base Plate	Structural base supporting the rotating assembly	1	10.00
5	Countersunk Screws	Fasteners used for securing plate and bearing components	16	8.00
6	Washers	Distributes fastening load and prevents loosening	16	3.00
7	Rubber Feet	Provides vibration damping and assembly stability	4	5.00
8	Rubber Drive Wheel	Transfers motor motion to the rotating plate through friction	1	2.00
9	Shaft Adapter	Couples motor shaft to the drive wheel	1	3.00
10	Motor Mount Bracket	Supports and aligns the motor assembly	1	5.00
11	Fasteners	Hardware used for securing motor and bracket components	1 set	4.00
12	Trigger Tab	Activates microswitch at the home position	1	1.00
13	Microswitch Mount Bracket	Holds microswitch securely in position	1	3.00
14	Load Cell Mounting Plate (Top)	Transfers load from rotating assembly to the load cell	1	3.00
15	Load Cell Mounting Plate (Bottom)	Secures load cell to the structural base	1	3.00
16	Spacer Standoffs	Provides spacing and support for accurate load sensing	4	4.00

Table 13.2: Rotating Plate System Mechanical Components List.

3.3. Morphological Box

Definition - A morphological box in a Bill of Materials (BOM) context is a systematic matrix used to identify, organize, and evaluate different component alternatives, design features, and configuration options for a product. It helps compare combinations of parts and functions during concept development and product design.

Most Suitable Option

System	Parameter	Possible variations		
		Option 1	Option 2	Option 3
Storage Unit and Breaking Mechanism System	Egg storage unit	Tilted Rectangular Storage	Vertical cartridge stack	Gravity-fed tray
	Egg transportation	Roller conveyor	Inclined gravity chute	Rectangular PP channel
	Gate actuation	Solenoid actuator	Servo gate actuator	Mini linear actuator
	Egg detection senor	Capacitive Sensor	IR optical sensor	Ultrasonic sensor
	Axle shaft material	Al 6061 rod	SS304 shaft	Carbon steel shaft
	Egg holder mechanism	Clamp holder	Integrated holder + disc	Rotary cup holder
	Flip actuation	Servo motor	Stepper motor	Pneumatic actuator
	Spear mechanism	Single piercing pin	Clamp puncture system	Dual SS304 spears
	Spring return	Compression spring	Torsion spring	Extension spring
	Platform actuation	Linear solenoid	Servo-driven slider	Cam mechanism

	Wedge opening mechanism	Servo wedge	Linear solenoid wedge	Spring-loaded wedge
	Outer shell material	Polycarbonate shell	Thermoformed ABS	Aluminum enclosure
	Thermal insulation	XPS foam	PU foam	Mineral wool
	Inner lining	Al 1050 sheet	SS304 sheet	HDPE lining
	Cooling airflow system	Axial fan	Blower fan	Passive ventilation
	Ventilation outlet	ABS grille	Metal mesh grille	Slotted vent panel
	Airflow ducting	Flexible duct	PP duct channel	Sheet metal duct
	Power supply	230V AC → 12V DC PSU	External adapter	Battery supply
	Solenoid Control	Relay module	Motor driver shield	MOSFET driver board
	Wiring system	Ribbon cable routing	Sleeved wire harness	PCB-integrated traces
	Fastening method	SS bolts + inserts	Rivets	Snap-fit assembly
Oil System	Storing System	Stainless Steel Tank	Food-Grade Plastic Tank (PP)	Flexible oil Pouch/Bladder
	Delivery System	Gravity Feed + Valve	Peristaltic Pump	Diaphragm Pump
	Flow Control	Relay Module	MOSFET Module	Manual Valve
	Drip Prevention	Inline Check Valve	Solenoid Valve	Anti-Drip Nozzle
	Tubing/Transport	Stainless Steel Tube	PTFE Tubing	Silicone Tubing
	Flow Regulation	PWM Control	Fixed Orifice	Needle Valve
	Spray Generation	Full Cone Nozzle	Flat Fan Nozzle	Drip Nozzle
	Nozzle Mounting	Fixed Bracket	Magnetic Mount	Adjustable Bracket

	Pump Mounting	Rubber Vibration Mount	Rigid Bracket	Foam Isolation Pad
Screen System	Display Type	TFT LCD	HDMI Display Module	SPI Display Module
	Touch Technology	Resistive Touch	Capactive Touch	No Touch (Buttons Only)
	Touch Interface	TSC2046 (SPI)	Capacitive Controller (I^2C)	Direct GPIO
	Processing Unit	STM32 (LTDC)	Raspberry Pi	ESP32-S3
	User Input Method	Touchscreen	Buttons+Encoder	Hybrid (Touch+Buttons)
	Display Interface	HDMI	Parallel RGB (TTL)	SPI
	Backlight Control	Fixed LED Driver	PWM+Transistor	PT4115 Constant Current
	Power Supply Method	DC Input	AC→DC	Battery Powered
	Voltage Regulation	Linear Regulator	Buck Converter	Integrated Regulator (on STM32 Board)
	Communication	UART	CAN Bus	RS485
	Mounting Method	Embedded Behind Glass	External Module	Panel Mount
	User Feedback	Visual Only	Visual+Buzzer	Visual+LEDs
	Protection Method	Fuse	Fuse + TVS diode	Full Protection (Fuse+EMI filter)
	Cooling Method	Passive Cooling	Heat Sink	Forced Airflow
	Enclosure Type	Plastic Housing	Metal Panel	IP-rated Sealed Enclosure
	Drive actuator	NEMA 23 stepper motor	NEMA 17 stepper motor	DC geared motor

Transportation System	Motor driver	TB6600 stepper driver	DM542 driver	A4988 driver
	Motion transmission	Timing belt & pulleys	Stainless Roller chain & sprockets	Gear train
	Conveyor support	Guide rail track	Stainless Support shafts with side guides	Sliding bed plate
	Shaft support	High-Temp Bearing blocks	Plain bushings	Pillow bearings
	Tensioning method	Slotted idler bracket	Spring tensioner	Turnbuckle tensioner
	Position feedback	Inductive Proximity Sensor	Rotary encoder	Timed open-loop only
	Plate attachment	Bolted plates	Magnetic mount	Pin-mounted removable plates
	Frame structure	Tubular frame	Stainless Side support plates	Sheet-metal frame
Containment Subsystem	Pocket plate style	Fixed egg tray	Removable egg pocket plates	Rotating cup holders Circular shape
	Pocket plate material	Stainless Steel (304/316)	Heat-safe polymer	Aluminum (Non-stick)
	Surface finish	Ceramic (Sol-Gel) coating	PTFE (Teflon) coating	Replaceable PTFE liner
	Mounting to conveyor	Bolted bracket	Plate mounting tabs	Sliding rail mount
	Removal method	Quick-release pins	Screw-fastened	Captive latch
	Service access	Lift-off top cover	Side access panel	Rear service hatch
	Hygiene sealing	Silicone gasket	High-temp gap strips	Open-frame design

	Drip management	Fixed drain channel	Disposable liner tray	Removable drip catch
	Food-safe fasteners	Plastic snap clips	SS food-safe fasteners	Permanent welds
Egg-Shell Storage System	Main container structure	SS304 rectangular drawer	Modular Cartridge bin	Folded sheet cassette
	Internal floor geometry	Single inclined plate	Double-slope floor	Curved trough floor
	Liner Material	TPU Liner	Silicone Liner	No Liner
	Liner Attachment	Magnetic strips	Snap-fit rails	Retaining Clips
	Waste inlet design	Side chute	Top funnel inlet	Hinged flap inlet
	Compression mechanism	Lever-actuated press	Screw-driven compactor	Manual sliding pusher
	Pusher reinforcement	Embossed ribs	Folded edge rib	Welded flat bar
	Motion guidance	Dual guide rods	Telescopic rails	Polymer slide tracks
	Bearing type	Ball bushings	Bronze bushings	PTFE slide blocks
	Return system	Extension springs	Compression springs	Gravity return
	Handle type	U-handle	Knob grip	T-bar handle
	Drawer movement	Stainless drawer slides	Roller tracks	Direct pull tray
	Drawer retention	Spring latch	Magnetic catch	Friction lock
	Shell Dust/Odor sealing	Brush seal	Silicone strip	Gasket seal
	Emptying method	Tilting drawer	Bottom hatch	Full drawer removal
	Cleaning strategy	Washable liner only	Fully removable internals	Tool-less disassembly
	Structural joining	Screws and nuts	Rivets	TIG welding

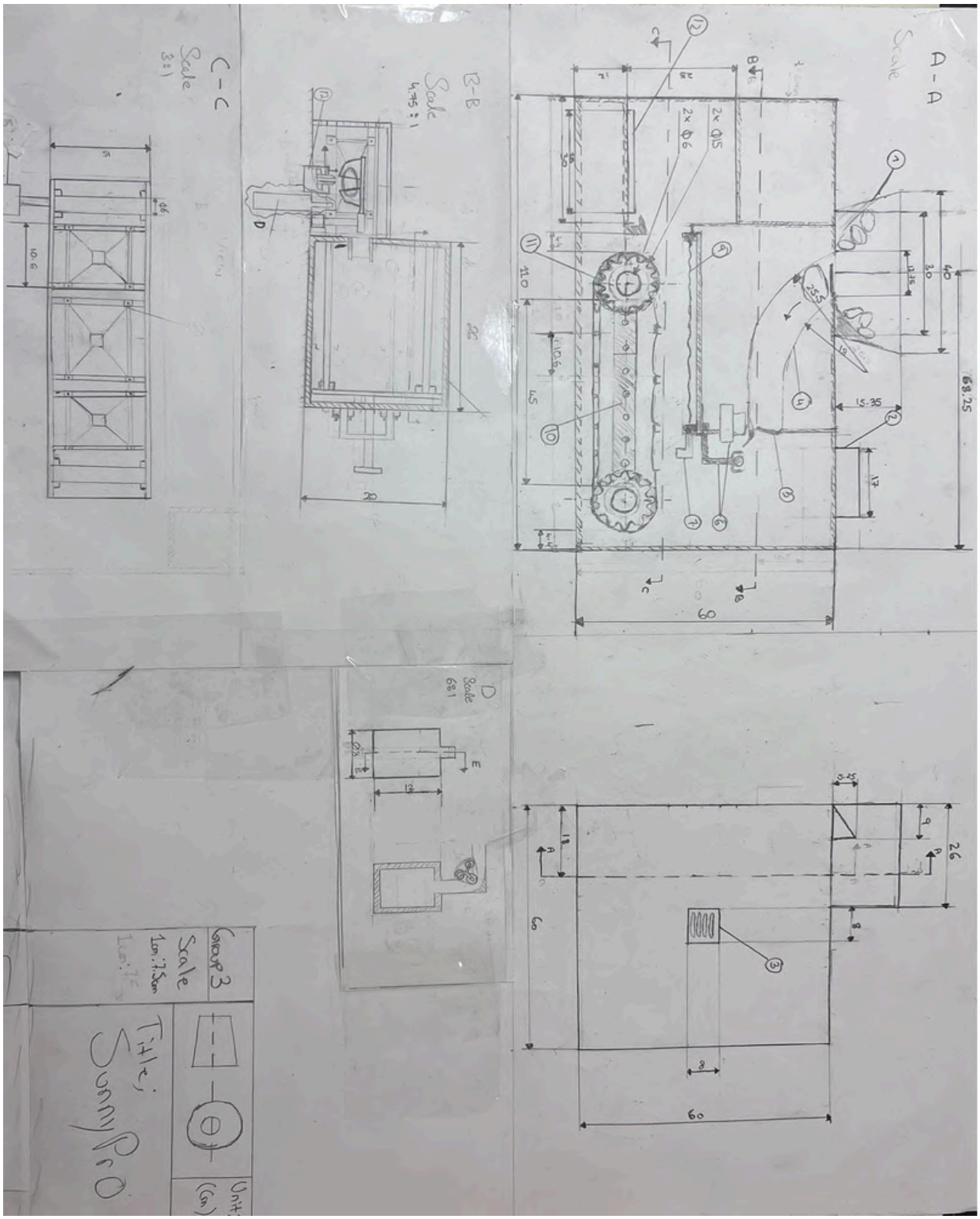
	Surface Finish	Polished finish	Brushed SS304	Bead blasted
	Sensor integration	None (MCU code dependent)	Fill-level sensor	Weight sensor
	User force method	Mechanical leverage	Ratchet compression	Direct push
	Mounting configuration	Cabinet integrated	Under-Counter	Freestanding
Bottom Heating Subsystem	Heating method	Tubular heater	Cartridge heaters	Silicone heater mat
	Temperature sensor	PT100 sensor	K-type thermocouple	NTC thermistor
	Temperature control	PID (Discrete Controller)	On/off thermostat	Microcontroller PID
	Power switching	Solid-state relay (SSR)	Mechanical relay	Triac dimmer
	Safety cut-off	Bimetal thermostat	Thermal fuse (Hard-wired)	Software-only
	Heating block material	Aluminum (Hard-anodized)	Stainless steel plate	Cast iron plate
	Insulation method	Air gap	Ceramic fiber / Mineral wool	Reflective foil
	Mounting method	SS Mounting brackets	Direct frame mount	Suspended mount
	Thermal interface	Dry contact	Mica sheet	High-temp Ceramic paste
	Lower protection	No shield	Stainless Steel heat shield	Cooling fans
	Top heat source	Quartz IR tube	Shielded IR heater	Ceramic IR emitter
	Power control	Mechanical relay	SSR + PID control	Phase-angle dimmer

Top IR Heating Subsystem	Safety sensing	Thermocouple	Thermal fuse /1 -shot cut-off	IR temperature sensor
	Reflector design	Parabolic reflector	Polished SS Reflector	Flat hood
	Protective cover	Ventilated SS guard	Wire mesh guard	Perforated shield
	Ventilation	Passive vents + Fan	Small fan only	No ventilation
	Support structure	Overhead frame	SS IR support brackets	Suspended arm
	Height adjustment	Fixed mount	Telescopic slider	Slotted
	User protection	Heat warning label	Transparent shield	External Cool-Touch Guard
	Cable protection	Loose routing	Cable chain	Fiberglass sleeving
Rotating Plate System	Rotating Surface Material	Stainless Steel Plate	Aluminum Plate	HDPE Food-Grade Plastic
	Plate Diameter	Ø250 mm	Ø280 mm	Ø300 mm
	Centering Method	Adjustable Guide Arms	Fixed Centering Ring	Rubber Alignment Pads
	Rotation Bearing	Deep Groove Bell Bearings	Slewing Ring Bearing	Lazy Susan Bearing
	Structural Base	Aluminum Base Plate	Steel Plate	Plywood Base
	Rotation Drive Method	Direct Shaft Coupling	Friction Wheel Drive	Belt & Pulley
	Motor Type	DC Gear Motor	28BYJ-48 Geared Stepper	NEMA 17 Stepper
	Motor Driver	ULN2003	A4988 Driver	DRV8825
	Motion Transfer Wheel	TPU Printed Wheel	Silicone Roller	Rubber Drive Wheel

	Shaft Coupling Method	Direct Press Fit	3mm-5mm Shaft Adapter	Adhesive Mount
	Motor Mounting	3D Printed Mount	Adjustable Slotted Bracket	Fixed Bracket
	Home Position Detection	Inductive Proximity Sensor	Lever Microswitch	Optical Endstop
	Trigger Mechanism	Metal Trigger Tab	Magnet Trigger	Printed Cam
	Plate Presence Detection	IR Sensor	Capacitive Sensor	Load Cell (5 kg)
	Load Signal Amplification	HX711 Module	INA125 Amplifier	ADS1232 Module
	Load Cell Integration	3-Point Support	4-Cell Corner Mount	Single-Point Mounting
	Vibration Isolation	Rubber Feet	Silicone Dampers	Foam Pads
	Fastening Method	Stainless Screws + Washers	Rivets	Adhesive Bonding
	Electrical Connection	Soldered Wiring	Ribbon Cable	Jumper Wires + Connectors
	System Position Reference	Encoder Feedback	Microswitch Homing	Open-Loop Timing Only

Table 14: Morphological Box demonstrating three options for each component.

3.4 Freehand Sketch



3.5 Bom for Free hand sketch

Bill of materials:

1. Storage unit
2. Touch screen
3. Vent
4. Tunnel
5. Tunnel gate
6. Breaking mechanism
7. Oil spraying mechanism
8. Egg shell storage unit
9. Top IR
10. Bottom Heating
11. Conveyor system
12. Rotating plate

4. Functional Structure

4.1. Objective

The objective of SunnyPro is to automate the preparation of sunny-side-up fried eggs as much as possible, making human interaction minimal for hotel breakfast areas, restaurants, and commercial kitchen environments. The machine stores raw eggs, separates and transfers them one by one, opens the eggs, removes the shells, transports the egg contents through a controlled heating zone, and delivers cooked eggs to the user. The system is designed to reduce manual labour, improve cooking consistency, and maintain hygienic separation between food material and waste material.

4.2. System Boundaries and Environmental Interaction

- System Boundary: The system boundary includes the internal mechanical, electrical, thermal, and control components of the machine. These include the egg storage unit, egg release mechanism, cracking mechanism, eggshell storage drawer, conveyor transport system, bottom heating system, top IR heating system, sensors, actuators, microcontroller, touchscreen interface, and safety components.

- External Environment: The external environment includes the user, the raw eggs added into the machine,the oil added to the machine , the cooked egg output , mains electrical power, ambient air, cleaning actions, and eggshell waste removal by the operator.

- Interactions:

The machine receives:

➢ Raw eggs from the operator
➢ Oil from the operator
➢ Electrical energy from the mains supply
➢ User input from the touchscreen
➢ Lining for Egg shell Storage Unit from Operator
➢ Operator settings input to the touchscreen
➢ Cleaning and maintenance actions from the operator

The machine outputs:

- ➢ Cooked sunny-side-up eggs
- ➢ Collected eggshell waste
- ➢ Heat losses to the surroundings
- ➢ Status information,Performance analytics, warnings, and error messages to the user

4.3. Working Principle (System Functionality Overview)

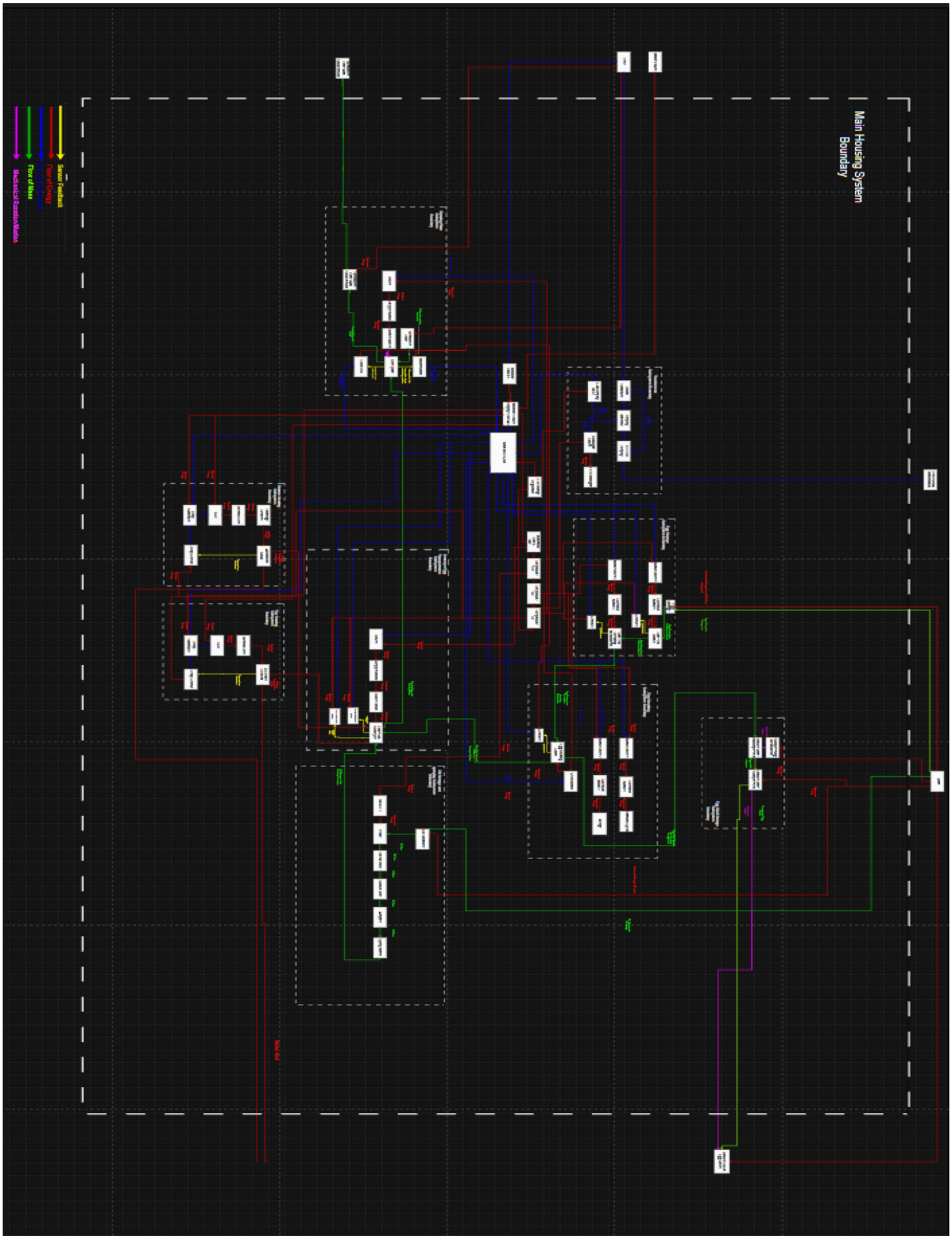
System Functionality Overview

1. **Preparation and Refilling:**
   Before operation, the kitchen staff fills the raw egg storage unit, checks and refills the oil container, inserts the eggshell storage drawer and drawer lining, and confirms that all removable food-contact parts are correctly installed.
2. **Startup:**
   The staff switches on the machine. The control system initializes the touchscreen, sensors, heating systems, conveyor, actuators, and safety components.
3. **Heating Settings / Doneness Selection:**
   The staff selects the required cooking setting through the touchscreen. This may include presets such as soft, medium, or well-cooked, or technical settings such as bottom heater temperature, top IR heater intensity, and conveyor speed. These settings determine how strongly and how long the egg is cooked.
4. **Preheating:**
   The bottom heating block and top IR heating system heat up to the selected operating temperature. Temperature sensors provide feedback to the controller. The machine only starts cooking when the required temperature range is reached.
5. **Customer / User Input:**
   The user selects the required number of eggs, from 1 to 4, through the touchscreen interface.The doneness setting is preset by staff for operation hours so as to not waste time waiting for temperature to go down or up between each user.
6. **Egg Storage and Release:**
   Raw eggs are stored in the egg storage unit. A gate release mechanism allows one egg at a time to move into the cracking position.
7. **Egg Detection:**
   Sensors confirm that an egg has reached the correct position before the cracking mechanism is activated.
8. **Egg Cracking:**
   The cracking mechanism opens the egg and separates the edible egg contents from the shell.
9. **Shell Removal:**
   The eggshells are directed into the eggshell storage drawer by the same breaking mechanism but through a different motor that basically tosses it to the side. The shell waste remains separated from the food path.

10. **Egg Transfer to Conveyor:**
    The egg contents are deposited into a removable egg pocket plate on the conveyor system.
11. **Oil Dosing / Surface Preparation:**
    A small amount of oil is supplied to the egg pocket automatically to reduce sticking and improve cooking quality. The oil container is refilled by the staff when required.
12. **Transport Through Heating Zone:**
    The conveyor moves the egg pocket through the cooking area at a controlled speed. The conveyor speed affects the cooking time.
13. **Bottom Heating:**
    The bottom heating block transfers heat to the underside of the egg pocket plate to cook the egg white from below.
14. **Top IR Heating:**
    The top IR heater applies radiant heat from above to support cooking of the upper egg surface while keeping the sunny-side-up appearance.
15. **Monitoring and Control:**
    Temperature sensors, home position sensors, egg detection sensors, oil level sensors, shell drawer position sensors, and safety cutoffs provide feedback to the control system.
16. **Egg Output:**
    The cooked egg leaves the heating zone and is transferred to the output area for the user or kitchen staff.
17. **Rotating Plate Positioning:**
    The rotating plate positions the cooked egg output correctly after each egg leaves the conveyor.
18. **Waste Removal and Refilling During Operation:**
    During longer operations, the kitchen staff can refill the raw egg storage unit and oil container. The eggshell storage drawer can be removed and emptied when the machine is paused or when the system indicates that the drawer is full.
19. **Shutdown / Standby:**
    The machine enters standby after the order is completed or shuts down if the staff ends service, or a safety fault is detected.

4.4. Functional Structure[17]

[17] Functional Structure Website.
https://www.drawio.com/


Main Housing System Boundary

4.5. Functional Structure Description

4.5.1. User Interaction

The user interacts with SunnyPro through the touchscreen interface. The touchscreen allows the user to select the required number of fried eggs, from 1 to 4.The kitchen staff may also select the cooking level, such as soft, medium, or well-cooked.

The touchscreen sends the selected input as an information signal to the microcontroller. The microcontroller then starts the required sequence for egg release, cracking, cooking, transport, and output positioning. During operation, the touchscreen can also display machine status, warnings, heating readiness, egg storage status, oil level warnings, shell drawer status, and fault messages.

4.5.2. Operator Interaction and Preparation

Before operation, the kitchen staff prepares the machine for use. The staff fills the raw egg storage unit, checks or refills the oil container, inserts the eggshell storage drawer, and confirms that the removable egg pocket plates are correctly installed on the conveyor.

The staff can also adjust heating settings through the touchscreen. These settings affect the final cooking result of the egg. For example, the machine may change the bottom heater temperature, the top IR heater intensity, or the conveyor speed depending on the selected cooking level. A longer heating time or higher heating intensity produces a more cooked egg, while a shorter time or lower intensity produces a softer sunny-side-up result.

During longer operations, the staff can refill the raw egg storage unit and oil container. The eggshell drawer can also be removed and emptied when the system is paused or when the machine indicates that the drawer is full.

During machine initialization, the conveyor system also performs a positioning reset. When the machine is switched on, the microcontroller activates the conveyor motor and moves the conveyor slowly until the first egg pocket plate is detected by the home/reference sensor. This allows the control system to define the starting position of the conveyor before any egg is released. Once the first plate is correctly aligned under the cracking mechanism, the conveyor stops and the system confirms that the machine is ready for egg deposition. This initialization step ensures that the raw egg contents land directly into the pocket plate and prevents splashing, misalignment, or egg loss at the beginning of operation.

The conveyor positioning system uses sensors to ensure that the egg pocket plates are correctly aligned during operation. Since there are always four removable egg pocket plates mounted on the conveyor, the system must know the position of the plates before the egg is released from the cracking mechanism

4.5.3. Egg Storage and Release

Up-to 260 raw eggs can be stored in the egg storage unit before cooking. The storage unit holds the eggs in a controlled position so that the eggs can be released one at a time into the transfer tunnel. The release system uses two gates to separate the eggs and prevent more than one egg from reaching the cracking mechanism at the same time.

The first gate is positioned between the egg storage unit and the transfer tunnel. When the machine receives an order, the microcontroller sends a signal to the gate actuator driver. The driver activates the first gate for a short controlled time, allowing one egg to leave the storage unit and enter the transfer tunnel. After the egg passes through, the first gate closes again to block the next egg.

A sensor in the transfer tunnel detects whether an egg has entered correctly. This sensor sends a signal back to the microcontroller. If one egg is detected, the system continues the sequence. If no egg is detected, the microcontroller stops the process and displays an error message on the touchscreen. If the tunnel sensors detect that two eggs are present in the tunnel, the microcontroller identifies this as a double-feed condition and prevents the second gate from releasing the eggs into the cracking mechanism until the problem is corrected by the program logic.

The second gate is positioned between the transfer tunnel and the egg cracking mechanism. This gate remains closed while the first egg is being separated from the storage unit. Once the transfer tunnel sensor confirms that one egg is correctly positioned and the cracking mechanism is ready, the microcontroller sends a signal to the second gate actuator driver. The second gate opens and allows the egg to move from the tunnel into the cracking position.

A second egg detection sensor near the cracking platform confirms that the egg has reached the correct position on the egg cracking mechanism or arm. This signal is sent to the microcontroller. Only after this confirmation does the microcontroller activate the cracking sequence.

4.5.4. Egg Detection and Position Confirmation

The egg detection system confirms that the egg is correctly positioned before the cracking mechanism is activated. This is important because the cracking mechanism should not move if the egg is missing, misaligned, or stuck.

The sensor detects the egg on the cracking platform or egg holder. The sensor sends an information signal to the microcontroller. The microcontroller checks this signal in the program logic. If the egg is in the correct position, the microcontroller continues the cracking sequence. If the egg is not detected, the machine pauses and sends a warning to the user interface.

4.5.5. Egg Cracking Mechanism

The egg cracking mechanism opens the egg and separates the edible egg contents from the shell. When the sensor confirms that an egg is positioned on the cracking platform, it sends a signal to the microcontroller. The microcontroller then sends control signals to the solenoid drivers.

The Solenoid motors move the platform, spear, and cracking pin into the egg. The spear and the platform move with a controlled force enough to first penetrate the egg shell while the pin pushes through the spear sides to open up the egg at a controlled time The mechanism is strong enough to open the shell but controlled enough to avoid destroying the yolk.

After the shell is opened, the egg contents flow downward into the removable egg pocket plate on the conveyor. This is the main food mass flow.After the egg contents have been released, the cracking mechanism returns to its original position. After a short delay, the microcontroller sends a signal to the servo motor connected to the shell-removal shaft. The shaft slowly flicks the remaining eggshell waste into the eggshell tunnel.The shaft is directly mounted with metal horns.

4.5.6. Eggshell Collection System

The eggshell collection system receives the broken eggshells after the cracking process. After the egg contents have been deposited onto the conveyor, the shell-removal shaft flicks the shell waste into the eggshell tunnel. The tunnel guides the shells into the eggshell storage drawer simply through gravity because it is inclined.

The eggshell drawer is separated from the food path to reduce contamination risk. The drawer can be removed by the operator for emptying and cleaning. Inside the drawer, an inclined floor helps guide the eggshells toward the collection area.

The drawer also includes a manual pressing mechanism. The operator can press a handle connected to a pusher plate. This pusher plate compresses the eggshells inside the drawer, reducing the volume of waste and allowing the drawer to store more shells before emptying. Return springs bring the pressing mechanism back to its starting position after use.

A removable inner liner is placed inside the drawer. This liner prevents shell residue and liquid egg remains from sticking directly to the stainless-steel drawer surface. It also makes cleaning easier because the liner can be removed, washed, or replaced.

The egg shell drawer should be emptied out at the end of every day's use to prevent things like rot and unpleasant smells, The advised threshold for emptying the egg drawer is at max of 260 eggs ; however it has the capacity for over 500 egg shells when compressed.

4.5.7. Conveyor Transport System

The conveyor transport system moves the egg contents through the cooking zone. After the cracking mechanism releases the egg contents, the egg falls into a removable egg pocket plate mounted on the conveyor.

The pocket plates are designed to hold the egg in a controlled shape with the centre of it being the peak of the concavity and prevent the egg from spreading too far while keeping the yolk slightly closer and centred . Each plate is attached to the conveyor using mounting tabs and quick-release pins. These quick-release pins allow the plates to be removed for cleaning, maintenance, or replacement.

The conveyor is driven by a NEMA-17 stepper motor. The microcontroller sends step and direction signals to the stepper motor driver. The driver supplies the required current to the motor. The motor rotates the drive shaft through the coupling and sprocket system. The sprockets move the roller chains, and the chains carry the egg pocket plates through the machine.

A home or limit sensor gives the conveyor a reference position. This allows the microcontroller to know where the egg pockets are located. The position reference is important for aligning the pocket plates with the cracking position, heating zone, and output position.

The conveyor positioning system uses sensors to ensure that the egg pocket plates are correctly aligned during operation. Since there are always four removable egg pocket plates mounted on the conveyor, the system must know the position of the plates before the egg is released from the cracking mechanism.

A home/reference sensor is used to detect the first egg pocket plate before the raw egg contents are deposited. This ensures that the pocket plate is correctly positioned under the cracking mechanism so the egg lands inside the pocket instead of missing the plate or splashing outside the food path.

At the end of the conveyor, another position sensor detects when the last pocket plate has passed through the cooking and output area. When this final plate is detected, the sensor sends a signal to the microcontroller. The microcontroller then commands the conveyor motor to continue or reset the conveyor position until the first pocket plate reaches the home/reference position again.

4.5.8. Bottom Heating System

The bottom heating system cooks the egg from below. It consists of cartridge heaters installed in an aluminium heating block. The heating block is positioned below the conveyor path so that heat is transferred to the underside of the egg pocket plates.

The temperature sensor measures the temperature of the heating block. This temperature information is sent to the temperature controller and microcontroller with bi-directional communication. The controller compares the measured temperature with the selected set point. If more heat is needed, the controller sends a signal to the solid-state relay (SSR). The solid-state relay (SSR) switches power to the cartridge heaters.

The bottom heating system mainly affects the cooking of the egg white from below. It must provide stable heat without burning the egg or overheating the conveyor components. A thermal fuse or safety cutoff is included as protection. If the heating block exceeds a safe temperature, the safety device interrupts the heating circuit or sends a fault signal.

4.5.9. Top IR Heating System

The Top IR Heating System cooks the upper surface of the egg without flipping it. This is important because the machine is designed to produce sunny-side-up eggs, where the yolk should remain visible and intact.

The Top IR Heater emits radiant heat downward toward the egg surface. A reflector shield directs the infrared radiation toward the conveyor pockets and reduces heat loss. A guard cover protects the user and other components from direct contact with the hot heater.

The Top IR Heater is controlled by the microcontroller and temperature controller through a solid-state relay (SSR). The selected cooking setting determines how much top heat is applied. For a softer egg, the IR intensity or active time can be reduced. For a more cooked egg, the IR heater can remain active for longer or operate at a higher intensity.

A safety cutoff or temperature sensor monitors the upper heating zone. If the temperature becomes unsafe, the controller stops the heater and sends a warning to the Touchscreen.

4.5.10. Oil Dosing and Surface Preparation

The Oil System prepares the cooking surface so the egg does not stick to the pocket plate. Oil is stored in an oil container that can be refilled by the staff. Before egg deposition, a controlled amount of oil (2ml) is supplied to the egg pocket plate.

The oil is delivered by a small peristaltic pump. This improves cooking quality and reduces waste.

The Touchscreen then displays a refill warning for the staff as the amount of oil per egg is set so after the set amount of eggs are used the touchscreen displays and oil warning level.

4.5.11. Power and Control System

The machine is powered from the AC mains supply. The AC mains input is first connected to the main protection and switching components, such as the main switch, fuse, emergency stop circuit, grounding connection, and other required safety protection elements.

From the AC mains input, the power system is divided into two main paths. The first path supplies the heating systems, which remain AC-powered. The bottom heating cartridges and the top IR heater are not supplied through step-down buck converters. Instead, they are controlled through relays or solid-state relays (SSR), which switch the AC power to the heaters according to the temperature control signals.

The second path supplies the low-voltage control and actuator systems. The AC mains power is converted by the main DC power supply. From this DC supply, separate step-down buck converters and DC-DC converters provide the required voltage levels for the different subsystems. These voltage levels include 24 V, 12 V, 6 V, 5 V, and 3.3 V, depending on the component requirements.

Each subsystem is connected only to the voltage level it is designed for, which prevents components from being damaged by incorrect voltage.

The microcontroller acts as the central control unit. It receives information from the touchscreen and sensors, processes the program logic, and sends control signals to the motor drivers, solenoid drivers, servo drivers, relays, oil dosing system, conveyor system, and output system. The heaters use AC power, but their switching is still controlled by the low-voltage control system through the relay or solid-state relay (SSR) interface.

4.5.12. Output System

After the egg has passed through the heating zone, the conveyor reaches the output end. At this point, the cooked egg slides off the removable egg pocket plate and onto the rotating serving plate.

Before the output sequence starts, the system must first confirm that a serving plate is actually placed in the output area. A load cell is installed under the rotating plate platform. When the operator places a serving plate on the platform, the load cell detects the added weight and sends a signal to the microcontroller. This signal acts as the permission condition for the output system. If no plate is detected, the microcontroller does not allow the egg output sequence to begin and can display a warning on the touchscreen.

Once the load cell confirms that a plate is present, the microcontroller starts the rotating plate initialization. The rotating plate is driven by a stepper motor. The motor transfers motion through a transmission shaft to a rubber wheel, which rotates the plate by friction. The plate has a small bottom tab or reference feature. During initialization, the stepper motor rotates the plate slowly until this bottom tab hits or triggers a micro switch. When the micro switch is triggered, it sends a signal to the microcontroller. This defines the home position of the rotating plate.

The timing is controlled in the microcontroller code.After the home position is confirmed, the microcontroller can rotate the serving plate according to the number of eggs selected by the user. If the user selects two eggs, the plate rotates in 180° steps. If the user selects three eggs, it rotates in 120° steps. If the user selects four eggs, it rotates in 90° steps. This ensures that the eggs are distributed evenly on the serving plate.

For example:

2 eggs: $\frac{360°}{2}$ = 180° per step

3 eggs: $\frac{360°}{3}$ = 120° per step

4 eggs: $\frac{360°}{4}$ = 90° per step

This output system prevents multiple eggs from falling onto the same location and allows the final serving plate to be automatically arranged based on the selected order size.

4.6. Relation to D-FMEA

The functional structure supports the later D-FMEA because it shows the important functions, interfaces, and dependencies of the machine. For example, if the egg detection sensor fails, the cracking mechanism may operate at the wrong time. If the conveyor home sensor fails, the egg pocket may not align with the cracking or output position. If the bottom heating sensor fails, the egg may be undercooked or the heating block may overheat. If the shell chute becomes blocked, eggshell waste may contaminate the food path. By identifying these functional relationships early, the D-FMEA can focus on the most critical failure modes and define suitable prevention and detection measures.

5. Human Machine Interface

5.1. Egg Geometry

Before any subsystem calculation can be performed, a consistent geometric model of the egg must be established. This model is referenced throughout all subsequent sections.

5.1.1. Volume

The target egg is a standard large egg of 60g. A fresh egg has a density in the range 1.03 - 1.10g/mL (midpoint 1.065g/mL), giving a volume of approximately 56.3cm3. Since the density is close to 1g/cm3, the volume is taken directly as 60cm3. This is a conservative upward rounding that slightly overestimates surface area and content volume, resulting in conservative outputs in the heat transfer section.

5.1.2. Sphere Model

V_{egg} = 60 cm^3 (simplified) The egg is modelled as a perfect sphere of equivalent volume. This simplifies all downstream area and geometry calculations.

$$r = \sqrt[3]{\frac{3V}{4\pi}} = \sqrt[3]{\frac{3 \times 60}{4\pi}} = \sqrt[3]{14.32} = 2.426 \text{ cm}$$

$$d_{\text{egg}} = 2r = 4.85 \text{ cm}$$

$$A_{\text{sphere}} = 4\pi r^2 = 4\pi(2.426)^2 = 73.93 \text{ cm}^2$$

5.1.3. Shape Correction Factor

A real egg is a prolate spheroid. Using a typical axis ratio a/b = 1.25 and the same volume, the true surface area is computed and compared to the sphere model.

$$V = \frac{5}{3}\pi b^3 = 60 \text{ cm}^3 \quad \Longrightarrow b = 2.256 \text{ cm}, \qquad a = 2.820 \text{ cm}$$

With eccentricity $e = \sqrt{1 - \left(\frac{b}{a}\right)^2} = 0.600$

$$A_{\text{spheroid}} = 2\pi b^2 \left(1 + \frac{a}{be} \arcsin e\right) = 74.86 \text{ cm}^2$$

$$k_{\text{shape}} = \frac{A_{\text{spheroid}}}{A_{\text{sphere}}} = \frac{74.86}{73.93} = 1.013$$

The sphere model introduces a surface area error of 1.3%. The correction factor k_{shape}= 1.013 is applied to all surface area terms in the heat transfer section.

5.1.4. Geometry Summary

Parameter	Value	Note
Egg mass	60 g	Standard large egg
Egg volume (simplified)	60 cm^3	Conservative upward rounding
Sphere diameter	4.85 cm	
Sphere radius	2.426 cm	
Surface area (sphere)	73.93 cm^2	
Shape correction k	1.013	Applied to surface area only
Shell mass (10%)	6 g	
Content mass	54 g	$m_{egg} - m_{shell}$

Table 15 : Geometric Summary of our egg dimensions.

5.2. Process Time

This section estimates the total preparation time before the egg enters the cooking stage. The preparation time includes oil pre-spray, gate release, egg transport, breaking mechanism actuation, egg draining, and spring reset. The calculated preparation time is then compared with the conveyor timing to verify whether the 30 s cooking-cycle target can be achieved.

5.2.1. Stage 0 - Oil Pre-Spray

The oil pre-spray stage prepares the cooking plate before the egg is released. This stage includes the MOSFET switching delay, pump rotor spin-up, oil delivery through the tube, check valve closure, and oil spreading over the heated plate.

Assumptions:
• MOSFET: IRLZ44N.
• Gate resistance: 10 Ω.
• Effective gate capacitance: 1.8 nF.
• Pump rotor mass: 5 g.
• Pump rotor radius: 8 mm.
• Pump torque: 0.005 Nm.
• Required oil volume: 2 mL.
• Pump flow rate: 50 mL/min.
• Oil tubing is pre-filled during system initialization.
• Check valve closure time: 1.0 ms.
• Estimated oil spreading time: 400.0 ms.
• Plate diameter: 15 cm.

a.MOSFET gate charge (IRLZ44N)

The MOSFET switching time is extremely small compared with the mechanical and fluid delivery times, so it is treated as approximately 0.00 ms in the total timing calculation.

$$\tau = RC = 10\ \Omega \times 1.8\ \mathrm{nF} = 18\ \mathrm{ns}$$
$$t_{\mathrm{MOSFET}} = 2.2\tau = 2.2(18\ \mathrm{ns}) = 39.6\ \mathrm{ns} \approx 0.00\ \mathrm{ms}$$

b.Pump rotor spin-up

The pump rotor spin-up time is calculated from the rotor moment of inertia and angular acceleration:

$$J = \frac{1}{2}mr^2 = \frac{1}{2}(0.005)(0.008)^2 = 1.6 \times 10^{-7}\ \mathrm{kg\,m^2}$$
$$J = 0.16\ \mu\mathrm{kg\,m^2}$$

$$\alpha = \frac{T}{J} = \frac{0.005}{1.6 \times 10^{-7}} = 31\,250 \text{ rad/s}^2$$

$$t_{\text{spin-up}} = \frac{\omega}{\alpha} = \frac{100}{31\,250} = 0.00320 \text{ s} = 3.20 \text{ ms}$$

c.Oil tubing travel (pre-filled at boot)

Since the oil tube is assumed to be pre-filled during system initialization, the oil tubing travel delay is taken as:

$t_{\text{tube}} = 0.00 \text{ ms}$

d.2 mL delivery @ 50mL/min

The oil delivery time is calculated from the required oil volume and the pump flow rate:

$$t_{\text{delivery}} = \frac{V}{Q} = \frac{2.0}{50/60} = 2.400 \text{ s} = 2400.0 \text{ ms}$$

e.Check valve closure (spring loaded)

After delivery, the spring-loaded check valve closes. This delay is estimated as:

$t_{\text{valve}} = 1.0 \text{ ms}$

f.Oil spreading on the plate

The oil film thickness over a 15 cm diameter plate is estimated from the delivered oil volume and the plate surface area:

$$\delta_{\text{film}} = \frac{V}{A} = \frac{2 \text{ mL}}{\pi(7.5 \text{ cm})^2}$$

$$\delta_{\text{film}} = \frac{2 \text{ cm}^3}{176.7 \text{ cm}^2} = 0.0113 \text{ cm} = 113 \ \mu\text{m}$$

At the heated plate temperature, the oil viscosity decreases significantly. The spreading delay is therefore estimated as:

$t_{\text{spread}} = 400.0 \text{ ms}$

The total oil pre-spray time is:

$$t_0 = t_{\text{MOSFET}} + t_{\text{spin-up}} + t_{\text{tube}} + t_{\text{delivery}} + t_{\text{valve}} + t_{\text{spread}}$$

$$t_0 = 0.00 + 3.20 + 0.00 + 2400.0 + 1.0 + 400.0$$

$$t_0 = 2804.20 \text{ ms} = 2.804 \text{ s}$$

Therefore, the oil pre-spray stage requires approximately 2804.20 ms, or 2.804 s. This stage is dominated by the oil delivery time, while the MOSFET switching delay is negligible.

5.2.2. Stage 1 - Gate Release

The gate-release stage begins when the first IR sensor detects that an egg is present at the release position. The sensor signal is read by the STM32 microcontroller, which then activates the solenoid driver. The solenoid retracts the gate and allows one egg to enter the transport channel.

Assumptions:
• IR sensor: GP2Y0A21.
• Sensor sampling frequency: 2 kHz.
• STM32 clock frequency: 64 MHz.
• EXTI interrupt latency: 3 clock cycles.
• GPIO switching instruction count: approximately 12 instructions.
• MOSFET gate resistance: 10 Ω.
• MOSFET effective gate capacitance: 1.8 nF.
• Solenoid coil inductance: 10 mH.
• Solenoid coil resistance: 5 Ω.
• Solenoid plunger begins moving at approximately 50% of maximum current.
• Gate solenoid stroke length: 10 mm.
• Solenoid force: 10 N.
• Opposing spring force: 0.5 N.
• Moving plunger mass: 15 g.
• Egg release lag due to inertia: 1.000 ms.

a.IR Sensor 1 (GP2Y0A21, 2 kHz)

$$t_{\text{sensor}} = \frac{1}{f} = \frac{1}{2000} = 0.0005 \text{ s} = 0.500 \text{ ms}$$

b.STM32 EXTI latency @ 64 MHz

The STM32 external interrupt latency is calculated from the controller clock frequency:

$$t_{\text{EXTI}} = \frac{3}{64 \times 10^6} = 46.9 \text{ ns} \approx 0.000 \text{ ms}$$

c.ISR + HAL_GPIO_WritePin

The time required for the interrupt service routine and GPIO output command is:

$$t_{\text{ISR}} = \frac{12}{64 \times 10^6} = 187.5 \text{ ns} \approx 0.000 \text{ ms}$$

d. MOSFET gate charge (solenoid driver)

The MOSFET gate charging time for the solenoid driver is:

$\tau = RC = 10\ \Omega \times 1.8\ \text{nF} = 18\ \text{ns}$

$t_{\text{MOSFET}} = 2.2\tau = 2.2(18\ \text{ns}) = 39.6\ \text{ns} \approx 0.000\ \text{ms}$

e. Solenoid coil energisation

The solenoid coil energisation time is calculated from the electrical time constant of the coil. The plunger is assumed to begin moving when the current reaches approximately 50% of its maximum value:

$\tau = \frac{L}{R} = \frac{10\ \text{mH}}{5\ \Omega} = 2.0\ \text{ms}$

$t_{\text{coil}} = -\tau \ln(1 - 0.5) = -2.0 \ln(0.5) = 1.386\ \text{ms}$

f. Gate solenoid mechanical stroke 10 mm

The solenoid mechanical stroke time is calculated using constant-acceleration motion:

$F_{\text{net}} = F_{\text{solenoid}} - F_{\text{spring}} = 10 - 0.5 = 9.5\ \text{N}$

$a = \frac{F_{\text{net}}}{m} = \frac{9.5}{0.015} = 633.3\ \text{m/s}^2$

$t_{\text{stroke}} = \sqrt{\frac{2L}{a}} = \sqrt{\frac{2(0.010)}{633.3}} = 0.005620\ \text{s} = 5.620\ \text{ms}$

g. Egg release lag (inertia response)

The egg release lag is estimated as:

$t_{\text{egg lag}} = 1.000\ \text{ms}$

The total gate-release time is:

$t_1 = t_{\text{sensor}} + t_{\text{EXTI}} + t_{\text{ISR}} + t_{\text{MOSFET}} + t_{\text{coil}} + t_{\text{stroke}} + t_{\text{egg lag}}$

$t_1 = 0.500 + 0.000 + 0.000 + 0.000 + 1.386 + 5.620 + 1.000$

$t_1 = 8.506\ \text{ms}$

Therefore, the gate-release stage requires approximately 8.506 ms. The dominant part of this stage is the mechanical solenoid stroke, while the microcontroller and MOSFET switching delays are negligible compared with the mechanical motion.

5.2.3. Stage 2 - Egg Transport

The egg transport stage begins after the release gate opens. The egg rolls through the inclined channel, passes through a curved fillet transition, settles into the aluminium holder, and is then detected by the second IR sensor before the breaking mechanism starts.

Assumptions:
• Egg mass: 60 g.
• Egg radius used for rolling model: 28 mm.
• Egg is approximated as a solid sphere for rolling inertia.
• Main transport channel angle: 30°.
• Main channel length: 20 cm.
• Kinetic friction coefficient on PP surface: 0.12.
• Fillet transition radius: 6 cm.
• Fillet arc angle: 30°.
• Velocity loss through fillet transition: 1.5%.
• Holder settling angle: 15°.
• Holder settling length: 5 cm.
• Holder friction coefficient: 0.15.
• Impact damping time in rubber-padded holder: 15.00 ms.
• IR Sensor 2, STM32, and MOSFET signal chain delay: 1.887 ms.

a.Egg moment of inertia

$$I = \frac{2}{5}mR^2 = \frac{2}{5}(0.060)(0.028)^2$$

$$I = 18.82 \times 10^{-6}\ \mathrm{kg\,m^2}$$

$$K = 1 + \frac{I}{mR^2} = 1.400$$

b.Rolling check

The rolling condition is checked for the 30° PP transport surface:

$$\tan(30^\circ) = 0.577$$

$$\frac{\mu_s}{1 + \frac{mR^2}{I}} = 0.043$$

$$0.577 > 0.043$$

Therefore, the egg is expected to roll rather than slide on the inclined PP surface.

c.Channel roll (θ=30°, μ_k=0.12, L=20cm)

The acceleration down the 30° channel is:

$$a = \frac{g(\sin\theta - \mu_k \cos\theta)}{K}$$

$$a = \frac{9.81(\sin 30° - 0.12\cos 30°)}{1.400}$$

$$a = \frac{9.81(0.500 - 0.104)}{1.400} = 2.775 \text{ m/s}^2$$

The time required to roll through the 20 cm channel is:

$$t_{\text{channel}} = \sqrt{\frac{2L}{a}} = \sqrt{\frac{2(0.20)}{2.775}}$$

$$t_{\text{channel}} = 0.37964 \text{ s} = 379.64 \text{ ms}$$

$$v_{\text{exit}} = at = 2.775(0.37964) = 1.054 \text{ m/s}$$

d.Fillet transition (R=6cm, 30° arc)

The egg then passes through the curved fillet transition. The arc length is:

$$s_{\text{arc}} = R\theta = 0.06\left(\frac{\pi}{6}\right) = 0.03142 \text{ m} = 31.42 \text{ mm}$$

The normal force through the fillet is estimated as:

$$N_{\text{fillet}} = m\left(g\cos 30° + \frac{v^2}{R}\right)$$

$$N_{\text{fillet}} = 0.060\left(9.81\cos 30° + \frac{1.054^2}{0.06}\right) = 1.62 \text{ N}$$

With an estimated 1.5% velocity loss through the fillet, the average velocity is taken as:

$$v_{\text{avg}} = 1.038 \text{ m/s}$$

$$t_{\text{fillet}} = \frac{s_{\text{arc}}}{v_{\text{avg}}} = \frac{0.03142}{1.038} = 0.03027 \text{ s} = 30.27 \text{ ms}$$

e.Settle into Al holder (θ=15°, μ=0.15, L=5cm)

The egg then settles into the aluminium holder. The rolling condition at the 15° holder angle is:

$$\tan(15°) = 0.268$$

$$0.268 > 0.043$$

Therefore, the egg is still expected to roll during the settling phase.

The settling acceleration is:

$$a = \frac{g(\sin 15^\circ - 0.15 \cos 15^\circ)}{1.400}$$

$$a = 0.798 \text{ m/s}^2$$

Using the entry velocity into the holder, the estimated settling time is:

$$v_{\text{entry}} = 1.038 \text{ m/s}$$

$$t_{\text{settle}} = \frac{v_0 - v_f}{a} = 353.92 \text{ ms}$$

f.Impact damping in Al holder

The impact damping time in the rubber-padded aluminium holder is:

$$t_{\text{damping}} = 15.00 \text{ ms}$$

g.IR Sensor 2 + STM32 + MOSFET chain

The sensor and control signal delay after the egg reaches the second IR sensor is:

$$t_{\text{IR2+control}} = 1.887 \text{ ms}$$

The total egg transport time is:

$$t_2 = t_{\text{channel}} + t_{\text{fillet}} + t_{\text{settle}} + t_{\text{damping}} + t_{\text{IR2+control}}$$

$$t_2 = 379.64 + 30.27 + 353.92 + 15.00 + 1.887$$

$$t_2 = 780.72 \text{ ms}$$

5.2.4. Stage 3 - Breaking Mechanism

3b. Spear Coil Energisation

Given:

- Inductance, **L = 12 mH**
- Resistance, **R = 4 Ω**

Time constant:

$$\tau = \frac{L}{R} = \frac{12 \times 10^{-3}}{4} = 3 \text{ ms}$$

Coil energisation time:

$$t = -\tau \ln(1 - 0.5)$$
$$t = -3 \ln(0.5)$$
$$t = 2.079 \text{ ms}$$

3c. Spear Stroke (12.5 mm) + Shell Fracture

Given:

- Net force:

$$F_{net} = 10 - 0.3 = 9.7 \text{ N}$$

- Mass:

$$m = 105 \text{ g} = 0.105 \text{ kg}$$

Acceleration:

$$a = \frac{F}{m} = \frac{9.7}{0.105}$$
$$a = 92.38 \text{ m/s}^2$$

Stroke distance:

$$s = 12.5 \text{ mm} = 0.0125 \text{ m}$$

Stroke time:

$$t_{stroke} = \sqrt{\frac{2s}{a}}$$

$$t_{stroke} = \sqrt{\frac{2(0.0125)}{92.38}}$$

$$t_{stroke} = 16.45 \text{ ms}$$

Additional shell fracture energy:

- Crack energy:

$$E_{crack} = 0.5 \text{ J}$$

- Crack displacement:

$$\delta = 1 \text{ mm}$$

Extra fracture time:

$$t_{extra} = 4.65 \text{ ms}$$

3d. Wedge Coil Energisation

Given:

- Inductance:

$$L = 8\ \text{mH}$$

- Resistance:

$$R = 4\Omega$$

Time constant:

$$\tau = \frac{L}{R} = \frac{8 \times 10^{-3}}{4}$$
$$\tau = 2\ \text{ms}$$

Energisation time:

$$t = -\tau \ln(1 - 0.5)$$
$$t = -2 \ln(0.5)$$
$$t = 1.386 \text{ ms}$$

3e. Wedge Stroke (6.5 mm)

Given:

- Net force:

$$F_{net} = 4.8 \text{ N}$$

- Mass:

$$m = 60 \text{ g} = 0.06 \text{ kg}$$

Acceleration:

$$a = \frac{F}{m} = \frac{4.8}{0.06}$$
$$a = 80 \text{ m/s}^2$$

Stroke distance:

$$s = 6.5 \text{ mm} = 0.0065 \text{ m}$$

Stroke time:

$$t = \sqrt{\frac{2s}{a}}$$
$$t = \sqrt{\frac{2(0.0065)}{80}}$$
$$t = 12.748 \text{ ms}$$

3f. Shell Halves Fall Open (Gravity, $\theta = 45°$)

Acceleration due to gravity component:

$$a = g\sin 45°$$
$$a = 9.81 \times 0.707$$
$$a = 6.937 \text{ m/s}^2$$

Opening length:

$$L = 25 \text{ mm} = 0.025 \text{ m}$$

Time for shell halves to fall open:

$$t = \sqrt{\frac{2L}{a}}$$
$$t = \sqrt{\frac{2(0.025)}{6.937}}$$
$$t = 84.90 \text{ ms}$$

Total Breaking Mechanism Time

$$T_{total} = 122.22 \text{ ms}$$

5.2.5. Stage 4 - Drain and Spring Reset

The drain and spring reset stage begins after the egg shell has opened and the egg contents are released from the breaking holder. This stage includes the draining of the egg white and yolk through the outlet, the surface-tension drip delay, and the spring-return motion of the spear and wedge mechanisms.

Assumptions:
• Egg white volume: 33 mL.
• Egg yolk volume: 13 mL.
• Total drained egg content volume: 46 mL.
• Egg white density: 1040 kg/m³.
• Egg yolk density: 1030 kg/m³.
• Effective drain height: 35 mm.
• Egg white dynamic viscosity: 200 mPa·s.
• Discharge coefficient after viscosity correction: 0.597.
• Outlet radius: 15 mm.
• Surface tension: 48 mN/m.
• Spear return spring stiffness: 365,741 N/m.
• Spear moving mass: 105 g.
• Wedge return spring stiffness: 200,000 N/m.
• Wedge moving mass: 60 g.

a.Volume of whites and yolks.

$$V_{\text{total}} = V_{\text{white}} + V_{\text{yolk}} = 33 + 13 = 46 \text{ mL}$$

b.Torricelli drain (viscosity-corrected)

The ideal drain velocity is estimated using Torricelli's law:

$$v_{\text{ideal}} = \sqrt{2gh} = \sqrt{2(9.81)(0.035)} = 0.829 \text{ m/s}$$

Because egg white is viscous, the ideal velocity is reduced using a discharge coefficient:

$$v_{\text{drain}} = C_d v_{\text{ideal}} = 0.597(0.829) = 0.495 \text{ m/s}$$

The outlet area is:

$$A_{\text{out}} = \pi r^2 = \pi(15 \text{ mm})^2 = 706.9 \text{ mm}^2$$

The drain time is calculated from the drained volume, outlet area, and corrected drain velocity:

$$t_{\text{drain}} = \frac{V}{A_{\text{out}} v_{\text{drain}}}$$

$$t_{\text{drain}} = \frac{46 \text{ mL}}{706.9 \text{ mm}^2 \times 0.495 \text{ m/s}} = 131.6 \text{ ms}$$

c.Surface tension drip delay

After the main volume has drained, a short surface-tension drip delay is included:

$$t_{\text{drip}} = 3\sqrt{\frac{\gamma}{\rho g d}}$$

$$t_{\text{drip}} = 37.6 \text{ ms}$$

d. Spear return spring

The spear return motion is modelled as a spring-mass return system:

$$\omega_{\text{spear}} = \sqrt{\frac{k}{m}} = \sqrt{\frac{365741}{0.105}} = 1866 \text{ rad/s}$$

$$t_{\text{spear}} = \frac{\pi}{\omega_{\text{spear}}} = \frac{\pi}{1866} = 1.683 \text{ ms}$$

e. Wedge return spring

The wedge return motion is calculated in the same way:

$$\omega_{\text{wedge}} = \sqrt{\frac{k}{m}} = \sqrt{\frac{200000}{0.060}} = 1826 \text{ rad/s}$$

$$t_{\text{wedge}} = \frac{\pi}{\omega_{\text{wedge}}} = \frac{\pi}{1826} = 1.721 \text{ ms}$$

The total drain and spring reset time is:

$$t_4 = t_{\text{drain}} + t_{\text{drip}} + t_{\text{spear}} + t_{\text{wedge}}$$

$$t_4 = 131.6 + 37.6 + 1.683 + 1.721$$

$$t_4 = 172.57 \text{ ms}$$

Time for processing and prep stage grand total is divided as follows:

Stage 0 — Oil pre-spray	2804.20 ms	72.1%
Stage 1 — Gate release	8.51 ms	0.2%
Stage 2 — Egg transport	780.72 ms	20.1%
Stage 3 — Breaking	122.22 ms	3.1%
Stage 4 — Drain + reset	172.57 ms	4.4%

t_prep = 3888.21 ms = 3.888 s

5.3. Cooking Time

5.3.1. Core Assumptions

- Egg mass: m = 60 g = 0.060 kg
- Egg volume: V ≈ 60 cm³
- Egg properties (effective white + yolk):
- Thermal conductivity: k = 0.52 W/(m·K)
- Density: ρ = 1040 kg/m³
- Specific heat: cp = 3600 J/(kg·K)
- Thermal diffusivity: α = k/(ρ·cp) = 1.3889×10^{-7} m²/s
- Entry temperature from storage: T_init = 20°C

5.3.2. Geometry In Heat Direction

Half-thickness plate → yolk centre: L = 5.95 mm = 0.00595 m

1D finite difference grid:

- Nodes: N = 20
- Spatial step: dx = $\frac{L}{N} = \frac{0.00595}{20}$ = 0.2975 mm
- Time step: dt = 0.005 s
- Fourier number: Fo = α·dt/dx² ≈ 0.00785 < 0.5 (stable)

5.3.3. Boundary and Initial Conditions

Bottom plate (with thin oil film):

- Plate temperature: T_plate = 200°C
- Oil film: δ ≈ 113 µm, k_oil ≈ 0.17 W/(m·K) → h_eff ≈ 1500 W/(m²·K)
- Approximated as Dirichlet: T(0, t) = 200°C

Top IR heater:

- IR temperature: T_IR = 120°C
- Heat transfer coefficient: h_top = 20 W/(m²·K)
- Mixed (Robin) boundary at top node N:

$$T_N^{(n+1)} = \frac{\frac{k}{\Delta x} T_{N-1}^{(n+1)} + h_{\text{top}} T_{\text{IR}}}{\frac{k}{\Delta x} + h_{\text{top}}}$$

- Initial condition (20°C entry):
- T(x, 0) = 20°C (uniform)

- Targets:
- Whites set: centre temperature T_centre = 65°C → defines t_cook

- Yolk limit: centre temperature T_centre = 68°C → defines t_limit

5.3.4. 1D Finite Difference Scheme

- Interior nodes (i = 1..N-1):

$$T_i^(n+1) = T_i^n + Fo\cdot (T(i+1)^n - 2\cdot T_i^n + T(i-1)^n)$$

- Boundary nodes:
  - Bottom node 0: $T_0^(n+1) = 200°C$
  - Top node N: mixed condition as in section 3

5.3.5. Cook Time Results (20°C Entry , 200°C Cooking Plate)

- From the simulation:
  - Time to centre 65°C (whites set):
    t_cook = 23.305 s (≈ 23.30 s)

  - Time to centre 68°C (yolk limit):
    t_limit ≈ 25.0–25.5 s (approximate band)

  - Cook window:
    Δt_window ≈ t_limit − t_cook ≈ 1.7–2.0 s

5.3.6. Total Time (First Egg)

- Total first-egg time:
  T_total = t_prep + t_cook
  ≈ 3.888 s + 23.305 s
  ≈ 27.193 s

First egg ready in ≈ 27.2 s (centre at 65°C).

5.3.7. Conveyor & Plate Timing

- Heated conveyor length under plates:
  L_heated = 450 mm

- Conveyor speed for t_cook over 450 mm:
  v = L_heated / t_cook
  = $450/23.305$
  ≈ 19.3 mm/s

- Plate geometry:
  Plate length = 106 mm
  Gap between = 15 mm
  Centre spacing = d_center = 106 + 15 = 121 mm

- Time between plate centres at loading position:
  T_plate = d_center / v
  = 121 / 19.3
  ≈ 6.27 s

- Prep vs plate timing:
  t_prep ≈ 3.888 s
  T_plate ≈ 6.27 s
  Buffer = T_plate − t_prep
  ≈ 6.27 − 3.888
  ≈ 2.38 s

The next egg is fully prepped about 2.4 s before its plate arrives.

5.4. Machine FlowChart

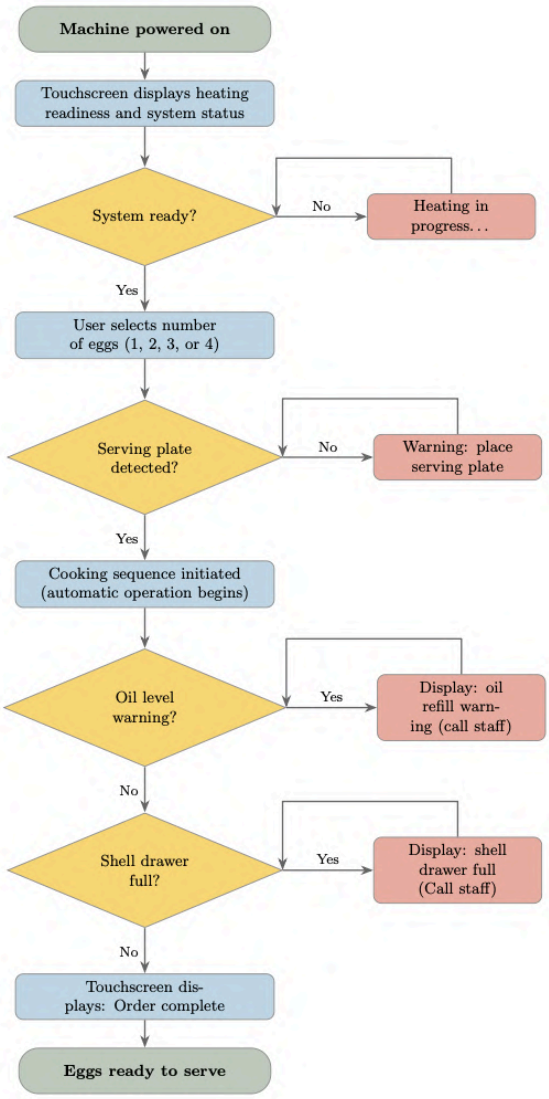


Figure 5 : SunnyPro - User Interface Flowchart.

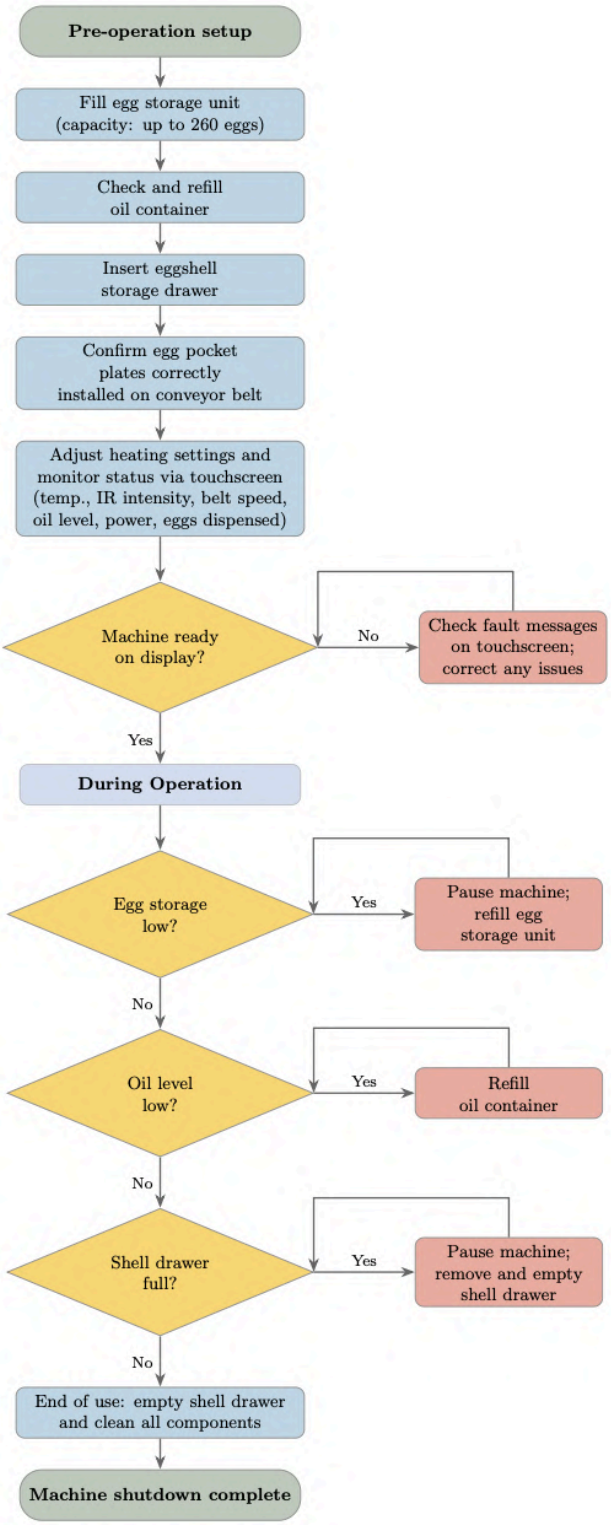


Figure 6 : SunnyPro - Staff Interface Flowchart.

5.4.1. Customer UI Design



Figure 7 : SunnyPro Customer UI.

The SunnyPro machine includes a touchscreen-based graphical user interface (GUI) designed to provide a simple and intuitive interaction experience for users. The interface allows users to select the desired number of eggs to be dispensed, ranging from one to four eggs.

The main screen displays four selection buttons, each representing a different egg quantity. Visual egg icons are used alongside numerical values to improve usability and make the selection process easier to understand. A high-contrast black-and-yellow color scheme was chosen to enhance readability and visibility under different lighting conditions.

The interface also includes:

- a greeting message,
- date and time display,
- machine branding (“SunnyPro”),
- and an administrator access button for monitoring and system configuration.

The design prioritizes:

- simplicity,
- large touch-friendly controls,
- minimal user confusion,
- and quick operation.

Once a quantity is selected, the machine processes the input and activates the egg frying mechanism accordingly.

5.4.2. Admin UI Design



Figure 8 : SunnyPro - Staff Admin UI Design.

The SunnyPro machine includes an integrated Admin Panel designed for system monitoring, operational management, and machine configuration. The panel allows authorized staff to supervise machine performance, monitor usage statistics, and adjust cooking-related settings through a centralized interface. The admin interface is organized into multiple sections, including System Analytics, Performance Monitoring, and Cooking Options, to improve maintenance efficiency and overall system management.

System Analytics

The System Analytics section helps staff monitor the usage and performance of the SunnyPro machine. It analyzes and displays the following information:

- **Total Eggs Fried** – The total number of eggs processed by the machine.
- **Machine Operating Time** – The total time the machine has been running.
- **Total Cooking Sessions** – The number of cooking operations completed.
- **Average Eggs per Order** – The average quantity of eggs selected by users per session.

This information helps administrators track machine activity, monitor productivity, and identify maintenance or refill requirements.

Performance Monitor

The Performance Monitor section provides real-time operational information about the SunnyPro machine. It helps staff monitor the machine's current status and resource levels to ensure proper operation and maintenance. The system monitors and displays the following data:

- **Oil Level Indicator** – Displays the current amount of oil available in the machine.
- **Egg Storage Level** – Shows the number of eggs remaining in storage.
- **Shell Drawer Status** – Indicates whether the shell waste drawer is full or available for use.
- **Power Consumption** – Monitors the amount of electrical power being used by the machine.
- **Machine Voltage** – Displays the current operating voltage of the system.

This information helps administrators maintain stable machine performance, monitor resource availability, and detect potential operational issues.

Cooking Options

The Cooking Options section allows administrators to adjust the cooking settings of the SunnyPro machine based on the desired cooking performance and egg preparation quality. The system provides configurable heating and cooking controls to optimize the frying process.

The available cooking options include:

- **Conveyor Speed Control** – Adjusts the speed of the conveyor system to control cooking duration.
- **Top IR Heat Control** – Regulates the intensity of the upper infrared heating element.
- **Bottom Heating Control** – Adjusts the heat applied from the lower heating surface.
- **Heat Level Selection** – Allows staff to select predefined cooking levels such as:
    - Low Heat
    - Medium Heat
    - High Heat

These settings help administrators customize the cooking process according to operational requirements and desired cooking results.

5.5. Inputs and Outputs

5.5.1. Machine Inputs

Signal / Input	Purpose	Signal Character	Hardware Implementation
Power ON signal	Start machine operation	Digital	Main power switch
User egg quantity selection	Select 1–4 eggs	Digital	Touchscreen buttons
Serving plate detection	Detect if plate is present	Analog	Load sensor
Oil level sensor	Monitor oil quantity	Analog	Float sensor
Shell drawer detection	Detect drawer presence/fullness	Digital	Limit switch or proximity sensor
Egg storage level sensor	Detect low egg inventory	Analog	Weight sensor or optical sensor
Temperature measurement	Monitor heating temperature	Analog	Thermocouple / RTD sensor

IR heater intensity setting	Control heating power	Analog / PWM	Heater driver interface
Conveyor belt speed feedback	Monitor/control belt speed	Analog / Pulse	Encoder sensor
Emergency stop	Immediate shutdown	Digital safety signal	Emergency stop pushbutton
Fault acknowledgment	Reset warnings/errors	Digital	Touchscreen command
Oil refill confirmation	Confirm refill completed	Digital	Touchscreen input
Shell drawer reinstalled confirmation	Confirm maintenance completed	Digital	Drawer switch sensor
Heating settings adjustment	Configure process parameters	Variable / Numeric	Touchscreen HMI
Machine ready confirmation	Internal readiness verification	Digital	PLC/controller logic

Table 16 : List of Inputs.

5.5.2. Machine Outputs

Signal / Output	Purpose	Signal Character	Hardware Implementation
Heating system activation	Cook eggs	Digital / PWM	IR heaters / heating elements
Conveyor motor control	Move egg plates	Digital / Variable speed	Motor driver + conveyor motor
Touchscreen status display	Show machine states	Text / Graphics	LCD touchscreen
“Heating in progress” message	Inform user	Text	Touchscreen display
“Order complete” message	Notify cooking finished	Text	Touchscreen display
Oil refill warning	Alert operator	Text + Visual	Display + warning LED
Shell drawer full warning	Alert operator	Text + Visual	Display + warning LED
Machine ready indicator	Show operational readiness	Digital visual	Green LED / touchscreen
Fault message display	Display system errors	Text	Touchscreen
Audible alarm	Alert operator/user	Audio	Buzzer
Egg dispensing mechanism	Release eggs	Digital actuator control	Solenoid / motor
Shutdown sequence	Safely power down	Digital control	Controller logic

Table 17 : List of outputs.

5.6. Safety Aspects

The SunnyPro machine includes several basic safety features to ensure safe and reliable operation during cooking and maintenance. These safety aspects help prevent overheating, improper operation, and mechanical issues while also protecting the user from hot surfaces and moving components. Sensors and warning systems are used to monitor important machine conditions such as oil level, serving plate presence, and shell drawer presence

Safety Aspect	Purpose	Suggested Implementation
Emergency stop	Immediate machine shutdown in emergencies	Emergency stop pushbutton
Overtemperature protection	Prevent overheating and fire risk	Temperature sensor + thermal cutoff
Plate presence detection	Prevent operation without serving plate	Optical/proximity sensor
Low oil detection	Prevent damage to heating system	Oil level sensor
Shell drawer full detection	Prevent overflow and blockage	Fullness sensor + warning message
Protective covers	Prevent contact with moving/hot parts	Mechanical safety enclosure

Table 18: Safety Aspects taken into consideration.

6. D-FMEA

6.1 Definition

Design failure mode and effect analysis (DFMEA) is a structured method that helps engineers understand the impact of potential risks associated with a design.[18] It helps determine how possible design failures may affect performance, safety and reliability.

The FMEA process considers three main factors:

1. Severity (S) – The potential impact of a failure on the system or user.
2. Occurrence (O) – The likelihood of the failure occurring.
3. Detection (D) – Assess how easy or difficult it is to detect the failure before it affects the system.

The Risk Priority Number (RPN) is calculated using the following formula:

*RPN = Severity * Occurrence * Detection*

A higher RPN indicates a higher level of risk, requiring immediate attention and appropriate corrective action.

RPN alone is not always enough, because it treats Severity, Occurrence, and Detection equally. In D-FMEA, some failures should receive attention mainly because their effect is severe, even if the RPN is not the highest. Action Priority (AP) helps show which failure modes need action first by considering Severity first, then Occurrence, then Detection.[19]

[18] Ansys. What is DFMEA? Ansys Blog. Published February 7, 2024. Accessed May 12, 2026. https://www.ansys.com/blog/what-is-dfmea
[19] Quality-One International. AIAG & VDA FMEA Handbook and Guide. Quality-One International. Accessed May 12, 2026.

6.2 D-FMEA Rating Scheme

Severity Scale		
Effect	**Criteria: Severity of Effect**	**Ranking**
Hazardous - Without Warning	May expose client to loss, harm or major disruption - failure will occur **without** warning	10
Hazardous - With Warning	May expose client to loss, harm or major disruption - failure will occur **with** warning	9
Very High	Major disruption of service involving client interaction, resulting in either associate re-work or inconvenience to client	8
High	Minor disruption of service involving client interaction and resulting in either associate re-work or inconvenience to clients	7
Moderate	Major disruption of service not involving client interaction and resulting in either associate re-work or inconvenience to clients	6
Low	Minor disruption of service not involving client interaction and resulting in either associate re-work or inconvenience to clients	5
Very Low	Minor disruption of service involving client interaction that does not result in either associate re-work or inconvenience to clients	4
Minor	Minor disruption of service not involving client interaction and does not result in either associate re-work or inconvenience to clients	3
Very Minor	No disruption of service noticed by the client in any capacity and does not result in either associate re-work or inconvenience to clients	2
None	No Effect	1

Table 19 : Severity Scale for Failure Modes and Effects.

Occurrence Scale			
Adapt as appropriate			
Probability of Failure	**Time Period**	**Per Item Failure Rates**	**Ranking**
Very High: Failure is almost inevitable	More than once per day	>= 1 in 2	10
	Once every 3-4 days	1 in 3	9
High: Generally associated with processes similar to previous processes that have often failed	Once every week	1 in 8	8
	Once every month	1in 20	7
Moderate: Generally associated with processes similar to previous processes which have experienced occasional failures, but not in major proportions	Once every 3 months	1 in 80	6
	Once every 6 months	1 in 400	5
	Once a year	1 in 800	4
Low: Isolated failures associated with similar processes	Once every 1 - 3 years	1 in 1,500	3
Very Low: Only isolated failures associated with almost identical processes	Once every 3 - 6 years	1 in 3,000	2
Remote: Failure is unlikely. No failures associated with almost identical processes	Once Every 7+ Years	1 in 6000	1

Table 20 : Occurrence Scale.

Detection Scale

Detection	Criteria	Ranking
Almost Impossible	No known controls available to detect failure mode	10
Very Remote	Very remote likelihood current controls will detect failure mode	9
Remote	Remote likelihood current controls will detect failure mode	8
Very Low	Very low likelihood current controls will detect failure mode	7
Low	Low likelihood current controls will detect failure mode	6
Moderate	Moderate likelihood current controls will detect failure mode	5
Moderately High	Moderately high likelihood current controls will detect failure mode	4
High	High likelihood current controls will detect failure mode	3
Very High	Very high likelihood current controls will detect failure mode	2
Almost Certain	Current controls almost certain to detect the failure mode. Reliable detection controls are known with similar processes.	1

Table 21 : Detection Scale.

Action Priority Scale

Priority	Criteria	Ranking
Low	Corrective action is not immediately required. The failure mode has a low effect, or the current controls are sufficient to manage the risk.	3
Medium	Corrective action should be considered. The failure mode has a moderate effect, or the currebr controls may not be fully sufficient.	2
High	Corrective action is required. The failure mode has a significant effect and the current controls are not sufficient to prevent or detect it.	1

Table 22: Action Priority Scale.

D-FMEA

Process/Product Name: Sunny Pro
Responsible: Group 3
Prepared By: Group 3
FMEA Date (Orig.):
(Rev.):

Component	Design Function	Failure Mode	Potential Failure Effects	Potential Causes	S (1 - 10)	O (1 - 10)	D (1 - 10)	RPN	Recommended Action	Action Priority (1-3)
What is the process step, change or feature under investigation?	In what ways could the step, change or feature go wrong?	What is the failure mode?	What is the impact on the customer if this failure is not prevented or corrected?	What causes the step, change or feature to go wrong? (how could it occur?)					What are the recommended actions for reducing the occurrence of the cause or improving detection?	
Power Supply System										
ON/OFF switch	Turns the machine on/off when user presses switch	no function	Machine cannot be switched on/off; user cannot operate the system	Defective switch; broken contact; loose wire; damaged connector	5	3	2	30	Use reliable switch and verify wiring connection	3
Power Supply	Provides required electrical power to the machine	No function	Machine does not turn on; user cannot operate the system	Defective power supply; blown fuse; loose connector	8	2	5	80	Have to replace power supply unit	2
	Provides stable voltage to controller and heating system	Under function	Machine works weakly or resets; heating may not reach required temperature	Low output voltage; overloaded supply; voltage drop; weak connector	6	2	5	60	Select power supply with sufficient margin; Check voltage under load and add voltage monitoring	2
	Supplies correct voltage and current level	Over function	Overvoltage may damage controller, display, sensors, or heater control	Regulator failure; short circuit in feedback path	9	2	5	90	Add overvoltage protection and monitoring	1
Main MCU STM	Controls machine logic and coordinates all subsystems	no function	Main MCU doesn't start; motors, sensor, the display, buzzer doesn't work; machine can't operate	Defective MCU	8	3	1	24	Replace MCU	2
		Under function	Some outputs work but others fail; display may update incorrectly, buzzer may not warn, rotating plate may not move correctly	Firmware error; wrong I/O configuration	8	4	9	288	Replace MCU	2
		Intermittent function	MCU resets or freezes during operation; stepper driver receives unstable commands	unstable power supply; overheating	7	4	8	224	Get a voltage regulator and voltage monitoring	2

Internal

		Unintended function/Malfunction	MCU sends wrong message; motor, buzzer, display and others parts may activate at the wrong time	Software logic error; communication problem	5	3	9	135	Individual subsytems tested with code	3
Storage Unit and Breaking Mechanism System										
Egg Storage Unit	Stores eggs safely	Under function	Egg cracks and causes mess; user must stop	Rough loading by staff	2	4	2	16	Staff has to remove eggs and clean	3
	Keep eggs availabe for cooking	No function	Unit runs empty; staff must reload eggs; causes delay in cooking cycle	Staff forgets to load eggs; storage capacity too small	1	3	1	3	There will be a warning pop-out on the screen - accessed by the staff	3
Gate Actuators	Opens/closes gate to release egg at correct time	Over function	More than one egg is released; eggs collide, crack	Actuator timing error; control signal error; gate doesn't close fast enough	5	3	6	90	Add me mechanical stopper for single egg release; improve gate timing	2
		No function	Gate doesn't open; eggs will cram at the opening; machine cycle can't start	Actuator failure; no power signal	7	3	3	63	Give error message if gate doesn't happen	2
Flow sensors	Detect wheter egg has passed through system	No function	Machine doesn't detect egg; cycle will stop unexpectly because no egg detection	Sensor gets blocked	6	2	7	84	Have it in the service manual to clean the sensor periodically	2
Flip Actuator	Flips eggshell into eggshell storage	Intermittent function	Actuator does not full required range	Power goes below threshhold	7	3	2	42	Voltage Monitoring System on Touch Screen	2
Spear Solenoid	Drives the wedge apart	Under function	Egg shell is not opened properly; egg shell may enter egg white and yolk	Desynchronization with wedge and platform solenoid so the wedge isnt spread apart to open egg shell properly	6	3	4	72	Synchronization through checks within the logic code and flagging desynchronization	2
Return spring	Returns spear mechanism to starting position	No function	Mechanism stays open or stuck, next egg may jam or release incorrectly	Spring fatigue; corrosion	8	2	7	112	Long-term service checks	1
Platform + Wedge Solenoid	Moves front/back to release egg	No function	Platform doesn't move into correct position; egg may drop incorrectly or machine cycle stops	Solenoid plunger jammed	7	3	6	126	Add sealed food-safe solenoid; position senor	2
Thermal Insulation	Reduces heat; transfers to outher shell	Under function	Outer shell becomes too hot; electronics may overheat	Insulation too thin; poor ventilation	8	3	6	144	Use higher-rated insulation	1

. Internal

	Prevents steam from affecting internal parts	Unintended Function/Malfunction	Moisture build up inside insulation area; insulation performance decreases; condensation may form near electronic parts	Steam from cooking trapped inside housing; insulation absorbs moisture	8	3	7	168	Use moisture-resistant insulation, add ventilation	1
Axial Fan	cools internal electronics	No function	Fan stops, electronics overheat; machine shuts down; electrical components can fail; steam or smoke can come out of machine	Fan motor failure; dust/grease buildup; poor power supply	9	3	2	54	Add temperature sensor, protective filter, make fan/filter area easy to clean	1
	removes heat from enclosure	Under function	Hot air is not removed efficiently; heat builds up inside machine; outer shell becomes warmer; cooking performance reduces	Fan too small; poor fan placement; dust/grease buildup	6	5	5	150	Improve fan placement; make fan/filter area easy to clean	2
Ventilation Grille	allows airflow while blocking debris	Under function	airflow is restricted; machine overheats; cooking performance decreases	Grease buildup; blocked by user placement	7	4	6	168	Put machine in an open area; add "do not block vents" label	2
Vent Duct Channel	Guides hot air put of machine	Under function	Hot air isn't removed properly; heat builds up in machine	Duct too narrow; grease blockage	7	4	6	168	Smooth duct path; increase duct size	2
Step down converter	Converts 24V DC to low-voltage DC power	Over function	Current is too high; electronics fail; solenoid overheat; machine stops working	Wrong converter rating; poor converter quality	7	2	4	56	Use certified converter and high current protection	2
	provides stable and safe electrical power to the system	Unintended function/ malfunction	PSU provides unstable power, electrical noise, short circuit: system becomes unsafe	Overheating	7	2	4	56	Check fuse for easy detection	2
Solenoid Driver	Controls current to solenoid at the correct time	Unintended function	It activates at wrong time or stays on; egg mechanism moves incorrectly	software error; short circuit	7	3	7	147	Add position feedback by using an encoder	2
Oil System										
Oil Tank	Stores oil	Unintended function	Oil leaks into machine; causes mess and cleaning delay, oil might leak onto hot surfaces	Overfilling	4	3	2	24	fill-level marking	3
Peristaltic Pump	Pumps oil from tank to noozle	No function	No oil flow; conveyer stays dry; egg may stick or burn	Defective pump; pump might be jammed	7	3	3	63	No function	2

Internal

		Under function	Too little oil is pumped and sprayed; egg may stick or burn, cooking quality reduces	Pump wear; low voltage; tube compression loss	6	4	4	96	Test the pump flow rate during assembly, controll run time through controller	2
		Over function	Too much oil is pumped and sprayed; too much oil on conveyor; egg will be too greasy and there's a risk of smoke	Pump runs too long; wrong software timing	8	4	2	64	Calibrate spray time	1
Inline Check Valve	Prevents backflow and dripping when pump stops	Unintended function / malfunction	Oil drips; conveyer becomes too oily; causes mess or smoke	Valve stuck open; oil residue buildup	4	4	4	64	Use a food-grade check valve	3
Silicone Tubing	Transfers oil between tank, pump, and noozle using flexible food-grade tubing	Under function	Reduced or stopped oil flow completly; uneven spraying, oil could leak into machine	Kinked tube; loose connection; heat damage	4	4	6	96	Use food-grade high-temperature tubing, secure it with clamps, avoid sharp bends	3
Flat Fan Spray Nozzle	Distributes oil evenly	No function	No oil reaches conveyor; egg may stick or burn	Noozle blocked; dried oil residue	5	5	5	125	Make the nozzle removable for cleaning	2
		Under function	Distributes oil unevenly or less then intended; some areas get too much oil while others too little, reduces cooking quality	Nozzle clogging	4	5	5	100	Make the nozzle removable for cleaning	3
MOSFET Driver Module	Controls pump operation electronically from controlsystem	No function	Pump doesn't run; no oil is sprayed onto conveyor	Defective MOSFET, no control signal	8	2	9	144	Get a verefied MOSFET	2
Screen System										
Touch Controller	Detects and processes user touch input	No function	User cannot operate the product; device will appear frozen	MCU failure	8	3	2	48	Use Power-on self-test and touch input check	2
	Sends selected command to the microcontroller	Intermittent function	Touch is sometimes detected; selection may fail or needs repeated touching	Unstable voltage; poor calibration, loose connector	4	4	2	32	Have firm connection	3
Display Interface	Transfers display data and control signal to TFT LCD Display	No function	No display data is transferred; screen remains blank or frozen	Broken interface line, wrong pin connection, missing clock/data signal, connector fault	6	4	3	72	No action	2
TFT LCD Display	Displays graphical user interface and system information to user	No function	Blank screen; user can't see menu; machine won't work	Defective display; damaged cable; broken by force; dead pixels	7	4	1	28	Periodic pixel checks	2

. Internal

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Backlight Driver	Controls current supplied to the display backlight LEDs	No function	Display is dark or unreadable; user cannot clearly see menu	Driver IC defect, broken LED line	4	3	3	36	Check backlight power supply	3
	Limits backlight	Over function	Backlight too bright; higher	Output current too high,	2	3	2	12	Get a verified	3
Display MCU	Processes display data and controls the graphical user interface	No function	Display does not show menu, cooking time, temperature, or warning messages; user cannot operate or monitor the machine correctly	Defective MCU, code crashes	7	3	3	63	Test the MCU periodically in the surface	1
Transportation System										
Driver for Stepper Motor	Controls function of stepper motor (Rotation and direction)	No function	Motor may not move/ get stuck, vibration without rotation	Insufficient voltage, inconsistent wiring	5	4	2	40	Verifying wiring polarity and coil connections	2
		Overfunction	Extra unintended steps, excessive heating, overtorque	Inadequate cooling, high voltage inducing spikes/resonance	6	3	1	18	Monitor LEDS, auto idle reduction if overheating occurs	2
Stepper Motor	Stepper motor advances conveyor belt to position egg plates precisely under heaters for timed cooking cycles.	No function	Complete belt stoppage; Upstream egg plates backup into loading zone; Overflow spills onto kitchen floor	Bearing seizure from grease dry-out; Coil windings shorted/burnt from overcurrent; Shaft key shear from belt jam torque	9	4	3	108	Add current signature monitoring for bearing wear detection, avoid resonance speeds	1
		Under function	Plates stop short of heater zone; Eggs partially cooked/misaligned Home sensor false trips repeatedly	Resonance, Acceleration ramp could be too aggressive	7	6	5	210	Install Auto-tensioners, encoders for slip detection	2
		Overfunction	Motor stalls with burning odor; Holding torque lost suddenly; Conveyor freewheels backward;	Stall current from continuous duty; No airflow; Conduction from nearby heat block	9	5	2	90	Installing a heatsink/fan, temperature monitor	1
Roller Chain	Transmit mechanical power and synchronized motion from the drive motor to the conveyor bed	No function	Catastrophic sprocket disengagement; Egg plates crash into heater block; Chain debris contaminates eggs	Fatigue, overload shock, Sprocket tooth shear transfers load spike;	10	3	3	90	Using an oversized chain, shock absorbers on the belt	1
		Under function	Slip/jam, product misalignment, Rotation part stops	Shaft bending; wear; lack of lubrication	7	3	6	126	Use strong corrosion-resistant shaft; add bearing support; use a stainless chain (15k hour life)	2

. Internal

		Overfunction	Binding/ Heat buildup; Sprocket galling locks system; Friction heat chars nearby eggs; Chain welds to sprocket	Excess lubricant, misalignment	7	4	4	112	Precision sockets, specified lubricant	2
		Unintended function	Vibration/noise, socket damage	Loose fit, imbalance	5	5	6	150	Chain guides, retainers	3
		Intermittent function	Erratic speed, intermittent slips	Wear debris, contamination	6	5	5	150	Improved seals/filters	2
Plates on Conveyer	Trasports the eggs while they cook on the belt	No function	Raw eggs/ contamination	Heater burnout, poor contact	9	4	3	108	Redundant heaters, fault alarm system	1
		Under function	Undercooked/sticky eggs	Thermal warp, uneven heating	8	5	4	160	Aluminum core for conduction, insulation below	2
		Overfunction	Burnt/charred eggs, fire risk	Overcooked, stuck eggs	10	4	3	120	PID temp control, overheat fuse	1
Home Centering Sensor	Detects egg presence and centering in plates to halt conveyor if misaligned.	No function	Empty conveyor plates proceed, eggs may not fall direclty onto the plate	Dirt blockage, emitter could fail	8	5	4	160	Cleaning with air blasts, redundant sensors	2
		Under function	Unnecessary stops, downtime	Ambient light interference, misalignment	6	6	5	180	IR filter lens, adjustable threshold	3
		Over function	Off-center eggs proceed, breakage	Reflective residue, range overshoot	8	4	4	128	Dual-beam verification, anti-reflect coating	2
End position sensor	IR end position sensor detects last plate passage to trigger conveyor reset for first plate home positioning.	No function	Conveyor never resets, jam	Dust block, LED fail	9	5	4	180	Auto clean sensor, redundant IR	1
		Under function	Premature reset, position error	Ambient light, misalignment	5	6	5	150	IR filter, auto-calibrate	3
		Over function	Skips reset, endless cycle	Reflective dust, range error	8	4	4	128	Polarized lens, dual check	2
Egg-Shell Storage System										
Main waste drawer tray	Removable stainless steel drawer used for shell waste collection	No function	Egg shells are not collected; shells may fall into machine	Drawer not inserted correctly	7	3	2	42	Improve drawer positioning and add clear insertion guide	2
Linear Ball Bearings / Bushings	Supports smooth sliding motion of guide rods	No function	Pusher mechanism cannot move; egg shells are not compressed; drawer fills too early	Bearing seized; dirt/shell fragments inside bearing	4	3	3	36	Check sliding movement during assembly	3

. Internal

Return Springs	Returns pusher mechanism to default position after operation	No function	Pusher stays in compressed position; new shells may not enter correctly; drawer may jam	Broken spring	6	3	3	54	Replace spring	2
		Under function	Pusher returns slowly or only partly; shell compression mechanism may block next cycle	Spring fatigue	4	4	4	64	Use spring with high lifetime margin and test return force	3
Bottom Heating System										
Temperature Controller	Regulates heating temperature using feedback control	No function	Raw/safe eggs, contamination	Power fail, thermocouple disconnect	9	4	3	108	Watchdog timer, backup controller	1
		Under Function	Undercooked eggs, bacteria risk	Sensor drift, poor tuning	8	5	4	160	Calibrated RTD/thermocouple, dual sensors	2
		Over Function	Burnt eggs, fire hazard	Heater stuck-on, setpoint error	10	4	3	120	Independent safety cutoff, thermal fuse	1
Thermal fuse	Provides over-temperature protection for the heating system	No function	Uncontrolled overheat, fire	Defective fuse, wrong rating	10	3	2	60	High-reliability rated fuse, testing of fuse	1
		Under function	Frequent shutdowns, downtime	Poor thermal coupling to block	6	5	3	90	Optimized block embed position	2
		Overfunction	Excessive block temp before cutoff; causes extreme heating issues	High thermal mass lag between actual overheating and detection by fuse	9	4	3	108	Dual fuses, closer to cartridges	1
Solid State Relay	Switches cartridge heater power	No function (Off)	No heat, raw eggs	Input signal loss,	7	4	4	112	Dual SSR parallel, signal watchdog	3
		Overfunction	Overheat/burn, fire	Fail-short from surge	8	3	2	48	Included a Thermal fuse for safety	2
Cartridge Heaters	Tube-shaped, heavy-duty, industrial Joule heating element	No function	Cold block, raw eggs	Coil burnout, vibration wire break	8	4	4	128	Redundant cartridges, vibration damp	2
		Under function	Slow/uneven cooking	MgO moisture ingress, partial short	6	5	4	120	Tight bore fit (H7/g6), moisture seal	3
		Over function	Local burning, fire risk, hot spots	Poor fit/voids, overload	10	3	3	90	Multi-zone wiring, thermocouples	1
Aluminium Block	Conductive block with drilled heater/sensor holes	No function	Hot spot failure in the block, uneven cook	Drill stress riser, thermal shock	7	3	3	63	Ultrasonic test, pilot drills	2
		Under function	Poor thermal contact, slow heat	Oversize holes because thermal expansion	7	5	4	140	Precision ream H7, thermal compound	2
		Over function	Block edge overheating	Oversize holes because thermal expansion	6	4	5	120	Optimize hole spacing/depth	2
Temperature Sensor	Used to check the temperature of the aluminium block	No function	No temp feedback, overheat risk	Wire break, connector fails	9	4	3	108	Dual sensors voting, continuity monitor	1
		Under function	Undercooking from false cold	Sensor drift, poor thermal contact	7	5	4	140	Spring-loaded sensor, annual calibration	2

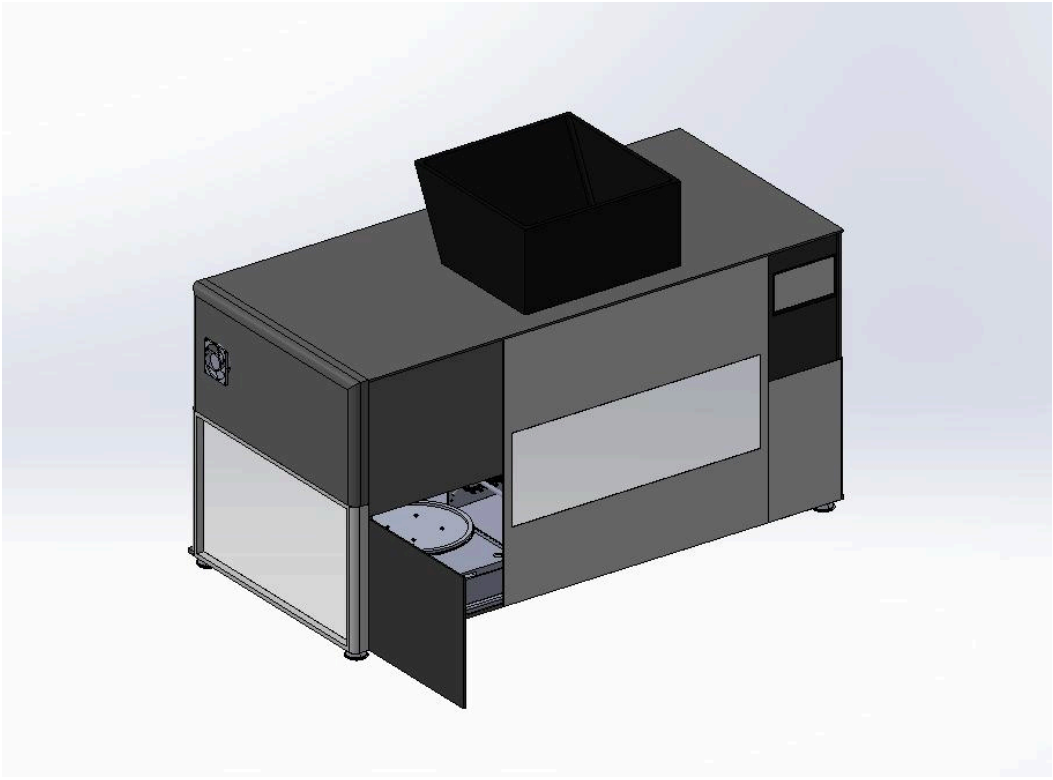
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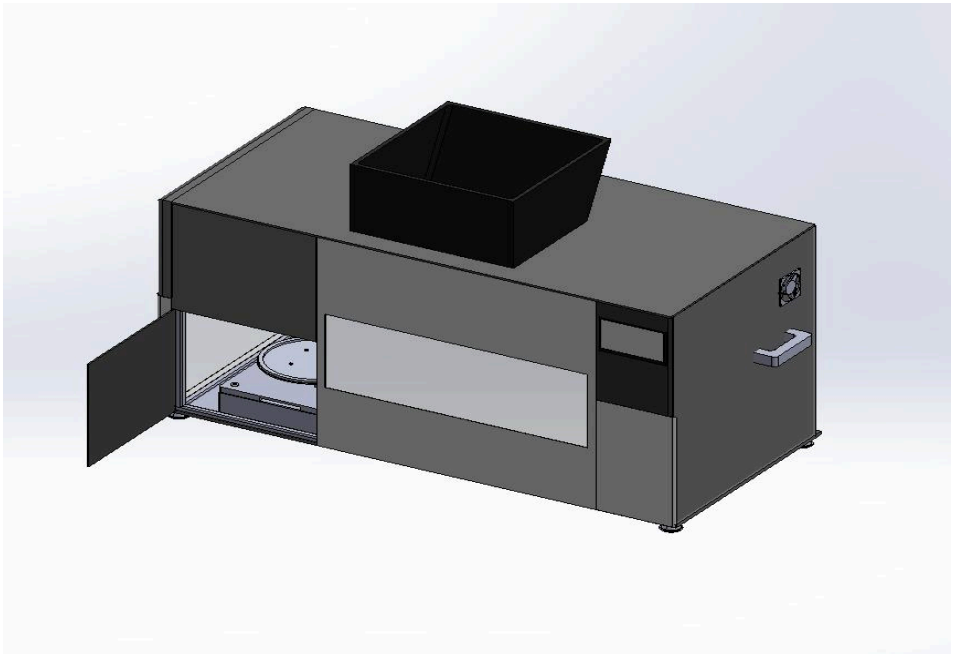
		Over function	Overcooling, wasted energy	Insulation short, moisture ingress	4	4	4	64	Moisture-proof connector, dielectric test	3
TOP IR Heating System										
IR Heating element	400–450 mm shielded IR heater, 300–600 W class, heats eggs from above	No function	Undercooked tops, raw yolks	Filament burn, power loss	6	4	4	96	Redundant lamps, auto-switch	3
		Under function	Inconsistent cooking	Element aging, voltage drop	4	5	4	80	Voltage regulator, spare element	3
		Over function	Scorched tops, smoke/fire	Overdrive power, reflector focus	9	5	3	135	Current limiter, IR sensor	1
Rotating Plate System										
Stepper Driver Module	Controls stepping sequence of motor	No function	Driver doesn't work; Motor won't work; Rubber wheel doesn't move; Plate won't rotate; eggs will overlap and damage the yolk	Defective driver, no power supply, driver overheating	4	4	2	48	Check power/enable signal	3
		Under function	Motor receives weak/incorrect signal; plate moves too slowly; eggs may overlap and damage the yolk	Motor current too low, wrong step setting, mechanical friction, overload from plate	3	4	2	48	Verify driver current limit	3
Stepper Motor	Drives rotation of plate via friction mechanism	No function	Motor doesn't work; Rubber wheel doesn't work; Plate does not rotate; eggs will overlap and damage the yolk	Defective motor, broken winding, no motor power	4	4	4	96	Verify motor power	3
		Under function	Motor torque is too low; rubber wheel slips or moves slowly; Plate does not reach correct position	Low motor torque, poor friction contact, mechanical wear, overloaded plate	6	4	4	96	Improve friction contact and test motor torque under load	3
Rubber Wheel	Transfers motion to plate through friction contact	No function	Rubber wheel doesn't transfer motion; plate doesn't rotate; eggs may overlap and damage the yolk	Wheel missing, broken, loose, no contact with plate	2	4	1	8	Long-term service checks	3
Microswitch	Detects home position (0° reference)	No function	Plate position is not detected; system may drop egg at wrong plate position	Defective microswitch, broken wire	7	3	3	63	Verify switch position	2
		Intermittent function	Plate position signal is unreliable; plate may stop at wrong position	Defective microswitch, broken wire, poor mounting	7	4	4	112	Improve switch mounting	2
Base Plate	Structural base supporting entire mechanism	No function	Plate does not rotate; eggs fall on top of each other; yolk may be damaged	Plate jammed, blocked bearing, plate not aligned	6	3	2	36	Verify plate alignment	3

. Internal

Milestone 2

7. Complete 3D Model (CAD)[20]





[20] Dassault Systèmes. SOLIDWORKS. *SOLIDWORKS*. https://www.solidworks.com

In the final assembly model, the custom printed circuit boards were not represented with detailed electronic components such as individual resistors, capacitors, ICs, terminals, and copper traces. Instead, simplified PCB models were created using the estimated overall dimensions of each board.

This approach allowed the assembly to accurately represent the required volume, mounting position, and space allocation inside the electrical box without overcomplicating the CAD model. Each PCB was modelled as a rectangular board with its approximate size and thickness, including clearance for mounting on M3 standoffs.

The placement of the Power Regulation PCB, Actuator and Sensor Interface PCB, Motion and Load Cell PCB, and Temperature Measurement PCB was arranged to ensure that enough space was available for wiring, airflow, connectors, and nearby electrical components.

8. Electric Drive

8.1. Drive 1 - Conveyor (NEMA17 stepper)

8.1.1. Description of movement/task and estimated load

The conveyor drive moves two parallel 06B-1-K1 roller chains carrying four egg pocket plates around a closed loop (top straight 666.51 mm, bottom return 666.51 mm, two 180° wraps around 18T sprockets). Movement is unidirectional, indexed motion: each plate must dwell at the egg-breaking station, frying zone, and ejection point in turn. Target linear speed is 30 mm/s, giving ≈ 22.2 s travel over a single 666.51 mm straight.

Load m = 3.27 kg

The 3.27 kg total linear moving mass is taken from CAD mass properties in SolidWorks with AISI 316 stainless steel assigned to all custom parts, plus catalogued supplier mass for the 06B-1-K1 chain. It is the sum of all parts that translate with the chain:

Part	Qty	Mass each	Total
Egg pocket plate (AISI 316 sheet)	4	383.80 g	1.5352 kg
Egg payload (assumed 60 g per egg, max load)	4	60.00 g	0.2400 kg
Chain side assembly incl. connecting links (06B-1-K1)	2	724.84 g	1.4497 kg
SmCo magnets (Ø8 × 3 mm, ρ ≈ 8.3 g/cm³)	4	1.25 g	0.0050 kg
Plate-to-chain fastener sets (A4 M3 screw + nut + washers)	16	1.25 g	0.0200 kg
Retained food-grade lubricant film (allowance)	—	—	0.0200 kg
Total linear moving mass			**3.2699 kg ≈ 3.27 kg**

Table 23 : Conveyor System Load Split.

8.1.2. Summary of technical requirements

Requirement	Value
Motion type	Continuous unidirectional with indexed dwell
Target speed	30 mm/s (10.5 rpm at sprocket, 283.5 rpm at motor)
Speed range	0 → 30 mm/s (slow start, dwell-capable, no reverse needed)
Acceleration	0.030 m/s² (1 s ramp; very gentle to avoid egg slip)
Required output torque (with SF 2)	0.50 N·m at sprocket shaft
Required motor torque	≥ 25 mN·m at 283.5 rpm
Positioning accuracy needed	≤ 1 mm at pocket (alignment with breaker, fryer, ejector)
Resolution required	171.45 mm/rev / N_steps ≤ 1 mm → ≥ 172 steps/rev at output
Control mode	Position control (indexed open-loop stepping with home reference)
Homing	One-shot at power-up via Hall sensor on idler shaft (existing rotating-plate sensor approach reused)
Duty cycle	Continuous during cook cycle, intermittent overall
Ambient at motor location	≤ 50 °C (motor must sit outside heated zone — verified by 1.5 in input table)
Direction reversibility	Not required (one-way conveyor)

Table 24 : Conveyor System Technical Requirements.

8.1.3. Type of motor and reasons for specific choice

Choice: 2-phase hybrid stepper motor, open-loop position control.

Reasoning, in order of weight:

1. **Indexed positioning is the core requirement.** The conveyor must repeatably stop pockets under the egg breaker, then under the heating zone, then over the ejector. A stepper gives intrinsic, gearbox-multiplied positional resolution (200 × 27 = 5400 output steps/rev → 0.032 mm/step at the plate) with no encoder. A DC motor would need an encoder + closed-loop controller for the same job.
2. **Load is light and predictable.** Required motor torque (25 mN·m) is ~3 % of holding torque far inside the safe operating envelope where stepper torque loss at speed and missed-step risk are not concerns. There is no risk of stall under normal operation.
3. **Low speed is native territory for steppers.** 283.5 rpm is squarely in the high-torque band of a NEMA 17. A brushless DC motor would need significant gear reduction added on top of an already-coarse reduction; a brushed DC would suffer at this low speed with cogging and brush wear.
4. **Inertia ratio (0.31) is excellent,** well below the 5:1 comfort threshold for steppers, so acceleration/deceleration is crisp and resonance-free.
5. **Single-direction, no regen,** no need for a 4-quadrant drive. Stepper drivers are the cheapest solution for this profile.
6. **Project consistency.** The rotating plate subsystem already uses a stepper (28BYJ-48) and the same STM32 step/dir control firmware can be extended. One driver family across subsystems simplifies BOM, firmware, and diagnostics.

Why not alternatives:

- Brushed DC + encoder + PID: more firmware effort, encoder cost, and brush maintenance in a food/heat environment.
- BLDC + encoder: overkill for 25 mN·m at 283 rpm; controller cost and complexity not justified.
- Servo: same as BLDC; no advantage at this load and speed.
- AC induction with VFD: minimum practical motor sizes are far too large; positioning resolution is poor without an encode.

8.1.4. Motor choice and corresponding driver

Motor: NEMA 17 hybrid stepper — AM17HDB410-02N

Parameter	Value	Margin vs. requirement
Holding torque	0.85 N·m	34× the 25 mN·m needed
Phase current	1.8 A/phase	Sets driver minimum rating
Rotor inertia	123 g·cm²	Load ratio 0.31 — good
Shaft	Double (10 mm bore gearbox + free end available for optional encoder)	Future-proofs closed-loop upgrade
Mass	0.6 kg	Fixed mount, not relevant
Ambient rating	−20 to +50 °C	Requires mounting outside heated zone

Table 25 : NEMA 17 Motor Requirements.

Driver: TB6600 external stepper driver module.

Parameter	Value	Justification
Phase current	0.5 – 4.0 A peak, DIP-selectable	Set to 2.0 A peak (≈ 1.4 A RMS) for the 1.8 A/phase motor — full nameplate torque with thermal margin
Microstepping	1/1, 1/2, 1/4, 1/8, 1/16, 1/32 (DIP-selectable)	Use 1/8 → 1600 steps/rev × 27 = 43 200 steps/rev output → 0.004 mm/step at plate (resolution is not the binding constraint)
Supply voltage	9 – 42 V DC	Powered from the 24 V rail in your power tree
Logic inputs	STEP / DIR / EN, opto-isolated	3.3 V STM32 GPIO drives the opto LEDs through a series resistor (typical 270 Ω for 3.3 V logic)
Pulse frequency	Up to ~200 kHz	At 283.5 rpm motor × 1600 steps/rev = 7.56 kHz — well inside the linear range
Enclosure	Aluminium heatsink case, DIN-rail / panel mount	Mounts off-PCB on the control panel; survives ambient heat better than a chip-on-board solution

Table 26 : NEMA 17 Driver TB6600.

8.2. Drive 2 - Rotating Plate (28BYJ-48 stepper)

8.2.1. Description of Movement and Estimated Load

Movement & Task: The drive rotates a Ø250 mm stainless-steel plate that receives fried eggs from a dispenser located above a fixed dispensing point. The plate starts at a homed 0° position; egg 1 is deposited, then the plate indexes 90° and waits for egg 2, and so on for a total of four eggs spaced 90° apart around the plate, with 15 s between dispense events. Motion is purely rotational about the vertical axis, supported by a 4-inch (Ø102 mm) lazy-Susan bearing. Power is transmitted from the motor to the plate by a Ø38 mm rubber wheel (25.4 mm inner bore) coupled to the motor shaft and pressed against the outer edge of the plate (friction drive). A lever microswitch provides the absolute 0° home reference at start-up; an HX711 + load cell under the assembly detects the arrival of each egg.

Motion pattern: Three discrete index moves of 90° each (between eggs 1→2, 2→3, 3→4), separated by 15 s windows that include both the move and a dwell during which the next egg is dispensed and lands. After egg 4, the plate has rotated 270° from home; a final 90° (or reverse 270°) returns it to 0°.

Mass breakdown:

Rotating components :

- Stainless steel plate (Ø250 × 3 mm, 6061-T6 aluminium): **0.716 kg**
- Lazy-Susan bearing top race (rotating half of 92 g total for the bearing pair): ≈ 0.046 kg
- Rotating-side screws + centering ring (estimated half of the 0.2 kg total fastener allowance): ≈ 0.1 kg
- Dish (900 g) + 4 eggs ($4 \times 60g = 240g$): 1.14 kg (worst case, after the 4th egg lands)
- **Total rotating mass (worst case): ≈ 2.0 kg**

Weight on the bearing:

$$W = 2.0 \cdot 9.81 \approx 19.6N$$

Polar moment of inertia (worst case, treated as a uniform disk of r = 0.125 m):

$$J \approx \frac{1}{2} \cdot m \cdot r^2 = 0.5 \cdot 2.0 \cdot 0.125^2 \approx 0.016kg \cdot m^2$$

Load torque at the plate:

4-inch lazy-Susan friction (μ_eff ≈ 0.04 at race radius ≈ 0.040 m):

$$T_f \approx 0.04 \cdot 19.6 \cdot 0.040 \approx 0.031N \cdot m$$

Acceleration torque for a gentle ramp (0 → 1.5 RPM in 1 s, α ≈ 0.16 rad/s²):

T_a = J · α ≈ 0.016 · 0.16 ≈ 0.003 N·m (negligible)

Required torque at the plate (with margin): ~0.1 N·m

Transmission: The Ø38 mm wheel is coupled rigidly to the motor shaft via the coupler, they rotate 1:1. The Ø38 mm wheel driving the Ø250 mm plate edge is therefore a single-stage friction reduction of $i_friction = 250/38 \approx 6.58 : 1$ (speed down, torque up at the plate). One motor revolution rolls $pi \cdot 38 \approx 119mm$ of arc on the plate edge against the plate circumference $pi \cdot 250 \approx 785mm$. Referred to the motor shaft:

$T_motor \approx 0.1/6.58 \approx 0.015N \cdot m \approx 15mN \cdot m$

8.2.2. Summary of Technical Requirements

Requirement	Value
Rotating mass (worst case)	≈ 2.0 kg
Weight on bearing	≈ 19.6 N
Required torque at the plate (incl. margin)	≥ 0.1 N·m
Required torque at motor shaft (after 6.58:1 reduction)	≥ 15 mN·m
Motion profile	Indexed: 90° step, then dwell
Cycle time (move + dwell)	15 s per egg

Move duration / dwell split (chosen)	~10 s move, ~5 s dwell
Plate speed during move	~1.5 RPM (9 °/s)
Motor shaft speed during move	~9.9 RPM
Step rate to driver (half-step)	~673 Hz
Half-steps per 90° plate move	~6738
Total travel across 4 eggs	270° (0° → 90° → 180° → 270°)
Positioning accuracy	≤ 1° at the plate (dispense alignment)
Control type	Open-loop position control with absolute home reference (microswitch)
Holding during dwell	hold position against egg-drop impulse (~few N·s)
Direction	Bidirectional (forward indexing; reverse for re-home)
Duty cycle	Intermittent — short moves, short dwells, no continuous running

Supply	5 V DC (already available on the board as +5V_FUSED)
Logic interface	4 digital lines from MCU (PLATE_IN1…IN4)
Environment	Indoor, low vibration, no high temperature

Table 27 : Rotating Plate Technical Requirements.

8.2.3. Type of Motor - Choice and Reasoning

Stepper motor (open-loop). Gives discrete angular steps directly from a step count, holds position when energised, no feedback loop required, and matches the slow / low-torque / indexed-with-dwell profile of this task exactly.

Reasons:

- **Positioning without an encoder.** After homing on the microswitch at start-up, the MCU knows the absolute angle simply by counting steps. There's no back-drive path under normal use to cause step loss.
- **Native indexed-then-dwell behaviour.** A stepper holds its commanded position with the coils energised (and with significant detent torque even when de-energised, thanks to the internal gearbox). This is exactly what the task asks for: move 90°, stop, wait, repeat.
- **Speed range matches the task.** The plate only needs ~1.5 RPM during the move. Steppers are at their best in the low-speed region where DC motors would need heavy gearing.
- **Torque margin.** Required motor torque (~15 mN·m) is far below the holding torque of common small steppers, so there's no risk of stalling or losing steps under the egg-drop impulse.
- **Cost and integration.** A small unipolar stepper + 4-channel Darlington driver is the cheapest electronic drive that meets the requirements.

8.2.4. Motor and Driver Selection

Motor: 28BYJ-48, 5 V geared unipolar stepper.

Parameter	Value
Rated voltage	5 V DC

Phases	4 (unipolar, 5 wires)
Internal gearbox	~1 : 64
Step angle at output shaft (half-step)	$5.625^\circ/64 \approx 0.088^\circ \rightarrow$ 4096 half-steps/rev
Holding torque (output shaft)	≈ 0.3 N·m
Typical no-load speed	~15 RPM at output shaft
Current per phase	~150–200 mA at 5 V

Table 28 : 28BYJ-48 Motor Requirements.

Sanity check against the new geometry:

- **Torque:** $0.3N \cdot m \times 6.58 \approx \ 2.0N \cdot m$ **at the plate**, against ~0.1 N·m required → ~20× margin.
- **Speed:** 9.9 RPM required vs ~15 RPM available → ~1.5× margin (more comfortable than with the 8.33:1 reduction).
- **Resolution at the plate:** $0.088^\circ/6.58 \approx 0.013^\circ$ **per half-step**, far finer than the ≤1° target.
- **Load cell:** rotating + stationary above the load cell

  $\approx 2.0kg + \ 1.4kg = 3.4kg$ total. Within a 5 kg / ~49 N rated load cell with margin.

Driver: ULN2003A Darlington array, shown as U1 on the schematic.

- 7 open-collector Darlington channels rated 500 mA / 50 V each, comfortably above the ~200 mA per phase the 28BYJ-48 draws.
- Built-in flyback diodes to COM (tied to +5V_FUSED on pin 9) clamp the inductive kickback from the coils, which is essential for a unipolar stepper.
- Four inputs PLATE_IN1…IN4 from the MCU map directly to outputs O1…O4 driving PLATE_COIL1…COIL4 , The MCU outputs the step sequence (half-step pattern), the ULN2003A buffers and sinks the coil current.
- Only 4 of the 7 channels are used; the rest are tied off (I5…I7 crossed out, O5…O7 unused), matching the schematic.

8.3. Drive 3 - Oil System
8.3.1. Description of Movement and Estimated Load

Movement / task: The drive is a small peristaltic pump that delivers a controlled volume of cooking oil through a flat-fan nozzle onto each egg pocket on the rotating plate, before the egg is dispensed. There is **no positioned motion**, The pump simply needs to deliver a defined volume of fluid per dose. One dose = **2 mL**, and there are **four doses per customer** (one before each egg), giving **2-8 mL** per customer cycle.

Mechanism:

A peristaltic pump uses rollers on a rotor to compress flexible silicone tubing against a curved housing, advancing a slug of fluid with each rotation. The drive motor is a **small brushed DC gear motor** built into the pump head. Volume delivered is essentially the rotor's swept volume × revolutions, independent of back-pressure within the small range present here.

Hydraulic / mechanical load:

- **Oil:** cooking oil, ρ ≈ 920 kg/m³, viscosity ≈ 60–80 mPa·s at room temperature (falls quickly as it warms).
- Static head from tank to nozzle: a few tens of centimetres → only a few mbar of pressure drop. Negligible for the pump.
- **Tubing:** 4 × 6 mm food-grade silicone, short run (< 1 m). Pressure drop at the chosen flow rate is small, on the order of tens of mbar.
- **Nozzle:** 1/8", 65° flat-fan, flow rating ~0.1 (Spraying Systems "01" convention, ~0.4 L/min at 3 bar). At the very low pressures and flows used here, the nozzle is operating well below its rated point it will spray, but barely, which is exactly what's wanted to coat a small pocket.
- **Timing constraint (binding):** the oil must be in the pocket *before* the egg arrives. Egg-break to pocket landing $(t_prep) = 3.88s$, so the pump dispense + oil settle must fit inside that window with margin.
- **Required dispense volume per dose: 2 mL ± ~10%** (cooking oil dosing tolerates modest variation).
- **Required dispense time per dose: ~2.5 s** (leaves ~1.4 s of the 3.88 s window for oil flight, nozzle clearing, and settle in the pocket).
- **Required flow rate:** $2mL/2.5s = 0.8mL/s \approx 48mL/min.$ Comfortably inside the 30–100 mL/min range of a 12 V mini peristaltic pump.
- **Pump motor draw:** ~150–250 mA at 12 V (typical for this class), ~2–3 W input power.

8.3.2. Summary of Technical Requirements

Requirement	Value
Function	Volumetric dosing of cooking oil
Volume per dose	**2 mL**
Doses per customer cycle	**1- 4** (one before each egg)
Inter-dose interval	~15 s
Available window per dose (egg-break → pocket)	**3.88 s**
Dispense time per dose (chosen)	**~2.5 s**
Required flow rate	**~48 mL/min (≈0.8 mL/s)**
Margin within 3.88 s window	**~1.4 s** (oil settle + safety)
Volume repeatability	±10% acceptable
Control type	**Open-loop time-based dosing** (volume = flow rate × on-time)
Direction	Single-direction (no reverse needed)
Speed control	Optional PWM; simple on/off + timed pulse is sufficient
Back-flow prevention	Inline check valve keeps tubing primed between doses
Supply	12 V DC (already available on the board as +12V_FUSED)
Logic interface	1 digital line from MCU (PUMP_CTRL)
Duty cycle	Very intermittent — ~2.5 s on per dose, 1 - 4 doses spaced 15 s apart per customer, idle between customers

Environment	Inside machine enclosure; ambient ~30–45 °C; food-grade silicone in wetted path; no direct splash exposure

Table 29: Technical Requirements Table for Oil System

8.3.3. Type of Motor - Choice and Reasoning

From the morphological box, three oil-delivery options were carried forward to evaluation:

- **Gravity feed + solenoid valve.** Oil flows from an elevated tank through a solenoid valve when energised; dose volume = flow rate × open time. Electrically the simplest option (one solenoid, one MOSFET), and there's no motor at all. The problem is volumetric repeatability: flow rate depends on the head of oil in the tank (which drops as oil is used), on oil viscosity (which falls as the cooking surface warms the cabinet), and on any air bubbles in the line. Across a 600 mL tank the head changes by several centimetres, easily a 20–30% swing in flow, which exceeds our ±10% dose tolerance. The design also forces the tank to sit *above* the nozzle, which conflicts with the compact cabinet layout. Rejected on accuracy and packaging.
- **Diaphragm pump.** A reciprocating diaphragm driven by a small motor delivers oil in pulsed slugs. Self-priming, can handle higher pressures than a peristaltic, and tolerates dry running better. But the wetted path includes the diaphragm, inlet valve, and outlet valve all in contact with oil, all needing to be food-safe and serviceable. Small 12 V food-grade diaphragm pumps exist but cost noticeably more (~€15–25 vs ~€3), and the pulsed output makes precise 2 mL dosing harder than a continuous-flow peristaltic. The higher pressure capability is wasted here because we're spraying at near-zero back-pressure. Over-specified and more expensive without a corresponding benefit. Rejected.
- **Peristaltic pump.** Rollers on a rotor compress a flexible silicone tube against a curved housing, advancing a defined slug of fluid per revolution. Volume per second is set by motor speed and tube geometry, so timed open-loop control gives repeatable doses. **Chosen.**

Choice: Peristaltic pump with brushed DC gear motor, switched on for a calibrated dose time per egg.

Reasons:

1. **Repeatable volumetric dosing without feedback.** Per-revolution displacement is fixed by tube ID and rotor geometry, independent of tank level. At a constant supply voltage and constant on-time, dose-to-dose variation easily stays within the ±10% target something the gravity option cannot achieve.
2. **Food safety by construction.** Only the silicone tube touches the oil. The motor, gears, rotor, and rollers never contact the fluid path, and the tube is the single replaceable wear part. This is much simpler than the diaphragm pump, where multiple wetted parts (diaphragm + valves) all need to be food-grade and serviceable.
3. **No back-flow and no drip.** When the rollers stop, they occlude the tube and seal the fluid path. Combined with the inline check valve , there is no oil drip from the nozzle between doses a problem both gravity feed and a stopped diaphragm pump can suffer from.
4. **Self-priming, no air-lock issue.** The pump primes itself from an empty tube on first run and tolerates the occasional air bubble. Gravity feed by contrast becomes unpredictable in the presence of bubbles.
5. **Speed fits the 3.88 s prep window.** At full 12 V the pump delivers ~48 mL/min exactly enough to complete a 2 mL dose in ~2.5 s and leave ~1.4 s of settle time before the egg lands. The diaphragm pump's pulsed delivery would meet the average flow but make precise timing of the *end* of the dose less clean.
6. **Simple drive electronics.** A brushed DC motor switched on for a fixed time delivers the right dose one MOSFET as a low-side switch is all that's required. No H-bridge (no reverse), no microstepping driver, no encoder.
7. **Cost and packaging.** The complete 12 V mini peristaltic pump including motor, gearbox, rotor, rollers, and tube section is significantly cheaper than the diaphragm pump alternative, and only marginally more than the solenoid option (which would have lost on accuracy regardless of price).

8.3.4. Motor (Pump) and Driver Selection

Pump: 12 V mini Peristaltic pump , shown as Oil_Pump1 on the schematic.

Parameter	Typical value for this class
Rated voltage	12 V DC
Motor type	Brushed DC + built-in gearbox
Typical flow rate	30–100 mL/min (depends on tube ID and head speed)
Running current	~150–250 mA

Stall current	~400–500 mA
Tube fitting	4 mm ID silicone
Self-priming	Yes
Back-flow when idle	Negligible (rollers occlude the tube), reinforced by check valve

Table 30: Pump Selection Table

Driver: IRLZ44N logic-level N- channel MOSFET as a low-side switch (BOM #7), shown as Q_PUMP1 on the schematic.

Parameter	Value	Relevance
VDS	55 V	>> 12 V supply, large margin
ID continuous	47 A	>> 0.5 A pump stall current, vastly oversized
RDS(on) at VGS = 5 V	~22 mΩ	Logic-level — switches fully on from a 3.3 V or 5 V MCU pin
Power dissipation at 250 mA	$I^2 \cdot RDS(on) \approx 1.4mW$	Negligible — no heatsink required
Switching speed	Fast	Slowed deliberately by R4 = 100 Ω gate resistor to limit EMI

Table 31: Driver Selection Table

8.4. Drive 4 - Fans

8.4.1. Description of Movement and Estimated Load

Movement / task. Two 80 mm axial fans provide continuous airflow through the machine cabinet: **FAN1 (intake)** draws cool ambient air in on one side, and **FAN2 (exhaust)** pushes warm air, water vapour, and fine oil mist out on the opposite side. The push–pull arrangement ensures a defined airflow direction across the cooking area, avoids dead zones, and keeps the cabinet at slight positive or balanced pressure rather than letting fumes leak through unsealed seams. There is no positioning, no speed regulation, and no on/off cycling — both fans run continuously whenever the machine is powered, as the schematic title makes explicit ("12V Always-On").

Purpose of the airflow.

- **Heat removal.** The frying surface dissipates heat into the cabinet during operation. The cabinet air must carry this away to keep ambient near room temperature.
- **Fume and oil-mist extraction.** Frying releases water vapour and a fine aerosol of cooking oil; without forced exhaust these would condense on the touchscreen, electronics, and load cell.
- **Electronics cooling.** The components all benefit from a moving air stream, even though none of them individually requires forced cooling.

8.4.2. Summary of Technical Requirements

Requirement	Value
Function	Continuous forced ventilation of machine cabinet
Heat load to remove from cabinet air	~100 W
Required net airflow through cabinet	~30 m³/h

Airflow per fan (free-air)	≥ 50 m^3/h
Static pressure capability	~1 mm H_2O against louvres / grease screen
Configuration	Intake + Exhaust (push–pull, opposite sides)
Speed control	**None — always on** with machine power
Control type	Uncontrolled DC supply
Direction	Single-direction axial flow per fan
Duty cycle	100% (continuous whenever +12V_FUSED is present)
Service life	≥ 30,000 h (matches expected machine life)
Acoustic noise	< 30 dB(A) at 1 m (kitchen-equipment-appropriate)
Supply	12 V DC (already available on the board as +12V_FUSED)
Logic interface	**None** — direct connection to +12V_FUSED and GND

Environment	Enclosed inside machine cabinet; ambient ~30–45 °C near the cooking surface; dust and oil-mist exposure on the exhaust fan

Table 31: Technical Requirements Table for Fans

8.4.3. Type of Motor - Choice and Reasoning

From the morphological box, three ventilation options were carried forward to evaluation:

- **Passive ventilation.** Relies on natural convection through louvres or vents — hot air rises and escapes through the top, cool air enters through the bottom. Requires no electrical components, no power consumption, and has no moving parts to fail. The problem is throughput: natural convection in a small cabinet typically delivers only a few m³/h, far below our ~30 m³/h heat-removal requirement, and it provides no defined direction for fume extraction. Hot air and oil mist would pool near the top of the cabinet and condense on the touchscreen and electronics. Acceptable only for waste-heat levels around ~10–20 W, which is well below our ~100 W load. Rejected on capacity.
- **Centrifugal blower fan (12 V).** A radial-impeller blower delivers high static pressure (~10 mm H_2O) at moderate flow. Excellent when air must be forced through long ducts, fine filters, or restrictive grease screens. In our cabinet the ventilation paths are short and open, so the extra pressure capability is wasted. Blowers are also physically larger, noisier at equivalent flow (the radial impeller creates more tonal noise than an axial), and significantly more expensive (~€15–25 vs ~€3–5 for an axial). The intake/exhaust pair would also be harder to package because of the 90° flow turn each blower introduces. Over-specified and worse on cost, noise, and packaging without a corresponding benefit. Rejected.
- **Axial fan with brushless DC motor (12 V).** Compact, quiet, long-life, runs directly off a 12 V rail, includes all commutation electronics inside the rotor hub so no external driver is required. Delivers exactly the right combination of high volume flow at low static pressure. **Chosen.**

Choice: 80 mm brushless DC (BLDC) axial fan, 12 V, two units in intake/exhaust configuration.

Reasons:

1. **Matches the operating point.** Cabinet needs high flow at low back-pressure — the sweet spot for axial fans. Passive can't reach the required flow; centrifugal is over-spec'd.

2. **Capacity with redundancy.** One fan already meets the ~30 m³/h target; two in intake/exhaust add directional fume control and graceful degradation if one fails.
3. **No external driver.** BLDC commutation is built into the fan hub — just +12 V and GND, no MOSFET, no PWM, no MCU pin.
4. **Long, maintenance-free life.** Ball-bearing BLDC fans rated 40,000+ hours (~4.5 years continuous). Brushed DC or AC alternatives wear out faster or draw more power.
5. **Quiet.** ~20–25 dB(A) at 1 m, below kitchen ambient. Centrifugal blowers at equivalent flow are noticeably louder.
6. **Cheap and standard.** ~€3–5 each, off-the-shelf 80 mm form factor, easy to mount.
7. **Always-on by hardware.** Direct connection to +12V_FUSED means no firmware path can accidentally disable ventilation during cooking or cool-down.

8.4.4. Motor (Fan) and Driver Selection

Fans: 2 × 80 mm 12 V DC axial fan (BLDC), shown as FAN1 (intake) and FAN2 (exhaust) on the schematic.

Parameter	Typical value for this class
Frame size	80 × 80 × 25 mm
Rated voltage	12 V DC
Motor type	Brushless DC, internal commutation electronics, ball bearing
Airflow (free-air)	50–80 m³/h (~30–47 CFM)
Static pressure (max)	~2 mm H_2O
Rotational speed	~2000 RPM
Running current	~150–250 mA
Acoustic noise	~20–25 dB(A) at 1 m
Rated life (L_{10}, 40 °C)	≥ 40,000 h
Connection	2-wire (+, −) — no tach, no PWM input

Table 32: Motor Selection for Fans

"Driver": none required — direct connection to +12V_FUSED.

This is genuinely the case, and the schematic reflects it: each fan has only two wires, + to +12V_FUSED and − to GND.

8.5. Drive 5 - Eggshell Flipper Servo

8.5.1. Description of Movement and Estimated Load

Movement / task. After the breaking mechanism (Stage 3) has cracked the egg open and the contents have drained onto the cooking plate (Stage 4), the **empty shell remains in the breaking holder**. A short rotary arm driven by the servo then sweeps the shell sideways into the shell drawer, clearing the holder before the next egg arrives. The motion is a single discrete swing — rest position → flip → return to rest — repeated once per egg.

Position in the process timeline (per §5.2 of the report).

Stage	Duration	Cumulative
0 — Oil pre-spray	2804.20 ms	2.804 s
1 — Gate release	8.51 ms	2.813 s
2 — Egg transport	780.72 ms	3.593 s
3 — Breaking mechanism	122.22 ms	3.716 s
4 — Drain + spring reset	172.57 ms	**3.888 s**

Table 33: Process timeline and cumulative duration up to shell removal

The eggshell flip is **not part of** t_prep , it happens *after* Stage 4 ends at

t = 3.888s, in parallel with the start of cooking. The deadline is set by the next egg arriving at the breaking holder.

Motion geometry.

- Estimated arm length (centre of servo shaft to shell contact point): ~200 mm
- Required angular travel: **~90°** (rest above holder → swept past the drawer edge)
- One actuation per egg → **1- 4 cycles per customer**

Load on the servo.

- Aluminium arm $(200 \times 15 \times 3mm) : m_arm \approx 25g$, centre of mass at 100 mm
- Empty eggshell at the tip: $m_shell \approx 6g$ at 200 mm
- Worst-case gravity torque (arm horizontal):
  $T_grav = (0.025 \cdot 9.81 \cdot 0.100) + (0.006 \cdot 9.81 \cdot 0.200) \approx 24.5 + 11.8 \approx 36mN \cdot m$
- Moment of inertia about the shaft:

$$J = \frac{1}{3} \cdot 0.025 \cdot 0.200^2 + 0.006 \cdot 0.200^2 \approx 3.3 \times 10^{-4} + 2.4 \times 10^{-4}$$ **≈ 5.7 × 10⁻⁴ kg·m²**

- For a deliberately gentle flip (90° in ~0.5 s, α ≈ 12.6 rad/s²):
  T_accel ≈ J · α ≈ 5.7 × 10⁻⁴ · 12.6 ≈ **7 mN·m**
- **Total worst-case torque at the shaft: ~45 mN·m** (dominated by gravity, not acceleration)

Timing budget (per §5.3 of the report).

- Earliest flip start: $t = 3.888s$ (end of Stage 4)
- Latest flip completion: $t \approx 6.27s$ (T_plate, next egg arrives at the breaking holder)
- **Available window: 2.38 s** (the buffer derived in §5.3)
- Required flip + return time: **~1.0 s** total (chosen slower than the servo's maximum, to keep the tip from launching the shell)
- Margin: **~1.4 s** inside the buffer

8.5.2. Summary of Technical Requirements

Requirement	Value
Function	Single-axis rotary flip of empty eggshell into drawer

Arm length	**200 mm**
Angular travel	**~90°** (rest ↔ flip)
Required torque at the shaft (incl. margin)	**≥ 45 mN·m**
Earliest flip start	**t = 3.888 s** (after t_prep completes)
Latest flip completion	**t ≈ 6.27 s** (before next egg arrives)
Available window for flip + return	**2.38 s**
Chosen flip time	**~0.5 s** (≈ 180 °/s peak)
Chosen return time	**~0.5 s**
Total servo activity per egg	**~1.0 s** (≈ 42% of the 2.38 s budget)
Positioning accuracy	~10 mm at the tip (mechanical geometry locates the arm at end stops)
Cycles per customer	**1 - 4** (one per egg)

Holding torque at rest	Must hold arm against gravity (**~36 mN·m**) during idle
Control type	**Open-loop position command** (servo's internal closed loop handles the actual positioning)
Direction	Bidirectional (flip + return)
Duty cycle	Very intermittent — ~1.0 s active per ~6.27 s cycle, ~16% peak duty
Supply	+6V_FUSED rail (6 V DC) for both logic and motor power
Logic interface	1 PWM line from MCU (SERVO_PWM, 50 Hz, 1–2 ms pulse)
Environment	Inside machine enclosure; ambient ~30–45 °C; possible incidental contact with egg residue on the arm tip (not on the servo body)

Table 34: Technical Requirements Table for Eggshell Flipper

8.5.3. Type of Motor — Choice and Reasoning

From the morphological box, three actuator options were carried forward for the flip motion:

- **Stepper motor.** Gives precise angular positioning and silent holding without feedback. But at this scale (~5 mN·m, 90° in 0.3 s) a stepper would need a microstepping driver, a homing routine, and step-counted commands — all to perform a motion that has only two meaningful end positions. The advantages of

step-level resolution and dead-reckoning positioning are wasted here, and the added driver electronics and firmware complexity outweigh the benefit. Rejected on complexity vs. value.
- **Pneumatic actuator.** A rotary or linear pneumatic cylinder gives fast, powerful, repeatable end-to-end motion and tolerates harsh environments well. The fatal problem is infrastructure: pneumatic actuation requires a compressor, reservoir, pressure regulator, solenoid valves, fittings, and an air supply — none of which exists anywhere else in SunnyPro and which would dominate the bill of materials and the cabinet volume for the sake of one flip motion per egg. Pneumatics also has poor end-of-stroke damping without flow control, producing a loud bang each cycle in a kitchen environment. Rejected on infrastructure and noise.
- **Hobby servo motor.** A DC motor + gearbox + internal potentiometer + closed-loop driver, all in one ~€3–5 package, commanded by a single PWM line. Gives defined start and end angles, holds position when commanded, and handles its own position feedback. **Chosen.**

Choice: high-torque hobby servo (MG996R), commanded by a single PWM line from the MCU.

Reasons:

1. **Two defined positions, native to the servo's interface.** "Rest" and "flipped" map directly to two PWM pulse widths. No homing, no step counting, no microstep driver — none of the overhead a stepper would impose.
2. **Self-contained drive electronics.** Motor , gearbox, position sensor, and H-bridge are all inside the servo housing. The "driver" is the MCU's PWM output — no MOSFET, no flyback diode, no microstep IC required externally.
3. **No infrastructure required.** Runs from the existing +5V_FUSED rail and a single MCU pin. The pneumatic alternative would have needed an entirely new air system that exists nowhere else in the machine.
4. **Controlled motion, not a bang.** The servo's closed loop produces a smooth, damped sweep with defined start and end. Pneumatics without expensive flow-control hardware would slam the arm into its end stops on every cycle — unsuitable for a customer-facing appliance.
5. **Speed and torque comfortably match the task.** Required ~45 mN·m vs ~1 N·m available → ~22× margin. The 200 mm arm at a deliberately moderated 0.5 s sweep keeps the tip speed at ~0.6 m/s, low enough that the shell drops cleanly rather than being thrown.
6. **Cost and integration.** ~€4 for the servo, one PWM pin on the MCU, no PCB additions. The stepper alternative is ~€10+ with its driver; the pneumatic alternative is in a different cost class entirely once the compressor and valves are included.

8.5.4. Motor and Driver Selection

Motor: Tower Pro MG996R high-torque servo (SERVO_1 on the EGG_BREAKING_STORAGE schematic), operated at 6 V.

Parameter	Value
Type	Brushed DC + metal gearbox + potentiometer + integrated PWM controller
Operating voltage (chosen)	**6.0 V DC**
Voltage range supported	4.8–7.2 V DC
Stall torque @ 6 V	**~1.08 N·m** (~11 kg·cm)
No-load speed @ 6 V	**~430 °/s** (0.14 s/60°)
Angular range	0–180°
Control	PWM, 50 Hz, 1.0–2.0 ms pulse → 0–180°
Idle current @ 6 V	~10 mA
Running current @ 6 V	~200–300 mA (no-load → moderate load)
Stall current @ 6 V	~1.5 A (transient only)

Local bulk capacitance at servo connector	**~1000 µF** (absorbs start-up transient)
Connector	Standard 3-pin: signal, V+, GND

Table 35: Technical Requirements Table for Motor and Driver

Driver: none required externally — the servo is its own driver.

What's on the schematic in support of the servo:

- **3-pin connector to the servo** (SERVO_1 = MG996R): signal, +6 V, GND.
- **SERVO_PWM signal from the MCU**: a single PWM-capable GPIO at 50 Hz. No level shifter needed — the MG996R accepts 3.3 V logic on the signal line even when its power rail is 6 V.

8.6. Drive 6 - Platform Breaker Solenoid

8.6.1. Description of Movement and Estimated Load

Movement / task. The platform solenoid drives the **spear** straight down onto the egg sitting in the breaking holder, puncturing the shell so its contents drain through the holder's outlet onto the cooking plate below. The motion is a single short linear stroke — energise → plunger pulls in (spear travels down) → shell punctures → de-energise → return spring pulls the spear back up. Repeated once per egg, four times per customer cycle. This is the heaviest mechanical job in the egg-breaking subsystem and the only place in SunnyPro that needs to deliver impact-class force.

Position in the process timeline (per §5.2 of the report).

The platform solenoid is the actuator of **Stage 3 — Breaking Mechanism** in the prep sequence:

Stage	Duration	Cumulative
0 — Oil pre-spray	2804.20 ms	2.804 s
1 — Gate release	8.51 ms	2.813 s
2 — Egg transport	780.72 ms	3.593 s
3 — Breaking mechanism	**122.22 ms**	3.716 s
4 — Drain + spring reset	172.57 ms	3.888 s

Table 36: Process timeline and cumulative duration for the platform breaker solenoid

Stage 3 covers the full breaking action: solenoid energise + plunger travel + shell puncture. Its spring return is then folded into Stage 4 along with the wedge return.

Motion geometry.

- Stroke length (plunger fully extended → fully retracted): **~10 mm**
- Direction: vertical, downward when energised (pull solenoid mounted above the egg holder)
- Single-shot per egg → **1 - 4 cycles per customer**

Mechanical load on the spear (per §5.2.4 / §5.2.5 of the report).

- Spear moving mass: **m_spear = 105 g**
- Spear return spring stiffness: **k = 365,741 N/m**
- Required puncture force on the shell: **~20–30 N** at the sharp tip (typical for a sharp-point puncture of an eggshell along its weak axis)
- Friction in the spear guide: estimated ~2 N
- **Total worst-case force the solenoid must produce at the start of the stroke: ~25–35 N**

Solenoid output (JF-0837B at 24 V, typical).

- Pull force at the start of the stroke (large air gap closing): ~25–40 N (datasheet-dependent, ~30 N nominal)
- Force rises as the gap closes — by the time the spear contacts the shell, the solenoid is already at full pulling force
- Pulse duration during Stage 3 is well within the JF-0837B's intermittent duty rating (~25%)

Timing budget.

- Stage 3 allocates **122.22 ms** to the whole breaking action: coil energisation + plunger travel + puncture
- Of this, the L/R coil energisation up to the ~50% trigger-current point is ~0.5–1 ms (negligible)
- Plunger travel through 10 mm under ~30 N net force on a 105 g mass: $t \approx \sqrt{(2x/a)} \approx \sqrt{(2 \cdot 0.010/(30/0.105))} \approx \ 26ms$
- The remaining ~95 ms in Stage 3 covers shell yielding under the spear tip, full plunger seating, and the deliberate hold for full puncture before release

8.6.2. Summary of Technical Requirements

Requirement	Value
Function	Electromagnetic solenoid driving the spear that punctures the eggshell

Motion	Single stroke down (energise) + spring return up (de-energise)
Required force at the spear tip	**≥ 25–30 N** at start of stroke
Stroke length	**~10 mm**
Moving mass (spear)	**105 g**
Return spring stiffness	**365,741 N/m** (per §5.2.5)
Stage 3 budget (full breaking action)	**122.22 ms**
Plunger travel time	**~26 ms** at full force
Cycles per customer	**1 - 4** (one per egg)
Inter-cycle interval	~6.27 s (T_plate per §5.3)
Control type	**Open-loop on/off** — pulse duration set in firmware
Direction	Single-direction pull (return by spring)
Duty cycle	**~25% max** (per JF-0837B intermittent rating) — actual usage ~2% (~120 ms per 6.27 s cycle)
Supply	**+24V_FUSED**
Logic interface	1 digital line from MCU (SOL_PLATFORM)
Environment	Inside machine enclosure; ambient ~30–45 °C; possible incidental egg residue on the spear (not on the solenoid body)

Table 37: Technical requirements of the platform breaker solenoid

8.6.3. Type of Motor - Choice and Reasoning

From the morphological box, three options were carried forward:

- **Servo-driven slider.** Hobby servo + lead-screw or rack drives the spear down. Too slow — a hobby servo needs ~50 ms just for the rotation, before screw conversion. Force at the spear is limited by stall torque ÷ mechanical advantage, much lower than a direct electromagnetic pull. Rejected on speed and force.
- **Cam mechanism.** Continuously rotating motor + cam lifts and drops the spear. Needs a motor, a clutch or brake, a precision cam, a follower, and bearings — mechanically the most complex option, and the cam profile gives whatever stroke shape it's machined for, not a sharp impulse. Rejected on complexity.
- **Linear solenoid.** Coil + plunger, sharp pull when energised, spring return. Force rises as the air gap closes, naturally matches "soft approach, hard hit" on the shell. **Chosen.**

Choice: Tubular pull solenoid (JF-0837B), 24 V, driven by an N-channel MOSFET low-side switch.

Reasons:

1. **Force profile matches the task.** Low force at start of stroke (safe approach), peak force at contact (clean puncture). Servo gives constant force; cam gives whatever the contour describes. Only the solenoid produces this shape naturally.
2. **Fast.** ~26 ms travel time fits inside the 122 ms Stage 3 budget with margin.
3. **24 V chosen over 12 V.** A 12 V JF-0530B delivers only ~10 N — borderline for puncture. The 24 V JF-0837B gives ~30 N, comfortably above the ~25 N threshold.
4. **Single off-the-shelf part.** No gearbox, no cam, no precision machining.
5. **Trivial drive electronics.** One MOSFET + flyback diode + gate network — identical to the other solenoids on the same sheet.

8.6.4. Solenoid and Driver Selection

Solenoid: JF-0837B tubular pull solenoid, 24 V variant (SOL_PLATFORM on the EGG_BREAKING_STORAGE schematic).

Parameter	Typical value at 24 V
Type	Tubular pull (linear, single-direction, spring return external)
Rated voltage	24 V DC

Coil resistance	~24 Ω
Steady-state current	~1.0 A
Coil inductance	~15–25 mH (typical)
L/R time constant	~0.6–1.0 ms
Steady-state power dissipation	~24 W (intermittent duty only)
Pull force at start of stroke	~25–40 N (≈30 N nominal)
Stroke length	~10 mm
Duty cycle rating	**≤ 25%** (intermittent)
Plunger mass	~15–20 g internal (drives the external 105 g spear assembly)
Body size	~37 × 22 × 16 mm

Table 38: Solenoid specification for Drive 6 – Platform Breaker Solenoid

Driver: IRLZ44N logic-level N-channel MOSFET as a low-side switch (Q on the schematic, identical topology to oil pump and other solenoids).

Parameter	Value	Relevance

V_DS	55 V	>> 24 V supply, large margin
I_D continuous	47 A	>> 1.0 A solenoid current, vastly oversized
R_DS(on) at V_GS = 5 V	~22 mΩ	Logic-level — switches fully on from a 3.3 V or 5 V MCU pin
Power dissipation at 1 A	$I^2 \cdot R_D S(on) \approx 22mW$	Negligible — no heatsink required
Switching speed	Fast, slowed deliberately by gate resistor to limit EMI	

Table 39: Specification of the IRLZ44N MOSFET driver

8.7. Drive 7 - Wedge Solenoid

8.7.1. Description of Movement and Estimated Load

Movement / task. After the spear has punctured the shell (driven by SOL_PLATFORM), the wedge solenoid drives a thin spreader into the puncture to **open the cracked shell** so the egg white and yolk can drain freely onto the cooking plate. Without the wedge, surface tension and the natural elasticity of the membrane would close the puncture and slow or block the drain. The motion is a single short linear stroke , energise → plunger pulls in (wedge spreads the cracks open) → egg contents drain → de-energise → return spring pulls the wedge back. Repeated once per egg, four times per customer cycle.

Position in the process timeline (per §5.2 of the report).

The wedge fires inside **Stage 3 — Breaking Mechanism**, immediately after the spear has seated:

Stage	Duration	Cumulative
0 - Oil pre-spray	2804.20 ms	2.804 s
1 - Gate release	8.51 ms	2.813 s
2 - Egg transport	780.72 ms	3.593 s
3 - Breaking mechanism (spear + wedge)	**122.22 ms**	3.716 s
4 - Drain + spring reset (incl. wedge return)	172.57 ms	3.888 s

Table 40: Process timeline for the wedge mechanism in the egg-breaking stage

The wedge's *forward* stroke (opening the shell) is part of Stage 3; its *spring return* is part of Stage 4.

Motion geometry.

- Stroke length: **~10 mm** (within the report's stated 10–25 mm preferred range)

- Direction: linear, into the shell crack (mechanically coupled to the spreader geometry)
- Single-shot per egg → **1 - 4 cycles per customer**

Mechanical load (per §5.2.5 of the report).

- Wedge moving mass: $m_w edge = 60g$
- Wedge return spring stiffness: **k = 200,000 N/m**
- Required force on the shell to spread the cracks: ~5–10 N (the puncture has already weakened the shell; the wedge only has to defeat the membrane and natural closing tension)
- Friction in the wedge guide: ~1–2 N
- **Required solenoid pull at the operating point: ~10 N**, within the JF-0530B's nominal output

Timing budget.

- Stage 3 allocates 122.22 ms total to spear + wedge actuation
- Coil energisation to 50% trigger current: ~1.4 ms
- Plunger travel through 10 mm under ~8 N net force on a 60 g assembly:

$$t \approx \sqrt{(2 \cdot 0.010/(8/0.060))} \approx \ 24ms$$

- Wedge fires immediately after the spear seats; both actions fit inside the 122 ms Stage 3 budget with margin
- Wedge return (Stage 4, ~50–100 ms) is comfortably inside the 172.57 ms Stage 4 budget

8.7.2. Summary of Technical Requirements

Requirement	Value
Function	Electromagnetic solenoid that drives the wedge to spread the cracked eggshell open for drainage
Motion	Single stroke into the shell (energise) + spring return (de-energise)
Required force at the wedge	**~10 N** at operating point
Stroke length	**~10 mm**
Moving mass (wedge)	**60 g**

Return spring stiffness	**200,000 N/m**
Stage 3 forward-stroke budget	inside 122.22 ms (shared with spear)
Plunger travel time	**~24 ms**
Response time	**≤ 100 ms** preferred
Cycles per customer	**1 - 4** (one per egg)
Inter-cycle interval	~6.27 s
Control type	**Open-loop on/off** — pulse duration set in firmware
Direction	Single-direction pull (return by spring)
Duty cycle	**Intermittent**, actual ~2% (~120 ms per 6.27 s cycle)
Supply	**+12V_FUSED**
Logic interface	1 digital line from MCU (SOL_WEDGE)
Environment	Inside machine enclosure; ambient ~30–45 °C; possible incidental egg residue on the wedge tip (not on the solenoid body)

Table 41: Technical requirements of Drive 7 – Wedge Solenoid

8.7.3. Type of Motor - Choice and Reasoning

From the morphological box, three wedge-opening options were carried forward:

- **Servo wedge.** Hobby servo + linkage drives the wedge. Slow (~50 ms min for full travel), force limited by horn × linkage geometry, adds a PWM channel for a two-state motion. Rejected on speed and complexity.
- **Spring-loaded wedge.** The preloaded spring releases the wedge mechanically when the spear trips a latch. No electrical drive , appealing for cost, but ties wedge timing to the spear stroke and needs a precision latch + reset mechanism. Rejected on coupling and reliability.

- **Linear solenoid wedge.** Coil + plunger gives a fast, independent, electrically commanded stroke. Same drive topology as the other solenoids on the sheet. **Chosen.**

Choice: tubular pull solenoid (JF-0530B), 12 V, driven by an N-channel MOSFET low-side switch.

Reasons:

1. **Fast and electrically independent.** ~25 ms forward stroke; firmware fires the wedge at any chosen offset after the spear, no mechanical linkage.
2. **Force matches the task.** Shell already punctured before wedge fires → only ~10 N needed, within the JF-0530B's nominal output.
3. **12 V chosen, not 24 V.** Wedge task is lighter than the platform spear; JF-0530B at 12 V is sized appropriately.
4. **Same drive topology as the other three solenoids on this sheet.** IRLZ44N + 1N4007 + 100 Ω gate / 10 kΩ pull-down — already validated pattern.
5. **Runs from the existing +12V_FUSED rail.** No new infrastructure.
6. **Single off-the-shelf part shared with the gate solenoids.** One BOM line covers three of the four solenoids.

8.7.4. Solenoid and Driver Selection

Solenoid: JF-0530B tubular pull solenoid, 12 V variant (SOL_WEDGE on the EGG_BREAKING_STORAGE schematic).

Parameter	Value at 12 V
Type	Tubular pull (linear, single-direction, spring return external)
Rated voltage	12 V DC
Coil resistance	~5 Ω
Steady-state current	~2.4 A
Coil inductance	~10 mH
L/R time constant	~2 ms (to 50% trigger current ~1.4 ms)

Steady-state power dissipation	~29 W (intermittent duty only)
Pull force at start of stroke	~10 N (rises as gap closes)
Stroke length	~10 mm
Duty cycle rating	Intermittent (≤ 25%)
Plunger mass	~15 g internal (drives 60 g wedge assembly)

Table 42: Solenoid specification for Drive 7 – Wedge Solenoid

Driver: IRLZ44N N-channel MOSFET as a low-side switch (one of Q1/Q2/Q3/Q4 on the schematic — same topology as the other three solenoids).

Parameter	Value	Relevance
V_DS	55 V	>> 12 V supply, large margin
I_D continuous	47 A	>> 2.4 A solenoid current, vastly oversized
R_DS(on) at V_GS = 5 V	~22 mΩ	Logic-level — switches fully on from a 3.3 V or 5 V MCU pin
Power dissipation at 2.4 A	$I^2 \cdot R_D S(on) \approx 127mW$	Negligible — no heatsink required

Table 43: Driver specification for Drive 7 – Wedge Solenoid

8.8. Drive 8 - Storage Gate Solenoids (GATE_A and GATE_B)

8.8.1. Description of Movement and Estimated Load

Movement / task. Two solenoid-driven gates control the release of eggs from the storage unit into the cracking mechanism, in sequence per egg (per §4.5.3 of the report):

- **GATE_A (first gate):** sits between the egg storage unit and the transfer tunnel. When energised, it retracts briefly to let exactly **one** egg drop from the storage stack into the tunnel, then closes again to hold back the next egg.
- **GATE_B (second gate):** sits between the transfer tunnel and the cracking mechanism. It stays closed while GATE_A is releasing. Once the tunnel sensor (IR_GATE_A) confirms one egg is in the tunnel and the cracking platform is empty, GATE_B opens to admit the egg into the cracking position.

The two-gate arrangement is a **double-feed prevention** mechanism: if the tunnel sensor reports zero eggs (release failed) or two eggs (double-feed), GATE_B refuses to open and the controller raises an error on the touchscreen. This protects the cracking mechanism from jamming and prevents two eggs landing in the same pocket plate.

Per-customer firing count. The user selects **1–4 eggs** on the touchscreen. Both gates fire once per egg in sequence (A then B). For the **maximum order of 4 eggs**, each gate fires **4 times** within a customer cycle.

Position in the process timeline (per §5.2 of the report). Per egg, the gate actuations are part of **Stage 1 — Gate Release**:

Stage	Duration	Cumulative
0 - Oil pre-spray	2804.20 ms	2.804 s
1 - Gate release	**8.51 ms**	2.813 s
2 - Egg transport	780.72 ms	3.593 s
3 - Breaking mechanism	122.22 ms	3.716 s
4 - Drain + spring reset	172.57 ms	3.888 s

Table 44: Timing sequence for the gate release stage

The §5.2.2 timing of 8.51 ms is the calculated chain time per gate firing (sensor read + MCU + MOSFET + coil energisation + plunger stroke + release lag). GATE_A and GATE_B each consume one such interval within a single egg's prep window, GATE_A fires at the start of Stage 1, GATE_B fires during Stage 2 once the tunnel sensor has cleared the egg.

Motion geometry (per §5.2.2).

- Stroke length: **10 mm**
- Direction: linear, retracting sideways to clear the egg path
- Each gate fires once per egg

Mechanical load on each gate (per §5.2.2).

- Plunger + gate moving mass: $m_plunger = 15g$
- Solenoid pull force: $F_sol = 10N$
- Opposing return-spring force: $F_spring = 0.5N$
- Egg release lag due to inertia: **1.0 ms**
- Required pull at the operating point: ~1–2 N (spring + light friction; the egg's weight doesn't load the solenoid because the gate moves *sideways*, not under the egg)

Timing breakdown per gate firing

Sub-step	Duration
IR break-beam (Adafruit 2167) electrical response time	~0.1 ms
STM32H743 EXTI latency @ 480 MHz	~0.02 µs
ISR + HAL_GPIO_WritePin (~12 instructions, M7 dual-issue)	~0.025 µs
MOSFET gate charge	~0.02 µs
Coil energisation to 50% trigger current	~1.4 ms
Mechanical stroke (10 mm under ~9.5 N net on 15 g)	~5.6 ms
Egg release lag (inertia)	1.0 ms
Total per gate	**~8.1 ms**

Table 45: Detailed timing breakdown for one gate release actuation

The dominant element is the mechanical stroke; the electronic switching delays are negligible.

8.8.2. Summary of Technical Requirements

Requirement	Value
Function	Electromagnetic solenoid that retracts a sliding gate to release one egg along the storage→tunnel→cracker path
Quantity	**2** (GATE_A: storage → tunnel; GATE_B: tunnel → cracker)
Motion	Single stroke retract (energise) + spring return close (de-energise)
Required force at the plunger	**~10 N**
Stroke length	**10 mm**
Moving mass	**15 g**
Opposing spring force	**0.5 N**
Stage 1 budget per gate firing	**8.51 ms**
Coil energisation time	~1.4 ms to 50% current
Mechanical stroke time	~5.6 ms
Egg release lag	1.0 ms
Firings per gate per customer	**up to 4** (one per egg; user selects 1–4 eggs)
Sequence per egg	**GATE_A → tunnel sensor check → GATE_B**
Inter-egg interval	~6.27 s (T_plate per §5.3)

Control type	**Open-loop on/off** — pulse duration set in firmware, gated by IR sensor confirmations
Direction	Single-direction pull (return by spring)
Duty cycle	**Intermittent**, actual ~0.15% per gate (~10 ms per 6.27 s egg cycle)
Supply	**+12V_FUSED** (existing rail)
Logic interface	2 digital lines from MCU (GATE_A, GATE_B)
Environment	Inside machine enclosure; ambient ~30–45 °C; dry path (no egg contact at the gate surfaces themselves)

Table 46: Technical requirements of the gate release mechanism

8.8.3. Type of Motor - Choice and Reasoning

From the morphological box, three options were carried forward for the gate actuation:

- **Servo gate actuator.** Hobby servo rotates an arm or slide to open the gate. Far too slow (~50 ms minimum for full travel) for the 8.51 ms Stage 1 budget per gate, and adds a PWM channel per gate plus a linkage. Rejected on speed.
- **Mini linear actuator.** A small DC motor driving a lead screw or rack. Gives controllable position, but typical units take seconds for a full stroke and cost an order of magnitude more than a solenoid. Massively over-specified for a two-state release motion. Rejected on speed and cost.
- **Solenoid actuator.** Coil + plunger retracts the gate when energised; spring closes it on release. Fast (~7 ms mechanical stroke), independently commandable per gate, no linkage. **Chosen.**

Choice: Tubular pull solenoid (JF-0530B), 12 V, driven by an N-channel MOSFET low-side switch - one per gate.

Reasons:

1. **Fast.** ~7 ms mechanical stroke fits inside the 8.51 ms Stage 1 budget. The other two options can't.
2. **Independent firmware control of each gate.** GATE_A and GATE_B fire from separate MCU pins, so the controller fires them in strict sequence and can refuse to open GATE_B if the tunnel sensor reports a double-feed or empty tunnel.
3. **Force vastly exceeds the requirement.** 10 N output vs ~1.5 N spring + friction → ~6× margin. Cannot stall.

4. **12 V chosen.** Gate task is light; JF-0530B at 12 V is correctly sized. Reuses the same part as the wedge solenoid → single BOM line covers three of the four solenoids in the egg-handling subsystem.
5. **Same drive topology as the other three solenoids on this sheet.** IRLZ44N + 1N4007 + 100 Ω gate / 10 kΩ pull-down , already-validated pattern repeated per channel.
6. **Runs from the existing +12V_FUSED rail.** No new infrastructure.

8.8.4. Solenoid and Driver Selection

Solenoid: 2 × JF-0530B tubular pull solenoid, 12 V variant (GATE_A and GATE_B on the EGG_BREAKING_STORAGE schematic).

Parameter	Value at 12 V
Type	Tubular pull (linear, single-direction, spring return external)
Rated voltage	12 V DC
Coil resistance	~5 Ω
Steady-state current	~2.4 A
Coil inductance	~10 mH
L/R time constant	~2 ms (to 50% trigger current ~1.4 ms)
Steady-state power dissipation	~29 W (intermittent duty only)
Pull force at start of stroke	~10 N
Stroke length	10 mm
Duty cycle rating	Intermittent (≤ 25%)
Plunger mass	~15 g

Table 47: Specification of the selected solenoids for Drive 8 – Gate Release

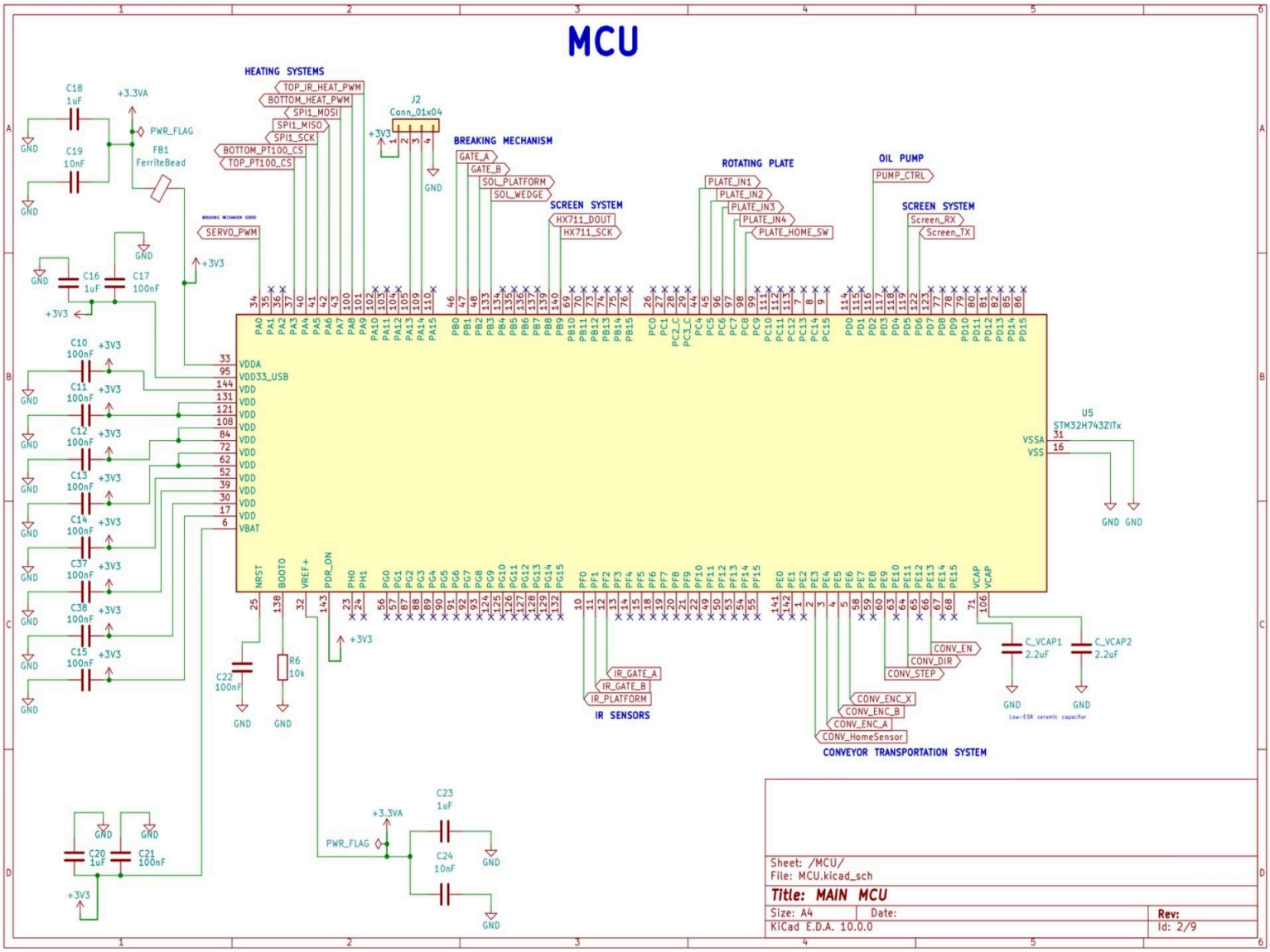
Driver: 2 × IRLZ44N N-channel MOSFET as low-side switches (one per gate on the schematic — same topology as SOL_WEDGE and SOL_PLATFORM).

Parameter	Value	Relevance
V_DS	55 V	>> 12 V supply, large margin
I_D continuous	47 A	>> 2.4 A solenoid current, vastly oversized
R_DS(on) at V_GS = 5 V	~22 mΩ	Logic-level — switches fully on from a 3.3 V or 5 V MCU pin
Power dissipation at 2.4 A	$I^2 \cdot R_D S(on) \approx 127mW$	Negligible — no heatsink required

Table 48: Driver specification for Drive 8 – Gate Release Solenoids

9. Circuit Diagram [21]

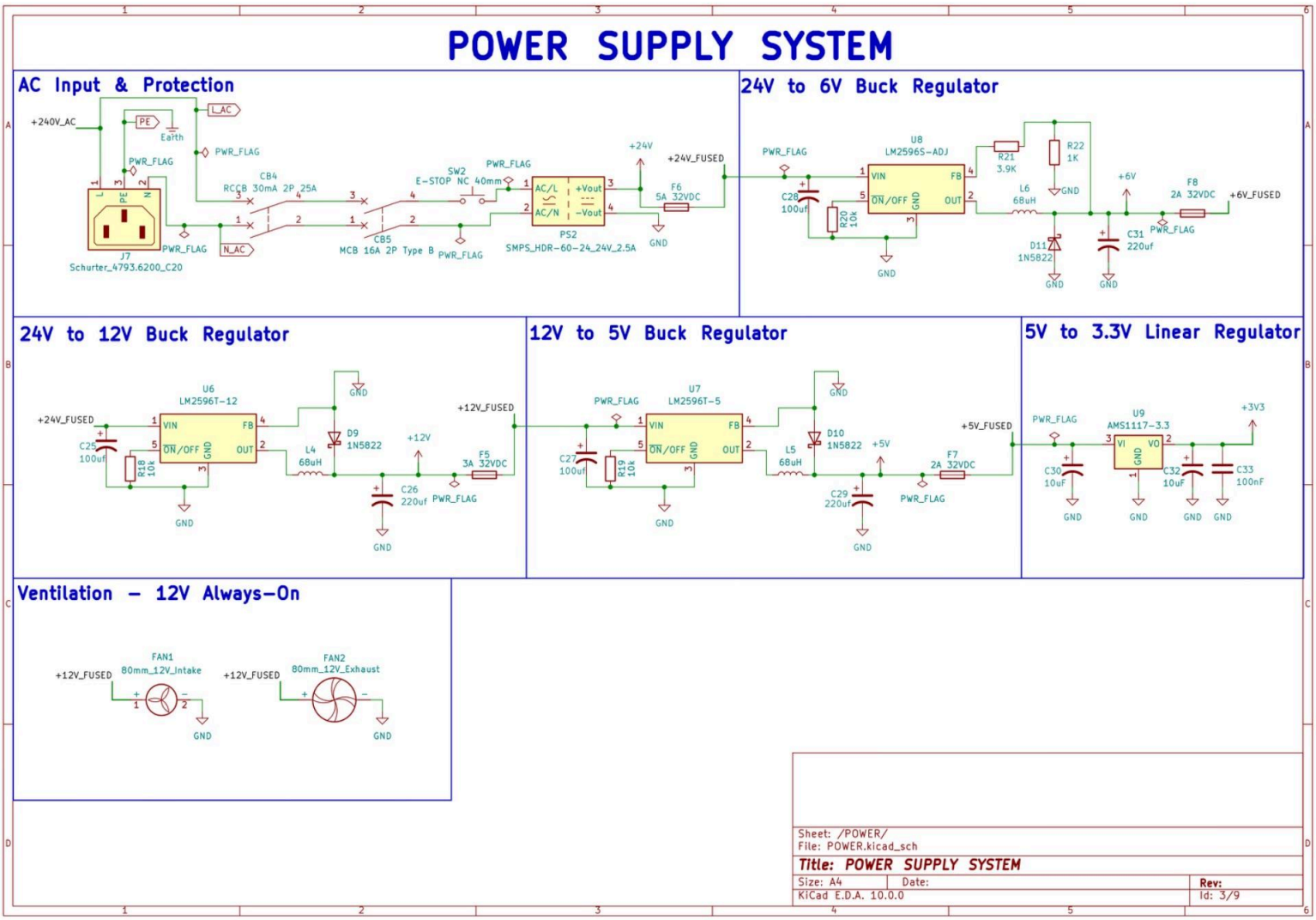
9.1. MCU



[21] KiCad Developers Team. KiCad EDA. *KiCad*. https://www.kicad.org

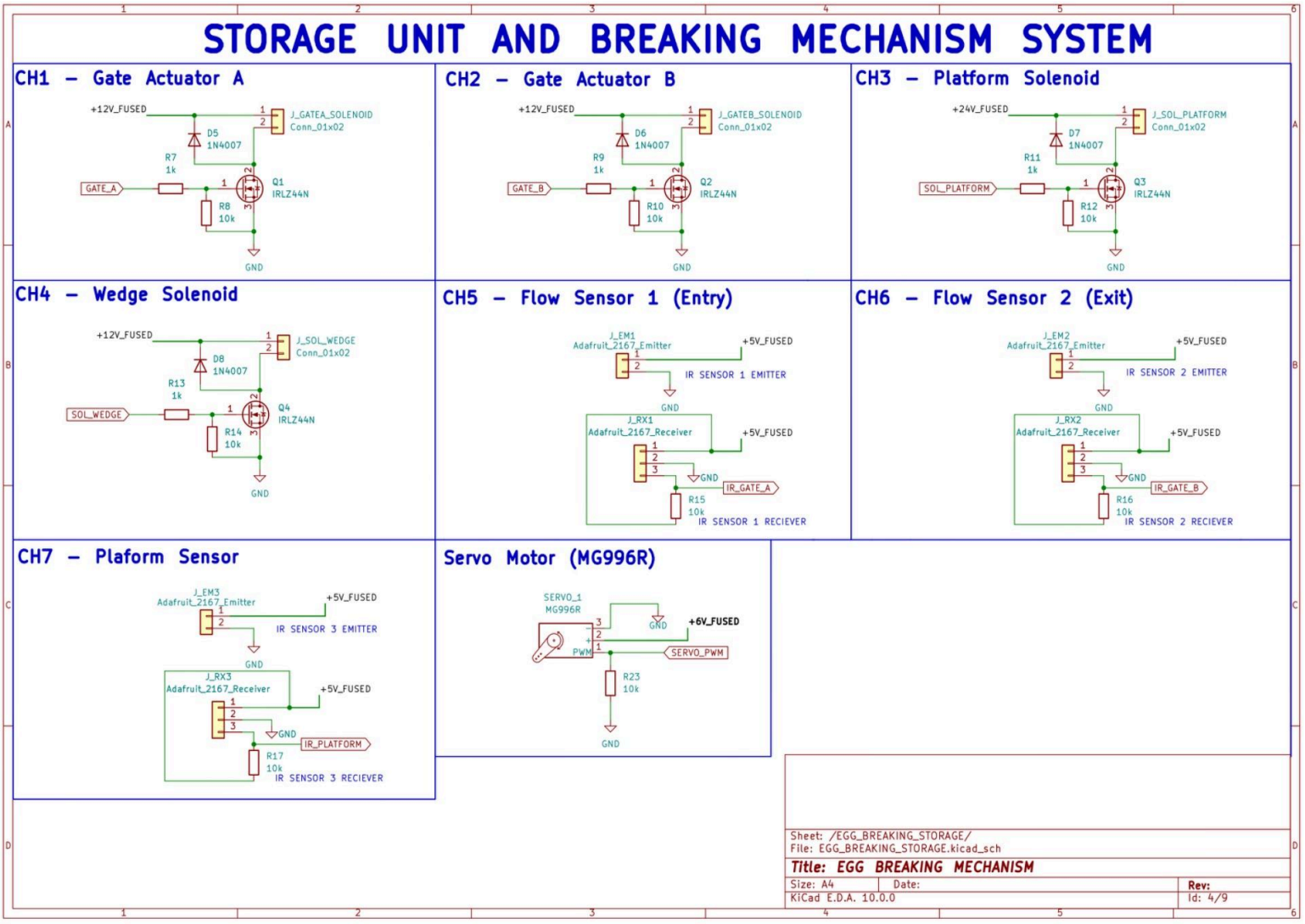
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9.2. Power Supply System



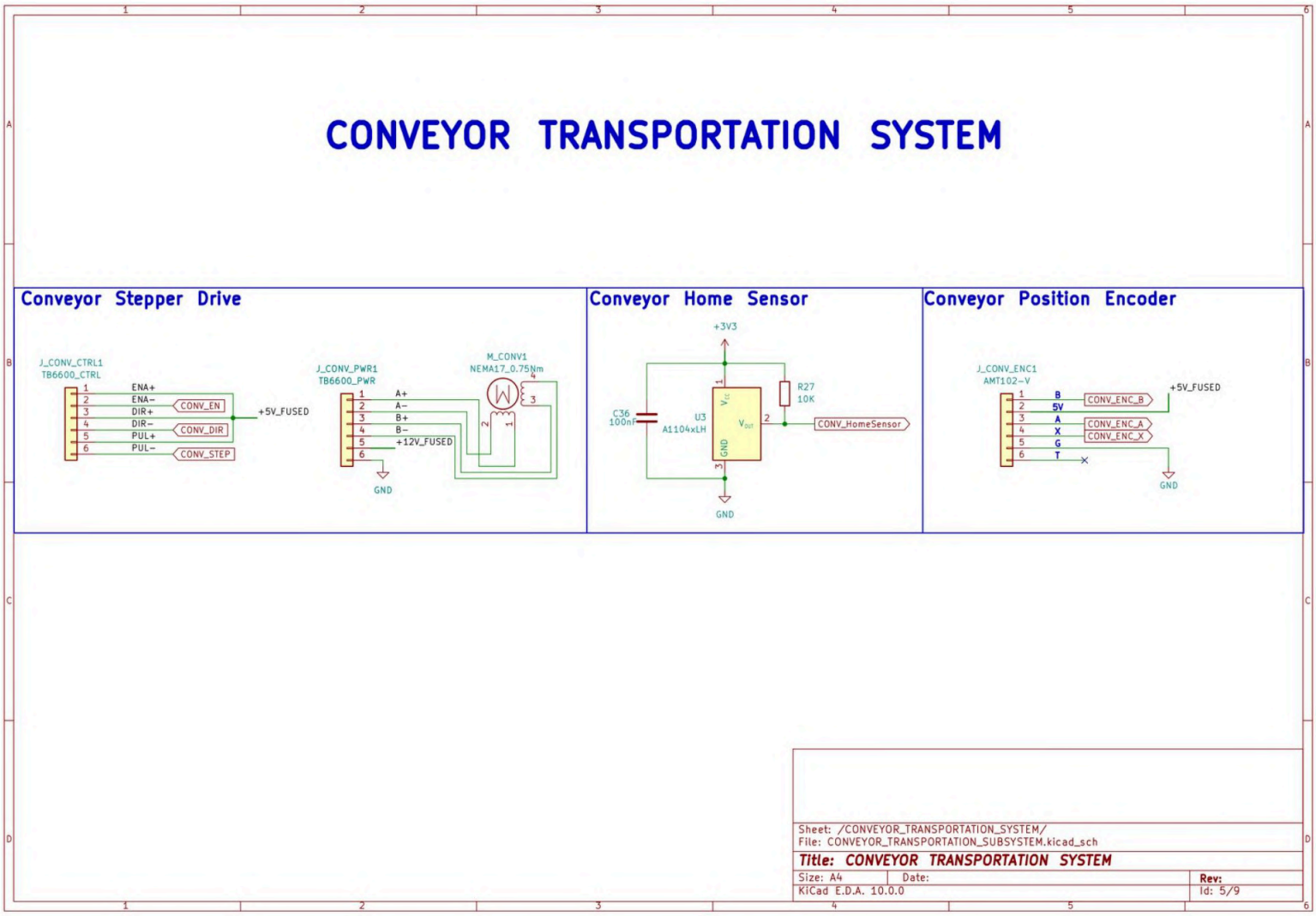
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9.3. Storage Unit and Breaking Mechanism System



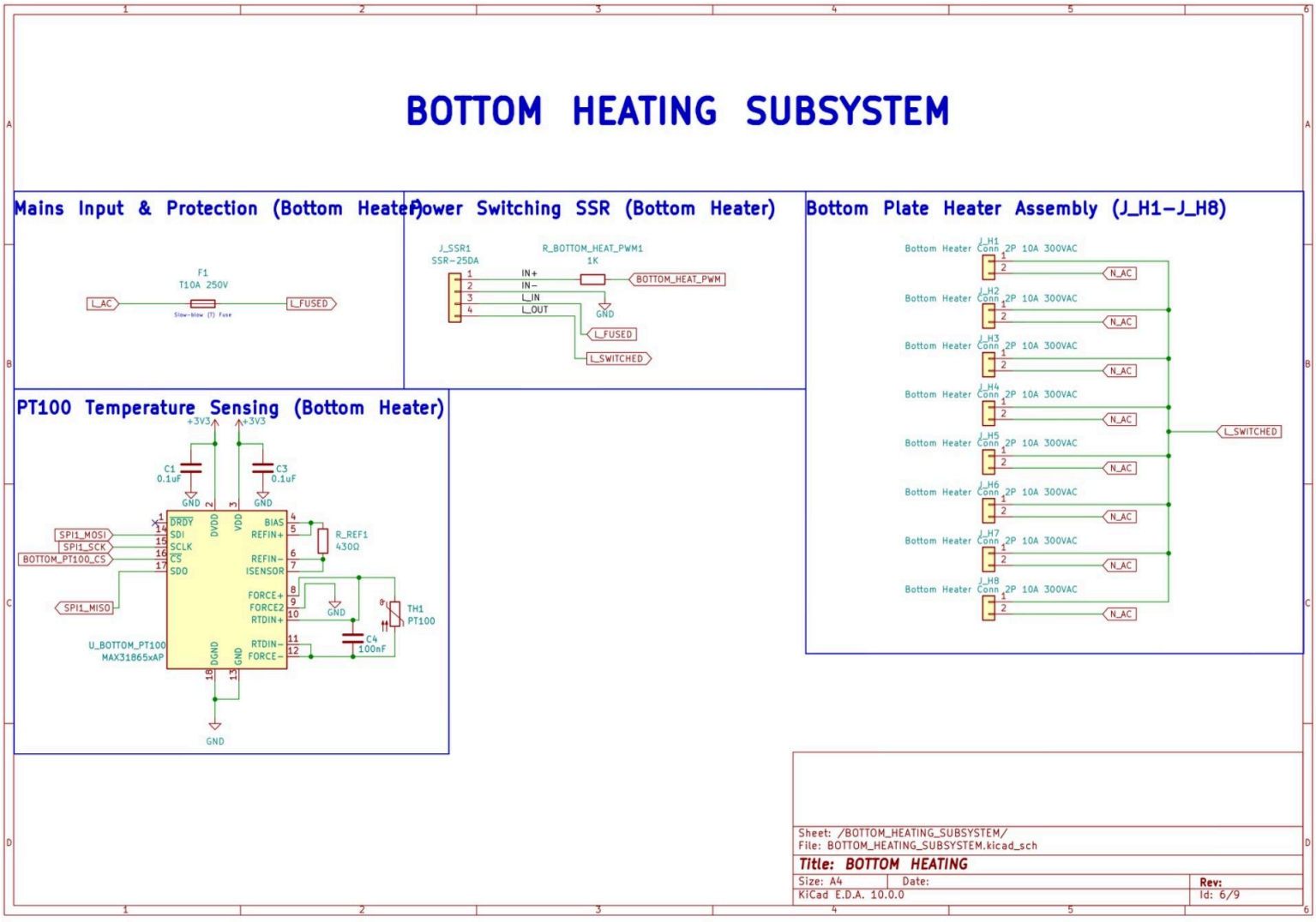
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9.4. Conveyor Transportation System



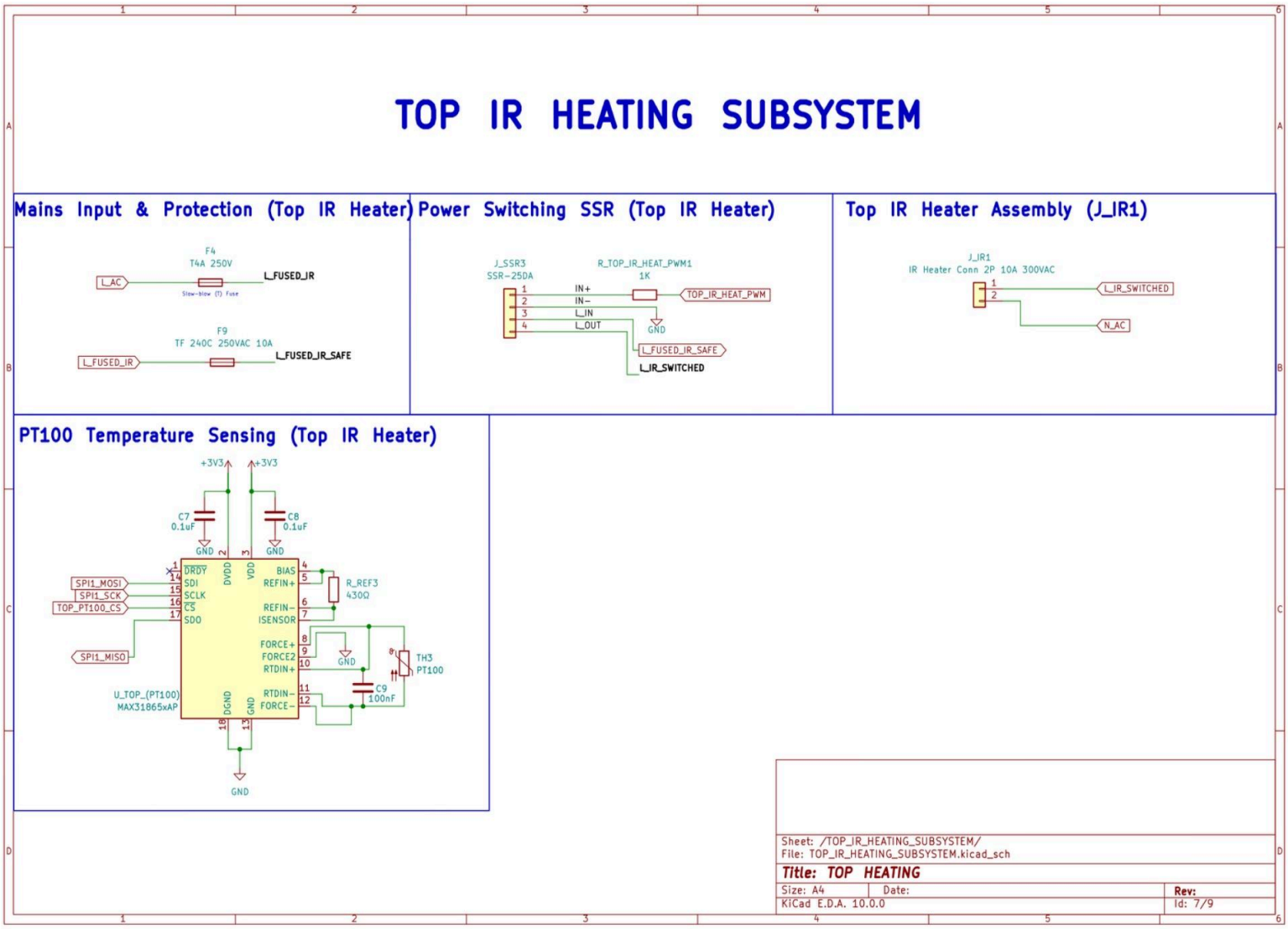
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9.5. Bottom Heating Subsystem



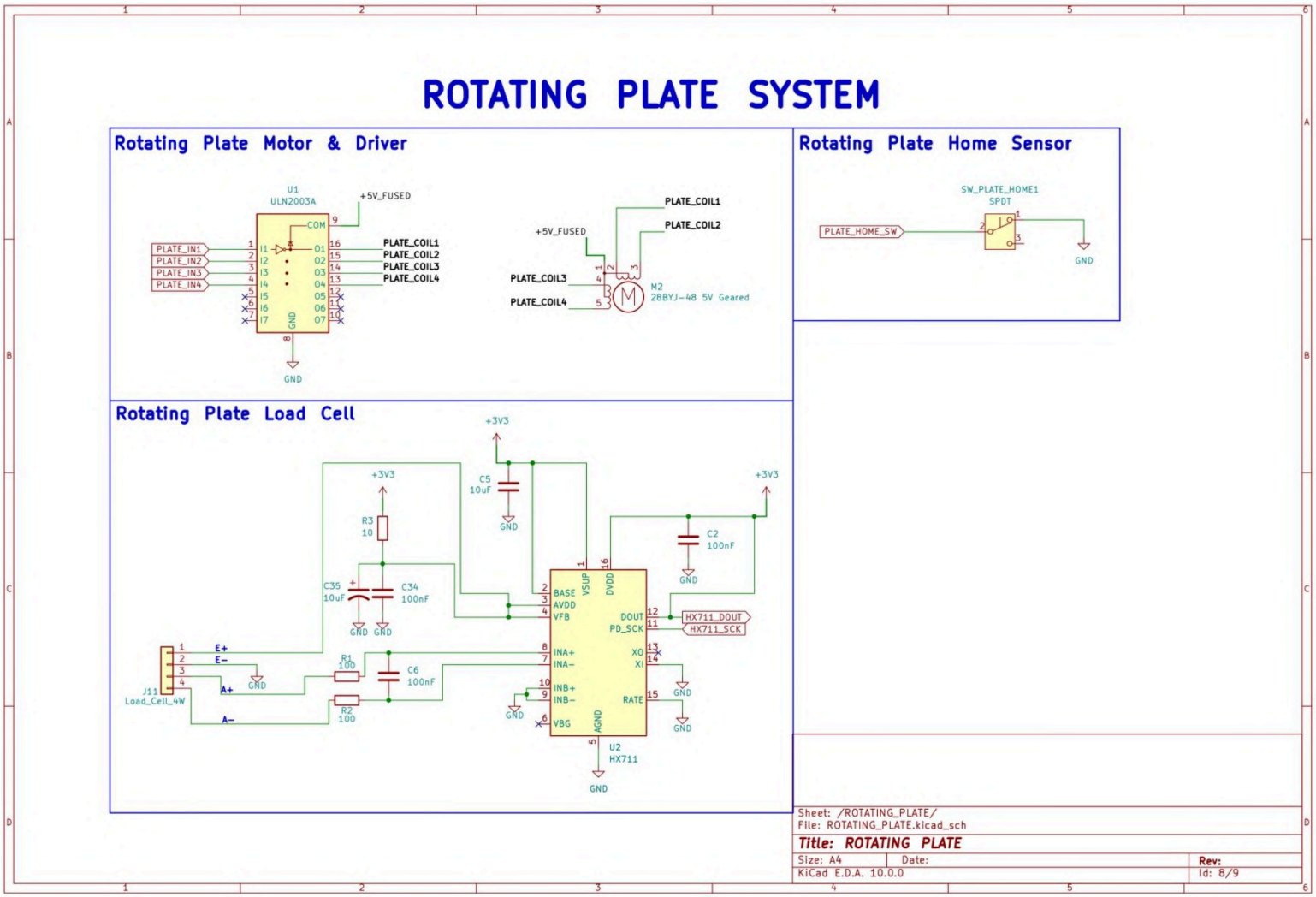
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9.6. Top IR Heating Subsystem



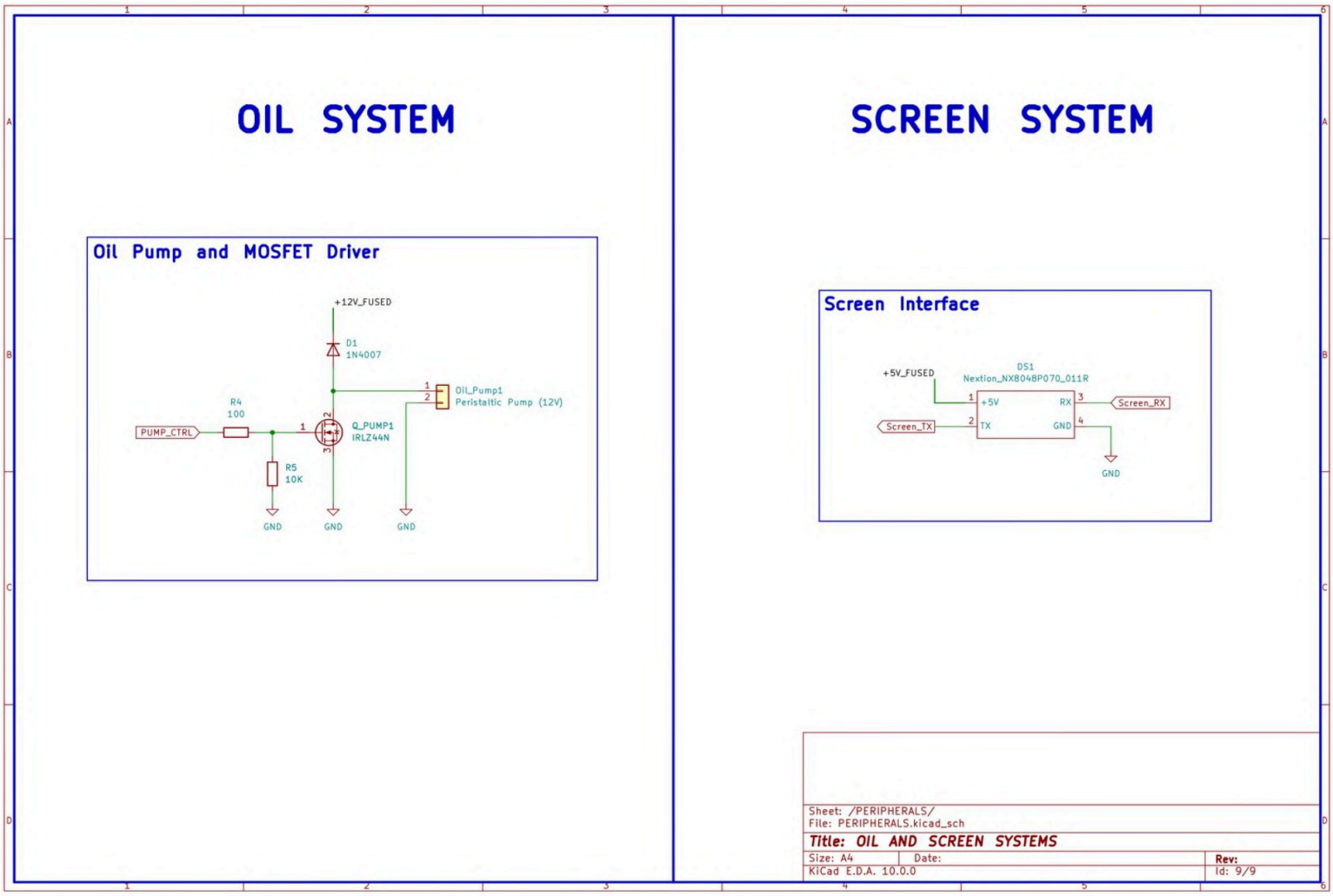
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9.7. Rotating Plate System



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9.8. Oil System and Screen System

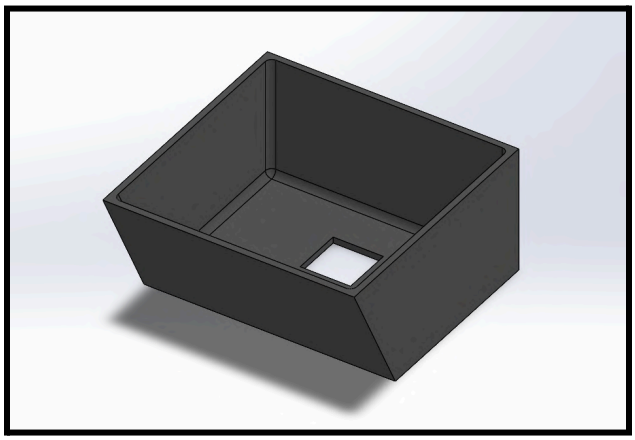


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10. Technology selection for „make“ parts

10.1. Egg Storage

10.1.1. Egg Storage Container



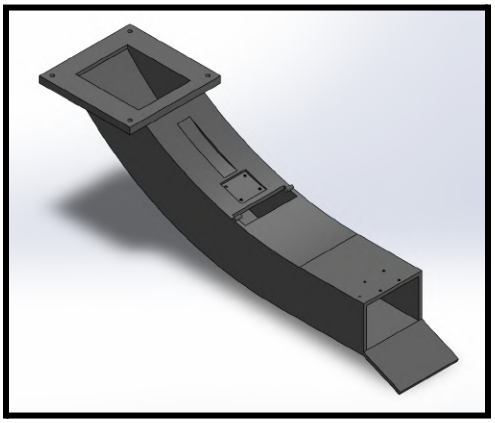
Material : PP-C HI / Polypropylene Copolymer

Injection moulding for the main raw egg storage container. The part is moulded as one PP-C plastic body, including the angled side walls, rounded internal fillets, reinforced top rim, and general outer shape.Moulded draft angles on the side walls.

A Core insert for the square middle cut-out. The square opening is formed directly in the injection mould using a fixed core insert, so it does not need to be cut out afterwards.
Stepped core pins The counterbore shape can be moulded directly using stepped core pins. If higher screw-seat accuracy is required, the holes are moulded slightly undersized and then finished using a drilling/counterboring operation.
Edge finishing using a manual deburring tool around the square cut-out, screw holes, top rim, and parting line. This removes plastic flash and sharp edges, making the part safer and easier to clean.

10.2. Egg Transportation

10.2.1. Egg Tunnel



Material name: PP-C HI / Polypropylene Copolymer

Manufacturing Technology:

Injection moulding as two PP-C half-shells for the curved egg transport tunnel. The tunnel is split into an upper half and lower half because a one-piece closed curved tube would be difficult to remove from the mould. The angled/curved egg path, rectangular tunnel shape, end flanges, and general outer profile are formed directly during moulding.

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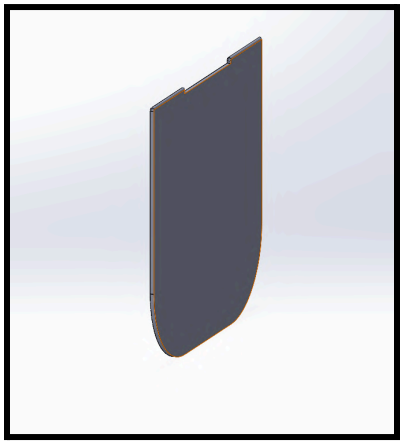
Steel injection mould with fixed core inserts to form the main internal egg path, square openings, end flanges, angled walls, and internal fillets. This gives a smooth food-contact surface and reduces secondary machining.
Draft angles on the tunnel walls and flange walls of approximately 1.5°–2° so both PP-C half-shells can be released from the injection mould without damage.

Side-action slides in the injection mould for the motor shaft hole and side sensor cut-outs. The slides move sideways before ejection, allowing the side holes and sensor windows to be moulded directly during production.

Core pins in the injection mould for the screw holes in the flanges. Hardened steel core pins form the screw clearance holes directly during moulding, so no drilling is required in normal production.

Ultrasonic welding machine used to join the upper and lower PP-C tunnel half-shells.
Heat-set threaded inserts The threaded fixing points are made by pressing brass heat-set inserts into moulded pilot holes after moulding. These inserts are placed in the flange and tunnel body where screws will be removed or tightened during assembly.
Mechanical deburring Done using a hand deburring tool around the drilled shaft hole, sensor cut-outs, square opening, and screw holes to remove burrs without creating sharp edges.

10.2.2. Gates for Transportation Tunnel



Material : POM-C / Acetal Copolymer

Justification: POM-C is selected because the gate is a rotating moving part and requires low friction, good stiffness, and good wear resistance at the shaft connection. It is preferred over PP because PP is lighter but softer and less wear-resistant. It is preferred over stainless steel because steel is unnecessarily heavy for this small low-load gate.

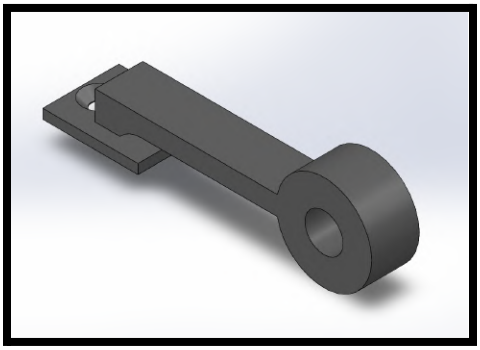
Manufacturing Technology:

Injection moulding for the rotating egg gate. The gate is moulded as one POM-C part, including the curved lower profile, flat gate face, top locating features, and general outer shape. The mould will be Polished Steel for a smooth surface.Draft angles of approximately 1°–2° on the gate faces and side edges so the gate can be released from the mould without damage.**Core pin in the injection mould for the motor shaft hole.** A hardened steel round core pin forms the shaft hole directly during moulding.

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Mechanical deburring around the parting line, shaft hole, top notches, and outer edges using plastic-safe deburring tools. This removes mould flash and sharp edges without damaging the food-contact surface.

10.2.3 Gate Shaft Holder for Tunnel



Material: POM-C / Acetal copolymer

Justification:
POM-C is used because the gate rotates inside the tunnel and needs low friction, good stiffness, and wear resistance around the shaft hole. It is preferred over PP because the hinge area and thin gate arm need better dimensional stability and lower wear.

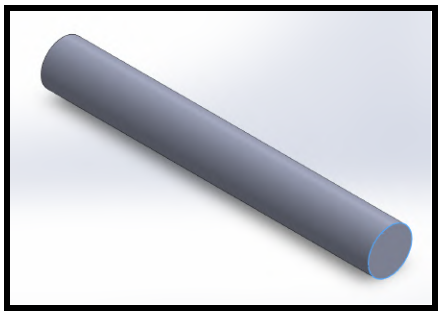
Manufacturing Technology:

- **Injection moulding:** The gate is moulded as one plastic part, including the cylindrical hinge hub, long gate arm, rectangular stop/actuator section, and flat rear plate.
- **Mould parting line:** The parting line is placed around the side edge of the flat gate arm and rear plate so the main sliding/contact faces stay clean.
- **Core pin for hinge hole:** The shaft hole through the cylindrical hub is formed by a core pin running through the hub axis. The hole is then reamed to final size so the gate rotates smoothly on the shaft.
- **Core pin for rear mounting hole:** The rear hole is formed through the flat end plate in the screw/fastener direction so it lines up with the actuator or linkage connection.
- **Draft direction:** Small draft is applied to the flat gate arm, rear plate, and hub side walls in the mould opening direction so the part releases without damaging the thin arm.
- **Gate position:** The injection gate is placed on the rear non-contact side of the part, away from the hinge hole and the tunnel-contact face.
- **Finishing:** The hinge hole, rear hole, parting line, gate mark, and outer edges are deburred after moulding.

Important manufacturing features:
The cylindrical hub is the critical feature because it controls the gate rotation. The hinge hole is finished after moulding to correct shrinkage and keep the shaft fit accurate. The parting line and gate mark are kept away from the sliding/contact face so the gate does not scrape inside the tunnel.

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10.2.4 Egg Tunnel Shaft



Material: EN 1.4301 / AISI 304 stainless steel precision round bar, Ø4 mm

Justification:
EN 1.4301 is used because the shaft is close to the egg tunnel and must resist humidity, raw egg residue, and cleaning. A Ø4 mm stainless shaft is stiff enough for the low gate load and gives a smooth rotating surface for the plastic gate hub.

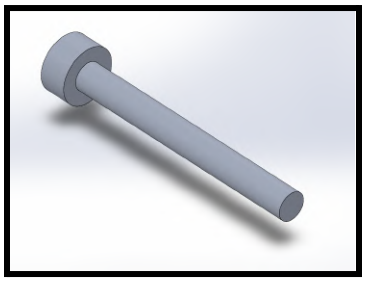
Manufacturing Technology:

- **Precision round bar cutting:** A Ø4 mm stainless steel precision ground rod is cut slightly longer than the final shaft length.
- **CNC turning / end finishing:** The shaft is clamped in a small collet chuck. Both ends are faced to the final 100 mm length and lightly chamfered so the shaft can slide through the gate hub and tunnel supports without damaging them.
- **Surface finishing:** The shaft diameter is polished if needed so the plastic gate rotates smoothly with low friction.
- **Deburring and passivation:** Both shaft ends and chamfers are deburred, then the shaft is passivated to restore corrosion resistance.

Important manufacturing features:
The Ø4 mm outer diameter is the functional bearing surface, so it must stay smooth and free from clamp marks. The end chamfers are important because the shaft passes through the plastic gate hub and should not scrape or crack the plastic during assembly. No threads are needed on the shaft itself; axial fixing can be done with small collars or clips outside the gate.

10.3. Egg Breaking Mechanism

10.3.1. Egg holding Shaft



Material: EN AW-6082 T6
Justification:
EN AW-6082 T6 is selected because the shaft rotates with the servo and must stay lightweight over its 200 mm length. Aluminium reduces rotating mass compared with stainless

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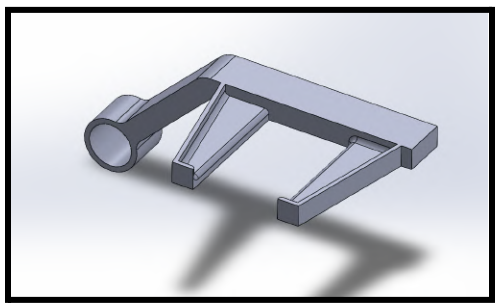
steel, while still being stiff enough for the egg holder load. It is also more suitable than plastic because the servo connection and long shaft need better stiffness and wear resistance.

Manufacturing Technology:

- **Automatic bar saw cutting:** The EN AW-6082 T6 round bar is cut slightly longer than the final shaft length to leave machining allowance on both ends.
- **Setup 1 - CNC turn-mill centre:** The raw bar is clamped in a collet chuck on the circular-head side, with extra stock left for gripping. The free shaft end is supported by a tailstock centre. In this setup, the long shaft diameter, shoulder, rear shaft end, and shaft-end chamfer are turned.
- **Setup 2- re-clamping in soft collet:** The part is flipped and clamped on the finished shaft diameter using a soft collet or protective split sleeve to avoid marking the shaft surface. The circular head face is then faced to final thickness.
- **C-axis milling on the head face:** The turn-mill C-axis locks the circular head in position. The servo horn pocket is milled into the front face of the circular head so the horn sits flat and aligned.
- **Hole machining on the head face:** The servo screw positions are spot drilled, drilled, and thread formed from the same head-face setup, so the pocket and screw holes stay aligned.
- **Final finishing:** The shaft end, shoulder, circular head edge, servo pocket, and threaded holes are deburred. The part is then clear anodised for corrosion resistance and cleanability.

Important manufacturing features:
The long shaft is supported during turning to prevent bending and vibration. The second setup clamps on the shaft diameter using soft tooling, not directly on the servo horn pocket or finished head face. The servo pocket and screw holes are machined in one setup on the circular head face to keep the servo horn accurately aligned with the shaft axis.

10.3.2. Egg Holding Arm



Material: EN AW-6082 T6

Justification:
EN AW-6082 T6 is used because the arm rotates with the shaft, so it must be lightweight but still stiff enough at the fork arms and ribs. Aluminium is preferred over stainless steel because it

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reduces servo load, and it is preferred over plastic because the hub and long arms need better stiffness and dimensional stability.

Manufacturing Technology:

Automatic band saw cutting to cut an EN AW-6082 T6 aluminium billet slightly larger than the final egg holder size.

5-axis CNC milling to machine the main arm shape, rectangular crossbar, open middle area, fork-like arms, triangular ribs, and outer profile. A 3-flute carbide end mill for aluminium is used for rough milling and profile cutting.

Finish milling on the 5-axis CNC milling centre to create the final egg-contact surfaces, fork-like tips, rib faces, and rounded edges. A 2-flute carbide ball-nose end mill for aluminium is used for the rounded surfaces and internal radii.

Circular milling of the cylindrical hub on the 5-axis CNC milling centre. A 3-flute carbide end mill for aluminium is used with circular interpolation to machine the outside diameter of the hub.

Spot drilling of the shaft hole position through the cylindrical hub using a carbide spot drill. This keeps the shaft hole centred and prevents the drill from walking.

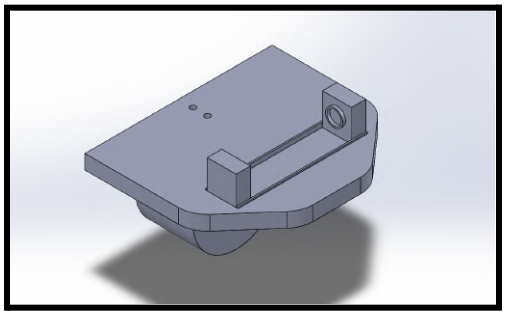
Drilling of the shaft hole through the hub using an HSS twist drill for aluminium.

Reaming of the shaft hole using a machine reamer to achieve an accurate fit between the holder and the rotating shaft.

c**lear anodising after machining** to improve corrosion resistance and cleanability in the raw egg/kitchen environment.

Final mechanical deburring using a hand deburring tool and fine abrasive pad around the shaft hole, fork tips, ribs, open middle area, and outer edges.

10.3.3. Base of Spear



Material:Clear anodised EN AW-6082 T6

Justification:

EN AW-6082 T6 is used because the base must keep the spear guide and shaft support accurate while staying lightweight. Clear anodising is added for corrosion resistance and easier cleaning in the raw egg area.

Manufacturing Technology:

Band saw cutting: An oversized aluminium billet is cut with extra stock for machining and clamping.

Setup 1 — top-side CNC milling: The billet is clamped from the bottom. The top reference face, guide rails, spear guide slot, rectangular cut-outs, curved outer profile, and small vertical mounting holes are machined from the top side.

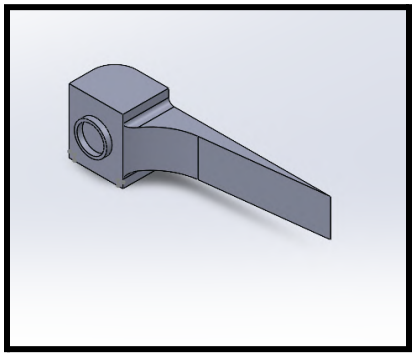
5-axis side machining: The machine indexes to the upright support block. The shaft/bushing hole is drilled and bored through the upright block in the shaft direction, so the spear shaft aligns with the guide slot.

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Setup 2 — bottom-side finishing: The part is flipped and held on the finished top face using soft jaws. The bottom reference face and lower rounded body are finished so the base sits flat in the assembly.
Finishing: The guide slot, mounting holes, shaft/bushing hole, rectangular cut-outs, upright block edges, and outer profile are deburred. The part is then clear anodised.

Important manufacturing features: The top face is the main datum because it controls the spear guide slot and mounting position. The shaft/bushing hole is machined after the upright block is finished so it stays aligned with the spear movement. The bottom face is finished last to make the base sit flat during assembly.

10.3.4. Spears : Left and Right



Material: Martensitic stainless steel, EN 1.4116 / X50CrMoV15, DIN EN 10088-3
Justification:
EN 1.4116 is used because the spear needs a hard, sharp cracking edge that can repeatedly contact eggshells without bending or wearing quickly. It is preferred over aluminium and AISI 304 because they are too soft for the cracking edge, while EN 1.4116 can be hardened, ground, polished, and passivated for the raw egg area.

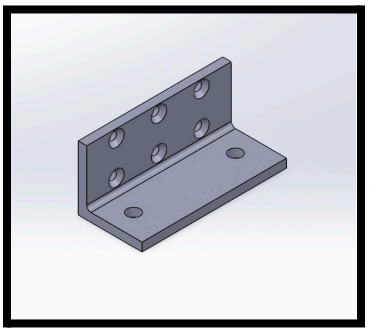
Manufacturing Technology:
Band saw cutting: Oversized EN 1.4116 billets are cut for the left and right mirrored spears, leaving machining allowance on the mounting block and tapered blade.
Setup 1 — mounting block milling: The billet is clamped on the bottom face. The rear mounting block, flat reference faces, curved transition, and rough tapered spear profile are CNC milled from the top side.
Hole and counterbore machining: The mounting block remains in the same setup. The shaft hole is spot drilled, drilled, bored, and reamed through the block. The counterbore is machined on the outside face of the block for the bushing or shaft seat.
Setup 2 — taper and underside finishing: The part is flipped and held on the finished mounting block faces using soft jaws. The underside of the spear body and final tapered blade thickness are milled.
Cracking edge grinding: The spear tip and lower cracking edge are ground along the long tapered blade. This edge is kept sharp and is not rounded during deburring.

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Heat treatment and finish grinding: The spear is vacuum hardened and tempered, then the cracking edge and shaft hole area are finish ground to restore final accuracy after heat treatment.
Finishing: Only the non-cutting edges, mounting block edges, counterbore edge, and shaft hole edge are deburred. The spear is electropolished and passivated for corrosion resistance and cleanability.

Important manufacturing features:
The mounting block is the main datum because it controls the shaft hole and spear position in the mechanism. The shaft hole and counterbore are machined from the same setup to keep them concentric. The cracking edge is ground after milling and protected during final deburring so the spear remains sharp.

10.3.5 Solenoid Motor Mounting Bracket



Material: EN 1.4301 / AISI 304 stainless steel sheet, 3.0 mm

Justification:
EN 1.4301 is used because the bracket is inside the machine and must resist humidity, oil vapour, and cleaning. A 3.0 mm stainless sheet gives enough stiffness to hold the solenoid motor and its casing without making the bracket unnecessarily heavy.

Manufacturing Technology:

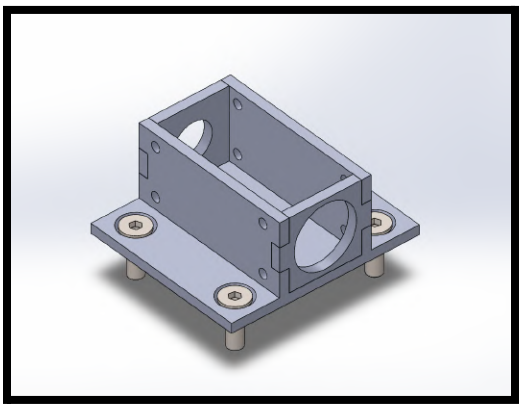
- **CNC fibre laser cutting:** The flat blank is cut from stainless steel sheet, including the outer profile, base mounting holes, vertical solenoid/casing holes, and bend reliefs.
- **Hole cutting:** The base holes and vertical flange holes are cut while the sheet is flat so the hole pattern is positioned accurately before bending.
- **Press brake bending:** The sheet is bent 90° along the bend line between the horizontal base flange and the vertical mounting flange. The inside radius is created by the press brake tool.
- **Bend direction:** The vertical flange is bent upward from the base flange. The base holes fix the bracket to the machine frame, while the vertical hole pattern mounts the solenoid motor and casing.
- **Hole finishing:** The hole edges are deburred after bending, especially around the vertical solenoid mounting holes and base holes, so the screws sit flat and the casing is not scratched.

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- **Surface finishing:** The outer edges, bend reliefs, hole edges, and flange corners are rounded. The part is brushed and passivated after cutting and bending.

Important manufacturing features:
The bend line controls the 90° angle between the base and vertical mounting face, so it controls the solenoid alignment. The vertical hole pattern is the critical feature because it locates the solenoid motor and casing. The base holes stay on the horizontal flange so the bracket can be fixed securely to the frame.

10.3.6 Solenoid Motor Casing



Material: Clear anodised EN AW-6082 T6 aluminium

Justification:
EN AW-6082 T6 is used because the casing must hold the solenoid motor accurately while staying lightweight and stiff. Aluminium is suitable because the part is not in direct food contact, is easy to machine, and provides enough strength for the solenoid mounting and frame connection.

Manufacturing Technology:

- **Block cutting:** An oversized aluminium billet is cut to leave machining allowance for the base flange, side walls, and front/rear faces.
- **Setup 1 — base and top machining:** The billet is clamped from the outer side faces. The bottom mounting face, base flange, open internal cavity, wall thickness, and top rim are CNC milled from the top side.
- **Large circular openings:** The front and rear circular solenoid openings are drilled and bored through the vertical end faces. These openings are machined from the solenoid axis direction so the motor body sits straight inside the casing.
- **Side holes and cut-outs:** The smaller side holes are drilled from the outside faces for fastening the solenoid cover, cable routing, or side-mounted components.
- **Base mounting holes:** The flange holes are drilled and counterbored from the top side of the base flange so the screw heads sit flat when the casing is fixed to the bracket or frame.

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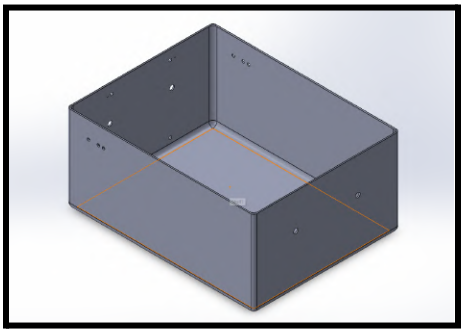
- **Setup 2 — bottom finishing:** The part is flipped and clamped on the finished top rim or side faces using soft jaws. The underside of the base is face milled so the casing sits flat on the mounting surface.
- **Finishing:** The large circular openings, side holes, base counterbores, top rim, internal cavity edges, and outside edges are deburred. The part is clear anodised after machining.

Important manufacturing features:
The bottom face is the main datum because it controls how the casing sits on the bracket. The large circular openings are the critical features because they locate the solenoid motor axis. The base counterbores are placed on the top side of the flange so the fixing screws sit flush and do not interfere with the solenoid body.

10.4. Egg Shell Storage

10.4.1. Egg Shell Storage (3 step manufacturing)

1.Main Drawer Box



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet, 1.0 mm
Justification:
EN 1.4301 is used because the drawer is exposed to wet eggshells, raw egg residue, cleaning water, and kitchen humidity. It gives enough corrosion resistance and cleanability for a non-direct food-contact storage part.

Manufacturing Technology:

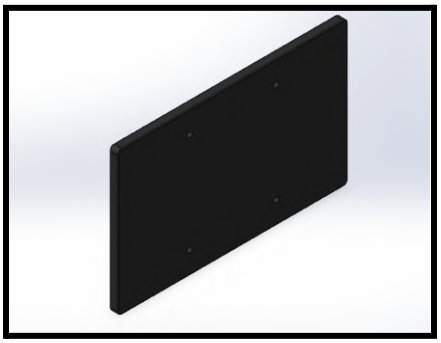
CNC fibre laser cutting: The flat sheet blank is cut with the outer profile, corner reliefs, shaft holes, handle holes, drainage/side holes, and top-edge geometry before bending.
Press brake bending: The bottom sheet is bent upward along the four drawer bend lines to form the front, rear, and side walls. The bend sequence starts with the shorter side walls, then the longer front and rear walls to avoid collision with the press brake tooling.
Top rim hemming: The upper edges of the drawer walls are folded outward/downward to remove sharp sheet edges and stiffen the open top rim.
TIG welding: The vertical corner seams are welded after bending to close the drawer corners and prevent egg residue from collecting in open gaps.
Weld finishing: The welded corner lines are ground smooth, especially on the inside

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corners, so the drawer can be cleaned easily.
Deburring and brushing: The laser-cut holes, top rim, front opening, welded corners, and outer edges are deburred. The visible stainless surfaces are brushed for a clean finish.

Important manufacturing features:
The bend lines define the drawer walls and must match the corner reliefs so the corners close cleanly after bending. The TIG weld seams are placed only on the vertical drawer corners. The inside welds are smoothed because this is where wet eggshell residue can collect.

2.Drawer Front Cover



Material: ABS-HI plastic
Justification:
ABS-HI is used because the front cover is a visible, non-food-contact part that needs good impact resistance, a clean surface finish, and enough stiffness around the screw holes. It is preferred over PP because it gives a better visible surface and is more dimensionally stable for a flat cover plate.

Manufacturing Technology:

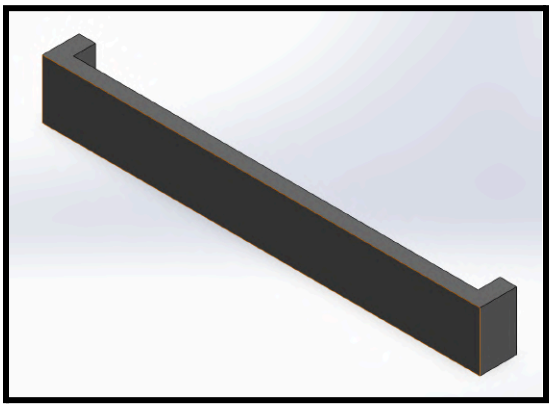
Injection moulding: The cover is moulded as one ABS part, including the flat front panel, rounded outer corners, raised/thickened border, and screw-hole bosses on the rear side.
Mould parting line: The parting line is placed around the outer edge of the cover so it is not visible on the main front face.
Core pins for holes: The visible screw holes and smaller locating holes are formed with core pins from the front-to-back direction. This keeps the holes aligned with the drawer mounting points.
Rear-side bosses: The area around the screw holes is locally thickened on the rear side to prevent cracking when the cover is screwed onto the drawer.
Draft direction: A small draft angle is applied on the outer edge and rear bosses in the mould opening direction so the cover can be released cleanly.

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Gate position: The injection gate is placed on the rear side of the cover so no gate mark appears on the visible front face.
Finishing: The outer edge, hole edges, and parting-line flash are trimmed and deburred after moulding.

Important manufacturing features:
The front face is kept free of gates, ejector marks, and parting lines because it is the visible user-facing surface. The holes are moulded in the front-to-back direction to match the screw direction during assembly. The rear bosses support the screw holes so the plastic cover does not deform or crack during tightening.

3.Drawer Handle



Material: ABS-HI plastic with brass threaded inserts

Justification:
ABS-HI is used because the handle is a visible user-contact part that needs good impact resistance, a clean surface finish, and low weight. Brass inserts are used for the threaded holes because repeated screw tightening would wear plastic threads.

Manufacturing Technology:

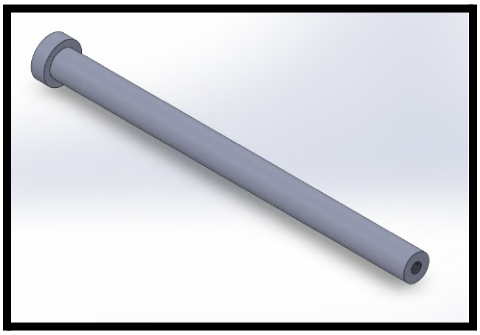
Injection moulding: The handle is moulded as one plastic part, including the long grip bar, thicker end blocks, rounded outer edges, and rear mounting bosses.
Mould parting line: The parting line is placed around the side edge of the handle so the front grip face stays clean and visible.
Core pins for mounting holes: The mounting holes are formed from the rear side of the handle, in the same direction as the screws used to attach it to the drawer cover.
Heat-set threaded inserts: Brass threaded inserts are pressed into the rear mounting holes after moulding. These create stronger threads for the drawer screws.
Draft direction: Small draft is applied on the side walls of the long grip and end blocks so the handle can release from the mould without marking the front face.
Gate position: The injection gate is placed on the rear side of the handle, away from

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the visible front grip surface.
Finishing: The parting-line flash, hole edges, and insert openings are trimmed and deburred after moulding.

Important manufacturing features:
The front grip face is kept free from gate marks and ejector marks because it is visible to the user. The threaded inserts are placed only in the rear mounting holes, where the handle is screwed to the drawer. The thicker end blocks support the inserts and prevent cracking when the handle is pulled.

4.Inner Shaft



Material: EN 1.4301 / X5CrNi18-10 stainless steel round bar

Justification:
EN 1.4301 is used because the shaft is inside the eggshell drawer and is exposed to wet eggshells, raw egg residue, and cleaning water. Stainless steel keeps the sliding surface corrosion-resistant and easy to clean.

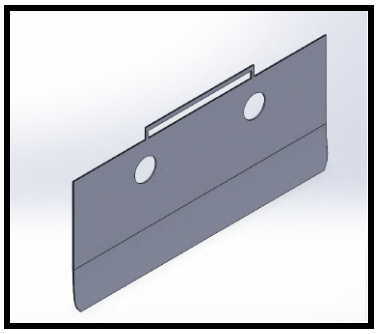
Manufacturing Technology:

- **Bar cutting:** The stainless steel round bar is cut slightly longer than the final shaft length to leave machining allowance on both ends.
- **Setup 1 — CNC turning:** The bar is clamped in a collet chuck and supported by a tailstock centre because the shaft is long. The long shaft diameter, end flange/stop collar, shoulder, and outer chamfers are turned in this setup.
- **Setup 2 — end machining:** The shaft is flipped and clamped on the finished shaft diameter using a soft collet. The end face is finished, then the axial hole is drilled and tapped from the shaft end.
- **Finishing:** The shaft ends, shoulder, flange edge, and threaded hole edge are deburred. The shaft surface is polished so the compression plate can slide smoothly.

Important manufacturing features:
The long shaft diameter is the functional sliding surface, so it is protected during re-clamping. The end thread is machined along the shaft axis so the fixing screw pulls straight into the shaft. The flange/stop collar controls the shaft position inside the drawer.

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5.Manual Compression Plate



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet

Justification:
EN 1.4301 is used because the plate moves inside the eggshell drawer and is exposed to wet eggshells, raw egg residue, and cleaning water. It gives enough stiffness, corrosion resistance, and cleanability for the sliding compression part.

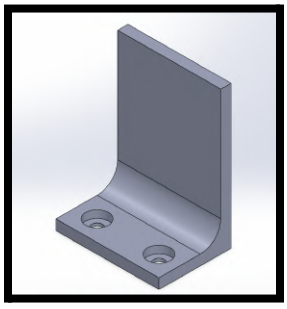
Manufacturing Technology:

- **CNC fibre laser cutting:** The flat sheet profile is cut first, including the outer contour, two shaft holes, long upper slot, top notch, and rounded lower corners.
- **Press brake bending:** The plate is bent along the horizontal bend line across the lower section. The bend gives the lower face its angled compression surface while keeping the shaft holes and upper slot in the flat upper area.
- **Hole finishing:** The two shaft holes are reamed after cutting so the plate can slide on the inner shafts without too much play.
- **Deburring and brushing:** The shaft holes, slot, notch, outer profile, rounded corners, and bend edge are deburred. The plate faces are brushed to improve cleanability.
- **Passivation:** The plate is passivated after cutting, bending, and finishing to restore corrosion resistance.

Important manufacturing features:
The shaft holes are kept in the flat upper section because they guide the plate along the drawer shafts. The bend line is placed below the shaft holes so the lower plate area can compress the eggshells without disturbing the sliding fit. All cut edges are deburred because the part is handled and cleaned regularly.

10.5. Oil Storage

10.5.1 Oil Storage Mount



Material: EN AW-6082 T6 aluminium
Justification:
EN AW-6082 T6 is used because the bracket must hold the oil system rigidly while staying lightweight. Aluminium is suitable here because the part is not in direct food contact, is easy to machine, and provides enough stiffness for the mounting load.

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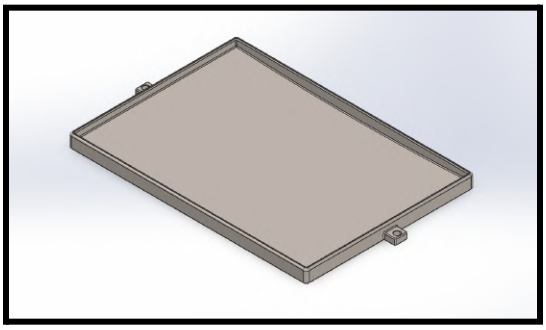
Manufacturing Technology:

- **Band saw cutting:** An oversized aluminium billet is cut to leave machining allowance for the L-shape, vertical wall, and mounting base.
- **Setup 1 — bottom/base machining:** The billet is clamped from the side faces. The bottom mounting face, outer base profile, and two counterbored mounting holes are machined from the top side. The counterbores are placed on the upper face so the screw heads sit flush.
- **Setup 2 — upright wall machining:** The part is rotated and clamped on the finished base face using soft jaws. The vertical mounting face, top edge, and side faces of the upright wall are milled.
- **Inside radius milling:** The curved internal transition between the base and upright wall is milled with a ball-nose end mill. This radius reduces stress concentration where the oil system load bends the bracket.
- **Finishing:** The counterbore edges, screw holes, inner radius, upright wall edges, and outer profile are deburred. The part can be clear anodised for corrosion resistance and cleanability.

Important manufacturing features:
The bottom face is the main datum because it controls how the bracket sits on the machine frame. The vertical wall face is machined perpendicular to the base so the oil system sits straight. The screw holes and counterbores are machined from the mounting side to keep the screw heads flush and accessible during assembly.

10.6. Conveyor System

10.6.1. Pocket Plate



Material: EN 1.4409 / GX2CrNiMo19-11-2 stainless steel precision casting

Justification:
EN 1.4409 is used because the plate is in direct contact with raw egg, oil, heat, and cleaning water. It gives high corrosion resistance and hygiene performance. Casting is preferred because the concave pocket, raised walls, rounded corners, and side tabs can be made as one part without welded joints in the food-contact area.

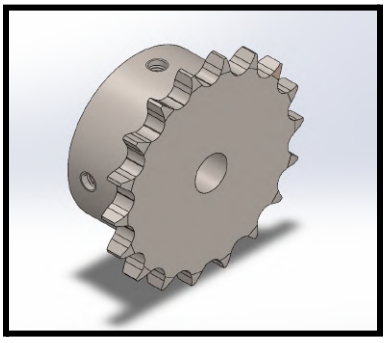
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Manufacturing Technology:

- **Wax pattern and investment casting:** The concave egg pocket, raised rim, rounded internal corners, side walls, underside shape, and integrated chain-mounting tabs are cast as one body.
- **Gate placement:** Casting gates are placed on the underside and outer non-food-contact areas, so the top egg-contact pocket remains uninterrupted.
- **Gate removal and blasting:** The gate marks are cut off from the underside. The cast surface is shot blasted to remove shell residue before machining.
- **CNC finish machining:** The plate is clamped on the underside and outer side walls. The concave pocket, upper rim, side walls, and underside reference face are finish milled so the egg-contact surface is smooth and the plate sits flat.
- **Tab hole drilling:** The mounting holes are drilled through the integrated side tabs for the plate-to-chain fasteners.
- **Polishing and passivation:** The concave pocket, raised rim, side walls, and reachable outer surfaces are polished. The part is passivated after machining and polishing.

Important manufacturing features:
The top concave pocket is the main food-contact surface, so casting gates and clamping marks are kept away from it. The side tabs are cast together with the plate to avoid welded joints near the cooking area. The underside is machined as the reference surface so the plate mounts flat on the conveyor attachments.

10.6.2. Sprocket for Conveyor Belt



Sprocket for Conveyor Belt
Material: EN 1.4301 / X5CrNi18-10 stainless steel round bar

Justification:
EN 1.4301 is used because the sprocket is close to the food and heating area, so it must resist oil, humidity, cleaning water, and corrosion. It also has enough strength and wear resistance to drive the 06B-1 roller chain.

Manufacturing Technology:

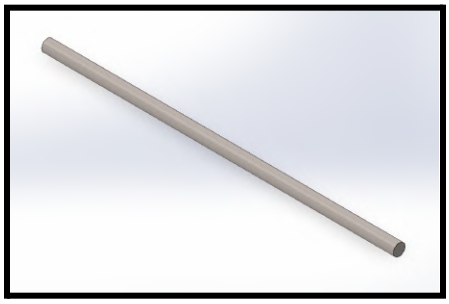
- **Band saw cutting:** A stainless steel round-bar blank is cut slightly thicker than the final sprocket width.
- **Setup 1 — CNC turning:** The blank is clamped in a chuck. Both side faces, the hub step, outside blank diameter, and central shaft bore are turned from the same axis so the bore and sprocket body stay concentric.

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- **Bore finishing:** The central bore is bored and reamed to the final shaft fit. This bore becomes the reference for the later tooth and locking-hole operations.
- **Sprocket tooth cutting:** The part is mounted on a mandrel through the finished central bore. The 06B-1 tooth profile is hobbed around the outside diameter so the teeth are concentric with the shaft bore.
- **Locking screw holes:** The sprocket is held on the central bore and indexed on a 4-axis CNC machine. The radial locking holes are drilled and tapped through the hub side so the sprocket can be fixed to the shaft.
- **Finishing:** The tooth edges, bore edges, side faces, hub edges, and threaded-hole openings are deburred. The visible and cleanable surfaces are polished, then the sprocket is passivated.

Important manufacturing features:
The central bore is the main datum because it controls sprocket rotation on the shaft. The teeth are cut after the bore is finished so the chain pitch circle stays concentric with the shaft. The locking screw threads are placed radially through the hub, not through the teeth, so the screws clamp directly onto the shaft without weakening the tooth profile.

10.6.3. Driving Shaft/Idler Shaft (Same Technology just one is slightly longer)



Material: EN 1.4301 / X5CrNi18-10 stainless steel round bar

Justification:
EN 1.4301 is used because the shafts are close to the food, oil, heat, and cleaning area. It provides corrosion resistance and enough strength to support the sprockets and keep the conveyor chains aligned.

Manufacturing Technology:

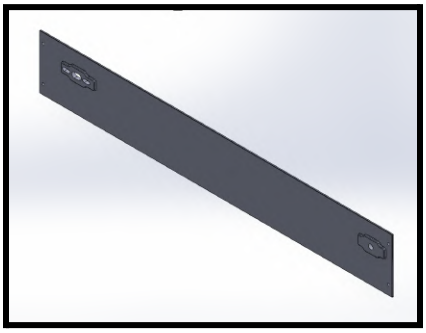
- **Bar cutting:** The stainless steel round bar is cut slightly longer than the final shaft length.
- **Setup 1 — CNC turning:** The bar is clamped in a collet chuck and supported with a tailstock centre because the shaft is long and slender. The shaft diameter, end face, and first end chamfer are turned from this setup.

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- **Setup 2 — end finishing:** The shaft is flipped and clamped on the finished diameter using a soft collet. The second end face is turned to final length and the second chamfer is added.
- **Bearing and sprocket seating surface:** The full shaft diameter is centreless ground or polished so the bearings and sprockets sit accurately and slide on without damaging the surface.
- **Finishing:** Both shaft ends, chamfers, and bearing/sprocket seating areas are deburred and passivated.

Important manufacturing features:
The shaft diameter is the main functional surface because it carries the bearings and sprockets. Soft clamping is used after turning so the finished shaft surface is not marked. The end chamfers help the shaft pass through the bearings and sprocket bores during assembly.

10.6.4. Conveyor Side Frame



Material: EN 1.4301 / X5CrNi18-10 stainless steel plate
Justification:
EN 1.4301 is used because the side frame is close to the conveyor, oil, heat, and cleaning area. It provides enough stiffness for the shafts and bearing mounts while giving suitable corrosion resistance for a non-direct food-contact structural part.

Manufacturing Technology:

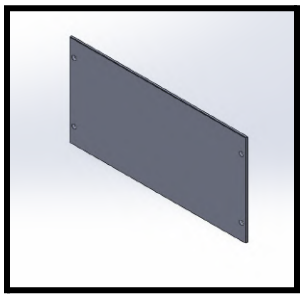
- **CNC fibre laser cutting:** The long rectangular outer profile, end mounting holes, bearing/shaft openings, and adjustment slots are cut from the stainless steel plate in the flat state.
- **Slot finishing:** The bearing adjustment slots are finished by CNC milling if tighter fit is needed. These slots allow the shaft/bearing position to be adjusted for chain tensioning.
- **Hole finishing:** The bearing mounting holes and frame fixing holes are drilled or reamed after laser cutting where accurate alignment is required.
- **Edge chamfering:** The long outer edges, slot edges, shaft openings, and mounting holes are lightly chamfered to remove sharp laser-cut edges.
- **Surface finishing:** The visible faces are brushed, and the part is passivated after cutting and machining.

Important manufacturing features:
The two end slot areas are the critical features because they control shaft and bearing

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alignment. The holes and slots are made while the part is flat so their spacing stays accurate along the full frame length. No bending is used for this part, so the flatness of the plate must be maintained during cutting and finishing.

10.6.5. Conveyor Back Frame



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet

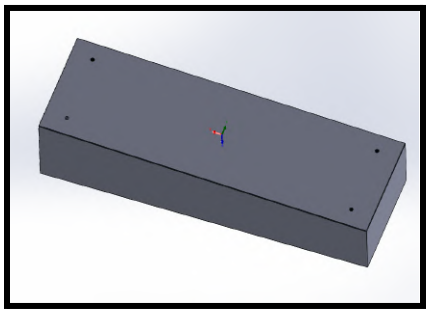
Justification:
EN 1.4301 is used because the back frame supports the conveyor structure and keeps the two side frames aligned. It is close to heat, oil, humidity, and cleaning water, so stainless steel is suitable for corrosion resistance and cleanability.

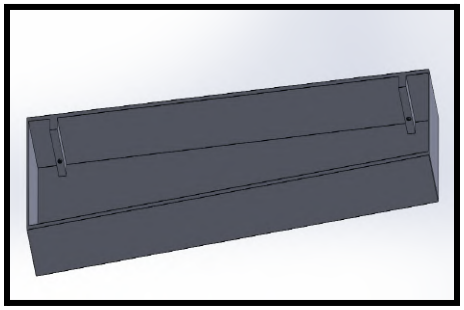
Manufacturing Technology:

- **CNC fibre laser cutting:** The rectangular outer profile and corner mounting holes are cut from a flat stainless steel sheet in one setup.
- **Mounting hole finishing:** The corner holes are drilled or reamed after laser cutting if tighter clearance is needed for the frame fasteners.
- **Edge finishing:** The outer profile and mounting holes are deburred and lightly chamfered to remove sharp laser-cut edges.
- **Surface finishing:** The visible faces are brushed, and the part is passivated after cutting and finishing.

Important manufacturing features:
The mounting holes are the critical features because they locate the back frame between the two conveyor side frames. All holes are made while the sheet is flat so the spacing stays accurate. No bending or welding is used, so the part must remain flat to keep the side frames aligned.

10.6.6. Frame under conveyor





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Conveyor Belt Hollow Support Frame

Material: EN AW-6082 T6 aluminium billet

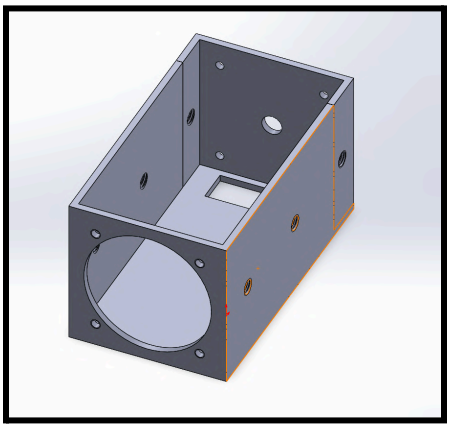
Justification:
EN AW-6082 T6 is used because the frame must support the conveyor while staying lightweight and stiff. The solid billet allows the underside to be hollowed out while leaving thick bosses for the M10 threaded mounting points.

Manufacturing Technology:

- **Block sawing:** An oversized aluminium billet is cut to leave machining allowance on the outer walls, top face, and bottom pocket.
- **Setup 1 — top-side machining:** The billet is clamped from the side faces. The top reference face, outer rectangular profile, and three M10 hole positions are machined from the top side.
- **M10 hole machining:** The three holes are drilled vertically through the top face using Ø8.5 mm core holes, then tapped M10×1.5 from the same top-side setup. This keeps the threads perpendicular to the conveyor mounting surface.
- **Setup 2 — underside pocket machining:** The part is flipped and clamped on the finished outer side faces using soft jaws. The underside cavity is milled out, while thicker internal bosses are left directly under the M10 threaded holes.
- **Edge finishing:** The outer profile, top edge, underside pocket edge, internal bosses, and M10 hole openings are chamfered and deburred.
- **Cleaning:** The hollow underside channel and threaded holes are washed to remove aluminium chips and cutting oil before assembly.

Important manufacturing features:
The top face is the main datum because it supports the conveyor assembly. The M10 threaded holes are machined before the underside pocket so their positions stay accurate. The underside pocket removes weight, while the internal bosses remain under the threaded holes to prevent thread stripping and local deformation during tightening.

10.6.7. Nema 17/Gearbox/Encoder Motor Housing



Material: EN AC-43500 / EN AC-AlSi10MnMg cast aluminium alloy

Justification:
EN AC-43500 is used because the housing must keep the NEMA 17 motor, gearbox, and encoder aligned

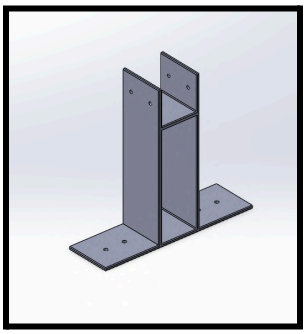
while remaining lightweight. Aluminium casting also allows the box shape, internal cavity, motor opening, and mounting features to be produced as one stiff casing.

Manufacturing Technology:

- **High-pressure die casting:** The housing body is cast with the outer box shape, open top, side walls, rear wall, front motor face, and internal cavity.
- **Parting line and draft:** The die-casting parting line is placed around the outer side of the housing. Draft is applied on the outer walls, inner cavity walls, and vertical ribs so the casing releases from the die.
- **Core inserts:** A main core insert forms the internal cavity for the motor, gearbox, and encoder. A second core forms the rough front gearbox opening.
- **CNC reference machining:** The front motor mounting face, top rim, bottom reference face, and side reference faces are milled after casting so the motor and gearbox sit square in the housing.
- **Front opening machining:** The front gearbox opening is bored to final diameter from the motor mounting face side, keeping it concentric with the motor position.
- **Mounting holes and threads:** The motor, gearbox, side-cover, rear-cover, and frame mounting holes are drilled and thread-formed after casting. These threaded holes are machined from their assembly side so screws enter straight during assembly.
- **Finishing:** The front opening, top rim, internal cavity edges, threaded holes, and outer edges are chamfered and deburred.
- **Powder coating:** The outside casing is powder coated for protection. The front motor face, bottom reference face, bored opening, and threaded holes are masked so coating does not affect assembly accuracy.

Important manufacturing features:
The front motor face is the main datum because it controls motor and gearbox alignment. The internal cavity is cast with a core insert instead of fully machined from solid material. Precision faces and threaded holes are machined after casting because cast surfaces alone are not accurate enough for the motor, gearbox, encoder, and cover alignment.

10.6.8 Nema Motor Mount



Material: EN 1.4301 / AISI 304 stainless steel sheet, 3.0 mm

Justification:
EN 1.4301 is used because the motor mount is close to the conveyor, oil vapour, humidity, and cleaning area. Stainless steel

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gives enough stiffness and corrosion resistance for holding the NEMA 17 motor and keeping the drive shaft aligned.

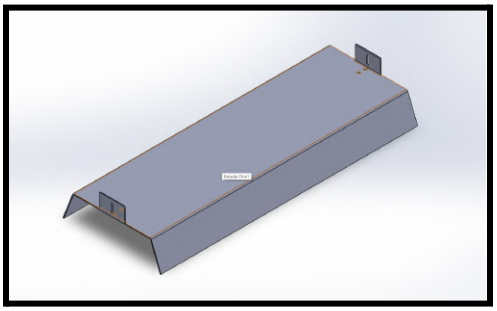
Manufacturing Technology:

- **CNC fibre laser cutting:** The base plate, two vertical motor support plates, and internal spacer/gusset plates are cut from stainless steel sheet. The base fixing holes, motor mounting holes, and motor shaft clearance opening are cut while the plates are flat.
- **Motor mounting hole finishing:** The NEMA 17 screw holes and central shaft opening on the vertical mounting plate are drilled or reamed after laser cutting if tighter alignment is required.
- **Welding fixture positioning:** The vertical plates are positioned perpendicular to the base plate in a welding fixture. The fixture keeps the motor face square to the base so the motor shaft aligns with the conveyor drive shaft.
- **TIG tack welding:** Tack welds are placed first at the base-to-vertical plate contact edges and at the spacer/gusset contact edges to lock the geometry before final welding.
- **TIG seam welding:** Weld seams are applied along the bottom edges of the vertical plates and along the spacer/gusset contact lines. These welds stiffen the motor mount against motor torque.
- **Edge finishing:** The base holes, motor holes, shaft opening, outer edges, and welded seams are deburred. The visible surfaces are brushed and passivated after welding.

Important manufacturing features:
The vertical motor plate is the critical reference because it controls the NEMA 17 motor position. The base plate holes fix the mount to the conveyor frame, while the motor holes locate the motor relative to the drive shaft. The spacer/gusset plates are welded between the vertical plates to reduce vibration and prevent the motor plate from bending under torque.

10.7. TOP IR System

10.7.1. TOP IR Shielding



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet, 2.0 mm

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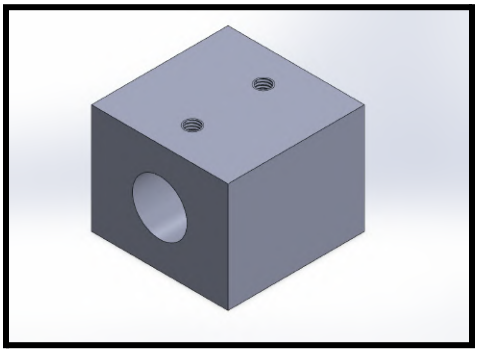
Justification:
EN 1.4301 is used because the shield is close to the IR heating zone and must resist heat, oil vapour, humidity, and cleaning. It is suitable because the part is a heat shield, not a direct food-contact cooking surface.

Manufacturing Technology:

- **CNC fibre laser cutting:** The flat sheet blank is cut with the outer profile, slotted mounting holes, circular screw hole, integrated tabs, bend reliefs, and front/back edge shape.
- **Slotted hole cutting:** The vertical slots are laser cut in the mounting tabs while the sheet is flat. These slots allow height adjustment when fixing the shield in the assembly.
- **Bend relief cutting:** Relief cuts are placed at the tab bases and side-wall transitions so the material does not tear or distort during bending.
- **Press brake bending — side walls:** The two long bend lines are bent downward to form the trapezoidal hood shape around the IR tube and guide the heat downward.
- **Press brake bending — mounting tabs:** The integrated tabs are bent upward from the top surface after the side walls are formed. The tabs are made from the same sheet, so no welded tab joints are needed.
- **Finishing:** The lower lips, slotted holes, circular hole, bend reliefs, tabs, and outer edges are deburred. The visible surfaces are brushed and then passivated.

Important manufacturing features:
The two long bend lines define the trapezoidal hood angle, so they control the final clearance around the IR tube. The tab bend lines control the mounting position of the shield. The bend reliefs are placed at the tab roots and side transitions to prevent cracking during press brake bending.

10.7.2. Ceramic Block for holding Quarts IR



Material: 96% aluminium oxide ceramic, Al_2O_3 C799

Justification:
Al_2O_3 C799 is used because the block sits directly beside the quartz IR tube and must resist high temperature while electrically insulating the tube from the metal shield. Stainless threaded inserts are used because ceramic threads would be too brittle for repeated screw tightening.

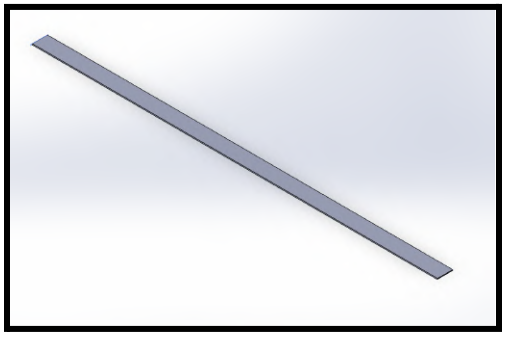
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Manufacturing Technology:

- **Ceramic powder pressing:** The rectangular block blank is pressed with extra allowance for sintering shrinkage.
- **Green-state machining:** The quartz-tube hole and top insert holes are machined while the ceramic is still in the softer green state. The tube hole is placed through the side of the block, while the insert holes are placed from the top face for the mounting screws.
- **Sintering:** The machined green block is fired in a ceramic furnace. The hole and outer dimensions include shrinkage allowance so the final ceramic block reaches the required size after firing.
- **Diamond finishing:** After sintering, the outer faces are diamond ground flat. The quartz-tube hole is diamond finished to its final diameter so the tube is supported smoothly without point contact.
- **Threaded insert placement:** A2 stainless threaded inserts are bonded into the top holes using high-temperature ceramic adhesive. These inserts provide the screw threads for mounting the ceramic block to the shield.
- **Edge finishing:** The block edges, tube-hole edges, and insert-hole edges are rounded with diamond abrasives to reduce chipping during assembly.
- **Cleaning:** The tube hole, insert holes, and outer surfaces are cleaned to remove ceramic grinding dust before assembly.

Important manufacturing features:
The side through-hole is the critical feature because it supports the quartz IR tube. The top insert holes are kept separate from the tube hole so the mounting screws do not load the quartz tube directly. All final sizing after sintering is done by diamond grinding because fired ceramic is too hard and brittle for normal drilling.

10.7.3. Side Lining for TOP IR



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet, 2.0 mm

Justification:
EN 1.4301 is used because the rail is close to the Top IR heating zone and must resist heat, oil vapour, humidity, and cleaning. It is suitable because the rail is a support part for the perforated mesh, not a direct food-contact surface.

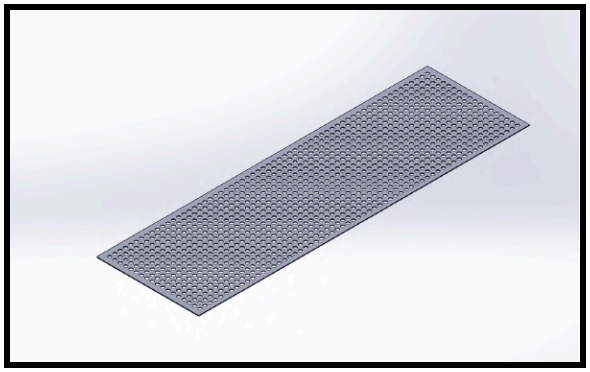
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Manufacturing Technology:

- **CNC fibre laser cutting:** The long flat strip, straight edges, square end, and slanted end are cut from stainless steel sheet in one flat operation.
- **Slanted end cutting:** The angled end is cut to match the sloped side wall of the Top IR shield, so the rail fits inside the trapezoidal hood without protruding.
- **No bending:** The part remains a flat strip. It has no bend lines because its function is only to act as an internal support ledge for the perforated mesh.
- **TIG tack welding:** The rail is first tack welded inside the Top IR shield at the required mesh height to hold its position before final welding.
- **TIG seam welding:** A weld seam is made along the contact edge between the rail and the inner wall of the Top IR shield. This fixes the rail permanently and creates a stable ledge for the mesh.
- **Finishing:** The long edges, slanted end, exposed square end, and welded edge are deburred. The weld area is cleaned and passivated after welding.

Important manufacturing features:
The slanted end follows the angled shield wall, so the rail sits flush inside the hood. The weld seam is placed along the rail-to-shield contact edge, not on the mesh-support surface. The rail must be aligned at the correct height before welding because it controls the final position of the perforated mesh.

10.7.4. Perforated Sheet



Material: EN 1.4301 / X5CrNi18-10 stainless steel perforated sheet, 1.0 mm
Justification:
EN 1.4301 is used because the mesh sits directly below the Top IR heater and must resist heat, oil vapour, humidity, and repeated cleaning. It is suitable because the part acts as a removable heat-protection and ventilation sheet, not as a direct cooking plate.

Manufacturing Technology:

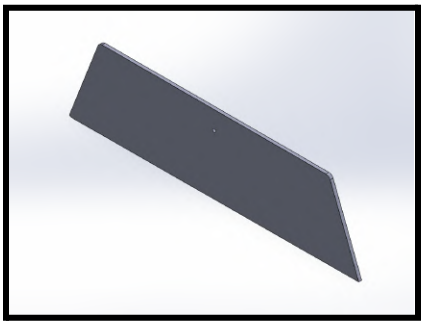
- **CNC turret punching:** The elongated ventilation slots are punched into the flat stainless steel sheet using an obround punch tool. The slot rows are kept evenly spaced so the sheet remains stiff and allows heat to pass through evenly.

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- **CNC fibre laser cutting:** After punching, the outer rectangular profile is laser cut to the final size. The outer border is aligned with the perforation field so the sheet fits correctly into the side lining rails.
- **Flatness correction:** The punched and cut sheet is roller-levelled to reduce warping caused by the large number of slots.
- **Deburring and edge rounding:** The slot edges and outer perimeter are deburred so the sheet can be handled and removed safely during cleaning.
- **Surface finishing:** The visible surface is brushed, and the sheet is passivated after punching, cutting, and deburring.

Important manufacturing features:
The slot pattern is the critical feature because it controls heat passage and prevents large food or shell pieces from reaching the IR tube. The outside perimeter must stay straight because the sheet slides into the welded side lining rails. No welding is used because the mesh must remain removable and replaceable for cleaning and maintenance.

10.7.5. TOP IR Back Frame



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet, 2.0 mm

Justification:
EN 1.4301 is used because the back frame is close to the IR heating zone and must resist heat, oil vapour, humidity, and cleaning. It also matches the Top IR shield material, which makes welding and corrosion protection consistent.

Manufacturing Technology:

- **CNC fibre laser cutting:** The flat back-frame profile is cut from stainless steel sheet, including the angled side edges, rounded corner transitions, lower edge, and central mounting hole.
- **Central hole cutting:** The central hole is laser cut in the same flat-sheet setup as the outer profile so it stays centred relative to the back-frame shape.
- **No bending:** The part remains a flat rear closing plate. It has no bend lines because it is welded directly to the rear opening of the Top IR shield.
- **TIG tack welding:** The back frame is tack welded to the rear of the Top IR shield first. It is aligned with the sloped side walls so the rear hood opening closes cleanly.

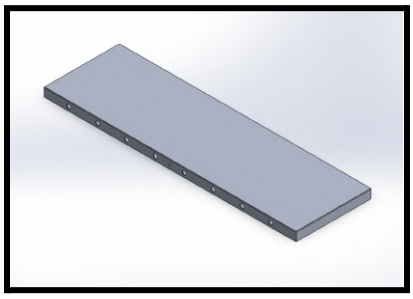
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- **TIG seam welding:** Weld seams are applied along the contact edges between the back frame and the Top IR shield. These seams permanently close the rear side and increase hood stiffness.
- **Finishing:** The angled edges, rounded corners, central hole, lower edge, and welded seams are deburred. The weld areas are cleaned, brushed, and passivated.

Important manufacturing features:
The angled side edges match the trapezoidal side walls of the Top IR shield, so the back frame fits without gaps before welding. The weld seams are placed around the rear contact edges, not across the central hole. The central hole is cut before welding because it is easier to position accurately while the sheet is flat.

10.8. Bottom Heating System

10.8.1. Heating block



Material: EN AW-6082 T651 aluminium plate

Justification:
EN AW-6082 T651 is used because the block must spread heat from the cartridge heaters evenly to the egg pocket plates. Aluminium is preferred over stainless steel because it has much better thermal conductivity and keeps the heating block lighter.

Manufacturing Technology:

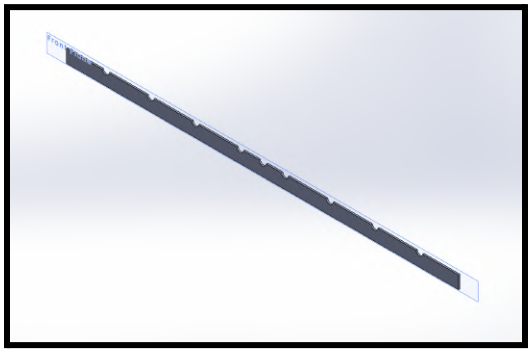
- **Band saw cutting:** An oversized aluminium plate blank is cut to leave machining allowance on the length, width, and thickness.
- **Setup 1 — top and outer machining:** The blank is clamped from the side faces. The top heating surface, rectangular outer profile, and side faces are CNC milled. The top surface is the main heat-transfer face, so it is finished flat.
- **Setup 2 — bottom reference machining:** The block is flipped and clamped on the finished side faces. The bottom reference surface is face milled so the block sits flat in the heating frame.

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- **Cartridge heater holes:** The Ø8 mm heater holes are deep-drilled from the long side face along the length of the block. The holes are kept parallel to the top heating surface so the heaters distribute heat evenly below the egg plates.
- **Temperature sensor hole:** The sensor hole is drilled from the side or bottom into the centre zone between the cartridge heater holes, so the sensor measures the actual block temperature instead of only the heater surface temperature.
- **Finishing:** The heater-hole openings, sensor hole, mounting edges, side edges, and machined corners are deburred. The top heating surface and visible side faces are brushed.

Important manufacturing features:
The top face is the critical heat-transfer surface and must remain flat and clean. The cartridge heater holes are drilled from the long side because the heaters are inserted lengthwise into the block. The sensor hole is positioned between heater holes so the temperature control represents the average heating block temperature.

10.8.2. Side Frame



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet, 2.0 mm

Justification:
EN 1.4301 is used because the strip is close to the bottom heating block and must resist heat, oil vapour, humidity, and cleaning. It is suitable because the strip is a support/clearance part, not a direct food-contact cooking surface.

Manufacturing Technology:

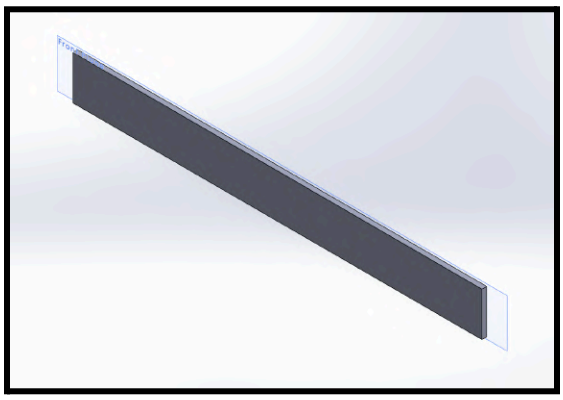
- **CNC fibre laser cutting:** The long strip profile, straight lower edge, end edges, and repeated semicircular notches are cut from flat stainless steel sheet.
- **Notch cutting:** The semicircular notches are laser cut along the upper edge with equal spacing. These notches clear or locate the repeated heater-related parts in the bottom heating assembly.
- **No bending:** The part remains a flat strip, so there are no bend lines or forming operations.
- **Edge finishing:** The long edges, end edges, and every semicircular notch are deburred and rounded so the strip does not damage nearby heater sleeves, wiring, or assembly parts.

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- **Surface finishing:** The visible faces are brushed, and the part is passivated after cutting and deburring.

Important manufacturing features:
The notch pattern is the critical feature because it must match the repeated heater positions. The strip is cut while flat so the notch spacing stays accurate. All notch edges are rounded because they are close to heater-related parts and wiring.

10.8.3. Front Frame



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet, 2.0 mm

Justification:
EN 1.4301 is used because the front frame is close to the bottom heating zone and must resist heat, oil vapour, humidity, and cleaning. It is suitable because the part is a welded support strip, not the main heat-transfer surface.

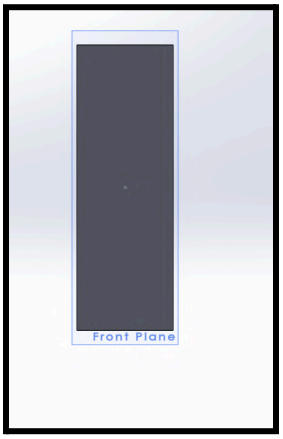
Manufacturing Technology:

- **CNC fibre laser cutting:** The flat front-frame strip is cut from stainless steel sheet, including the long edges and end edges.
- **No bending:** The part remains a flat strip, so there are no bend lines or forming operations.
- **Edge finishing:** The long edges and end edges are deburred and lightly rounded so the strip is safe to handle and fits cleanly against the side and bottom frame parts.
- **Welding fixture positioning:** The front frame is placed in a welding fixture together with the side frames and bottom frame. The fixture keeps the strips square and aligned before welding.
- **TIG tack welding:** Tack welds are placed at the corner/contact points first to hold the front frame in position.
- **TIG seam welding:** Weld seams are applied along the contact edges between the front frame, side frames, and bottom frame to create the welded bottom-heating frame.
- **Weld finishing:** The welded edges are cleaned, deburred, brushed, and passivated after welding.

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Important manufacturing features:
The front frame is cut as a flat strip and then becomes part of the welded heating-frame structure. The weld seams are placed only along the contact edges with the side and bottom frame parts. The welding fixture is important because it keeps the rectangular frame aligned before final welding.

10.8.4. Bottom Frame



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet, 2.0 mm

Justification:
EN 1.4301 is used because the bottom frame is close to the heating zone and must resist heat, oil vapour, humidity, and cleaning. It is suitable because the part is a welded support strip, not the main heat-transfer surface.

Manufacturing Technology:

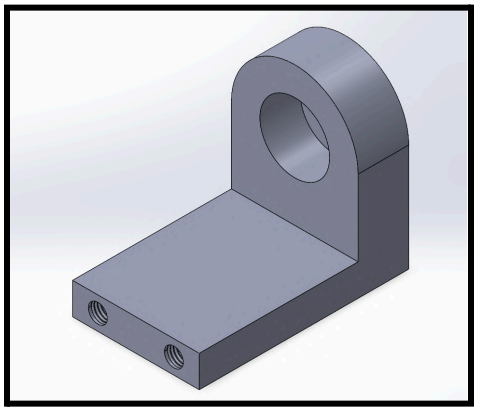
- **CNC fibre laser cutting:** The flat bottom-frame strip is cut from stainless steel sheet, including the long edges and end edges.
- **No bending:** The part remains a flat strip, so there are no bend lines or forming operations.
- **Edge finishing:** The long edges and end edges are deburred and lightly rounded so the strip fits cleanly against the side frames and front frame.
- **Welding fixture positioning:** The bottom frame is placed in a welding fixture with the side frames and front frame. The fixture keeps the heating-frame structure square before welding.
- **TIG tack welding:** Tack welds are placed at the corner/contact points first to hold the bottom frame in position.
- **TIG seam welding:** Weld seams are applied along the contact edges between the bottom frame, side frames, and front frame to create the final welded frame structure.
- **Weld finishing:** The welded edges are cleaned, deburred, brushed, and passivated after welding.

Important manufacturing features:
The bottom frame is cut as a flat strip and becomes the lower support member of the welded heating-frame assembly. The weld seams are placed along the contact edges

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with the side and front frame parts. The welding fixture is important because it keeps the frame straight and square during tack welding and final seam welding.

10.8.5 Mounting Brackets for Bottom Heating Block



Material: EN 1.4301 / AISI 304 stainless steel

Justification:
EN 1.4301 is used because the bracket is close to the bottom heating block and must resist heat, oil vapour, humidity, and cleaning. Stainless steel also reduces heat transfer to the surrounding frame compared with aluminium and gives strong threaded fixing points.

Manufacturing Technology:

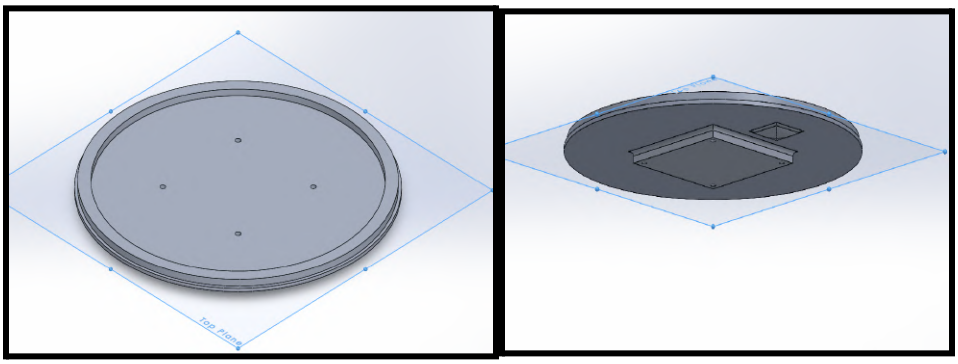
- **Block cutting:** A stainless steel billet or flat bar blank is cut oversized for the base, upright lug, and mounting faces.
- **Setup 1 — base machining:** The blank is clamped from the side faces. The bottom reference face, rectangular base profile, top base surface, and front face are CNC milled.
- **Threaded front holes:** The two front holes are drilled and tapped from the front face. These threads are used to fasten the bracket to the heating-block support frame or adjustment plate.
- **Setup 2 — upright lug machining:** The part is rotated and clamped on the finished base face. The upright lug profile, rounded top shape, and side faces are milled.
- **Large support hole:** The large hole through the upright lug is drilled and bored from the side face. This hole is used for the support pin, spacer, or fixing shaft that holds the heating block position.
- **Finishing:** The large lug hole, front threaded holes, base edges, lug edges, and outer profile are deburred. The part is brushed and passivated after machining.

Important manufacturing features:
The bottom face is the main datum because it controls how the bracket sits under the conveyor. The large lug hole is the critical alignment feature because it positions the heating block support point. The front threaded holes are machined perpendicular to the front face so the fixing screws enter straight and clamp the bracket securely.

10.9. Rotating Plate

10.9.1. Aluminum Plate

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Material: Clear anodised EN AW-6061 T6 aluminium plate

Justification:
EN AW-6061 T6 is used because the rotating plate is supported by a 5 kg load cell, so the plate must stay lightweight. Aluminium also gives enough stiffness for the rotating support surface. Clear anodising improves corrosion resistance, cleanability, and surface protection where cooked eggs may contact the plate.

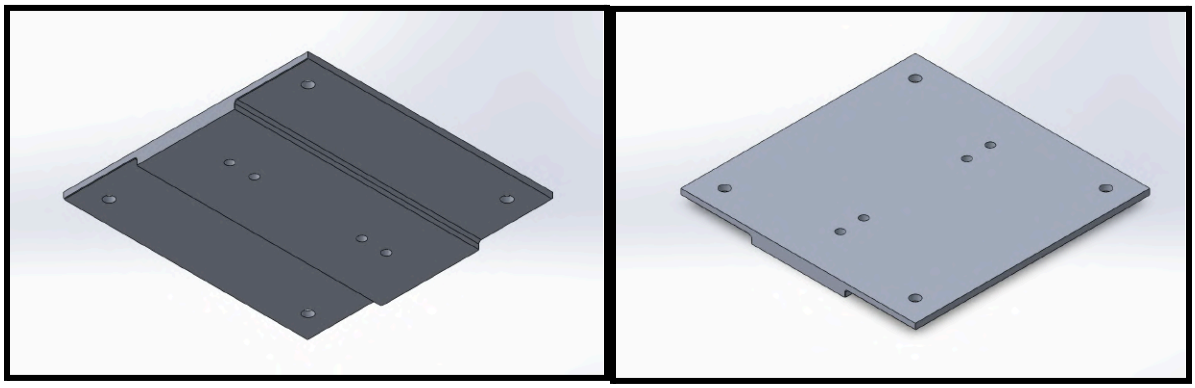
Manufacturing Technology:

- **CNC fibre laser cutting / saw cutting::** A circular aluminium blank is cut slightly oversized from plate material. The blank includes extra allowance for final turning of the outside diameter.
- **Setup 1 — top-side turning:** The circular blank is clamped on the outer diameter and underside. The top serving surface, outer diameter, raised rim, and circular groove/step are turned concentrically.
- **Setup 2 — bottom-side turning:** The plate is flipped and clamped on the finished outer diameter using soft jaws. The bottom reference face is turned flat so the plate sits level on the bearing and load-cell support.
- **Underside CNC milling:** The finished top surface is protected using a vacuum fixture or soft support. The square mounting pocket, underside support pad, rectangular sensor cut-out, and curved microswitch actuator tab are milled on the underside.
- **Microswitch tab finishing:** The curved actuator tab is rounded on its contact edge so it presses the microswitch lever smoothly during rotation.
- **Mounting holes and countersinks:** The mounting holes are drilled through the plate. The countersinks are machined from the top side so flat-head screws sit flush with the egg-contact surface.
- **Finishing:** The outer rim, circular groove, mounting holes, countersinks, underside pocket, rectangular cut-out, and microswitch tab are deburred. The part is then clear anodised.

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Important manufacturing features:
The top serving surface is protected from clamp marks because it may contact cooked eggs. The outer rim and circular groove are turned from the same axis so the plate rotates concentrically. The underside pocket and microswitch tab are machined after turning so they align with the rotating plate position. The countersinks are placed on the top face to keep screw heads flush with the egg-contact surface.

10.9.2. Top Load Cell Plate



Material: Clear anodised EN AW-6061 T6 aluminium plate

Justification:
EN AW-6061 T6 is used because the plate is part of the weighing structure and must stay lightweight so the 5 kg load cell is not overloaded. It also provides enough stiffness for supporting the rotating plate and bearing interface.

Manufacturing Technology:

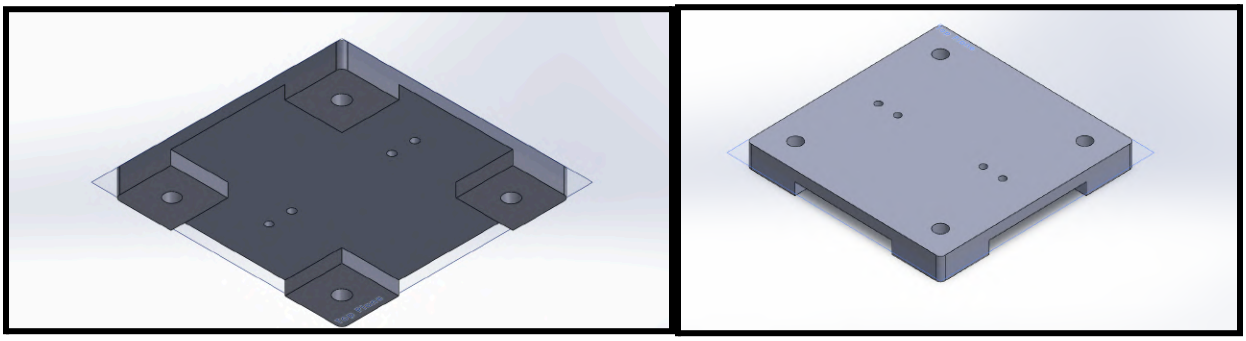
- **CNC fibre laser cutting / saw cutting::** A rectangular aluminium blank is cut slightly oversized, including allowance for the final outer profile and front notch.
- **Setup 1 — top-side machining:** The blank is clamped from the outer side stock. The top reference face, final outer profile, straight edges, front notch, and mounting-hole positions are CNC machined from the top side.
- **Mounting holes and counterbores:** The screw holes are drilled from the top side. Required counterbores are machined on the same side so screw heads or washers sit flat against the mounting surface.
- **Setup 2 — underside pocketing:** The part is flipped and held on the finished top face using a vacuum fixture or soft support. The underside pocket is milled while leaving the central raised rib and thicker mounting pads as part of the plate.
- **Bottom reference face:** The lower contact areas are face milled so the plate sits flat on the load cell interface.

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- **Finishing:** The mounting holes, counterbores, front notch, underside pocket, raised rib, and outer edges are deburred. The plate is then clear anodised.

Important manufacturing features:
The top face is the main datum because it supports the rotating plate and bearing parts. The underside pocket removes weight, while the central rib and mounting pads remain to keep stiffness around the load path. The counterbores are placed only where screw heads or washers need a flat seating surface.

10.9.3. Bottom Load Cell Plate



Material: Clear anodised EN AW-6082 T6 aluminium plate

Justification:
EN AW-6082 T6 is used because the base plate must support the load cell, bearing structure, rotating plate assembly, and motor-side forces with high stiffness. Its mass is below the load cell, so it does not affect the 5 kg measured load, but it must reduce vibration and movement during rotation.

Manufacturing Technology:

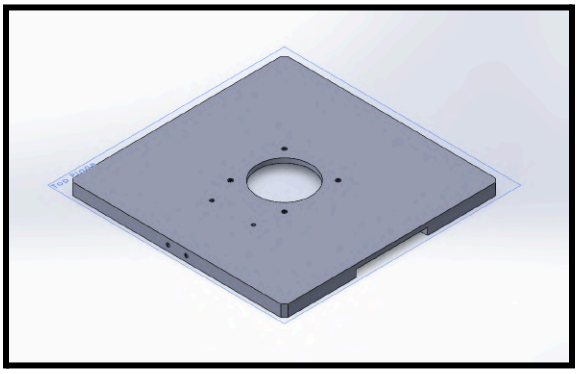
- **CNC fibre laser cutting / saw cutting:** An aluminium plate blank is cut slightly oversized. This step only prepares the raw blank. The final profile, holes, counterbores, threads, reference faces, underside pocket, ribs, and raised pads are produced by CNC machining afterwards.
- **Setup 1 — top-side machining:** The blank is clamped on the outer edges. The top reference surface, final square outer profile, side notches, corner radii, and mounting-hole positions are CNC machined from the top side.
- **Mounting holes and counterbores:** The holes for the load cell, bearing support, motor-side parts, and frame fixing are drilled from the top side. Required counterbores are machined on the same side so screw heads and washers sit flat.

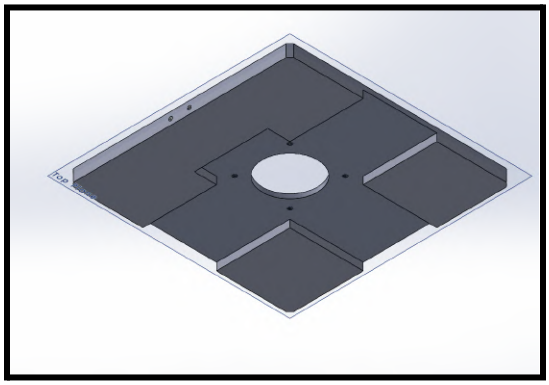
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- **Setup 2 — underside pocketing:** The part is flipped and held on the finished top face using a vacuum fixture or soft support. The underside pocket is milled while leaving the perimeter rim, corner support feet, and central raised support area.
- **Bottom reference areas:** The lower contact pads are face milled so the base plate sits flat on the machine frame and does not twist the load cell structure.
- **Finishing:** The mounting holes, counterbores, side notches, underside pocket, raised pads, and outer edges are deburred. The part is then clear anodised.

Important manufacturing features:
The top surface is the main datum because it controls the load cell and rotating assembly position. The underside pocket reduces weight, while the perimeter rim, corner feet, and central support area remain to keep the plate stiff. The counterbores are placed only where screw heads or washers need a flat seating surface.

10.9.4. Base Plate





Material: Clear anodised EN AW-6082 T6 aluminium plate

Justification:
EN AW-6082 T6 is used because the common base plate supports the rotating plate subsystem, load cell structure, motor housing, and frame connections while keeping the machine weight reasonable. It also provides enough stiffness to reduce vibration during rotation.

Manufacturing Technology:

- **CNC fibre laser cutting / saw cutting:** An aluminium plate blank is cut slightly oversized. This step only prepares the raw blank. The final profile, holes,

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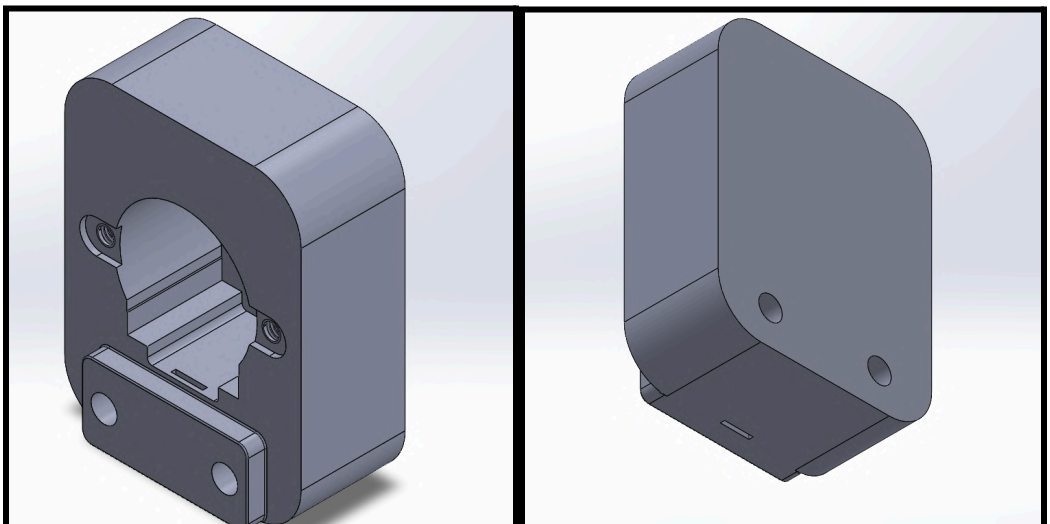
counterbores, threads, reference faces, underside pocket, ribs, and raised pads are produced by CNC machining afterwards.

- **Setup 1 — top-side machining:** The blank is clamped on the outer edges. The top reference face, final square profile, rounded corners, side notches, central through-hole, and top mounting-hole positions are CNC machined from the top side.
- **Top threaded holes and counterbores:** The top holes for the load cell support, motor housing, and bearing-related parts are drilled and thread-formed from above. Required counterbores are machined on the top face so screw heads or washers sit flat.
- **Setup 2 — side threaded holes:** The plate is indexed or re-clamped on its finished top/bottom faces. The side holes are drilled and thread-formed perpendicular to the side faces so the common base plate can be screwed into the surrounding frame.
- **Setup 3 — underside pocketing:** The part is flipped and held on the finished top face using a vacuum fixture or soft support. The underside pocket is milled while leaving the perimeter rim, central support area, raised pads, and ribs for stiffness.
- **Bottom reference areas:** The lower contact pads are face milled so the plate sits flat inside the main frame.
- **Finishing:** The central hole, top threaded holes, side threaded holes, counterbores, underside pocket, raised pads, ribs, side notches, and outer edges are deburred. The part is then clear anodised.

Important manufacturing features:
The top face is the main datum because it controls the load cell, motor housing, and rotating assembly position. The central through-hole is machined from the same top-side setup so it stays aligned with the mounted rotating parts. The side threaded holes are machined perpendicular to the side faces because they connect the plate to the main frame. The underside pocket removes weight, while the perimeter rim, ribs, and raised pads keep the plate stiff.

10.9.5. Stepper Motor Housing

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Material: Clear anodised EN AC-43500 / EN AC-AlSi10MnMg cast aluminium alloy

Justification:
EN AC-43500 is used because the housing must hold the stepper motor and gearbox accurately while keeping the rotating plate subsystem lightweight. Aluminium also helps dissipate heat from the motor area.

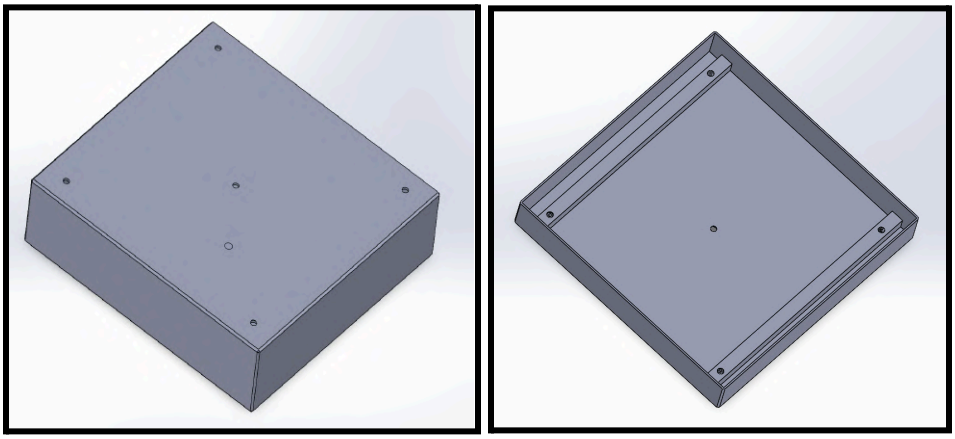
Manufacturing Technology:

- **High-pressure die casting:** The main housing body is cast with the rounded outer casing, front motor face, rear face, side walls, bottom wall, lower mounting flange, and internal motor cavity.
- **Parting line and draft:** The casting parting line is placed around the housing mid-plane. Draft is applied to the outer walls, cavity walls, screw bosses, and flange faces so the housing can be released from the die.
- **Core inserts and side slides:** A fixed core insert forms the internal motor/gearbox cavity, and another core forms the rough front circular opening. Side slides form the side recesses and side-access features that cannot release in the main die-opening direction.
- **CNC reference machining:** The front motor mounting face, rear reference face, bottom mounting face, and lower flange faces are milled after casting so the motor housing sits accurately on the common base plate.
- **Front opening and internal seats:** The front circular motor/gearbox opening is bored to final size from the front face. Internal seating surfaces are pocket milled where the motor and gearbox sit inside the housing.
- **Holes and threads:** The front motor mounting holes are drilled and thread-formed from the front face. The lower flange holes are drilled as clearance holes and counterbored from the mounting side so the screw heads sit flat on the flange.
- **Bottom cable slot:** The small rectangular bottom slot is milled after casting to provide cable clearance from the motor area.
- **Finishing:** The circular opening, threaded holes, counterbores, bottom slot, internal cavity edges, flange edges, and outer housing edges are deburred. The part is clear anodised after machining.

Important manufacturing features:
The front motor face is the main datum because it controls motor and gearbox alignment. The bottom mounting face and lower flange control how the housing sits on the common base plate. The front threaded holes are machined after casting so the motor screws align accurately, while the flange counterbores are placed on the mounting side for stable screw seating.

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10.9.6. Bottom Base for Rotating Plate



Material: EN 1.4301 / AISI 304 stainless steel sheet and flat bar

Justification:
EN 1.4301 is used because the base is inside the food machine and must resist oil, heat, humidity, and cleaning. A welded hollow sheet-metal box is used instead of a solid block to reduce weight and material cost, while internal flat bars provide enough thread depth for the M8 mounting holes.

Manufacturing Technology:

- **CNC laser cutting:** The top plate and four side wall panels are cut from stainless steel sheet. The outer profiles are cut flat before welding.
- **Flat bar cutting:** Two stainless steel flat bars are cut to length. These bars are placed under the top plate where the M8 threaded holes are needed.
- **Outer box welding:** The four side panels are positioned perpendicular to the top plate in a welding fixture. TIG weld seams are applied along the top-to-side panel contact edges and vertical corner edges to create the hollow box frame.
- **Internal support bar welding:** The box is flipped, and the two flat bars are clamped against the underside of the top plate. They are welded directly below the future M8 hole positions to create thick threaded zones.
- **Hole machining after welding:** The welded frame is clamped right-side up. The four corner M8 holes and central through-holes are spot drilled after welding so their positions remain accurate even if welding causes slight distortion.
- **Threaded holes:** The M8 core holes are drilled vertically through the top sheet and into the welded internal flat bars, then tapped M8×1.25 to give full thread engagement.
- **Central holes:** The centre holes are drilled through the top sheet only, where clearance or alignment is needed for the mounted components.

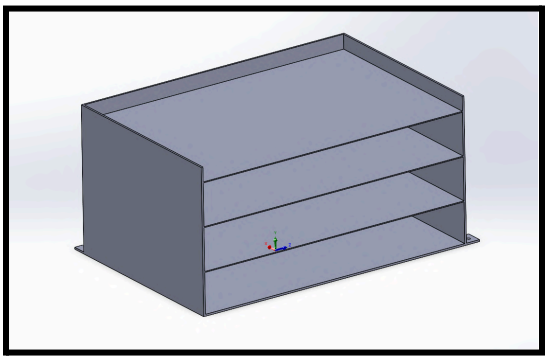
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- **Weld finishing:** The outside corner welds and top-edge weld seams are ground flush. The visible surfaces are brushed for a clean hygienic finish.
- **Final finishing:** The M8 threaded holes, central holes, top edges, side edges, and welded corners are deburred and passivated.

Important manufacturing features:
The internal flat bars are positioned directly under the M8 holes because the thin sheet alone would not provide enough thread depth. The mounting holes are drilled and tapped after welding so the hole positions match the final welded frame, not the pre-weld sheet parts. The weld seams are placed along the outer box edges and are ground smooth so residue cannot collect easily.

10.10. Electrical Box

10.10.1 Electrical Box Housing



Material: EN 1.4301 / AISI 304 stainless steel sheet, with thicker EN 1.4301 mounting tabs

Justification:
EN 1.4301 is used because the electrical housing must be rigid, corrosion-resistant, and easy to clean inside the machine. Sheet metal keeps the housing lighter than a solid block, while thicker mounting tabs provide stronger fixing points to the machine frame.

Manufacturing Technology:

- **CNC fibre laser cutting:** The back panel, side walls, top and bottom panels, internal shelves, and external mounting tabs are cut from flat stainless steel sheet. Any cable openings, ventilation holes, and fastening holes are also cut while the sheets are flat.
- **Mounting tab holes:** The external tabs are drilled and countersunk before welding. The countersinks are made on the outer face so flat-head screws sit flush when the housing is fixed to the machine frame.
- **Main shell welding:** The side walls, top panel, bottom panel, and back panel are aligned in a welding fixture. TIG weld seams are placed along the rear vertical edges and the top/bottom contact edges to form the open-front housing box.
- **Internal shelf welding:** The horizontal shelves are positioned inside the housing and TIG welded along their contact lines with the side walls and back panel. These weld lines hold the shelves and also stiffen the housing.

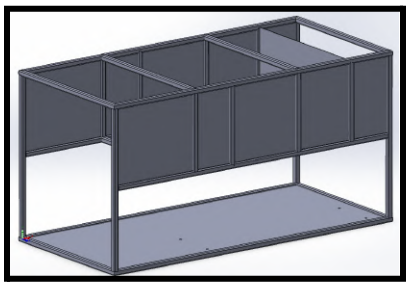
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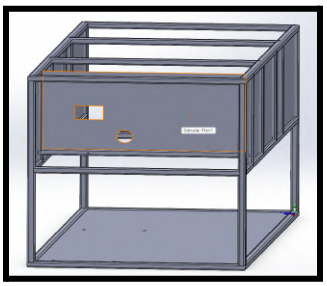
- **Mounting tab welding:** The prepared tabs are welded to the outside lower side edges of the housing. The tab welds are placed along the tab-to-wall contact edges so the screw loads transfer into the side walls.
- **Weld finishing:** External welds are ground smooth where they are visible or where the housing must sit against the frame. Internal shelf welds are cleaned but not over-ground, so shelf strength is not reduced.
- **Edge finishing:** All sheet edges, shelf fronts, cable openings, ventilation holes, countersinks, and welded corners are deburred so wires cannot be damaged during installation.
- **Surface finishing:** The visible outer surfaces are brushed, and the welded areas are passivated to restore corrosion resistance.

Important manufacturing features:
The internal shelves are welded after the main box is squared in a fixture so the housing does not twist. The mounting tabs are made from thicker stock because they carry the fixing loads. All cable openings and shelf front edges must be rounded because sharp stainless edges could cut insulation during assembly or maintenance.

10.11 Main Frame

10.11.1 Main Frame housing





Material: EN 1.4301 / AISI 304 stainless steel square hollow sections and stainless steel sheet

Justification:
EN 1.4301 is used because the frame is inside a food-machine environment and must resist heat, oil vapour, humidity, and cleaning. Square hollow sections provide the main stiffness with low weight, while sheet-metal panels close the frame and support internal components.

Manufacturing Technology:

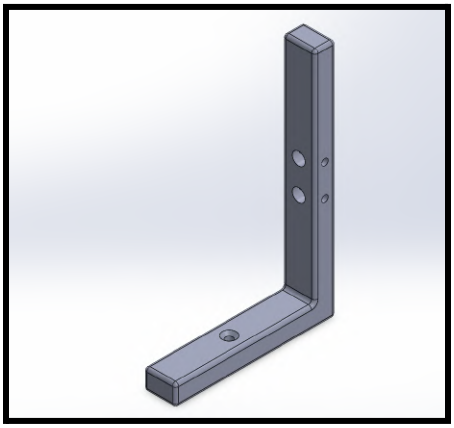
- **Tube cutting:** The stainless steel square hollow sections are cut to length for the vertical posts, bottom rails, and top rails. The tube ends are cut square or mitred depending on the frame corner connection.
- **Sheet laser cutting:** The base plate, side panels, and internal divider panels are laser cut from stainless steel sheet. The square cut-out, circular hole, cable openings, and mounting holes are cut while the panels are still flat.

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- **Base plate countersinks:** The base-plate mounting holes are countersunk from the top side so flat-head screws or levelling feet sit flush and do not interfere with internal parts.
- **Frame hole machining:** Small fastening holes are drilled or tube-laser cut into the square tubes where external covers, cable clips, or internal brackets are later attached.
- **Frame welding:** The cut square tubes are held in a welding fixture and TIG welded at the corner joints and tube intersections. The fixture keeps the frame square during welding.
- **Sheet panel welding:** The base plate, side panels, and internal divider panels are positioned against the tube frame. TIG tack welds are added first, then short stitch welds are used along the panel-to-tube contact edges to reduce sheet distortion.
- **Weld finishing:** External tube-frame welds are ground smooth where the frame is visible or where panels must sit flush. Internal welds are cleaned but not over-ground so the frame strength is maintained.
- **Edge finishing:** The laser-cut panel edges, square cut-out, circular hole, cable openings, countersinks, and drilled tube holes are deburred to remove sharp edges.
- **Surface finishing:** The visible stainless surfaces are brushed, and the welded areas are passivated after welding and grinding.

Important manufacturing features:
The square hollow tubes form the load-bearing skeleton, so their weld joints are made in a fixture to keep the frame straight. The sheet panels are stitch welded instead of fully seam welded to reduce heat distortion. The square and circular cut-outs are made before welding because they are easier to cut accurately while the sheet is flat. All cable and pipe openings are deburred so wiring, tubing, and cleaning access are not damaged by sharp edges.

10.11.2. L bracket for servo motor



Material: EN 1.4301 / AISI 304 stainless steel sheet, 2.0 mm

Justification:
EN 1.4301 is used because the bracket is inside the machine and must resist humidity, oil vapour, heat, and cleaning. Stainless steel also provides enough stiffness for a small structural mounting bracket.

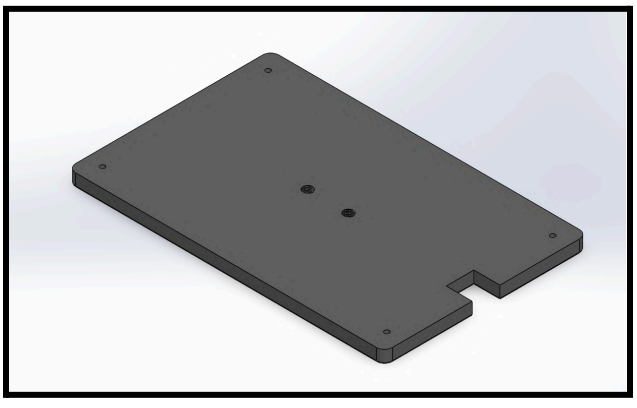
Manufacturing Technology:

- **CNC fibre laser cutting:** The flat sheet blank is cut first, including the outer profile, base mounting hole, vertical mounting holes, side holes, and bend reliefs.
- **Hole cutting:** All circular holes are cut while the sheet is flat so their positions stay accurate before bending.
- **Press brake bending:** The sheet is bent 90° along the bend line between the horizontal base flange and the vertical upright flange. The inside radius is created by the press brake tool.
- **Bend direction:** The vertical flange is bent upward from the base flange. The base hole remains on the horizontal mounting side, while the vertical holes are used for fastening the supported part.
- **Edge finishing:** The outer edges, hole edges, bend reliefs, and flange corners are deburred and lightly rounded.
- **Surface finishing:** The visible surfaces are brushed, and the part is passivated after cutting and bending.

Important manufacturing features:
The bend line controls the 90° angle between the base flange and upright flange. The holes are cut before bending because they are easier to position accurately in the flat blank. The vertical holes must stay aligned after bending because they locate the mounted component.

10.12 Display Casing

10.12.1 Plastic Back Plate



Display Back Plate
Material: ABS-HI plastic with brass threaded inserts

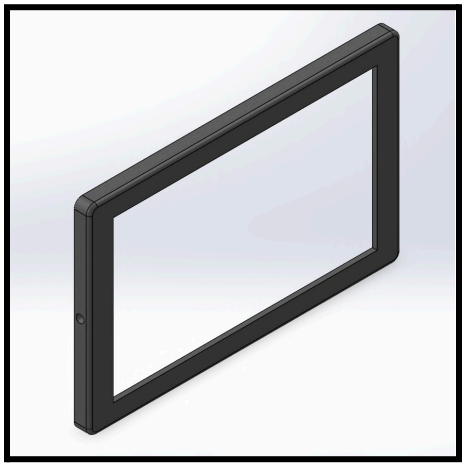
Justification:
ABS-HI is used because the back plate is a non-food-contact display support part that must stay lightweight, impact-resistant, and dimensionally stable. Brass threaded inserts are used for the display mounting points because repeated screw tightening would damage plastic threads.

Manufacturing Technology:

- **Injection moulding:** The back plate is moulded as one plastic part, including the flat rectangular plate, rounded corners, side relief notch, rear stiffening ribs, and local screw bosses.
- **Mould parting line:** The parting line is placed around the outer edge of the plate so the main flat mounting face stays clean.
- **Core pins for corner holes:** The four corner fixing holes are formed with core pins through the plate thickness. These holes are used to fasten the back plate to the display housing or frame.
- **Heat-set threaded inserts:** Brass inserts are pressed into the two central display mounting bosses after moulding. These create stronger threads for attaching the display module or bracket.
- **Relief notch:** The side notch is formed directly in the mould cavity to give clearance for nearby cables, connectors, or housing features.
- **Gate position:** The injection gate is placed on the rear side of the plate so no gate mark appears on the visible or functional mounting face.
- **Finishing:** The outer edge, corner holes, notch edges, insert openings, and parting-line flash are trimmed and deburred.

Important manufacturing features:
The central bosses are the critical features because they hold the threaded inserts for the display screws. The corner holes are moulded through the plate so the back plate can be fixed to the cover or frame. The main mounting face must remain flat so the display sits evenly without bending or twisting.

10.12.2 Front Frame for Casing



Material: ABS-HI plastic

Justification:
ABS-HI is used because the front cover is a visible user-facing part that needs good impact resistance, a clean surface finish, and enough stiffness around the display opening. It is suitable because the part is not

exposed to direct food contact or high heating surfaces.

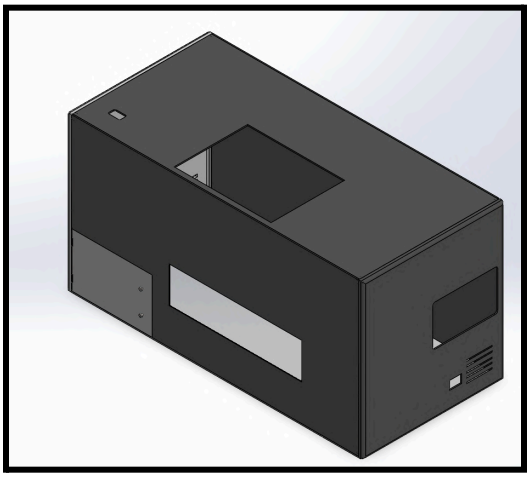
Manufacturing Technology:

- **Injection moulding:** The front cover is moulded as one plastic bezel, including the rectangular frame, rounded outer corners, inner display opening, side wall, and side screw hole boss.
- **Mould parting line:** The parting line is placed around the outer side edge of the bezel so the visible front face remains clean.
- **Core insert for display opening:** The large rectangular display window is formed directly by a core insert in the mould. This keeps the opening shape accurate and avoids cutting the window after moulding.
- **Core pin for side hole:** The side mounting hole is formed by a side core pin or slide because the hole direction is perpendicular to the main mould opening direction.
- **Rear-side bosses/ribs:** Local thickened areas are moulded on the rear side around screw positions and frame edges to support the display and prevent the bezel from flexing.
- **Gate position:** The injection gate is placed on the rear side of the cover so no gate mark appears on the visible front surface.
- **Finishing:** The inner display opening, outer edges, side hole, and parting-line flash are trimmed and deburred after moulding.

Important manufacturing features:
The front face is the visible user-facing surface, so gate marks, ejector marks, and parting lines are kept away from it. The display opening is the critical feature because it must fit the screen cleanly. The side hole is made with a side slide/core pin because it cannot be formed in the normal mould opening direction.

10.13 Main Case/Chassy

10.13.1 All Chassy Parts are made in the Same Manner



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Material: ABS-HI plastic sheet panels, with purchased glass window insert

Justification:
ABS-HI is used because the outer case must be lightweight, impact-resistant, and easy to clean. It is suitable for the external housing because it is not a direct food-contact or heating surface.

The glass window is bought separately because it needs optical clarity and scratch resistance.

Manufacturing Technology:

- **CNC routing of plastic panels:** The six case sides are cut from ABS-HI sheet: top panel, bottom panel, front panel, back panel, left side panel, and right side panel.
- **Panel cut-outs:** The front viewing opening, top square opening, side display opening, ventilation slots, button/sensor holes, and service openings are CNC routed while the panels are flat.
- **Corner radii and edge rounding:** The outside panel corners and visible edges are rounded with a router radius tool or edge-finishing tool so the case has safer, smoother edges.
- **Panel joining:** The six plastic panels are aligned in an assembly fixture and joined along the inside corner seams using plastic solvent bonding or plastic welding. The outside faces are kept clean and free from visible bonding marks.
- **Glass window installation:** The glass is a purchased part and is installed into the front rectangular opening using a rubber gasket or heat-resistant adhesive frame. The glass is not manufactured in-house.
- **Ventilation and access features:** The ventilation slots are placed on the side panel to allow airflow for the electronics area. The display and button cut-outs are made before assembly so their positions stay accurate.
- **Finishing:** All panel edges, cut-out edges, ventilation slots, glass opening, and display opening are deburred and lightly rounded before final assembly.

Important manufacturing features:
Each cut-out is made while the panel is flat because it is more accurate than cutting after the box is assembled. The glass opening must have a clean, flat edge so the gasket or adhesive frame seals properly. The corner seams are joined from the inside so the outside case surface stays clean and presentable.

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11. Requirements Manual for “buy” (CotS) Parts

For our sunny-side-up egg frying machine project, a Make-or-Buy analysis was conducted to determine which components should be manufactured in-house and which components should be bought externally as Commercial off-the-Shelf parts. In this strategic decision-making process we considered factors such as reliability in operation, design complexity, hygienic design, regulatory requirements and easy maintenance in a commercial kitchen environment.

Based on this analysis, several components were identified as suitable for external procurement. These parts are already available on the market with defined technical specification, proven performance and standardized interfaces. Buying them reduces development time, lowers manufacturing complexity and supports faster assembly of the machine.

The main criteria for selecting Buy parts are:

- Standardization: Using standardized components reduces design complexity and makes it easier to replace and maintain these parts.
- Proven reliability: Commercially available components are already tested and thus reduce the risk of failure during operation.
- Cost-Effectiveness: Mass-manufactured parts benefit from economies of scale.
- Regulatory Compliance: Pre-certified parts comply with legal safety standards.
- Technical suitability: The selected parts have to meet the technical requirements necessary to ensure reliable operation of the SunnyPro egg-frying machine.

This manual presents the Buy Requirements Table for the selected purchased components. Its purpose is to support the purchasing department in making a possible decision on purchasing a part based on the complete technical requirements and short explanations of their origin and reason.

11.1. List of Components

11.1.1. Power Supply System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin**
1	IEC C20 Power Inlet	1. Connector type: IEC C20 male inlet 2. Current rating: ≥16A 3. Voltage rating: ≥250V AC	Required as standardized detachable mains connection for a commercial 230 V machine.

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2	16A MCB- TYPE B Circuit Breaker	1. Current rating: 16A 2. Voltage rating: 230/400V AC 3. Pole configuration: 2P 4. Suitable standard certification	Protects the mains input wiring and allows safe disconnection of live and neutral conductors.
3	RCCB / RCD-TYPE A (30mA)	1. Residual current sensitivity: 30mA 2. Rated current: ≥16A 3. Voltage rating: 230V AC 4. Pole configuration: 2P 5. Standard compliance: IEC/EN 61008-1	Provides personal protection against electric shock on the AC mains side.
4	Emergency Stop Switch	1. Normally closed contact 2. Red mushroom-head actuator 3. Latching function with manual reset 4. Contact rating suitable for 24V DC control circuit	Needed so kitchen staff can immediately stop the machine in an unsafe situation.
5	DIN Terminal Blocks (L/N/PE)	1. Separate terminals for L, N, and PE 2. Rated current ≥16A 3. Rated voltage ≥250V AC 4. Compatible with 2.5 mm² wire , The selected AC wiring size is 2.5 mm².	Organizes and safely distributes mains wiring inside the electrical cabinet.
6	Protective Earth Terminal Block	1. Compatible with ≥2.5 mm² conductor 2. Bright color identification 3. Low-resistance connection 4. Secure screw or spring connection	Required to connect exposed conductive parts to protective earth for fault protection.
7	AC Wire	1. Conductor cross-section: 2.5 mm² 2. Voltage rating suitable for 230V AC 3. Color coding: brown, blue, green/yellow 4. Heat-resistant insulation	Carries the 16 A mains supply safely and allows clear identification of L, N and PE.
8	24V Industrial SMPS 60W (2.5A at 24V)	1.Input voltage: 230V AC 2.Output voltage: 24V DC, 2.5A 3.Overload and short-circuit protection	Provides the main industrial control voltage for the machine.
9	24V → 12V LM2596T-12	1. Input voltage: 24V DC 2. 3 A continuous (LM2596 class) 3. High efficiency 4. Overload and short-circuit protection	Supplies 12 V loads such as fans or actuators from the 24 V control supply.
10	12V → 5V LM2596T-5	1. Input voltage: 12V DC 2. Output voltage: 5V DC, 2A continuous 3.Output ripple/noise: ≤50 mV peak-to-peak preferred, ≤100 mV peak-to-peak acceptable	Supplies 5 V sensors, logic modules and interface electronics reliably.

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11	24V → 6V LM2596S-ADJ (Adjusted to 6V)	1. Input voltage: 24V DC 2. Output voltage: 6V DC, 2A continuous 3. Low heat generation / high efficiency	Supplies the 6 V servo/actuator circuit without overloading the main 24 V supply.
12	5V → 3.3V Regulator AMS1117-3.3	1. Input voltage: 5V DC 2. Output voltage: 3.3V DC 3. Output current : 800mA 4. Output ripple/noise: ≤50–100 mV peak-to-peak 5. Operating temperature: at least 0°C to 70°C	Supplies 3.3 V controller and logic components.
13	DC Terminal Blocks	1. Separate terminal groups for 24V, 12V, 6V, 5V, 3.3V, and 0V/GND 2. Current rating for 24V/12V actuator branches: ≥10A per terminal 3.Current rating for 5V/3.3V logic branches: ≥2A per terminal 4.Voltage rating: ≥24V DC 5. Wire compatibility: at least 0.5–2.5 mm²	Prevents mixing of different DC voltage levels and simplifies maintenance.
14	Inline Blade Fuse Holders	1. Fuse type: standard automotive blade fuse, ATO/ATC type 2. Voltage rating: ≥32V DC 3. Current rating: 3–5 A for actuator/fan branches; 1- 3 A for logic/sensor branches sized 4. Wire size compatibility: approx. 0.75–2.5 mm²	Allows simple branch protection for DC actuator, fan and logic circuits.
15	Blade Fuse Assortment	1. Fuse type: ATO/ATC blade fuses 2. Voltage rating: ≥32V DC 3. Fuse ratings: 1A, 3A, 5A, 10A	Provides replaceable protection for different DC load branches.
16	Shielded Signal Cable	1.Cable type: shielded multi-core signal cable 2.Number of cores: minimum 3 cores 3. Conductor size: approx. 0.14–0.5 mm 4. Voltage rating: ≥30V DC	Reduces interference on sensor and communication signals near motors, solenoids and converters.
17	Ferrules and Crimp Terminals	1. Ferrule size range: 0.14–2.5 mm² 2. Insulated ferrules 3. Crimp terminals compatible with screw/spring terminal blocks 4. Material: tinned copper conductor sleeves	Ensures secure and serviceable wire connections in the electrical cabinet.

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18	PCB Screw Terminal Connectors	1.Connector type: PCB mount screw terminal block, wire-to-board 2. Pitch: 5.08 mm 3. Voltage rating: ≥30V DC 4. Current rating: ≥5A per contact 5. Wire size compatibility: 0.14–1.5 mm²	Connects external wires to the control PCB safely and replaceably.
19	Electrical Cabinet Cooling Fan	1. Operating voltage: 12V DC 2. Quantity: 2 fans 3. Fan size: 80 x 80 x 25 mm 4. Airflow: approx. 58 m³/h per fan 5. Operating temperature: ≥ -10°C to +70°C 6. Protection: fan guard required	Connects external wires to the control PCB safely and replaceably.
20	Main Microcontroller STM32H743ZITx	1.Logic voltage: 3.3V 2.GPIO available: 114 (LQFP144) sufficient 3.PWM channels needed : ≥ 4 4.UART : ≥ 2 5.SPI : ≥ 2 6.I²C : ≥ 1 7.Operating Temp: 0–70 °C commercial; consider industrial -40 to +85 °C if cabinet ambient gets to ~50 °C	Controls sensors, actuators, HMI communication and machine timing.
Mechanical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	DIN Rail	1. Rail type: TS35 / 35 mm DIN rail 2. Material: zinc-plated steel or aluminum	Mounting base for protection devices, terminal blocks and power supply components.
2	Cable Glands / Strain Reliefs	1. Thread sizes: M12, M16, and M20 selected 2. Protection rating: IP54 minimum, IP65 preferred 3. Strain relief function included	Protects cable entries against pulling forces, dust, moisture and grease.
3	Cable Duct / Wire Raceway	1. Type: slotted wiring duct with removable cover 2.Material: flame-retardant PV 3. Separate routing possible for power and signal cables	Keeps cabinet wiring organized and reduces interference and service errors.
4	Cable Ties + Mounts	1.Cable tie material: nylon PA66 2. Tie size matched to cable bundle diameter 3. Temperature rating: at least -10 - +70°C	Fixes cable bundles safely inside the electrical cabinet.

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5	Pan Head Cross Recess Screw	1. Component type: pan head cross recess screw 2. Standard: ISO 7045 3. Screw size: M4 × 8 mm 4. Drive type: cross recess Z	Used for fastening components in the electrical cabinet.

11.1.2. Storage Unit and Breaking Mechanism System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Gate Actuators	1. Actuator type: Tubular pull solenoid, JF-0530B 2. Rated voltage: 12V DC 3. Stroke length: 10-20 mm preferred 4.Actuation Force: ≥10 N pull force 5. Duty cycle: Rated intermittent duty ≤25%	Used to release one egg at a time and prevent double-feeding in the storage mechanism.
2	Flow Sensors	1. Sensor type: IR break-beam sensor pair (emitter + receiver), Adafruit 2167 2. Supply voltage: 3.3-5V DC 3. Detection function: detects individual egg movement 4. Detection range: Detection range up to ~100 mm 5. Response time: fast	Required to confirm egg movement and control the gate sequence reliably.
3	Flip Actuator	1. Actuator type: high-torque metal-gear hobby servo, MG996R 2. Operating voltage: 4.8V-7.2V DC 3. Torque rating: approx. ≥1.0 N·m at 6 V 4. Control method: PWM servo control 5. Rotation range: ≥90° usable travel	Provides controlled angular movement for the flipping/cracking mechanism.
4	Platform Solenoid	1. Actuator type: Tubular pull solenoid, JF-0837B (24 V variant) 2. Stroke length: ~10 mm 3. Force output: ≥25 N pull force 4. Duty cycle: Rated intermittent duty ≤25% 5. Return action: spring return or compatible external return spring	Drives the spear vertically downward to puncture the eggshell, releasing contents onto the cooking plate.

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5	Wedge Solenoid	1. Actuator type: Tubular pull solenoid, JF-0530B (12 V variant) 2. Stroke length: ~10 mm 3. Force output: ≥10 N pullforce 4. Response time: Mechanical stroke time ~25 ms 5. Duty cycle: intermittent operation	Opens the cracked eggshell so the egg contents can drain onto the cooking plate.
6	Solenoid Driver	1. Driver type: multi-channel MOSFET driver module or equivalent switching circuit 2. Load voltage: 12V 3. Logic input: 3.3V or 5V compatible 4. Current rating: minimum 2A continuous per channel, 5A peak per channel preferred	Required because the microcontroller cannot directly switch the higher current needed by solenoids.
7	Wire Harness	1. Harness type: sleeved mixed-gauge wiring assembly 2. Power conductors: 0.75–1.5 mm² 3. Signal conductors: 0.14–0.5 mm² 4. Insulation: flexible insulated wire, rated at least 60V DC and 70°C	A sleeved harness keeps the subsystem wiring grouped, organized, protected and easier to install.
Mechanical Components			
No	**Component**	**Technical Requirements**	**Origin + Reason**
1	Axle Shaft	1. Material: Aluminum 6061 rod 2. Diameter: Ø10 mm 3. Length: 100 mm 4. Straightness: maximum runout/straightness error ≤0.2 mm over the shaft length preferred	Supports rotating or guided mechanism parts with low weight and sufficient stiffness.
2	Spear	1. Material: SS304 stainless steel spear assembly 2. Shaft diameter: Ø3 mm stock shaft 3. Length: 15 mm 4. Tip geometry: conical or tapered tip with nominal tip width of 0.2 mm	Punctures the eggshell and must be corrosion-resistant, cleanable and food-contact suitable.
3	Return Spring	1. Spring type: stainless steel compression spring 2. Spring rate: 0.5–2.0 N/mm 3. Working stroke: 10–30 mm, 4. Fatigue life: minimum 100,000 cycles	Returns the moving mechanism to its starting position after actuation.

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4	Thermal Insulation	1. Material: XPS foam insulation layer 2. Thermal conductivity: ≤0.035 W/m·K preferred 3. Thickness: approx. 20 mm 4. Temperature suitability: at least till 70°C 5. Moisture resistance: closed-cell structure with water absorption	Reduces heat transfer from the cooking area to the storage and breaking mechanism.
5	Inner Lining	1. Material: food-safe aluminum sheet lining 2. Thickness: 0.8–1.5 mm 3. Fabrication: bendable sheet with bend radius of at least 1–2× sheet thickness 4. Corrosion resistance: suitable for humidity	Creates a cleanable inner surface and protects insulation and housing parts.
6	Ventilation Grille	1. Material: ABS ventilation grille 2. Size: 80 mm standard fan grille 3. Open area ratio: minimum 60% open area 4. Slot or mesh spacing: maximum opening width ≤8 mm 5. Thickness: approx. 2–4 mm	Allows airflow while protecting the fan area from accidental contact and larger debris.
7	Fasteners Kit	1. Material: stainless steel fasteners, preferably A2 / AISI 304 2. Screw sizes: mainly M3-M6 3. Thread standard: ISO metric coarse threads 4. Serviceability: removable screw connections preferred	Used for mounting sensors, actuators, panels and mechanism parts while allowing service and cleaning.

11.1.3. Oil System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Peristaltic Pump	1. Pump type: electrically driven peristaltic pump suitable for food-grade oil transfer 2. Rated voltage: 12V DC 3. Flow rate: ~30–60 mL/min 4. Tubing material: compatible with 4 mm ID × 6 mm OD food-grade silicone tubing 5. Duty cycle: intermittent operation	To dispense small, controlled amounts of oil onto the conveyer belt.

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2	MOSFET Driver Module	1. Driver type: MOSFET switching module 2. Load voltage: 12V DC 3. Logic input voltage: 3.3V or 5V 4. Current rating: minimum 2A continuous 5. Control capability: supports ON/OFF switching	A MOSFET driver is needed because the microcontroller cannot directly supply the current required by the pump motor.
Mechanical components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Oil Tank	1. Material: food-grade stainless steel, preferably AISI 304 or 316 2. Tank capacity: 600mL 3. Opening: removable lid preferred 4. Outlet connection: bottom outlet with food-safe fitting; vented cap on top	Stores cooking oil safely and allows refilling, cleaning and reliable oil supply to the pump.
2	Inline Check Valve	1. Valve type: inline one-way check valve for oil line 2. Material: food-safe and oil-resistant body 3.Tubing connection: 4 mm push-fit or barbed 4. Flow direction marking: clear arrow or marking on valve body	1-Prevents oil backflow into the tank when the pump stops, and prevents drip from the nozzle between doses (which would otherwise cause the next egg to land in a pool of stale oil).
3	Silicone Tubing	1. Material: food-grade silicone tubing 2.Inner diameter 4 mm, outer diameter 6 mm 3. Oil compatibility: resistant to cooking oil 4. Flexibility: bend radius approx. 3x	Transports cooking oil safely from tank to pump and nozzle without contamination or leakage.
4	Flat Fan Spray Nozzle	1. Nozzle type: flat fan oil spray nozzle 2. Spray angle: 65° 3. Flow rate: ~48 mL/min 4. Material: stainless steel or food-grade plastic 5. Orifice size: approx. 0.2–0.5 mm	Spreads the oil evenly over the frying surface instead of dispensing it as a single stream.
5	Hose Adapter	1. Fitting type: hose adapter 2. Material: stainless steel or food-grade plastic 3. Tubing compatibility: 4 mm ID silicone tubing 4. Nozzle connection: "1/8" NPT 5. Leak sealing: compatible with O-ring	Connects the flexible oil tubing to the spray nozzle without leakage.

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6	Nozzle Mounting Bracket	1. Bracket type: adjustable mounting bracket 2. Material: stainless steel or aluminum 3. Angle adjustment: allows spray angle adjustment of approximately ±15° to ±30° 4. Mounting holes: fixing holes compatible with M3–M5 screws	Holds the nozzle in the correct position for even oil distribution over the frying area.
7	Pump Vibration Mounts	1. Mount type: rubber vibration isolators or rubber grommets 2. Material: oil-resistant rubber, e.g. nitrile rubber NBR or EPDM 3. Load capacity: rated for at least 1–3 kg 4. Deflection: approximately 1–3 mm compression deflection under pump load preferred	Rubber vibration mounts are used because the pump motor can create vibration during operation that should not transfer directly into the machine frame.

11.1.4 Screen System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Touchscreen Display Module — Nextion NX8048P070_011R	1.Type: 7" HMI touchscreen module 2.Display: 7", 800 × 480 px, 16-bit colour 4.Supply voltage: 5 V DC 5.Supply current: ~430-510 mA 6.Communication: TTL UART (5 V logic) 7.Operating temperature: -20 °C to +70 °C	Single integrated HMI module chosen over a discrete-component display because the Nextion handles GUI rendering, touch decoding, and backlight management internally, the main MCU only needs to send/receive text commands over UART. This eliminates the need for a second MCU, FPC connectors, touch controller, backlight driver, and etc.
2	JST-XH 4-pin Connector	1.Type: 4-pin JST-XH connector, 2.5 mm pitch 2.Pin assignment: 1 = +5 V, 2 = TX (Nextion-side), 3 = RX (Nextion-side), 4 = GND 3.Current rating: ≥ 1 A per contact 4. Keyed/polarised connector preferred	Connects the touchscreen module to the main PCB for power supply and UART communication.
Mechanical Components			

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No.	Component	Technical Requirements	Origin + Reason
1	Mounting Hardware	1.Type: M3 screws + standoffs (brass or stainless) 2.Screw length: 6–10 mm 3.Standoff length: 5–10 mm 4.Material: brass or stainless preferred	Fixes the touchscreen securely to the front panel while keeping clearance behind the display PCB.

11.1.5. Transportation System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Stepper Motor	1. Motor type: NEMA 17 bipolar stepper motor, double-shaft type 2. Holding torque: approx. 0.75–0.85 Nm 3. Step angle: 1.8° 4. Rated current: approximately 1.0–2.0 A per phase 5. Shaft diameter: 5 mm double shaft	A double-shaft stepper motor is used because the front shaft can drive the conveyor while the rear shaft remains available for an encoder, reference disc, hand adjustment, or future feedback function.
2	Stepper Motor Driver	1. Driver type: external bipolar stepper motor driver 2. Supply voltage: 24V DC 3. Output current: adjustable, minimum 2.0 A per phase 4. Control input: STEP/DIR input compatible with 3.3V logic signals	An external stepper driver is needed because the microcontroller cannot directly supply the motor phase current.
3	Control Wiring and Connectors	1. Wiring type: mixed-gauge wiring 2. Motor conductors: 0.5–0.75 mm²motor phase wires 3. Sensor/control conductors: 0.14–0.34 mm² signal wires 4. Voltage rating: ≥30V DC 5. Insulation temperature rating: ≥ 70°C	Connects motor, driver and sensors safely while preventing wiring errors, heat damage and loose connections.

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4	Hall Effect Sensor	1. Sensor type: A1104-class Hall effect switch sensor, 3-pin type 2. Supply voltage: 5V DC 3. Output signal: compatible with 3.3V controller input 4. Detection distance: approximately 2–5 mm from the 8 x 3 mm SmCo magnet 5. Operating range: ≥ 0-100°C	A Hall effect sensor is used because it detects the home/reference position without physical contact or mechanical wear.
5	SmCo Magnet	1. Magnet type: samarium cobalt SmCo permanent magnet 2. Magnet size: 8 x 3 mm 3. Temperature resistance: ≥150°C 4. Corrosion protection: e.g. coated or protected surface	Acts as the magnetic reference marker for the Hall sensor near the heating zone.
6	Incremental Rotary Encoder	1. Encoder type: incremental rotary encoder with A/B quadrature outputs 2. Resolution: approx. 600 P/R 3. Supply voltage: 3.3V DC 4. Shaft/coupling compatibility: matched to drive shaft diameter (5 mm)	Provides feedback about conveyor shaft movement, speed and direction.
Mechanical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Roller Chain with Bent Attachments	1. Chain type: roller chain with K1 bent attachments, DIN ISO 606 / DIN 8187 compatible 2. Pitch: 9.525mm 3. Material: special chain steel or equivalent wear-resistant steel 4. Attachment configuration: one-sided 5. Length: approx. 1500 mm	1- A standardized roller chain is used because the conveyor moves the egg carrier plates through the heating zone and provides mounting points for the cooking plates.
2	2-Bolt Flange Bearing Block	1. Component type: 2-bolt flange bearing block 2. Bearing type/size: KFL000 type or equivalent 3. Bore diameter: 10 mm 4. Material: zinc alloy, cast housing or corrosion-resistant housing 5. Rotation: low-friction bearing	Supports the conveyor drive and idler shafts while keeping the sprockets aligned.

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3	Flexible Clamp Motor Coupling	1. Component type: flexible clamp shaft coupling 2. Bore sizes: 10 mm 3. Material: aluminum or steel coupling body preferred 4. Flexibility: able to compensate small shaft misalignment 5. Torque capacity: ≥1 Nm 6. Length: approx. 20 mm	Transfers torque from the motor/gearbox to the conveyor shaft while tolerating small assembly misalignment.
4	Planetary Gearbox for Motor	1. Component type: planetary gearbox for NEMA 17 stepper motor 2. Gear ratio: 1:27 3. Motor compatibility: compatible with NEMA 17 4. Input/Output shaft compatibility: 5 mm / 10mm 5. Rated output torque: ≥2–3 Nm preferred	1- A planetary gearbox is used to reduce motor speed and increase torque for slow, controlled conveyor movement through the heating zone.

11.1.6. Containment Subsystem

Mechanical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Non-stick Surface	1. Component type: food-safe non-stick coating 2. Food safety: suitable for direct food contact 3. Temperature resistance: ≥150°C 4. Thickness: coating thickness approx. 20–50 µm, or replaceable liner thickness approx. 0.1–0.5 mm	1- A non-stick surface prevents the egg from sticking in the frying or transfer area and allows easier cleaning after operation.
2	Food-grade Seals / Gap Strips	1. Component type: food-grade sealing strip or flexible gap cover 3. Temperature resistance: ≥150°C 4. Strip thickness: approx. 1–3 mm 5. Strip width: approx. 10–20 mm	1- A sealing strip or gap cover closes small gaps where egg, oil or food residue could collect, improving hygiene and cleanability.

11.1.7. Egg-Shell Storage

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Mechanical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Non-disposable Inner Liner	1. Component type: removable non-disposable protective liner 2. Thickness: approx. 1.0 mm 3. Fit: cut to fit 4. Temperature range: at least 0-100°C	A removable liner protects the drawer interior and simplifies cleaning after shell waste collection.
2	Liner Retaining Strips / Clips	1. Component type: retaining clips or strips 2. Material: SS304 stainless steel or food-grade PE 3. Size: for strips approx. 80-100 mm long and approx. 0.8-1.5 mm thick, for chips approx. 15-25 mm (min. Needed 4-6 per machine)	1- Retaining clips or strips keep the silicone liner in position, while still allowing drawer removal during cleaning.
3	Rod End Collars / Retaining Stops	1. Component type: shaft collar or retaining stop for guide rods 2. Rod compatibility: suitable for Ø8 mm guide rods 3. Material: SS304 stainless steel	Retaining stops are needed to prevent the guide rods or pusher assembly from moving beyond the intended travel range.
4	Drawer Latch / Stop	1. Component type: stainless steel drawer latch, spring latch, or mechanical stop 2. Holding force: approx. 10–30 N 3. Size: compact latch body, approx. 20–50 mm length 4. Durability: suitable for repeated opening and closing cycles, ≥10,000 cycles preferred	Keeps the shell-waste drawer securely closed during operation while still allowing manual opening for emptying.
5	Front Sealing Strip / Gap Cove r	1. Component type: food-grade silicone sealing strip 2. Strip thickness/width: approx. 1–3 mm / approx. 10–20 mm 3. Length: cut to fit drawer front or inlet gap; approx. 260 mm	A sealing strip or gap cover is needed to reduce shell dust, residue, smell, and small splashes escaping through the drawer front or inlet gap.
6	Stainless Fasteners Kit	1. Material: SS304 / A2, stainless steel fasteners 2. Screw sizes: mainly M3-M5 3. Thread standard: ISO metric coarse threads 4. Length range: approx.6–16 mm	A fasteners kit is used to assemble drawer rails, brackets, handle and support parts while allowing maintenance and replacement.

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		5. Serviceability: removable connections preferred	

11.1.8. Bottom Heating Subsystem

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Cartridge Heaters	1. Component type: electrical cartridge heater 2. Size: Ø8 × 160 mm 3. Power rating: 200–250 W per heater 4. Supply voltage:230V AC 5. Sheath material: stainless steel sheath (Incoloy 800 preferred) 6. Temperature capability: ≥300°C	A cartridge heater is used because the bottom heating block needs compact, direct heat input from inside the block.
2	Temperature Sensor	1. Sensor type: PT100 2. Measurement range: 0-250°C 3. Accuracy: ±1–2°C in cooking range 4. Probe diameter: approx. Ø3–5 mm 5. Probe length: ~30–50 mm	A temperature sensor measures the heating block temperature so the controller can regulate the frying temperature.
3	Temperature Sensor Amplifier	1. Part: MAX31865xAP or equivalent PT100 interface 2. Supply: 3.3 V DC 5. Interface: SPI (SCLK, SDI, SDO, CS) to STM32H743, 3.3 V logic 6. Wiring mode: 3-wire or 4-wire PT100	Converts the PT100 signal into digital temperature data for the main controller.
4	Solid-state Relay	1. Component type: Zero-cross AC SSR; DC-controlled and AC-switched 2. Load voltage: 230V AC output switching 3. Load current rating: preferably 25A 4. Input control voltage: compatible with 3.3V or 5V control signal 5.Heatsink: mandatory.	Allows the controller to switch the heater power safely without directly handling mains voltage.
5	Heater Branch Fuse	1. Type: T10A 250 V slow-blow cartridge fuse, 5 × 20 mm 2. Breaking capacity: ≥1500 A HBC (high breaking capacity)	Protects the heater branch wiring against overcurrent or short-circuit faults.

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6	High-temperature Wiring	1. Component type: heat-resistant heater wiring 2. Conductor cross-section: ≥1.5 mm² 3. Voltage rating: ≥ 300V AC 4. Temperature rating: ≥180°C 5. Insulation material: PTFE or fiberglass-sleeved insulation	High-temperature wiring connects the heaters safely in the hot area where normal PVC wiring could degrade.
7	Grounding Hardware	1. Component type: protective earth bonding hardware 2. Grounding connection: ring terminal or grounding lug 3. Conductor compatibility: 4 mm² PE . Electrical resistance: low-resistance bond, preferably ≤0.1 Ω between metal part and PE connection	Bonds the heating block and metal frame to protective earth for electrical safety.
Mechanical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Thermal Compound	1. Component type: heat-conductive thermal paste 2. Temperature range: approx. 0-250°C 3. Thermal conductivity: ≥1 W/m·K 4. Electrical property: electrically insulating preferred	Thermal paste is needed to improve heat transfer where small air gaps may exist between the sensor and aluminum heating block.

11.1.9. TOP IR Heating System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	IR Heater Element	1. Heater type: shielded IR heater or quartz/ceramic IR element 2. Heated length: approx. 400–450 mm 3. Power rating: 300–600 W 4. Supply voltage: 230V AC	An IR heater element provides top-side radiant heat so the egg can be cooked evenly from above while moving through the heating zone.
2	IR Power Control / Relay	1. Component type: relay or solid-state relay 2. Load voltage: 230V AC 3. Load current rating: ≥ 10A 4. Input control voltage: 5V DC control signal	A relay or SSR is needed so the controller can switch the top IR heater power safely.

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3	Thermal Cutoff / Safety Sensor	1. Component type: thermal cutoff switch, thermal fuse or safety temperature sensor 2. Cutoff / alarm temperature: approx. 180–250°C 3. Signal type: normally closed contact preferred, opens during over-temperature fault 4. Reset type: manual reset	A thermal cutoff or safety sensor protects the top heating area if the IR heater overheats or the control system fails.
4	PT100 Sensor	1. Sensor type: PT100 resistance temperature sensor 2. Measurement range: 0-250°C 3. Accuracy class: Class B minimum 4. Probe diameter: approx. Ø3–5 mm 5. Connection type: 3-wire preferred	A PT100 sensor monitors the temperature near the IR heater module for temperature control and safety supervision.

11.1.10. Interface/Integration System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	SSR Heat Sink	1. Component type: aluminum heat sink for SSR 2. Compatibility: approx. 58 × 45 mm 3. Thermal resistance:≤2.5–3.0 °C/W 4. Operating temperature: ≥70°C	Removes heat from the SSR during heater switching and prevents overheating in the electrical enclosure.
2	Ceramic Terminal Block	1. Component type: ceramic high-temperature terminal block 2. Voltage rating: ≥250V AC 3. Current rating: ≥10A 4. Number of poles: 2-pole 5. Wire compatibility: suitable for approx. 0.75–1.5 mm² heat-resistant wires 6. Temperature resistance: ≥ 200°C	Provides a heat-resistant connection point for 230 V heater wiring near the heating area.
3	Low-voltage Terminal Blocks	1. Component type: low-voltage terminal blocks for sensor and control wiring 2. Voltage rating: ≥30V DC 3. Current rating: ≥2A per terminal 4. Number of poles: 4 pole 5. Wire compatibility: suitable for approx. 0.14–0.75 mm² signal/control wires	Organizes low-voltage sensor, control and driver wiring and reduces wiring mistakes during assembly or maintenance.

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11.1.11. Nuts/Bolt/Washers for Conveyer System, Top/Bottom Heating and Main Housing System

Mechanical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Countersunk Raised Head Cross Recess Screw	1. Standard: ISO 7047 2. Screw size: M10 x 25 mm and M10 x 60 mm 3. Drive type: cross recess Z 4. Material: stainless steel or corrosion-resistant steel preferred	A countersunk raised head screw is used where a low-profile screw head is required for strong fastening points in the housing or heating/conveyor structure.
2	Pan Head Cross Recess Screw	1. Standard: ISO 7045 2. Screw size: M8 x 16 mm, M4 x 30 mm and M3 x 20 mm 3. Drive type: cross recess Z 4. Material: stainless steel or corrosion-resistant steel preferred	Used for general fastening points where a flush countersunk head is not required.
3	Socket Head Cap Screw	1. Standard: ISO 4762 2. Screw size: M6 x 20 mm and M3 x 10 mm 3. Material: stainless steel preferred 4. Head type: cylindrical socket head / Allen head	Used for stronger or more compact mechanical fastening points, especially in the conveyor and support structure.
4	Prevailing Torque Hex Nut	1. Standard: ISO 7042 2. Thread size: M6 3. Thread standard: ISO metric coarse thread M6 × 1.0 4. Locking function: self-locking design	Prevents loosening of M6 screw connections caused by conveyor movement or vibration.
5	Plain Washer	1. Standard: ISO 7089 2. Washer size: M6 3. Inner diameter: suitable for M6 screw clearance	A plain washer is used to distribute the fastening load under the screw head or nut.
6	Socket Set Screw, Cup Point	1. Standard: ISO 4029 2. Screw size: M5 × 20 mm 3. Thread standard: ISO metric coarse thread M5 × 0.8	A socket set screw is used because the sprocket or hub must be fixed to the shaft without using a large external fastener.
7	Hex Thin Nut	1. Standard: ISO 4036, Grade B 2. Thread size: M3 and M4	A hex thin nut is used where a compact M3 fastening connection is needed.

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		3, Thread standard: ISO metric coarse thread	

11.1.12. Rotating Plate System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Stepper Motor	1. Motor type: 5V geared stepper motor, 28BYJ-48 class or equivalent 2. Step angle / gearbox: geared stepper motor with small step movement 3. Output torque: ≥30–40 mN*m 4. Speed range: low-speed rotation, approx. 1–10 rpm 5. Duty rating: intermittent operation	A geared stepper motor is used because the serving plate needs controlled step-by-step rotation, so that each cooked egg can be placed in a defined position.
2	Stepper Driver Module	1. Driver type: ULN2003A class or equivalent 2. Motor supply voltage: 5V DC 3. Logic input voltage: compatible with 3.3V controller signals 4. Output current: ≥500 mA total 5. Channels: 4 output channels	Allows the microcontroller to control the geared stepper motor safely.
3	Microswitch	1. Switch type: lever-type microswitch with NO/NC/COM contacts 2. Contact rating: ≥5V DC signal switching 3. Actuation force: approx. 0.5–2 N 4. Mechanical life: ≥ 100,000 cycles	A lever microswitch is used because it gives a simple physical signal when the rotating plate reaches the home position.
4	Load Cell	1. Sensor type: single-point bar load cell 2. Rated capacity: 5 kg 3. Supply / excitation voltage: 5V DC 4. Output signal: millivolt bridge signal approx. 1.0–2.0 mV/V sensitivity 5. Accuracy: ±1–5 g resolution after calibration preferred	A single-point load cell is used because the system needs to detect whether the serving plate is present by measuring weight.

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5	HX711 Amplifier Module	1. Module type: load-cell amplifier and ADC module 2. Supply voltage: 3.3V DC 3. Input compatibility: must be compatible with load cell 4. Data interface: DT/DOUT and SCK interface 5. Sampling rate: selectable or fixed 10–80 samples/s	The HX711-type amplifier is needed to convert the small load-cell signal into readable data for the controller.
6	Wiring and Connectors	1. Wiring type: jumper cables and connector wiring for motor, microswitch, load cell and driver module 2. Motor wiring: 4-core wiring, approx. 0.14–0.34 mm² 3. Sensor wiring: 2–4 core signal wiring, approx. 0.14–0.25 mm², 4. Voltage rating: ≥30V DC	Wiring and connectors are needed to connect all electrical components of the rotating plate system safely and replaceable.
Mechanical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Lazy Susan Bearing	1. Component type: turntable bearing 2. Outer diameter: Ø120 mm 3. Load capacity: ≥5 kg 4. Material: corrosion-resistant material preferred 5. Rotation: smooth 360° rotation	A Lazy Susan turntable bearing is needed because the serving plate must rotate while being supported from below and allowing smooth rotation.
2	Rubber Drive Wheel	1. Component type: rubber or polyurethane friction wheel 2. Outer diameter: Ø38.1 mm 3. Bore diameter: 5 mm 4. Wheel width: 24.4 mm 5. Material hardness: approx. 50–80 Shore A	A friction drive wheel is needed to transfer motor rotation to the serving plate without a direct central shaft connection.
3	Countersunk Flat Head Cross Recess Screws	1. Standard: ISO 7046-1 2. Screw size: M4 x 16 mm and M4 x 8 mm 3. Thread standard: ISO metric coarse thread M4 x 0.7 4. Material: stainless steel, preferably A2 / SS304	Countersunk screws are required so the screw head does not protrude above the bearing or mounting surface.

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4	Socket Head Cap Screw	1. Standard: ISO 4762 2. Screw size: M4 x 10 mm and M6 x 20 mm 3. Thread standard: ISO metric coarse thread 4. Material: stainless steel	A socket head cap screw is used for strong mechanical fastening points for the motor and load-cell plate connections.
5	Pan Head Cross Recess Screw	1. Standard: ISO 7045 2. Screw size: M2.5 x 16 mm M5 x 12 mm 3. Thread standard: ISO metric coarse thread 4. Material: stainless steel	Suitable because the microswitch and the bracket mounting connection does not require a flush countersunk head.
6	Socket Countersunk Head Screw	1. Standard: ISO 10642 2. Screw size: M4 × 30 mm 3. Thread standard: ISO metric coarse thread M4 × 0.7 4. Material: stainless steel	A socket countersunk screw is used because the top load cell plate needs a flush screw head with reliable tightening.
7	Hex Thin Nut	1. Standard: ISO 4036, Grade B 2. Thread size: M2.5, M4 and M6 3. Thread standard: ISO metric coarse thread 4. Material: stainless steel	A hex thin nut is used where the fastening area has a limited height and a normal nut may interfere with nearby parts.

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Milestone 3

12. Technical Drawings for Main Assembly

READ ME

Technical Note Regarding Assembly Drawing File

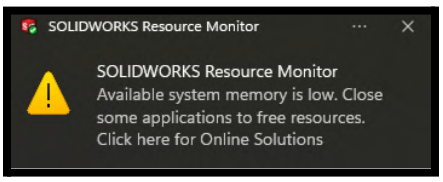
During the preparation of the full assembly drawing, some imported supplier CAD models caused SolidWorks section-view rebuild warnings. The warning mainly appears when SolidWorks tries to section complex imported components such as the standard nozzle/check-valve model and related downloaded components. These parts are buy components and are only required for assembly positioning, not for internal manufacturing representation.

When opening the drawing, SolidWorks may display the section warning several times while loading different drawing views. The warnings can be confirmed by clicking OK ALOT of times , 15+ times . The drawing then continues loading normally. Due to the size of the complete assembly, the number of section/detail views, and the imported CAD components, the drawing file can take approximately 12 minutes or more to open.

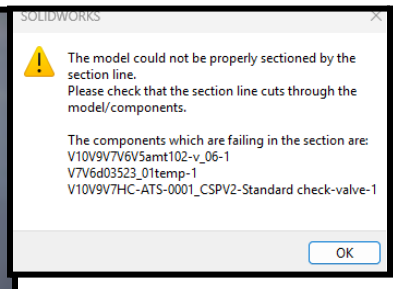
Occasionally, some drawing views may temporarily appear as incomplete or "phantom" views after opening or rebuilding. This is a display/rebuild issue in SolidWorks and not an intentional part of the drawing. Closing and reopening the drawing usually refreshes the views correctly. The PDF export should be used as the reference for checking the final drawing layout and view content.

No design changes are intended by these warnings. They are caused by rebuild/sectioning limitations of imported CAD geometry in the SolidWorks drawing environment.

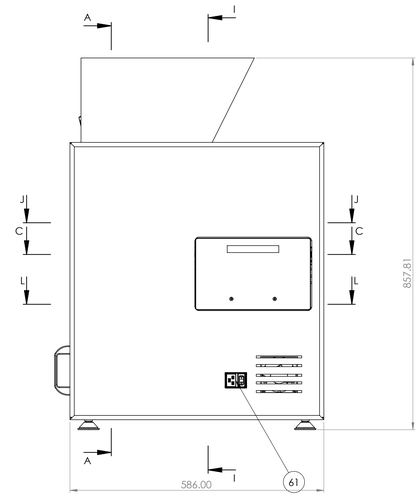
All 3 Problems Displayed Below

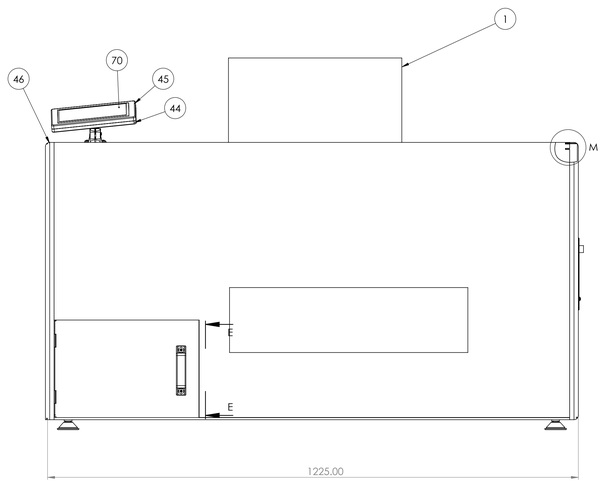


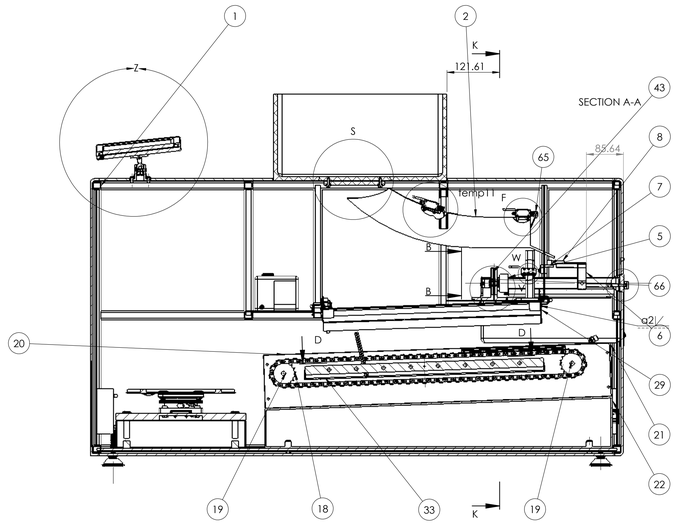




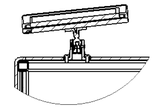
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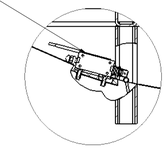




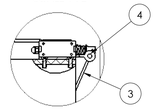
DETAIL Z
Display Casing Mounting
SCALE 1 : 4



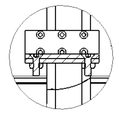
DETAIL Y
Gate Motor 1 To Tunnel Mounting
SCALE 1 : 2



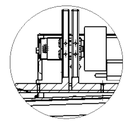
DETAIL X
Gate Motor 2 to Tunnel Mounting
SCALE 1 : 2



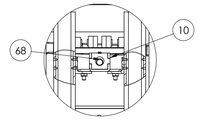
DETAILW
Egg Breaking Mechansim Motor Casing Mount
SCALE 1 : 1



DETAILV
Servo L Brackets To Main Frame
SCALE 1 : 2

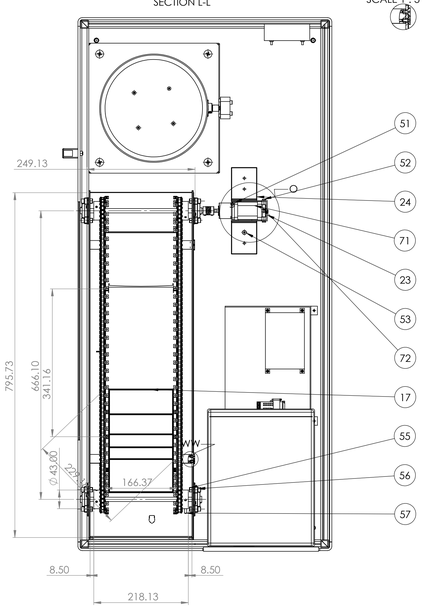


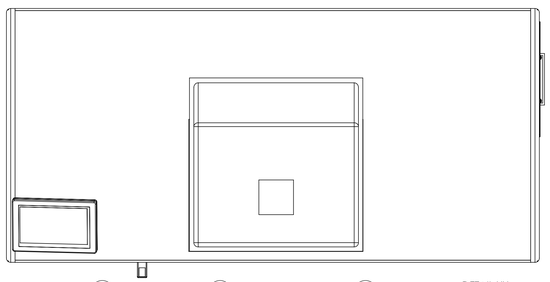
DETAIL U
Motor Bracket to frame
SCALE 1 : 2



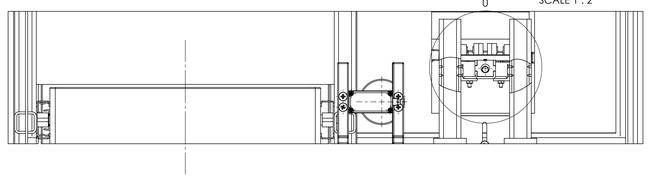
DETAIL WW
Bottom Heating Mounting Brackets to Conveyor Frame
SCALE 1 : 3

SECTION L-L

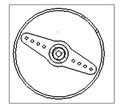




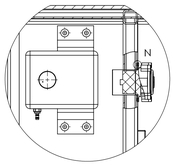
SECTION B-B
Servo Motor Mount to Brackets
Breaking Mechanism Motor Mount to Frame
U
SCALE 1 : 2



SECTION K-K
Servo Motor Horn to egg breaking Shaft
SCALE 1 : 1



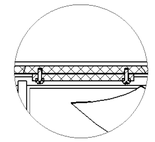
DETAIL T
Oil Tank+Pump to Frame Mount
SCALE 1 : 3



DETAIL N
Pump to Frame
SCALE 1 : 1



DETAILS
Storage Tunnel and Storage Unit to Frame
SCALE 1 : 3



DETAIL R
Front Cover To Egg Shell Drawer
SCALE 1 : 3



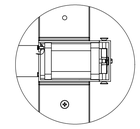
DETAIL Q
Compression Shafts to Drawer
SCALE 1 : 3



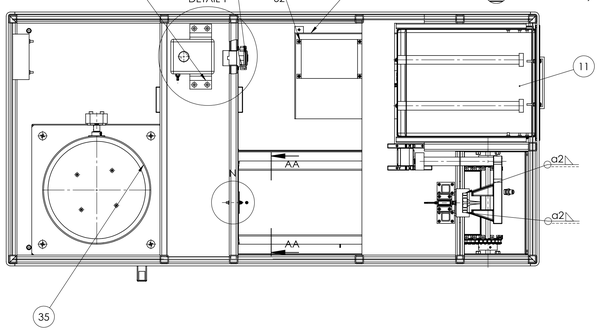
DETAIL P
Drawer Handle to Front Cover
SCALE 1 : 3



DETAIL O
Motor to Motor Mount
Motor Mount to Frame
Planetary Gearbox to Motor Casing
SCALE 2 : 5



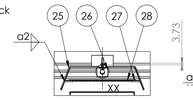
SECTION J-J
Breaking Mechanism,Oil Storage , Top Rotating Plate Mounting



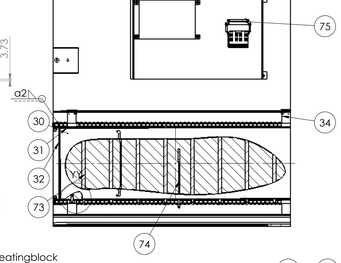
DETAIL XX
Top Frame into Quartz Block
SCALE 1 : 3



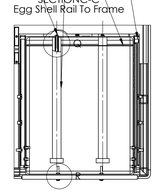
SECTION AA-AA



SECTION D-D
Heating Block Assembly



SECTIONC-C
Egg Shell Rail To Frame



DETAIL N
Top IR to Frame
: 4



DETAIL M
Chassy Mounting to each other
SCALE 1 : 3



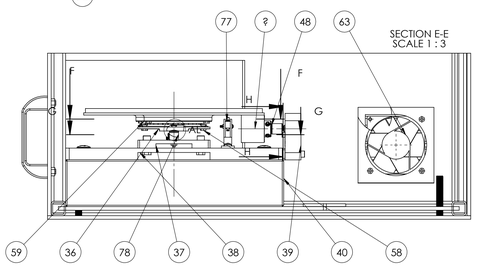
DETAIL YY
Mounting Brackets to Heatingblock
SCALE 1 : 3



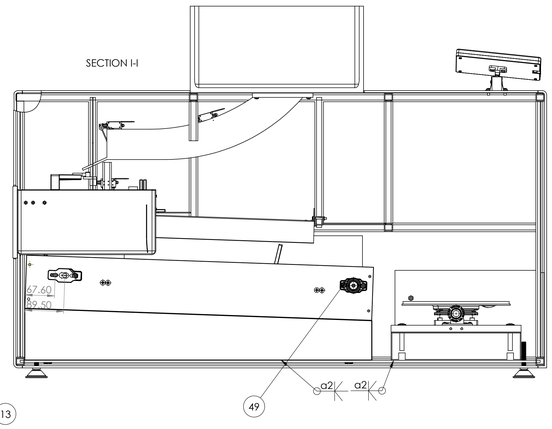
DETAIL ZZ
Load Cell to Top Load Cell Plate
SCALE 2 : 3



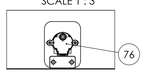
SECTION E-E
SCALE 1 : 3



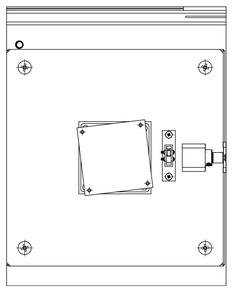
SECTION H-I



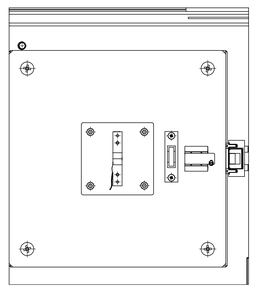
SECTION H-H
Motor to Motor housing
Motor Housing to Base Plate
SCALE 1 : 3



SECTION F-F
SCALE 1 : 3



SECTION G-G
SCALE 1 : 3





GENERAL NOTES:
1. I.ALL DIMENSIONS ARE IN mm UNLESS OTHERWISE STATED.
2. DO NOT SCALE DRAWING.
3. FIRST ANGLE PROJECTION.
4. THIS IS AN OVERALL ASSEMBLY DRAWING. MANUFACTURING DIMENSIONS ARE NOT SHOWN.
5. ITEM NUMBERS REFER TO THE BILL OF MATERIALS.
6. STANDARD PARTS SHALL BE ACCORDING TO THE STANDARD SPECIFIED IN THE BOM.
7. WELDS SHALL BE FILLET WELDS a2 UNLESS OTHERWISE SPECIFIED.
8. WELDED EDGES IN FOOD-CONTACT OR CLEANING AREAS SHALL BE DEBURRED AND SMOOTHED.
9. REMOVABLE PARTS SHALL REMAIN ACCESSIBLE FOR CLEANING AND MAINTENANCE.

general tolerances acc. to ISO 2768-N/A-assembly drawing
surface finish acc. to ISO 1302
material: N/A assembly
profile: N/A
weight: kg
edge finish acc. to ISO 13715
coating: acc. to make parts list

faculty or department: Technology & Bionics	technical reference	drawn by (last saved by)	drawing date: 7/4/2026
HOCHSCHULE RHEIN-WAAL, Rhine-Waal University of Applied Sciences	title, additional title: SunnyPro-Full Assembly Automatic Egg Frying Machine		projection method; scale: 15
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	drawing no.: Group 3 Milestone 3 Step 12	rev.: 03	date of issue: 7/7/2026
			sheet: 1 /1

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1	Egg Storage Container	PP-C HI / Polypropylene Copolymer		1	Yield: ~22 MPa | Density: 0.90 g/cm³ | Temp: -20 to 100°C	1
2	Egg Tunnel	PP-C HI / Polypropylene Copolymer		1	Yield: ~22 MPa | Density: 0.90 g/cm³ | Temp: -20 to 100°C	2
3	Gates for Transportation Tunnel	Acetal Copolymer (POM-C)		2	Yield: ~62 MPa | Density: 1.41 g/cm³ | Temp: -40 to 100°C	4
4	Gate Shaft Holder for Tunnel	POM-C / Acetal Copolymer		4	Yield: ~62 MPa | Density: 1.41 g/cm³ | Temp: -40 to 100°C	5
5	Egg Tunnel Shaft	EN 1.4301 / AISI 304		4	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	6
6	Egg Holding Shaft	EN AW-6082 T6		1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	7
7	Egg Holding Arm	EN AW-6082 T6		1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	8
8	Base of Spear	EN AW-6082 T6 (clear anodised)		1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	9
9	Spears (Left and Right)	EN 1.4116 / X50CrMoV15 (hardened)		2	Yield: ~800–1000 MPa | Density: 7.70 g/cm³ | Temp: -20 to 200°C	10
10	Solenoid Motor Mounting Bracket	EN 1.4301 / AISI 304		2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	11
11	Solenoid Motor Casing	EN AW-6082 T6 (clear anodised)		3	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	12
12	Main Drawer Box	EN 1.4301 / X5CrNi18-10		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	13
13	Drawer Front Cover	ABS-HI plastic		1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	14
14	Drawer Handle	ABS-HI plastic w/ brass inserts		1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	15
15	Inner Shaft (Drawer)	EN 1.4301 / X5CrNi18-10		2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	16
16	Manual Compression Plate	EN 1.4301 / X5CrNi18-10		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	17
17	Oil Storage Mount	EN AW-6082 T6		2	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	18
18	Pocket Plate	EN 1.4409 / GX2CrNiMo19-11-2 (cast)		4	Yield: ~185–205 MPa | Density: 7.90 g/cm³ | Temp: -196 to 400°C	19
19	Sprocket for Conveyor Belt	EN 1.4404 / X2CrNiMo17-12-2		4	Yield: ~205 MPa | Density: 8.00 g/cm³ | Temp: -196 to 400°C	20
20	Driving Shaft / Idler Shaft	EN 1.4404 / X2CrNiMo17-12-2		2	Yield: ~205 MPa | Density: 8.00 g/cm³ | Temp: -196 to 400°C	21
21	Conveyor Side Frame	EN 1.4301 / X5CrNi18-10		2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	22
22	Conveyor Back Frame	EN 1.4301 / X5CrNi18-10		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	23
23	Conveyor Belt Hollow Support Frame	EN AW-6082 T6		1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	24
24	Nema 17/Gearbox/Encoder Motor Housing	EN AC-43500 / AlSi10MnMg (cast)		1	Yield: ~140–180 MPa | Density: 2.68 g/cm³ | Temp: -40 to 150°C	25
25	Nema Motor Mount	EN 1.4301 / AISI 304		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	26
26	TOP IR Shielding	EN 1.4301 / X5CrNi18-10		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C (shield, near IR zone)	27
27	Ceramic Block for Holding Quartz IR			1	Flexural: ~300 MPa | Density: 3.70 g/cm³ | Temp: up to 1600°C	28
28	Side Lining for TOP IR	EN 1.4301 / X5CrNi18-10		2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	29
29	Perforated Sheet	EN 1.4301 / X5CrNi18-10		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	30
30	TOP IR Back Frame	EN 1.4301 / X5CrNi18-10		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	31
31	Heating Block	EN AW-6082 T651		1	Yield: ~250 MPa | Density: 2.70 g/cm³ | Temp: -40 to 200°C	32
32	Side Frame (Bottom Heating)	EN 1.4301 / X5CrNi18-10		2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	33
33	Front Frame (Bottom Heating)	EN 1.4301 / X5CrNi18-10		2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	34
34	Bottom Frame (Bottom Heating)	EN 1.4301 / X5CrNi18-10		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	35
35	Mounting Brackets for Bottom Heating Block	EN 1.4301 / AISI 304		4	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	36
36	Aluminum Plate (Rotating Plate)	EN AW-6061 T6 (clear anodised)		1	Yield: ~275 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	37
37	Top Load Cell Plate	EN AW-6061 T6 (clear anodised)		1	Yield: ~275 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	38
38	Bottom Load Cell Plate	EN AW-6082 T6 (clear anodised)		1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	39
39	Base Plate (Rotating Plate)	EN AW-6082 T6 (clear anodised)		1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	40
40	Stepper Motor Housing	EN AC-43500 / AlSi10MnMg (cast, clear anodised)		1	Yield: ~140–180 MPa | Density: 2.68 g/cm³ | Temp: -40 to 150°C	41

Item no.	Part Name	Material specification	Link/Brand	Qty.	Technical Specification	Position in Technical assembly

faculty or department	technical reference	drawn by (last saved by)	drawing date	released by
Technology & Bionics		ahmad	09/12/2020	

HOCHSCHULE RHEIN-WAAL
Rhine Waal University of Applied Sciences

title, additional title	projection method	scale
SunnyPro Group3		

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drawing no. prefix	type of document
-ME.6.2701.03	bill of material

drawing no.	rev.	date of issue	sheet
260707-01-001	00	07/07/2026	1 /1

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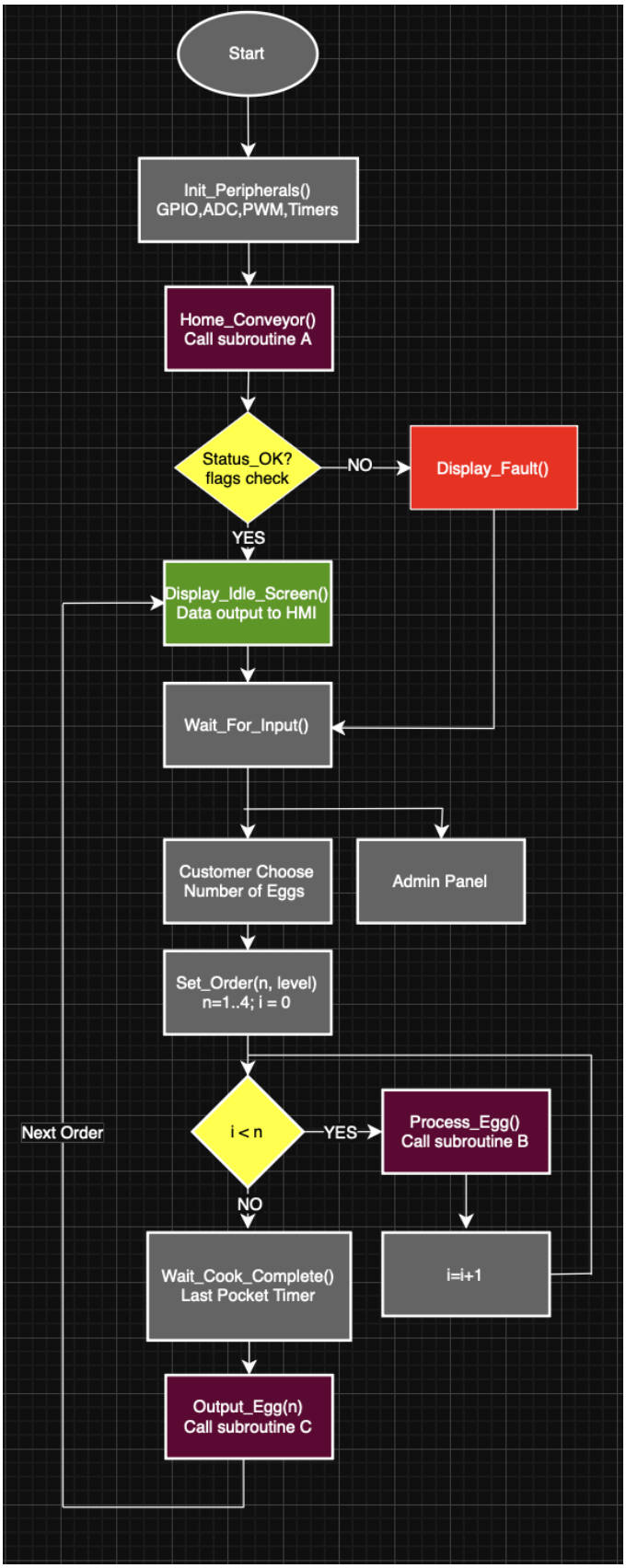
41	Bottom Base for Rotating Plate	EN 1.4301 / AISI 304		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	42
42	Electrical Box Housing	EN 1.4301 / AISI 304		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	43
43	Main Frame Housing	EN 1.4301 / AISI 304		1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	44
44	L Bracket for Servo Motor	EN 1.4301 / AISI 304		4	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	45
45	Plastic Back Plate (Display)	ABS-HI plastic w/ brass inserts		1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	46
46	Front Frame for Casing (Display)	ABS-HI plastic		1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	47
47	Main Case/Chassy Panels	ABS-HI plastic + purchased glass		1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	48
48	DIN Rail	zinc-plated steel	Hutschiene TS 35 x 7,5mm gelocht - L=326mm	2	35 x 7,5 x 326 mm	49
49	Axle Shaft	Aluminum	Aluminum shaft	1	Ø 10 mm x100 mm	50
50	2-Bolt Flange Bearing Block	Aluminum	Stehlagereinheiten / Flanschlagereinheiten 2- Loch KFL000	4	Ø 10mm length 66mm total diameter 40 mm	51
51	Planetary Gearbox for Motor	Stainless Steel	Planetengetriebe Schrittmotor 27:1	1	Planetary gear 27:1	52
52	Countersunk Raised Head Cross Recess Screw	A2 Stainless Steel	Senkkopfschrauben M10x25	38	M10x25 / M10x60	53
53	Pan Head Cross Recess Screw	A2 Stainless Steel	Senkkopfschrauben M10x60	40	DIN 7985 (ISO 7045) / M8x16 / M3x12	54
54	Socket Head Cap Screw	A2 Stainless Steel	Linsenschrauben M8x16	12	M3x25 / M3x10mm	55
55	Pan Head Cross Recess Screw	A2 Stainless Steel	DIN 912/ISO 4762 A2 – M3 x 25 mm 100 Stück	26	ISO 7045 / M4x8	56
56	Prevailing Torque Hex Nut	Stainless Steel	DIN 912/ISO 4762 A2 – M3 x 10 mm 100 Stück	18	M6	57
57	Plain Washer	Polyoxymethylene	Flachkopfschraube ISO 7045 PZ M4x8-Z 4.8 galv.verz. 2000St.	10	M6	58
58	Socket Set Screw, Cup Point		Sechskantmuttern M6 DIN 934 Edelstahl A2	12	M5x20 / M5x30	59
59	Hex Thin Nut		DIN 439/ISO 4035 Sechskantmutter Edelstahl A2, M4 - 100 Stück	52	M3 / M4	60
60	Lazy Susan Bearing		Blickle 942239 DK-L-3 Stahlblech-Drehkranz 1 St.	2	140 x 110 mm	61
61	Rubber Drive Wheel		Freies Verschiffen 10PCS 8*32*16 flache typ PA nylon	3	8x32x16mm	62
62	IEC C20 Power Inlet		IEC 20 Power Inlet	1	16A, 250VAC	63
63	PCB Screw Terminal Connectors		20Paar 5.08mm Pitch Male Buchse PCB Screw Terminal Block | eBay.de	1	25x10x15mm, AC 300V, 15A	64
64	Axial Fan 8414NGML		EBM Papst 8414NGML Axiallüfter 24 V/DC 42 m³/h (L x B x H) 80 x 80 x 25mm | » Freue dich auf voelkner	2	80x80x25mm, 42m³/h, 24V DC	65
65	Gate Actuators JF-0530B		JF-0530B DC12V 300mA 5N/10mm Pull-Push-Type Solenoid Electromagnet	2	DC12V, 300mA, 5N/10mm, 30x16x15mm	66
66	Flip Actuator MG996R		MG996R Digital Servo Motor mit Metall Getriebe	1	0.17 sec/60° @ 4.8V, 40x19x43mm, 4.8-7.2V	67
67	Platform Solenoid JF-0837B		JF-0837B DC 24V 1A 15N 10mm Push Pull Type Open Frame Linear Motion Solenoid Electromagnet | Harfington	1	DC24V, 1A, 15N, 37x20x26mm	68
68	Peristaltic Pump		DC 12V Dosing Pump Peristaltic Pump	1	DC12V, Three roller, 11ml/min	69
69	Touchscreen Display Module - Nextion NX8048P070_011R		NEXTION 4.3/5.0/7.0" Intelligent LCD Touch Display	2	108x181x9.3mm	70
70	Stepper Motor		NEMA 17	1	2.8V, 5mm shaft, 2 phase	71
71	Incremental Rotary Encoder		AMT102-V	1	3.6V to 5.5V	72
72	Cartridge Heaters		Cartridge Heaters	4	250W, 230V AC	73
73	Temperature Sensor		PT100 temp. Sensor	1	-	74
74	Solid-State Relay		solid state relay 25A SSR-25AA , ZG3NC-325A Single phase	1	25A, 90-480VAC	75
75	Stepper Motor		FREI Standard-Schrittmotor, 5V	1	5V DC, 4 phase, 34.3 mN·m	76
76	Microswitch		5Pcs Mini Micro Limit Switch NO NC 3 Pins PCB Terminals SPDT 5A 125V 250V 29mm Roller Arm lever Snap Action Push Microswitches		5A, 125V/250V	77
77	Load Cell and HX711 Amplifier Module		Stemedu 5 kg Load Cell Digital Weight Sensor for Portable Electronic with HX711 Ad Amplifier Module, 11 lbs Weighing Sensor Arduino Raspberry Pi	1	4-wire, 5-10V, 12.7x12.7x80mm, voltage range 2.6-5.5V	78
Item no.	Part Name	Material specification	Link/Brand	Qty.	Technical Specification	Position in Technical assembly

faculty or department: Technology & Bionics	technical reference:	drawn by (last saved by): ahmad	drawing date: 09/12/2020	released by:	
HOCHSCHULE RHEIN-WAAL Rhine Waal University of Applied Sciences		title, additional title: SunnyPro Group 3		projection method	scale
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		drawing no.: 260707-01-001	rev.: 00	date of issue: 07/07/2026	sheet: 1 /1

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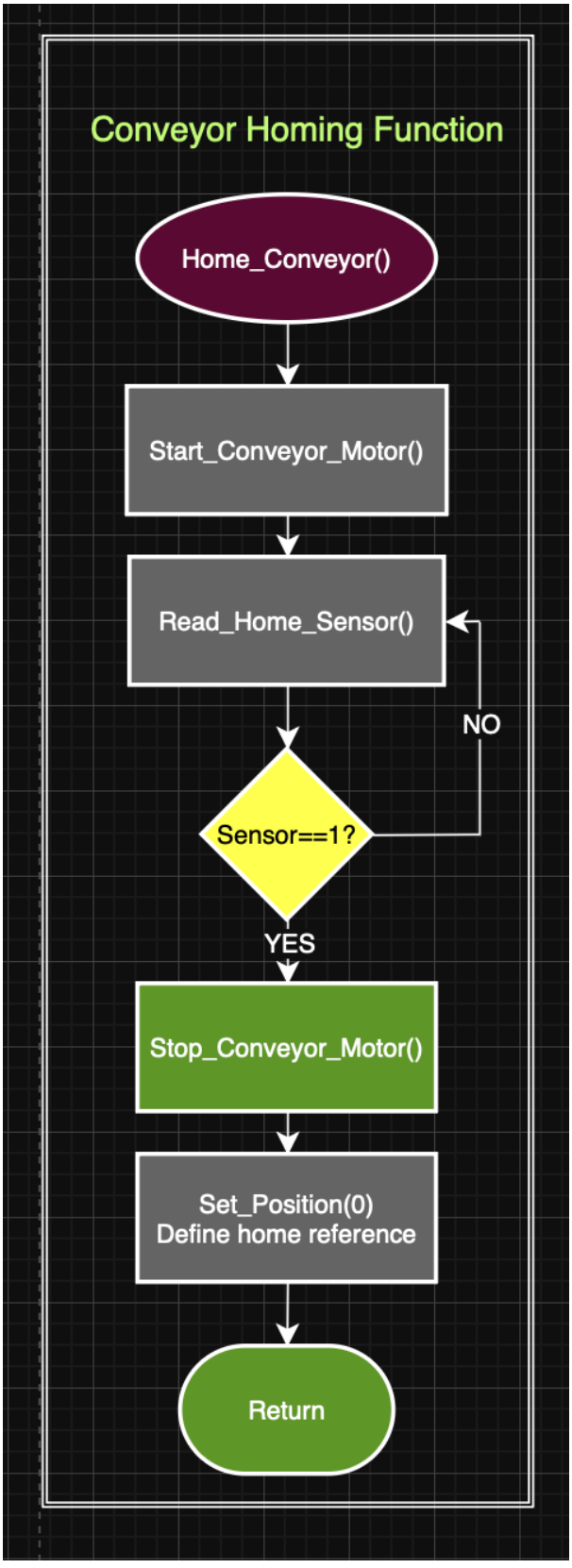
13. Flowchart

13.1. Customer Flowcharts



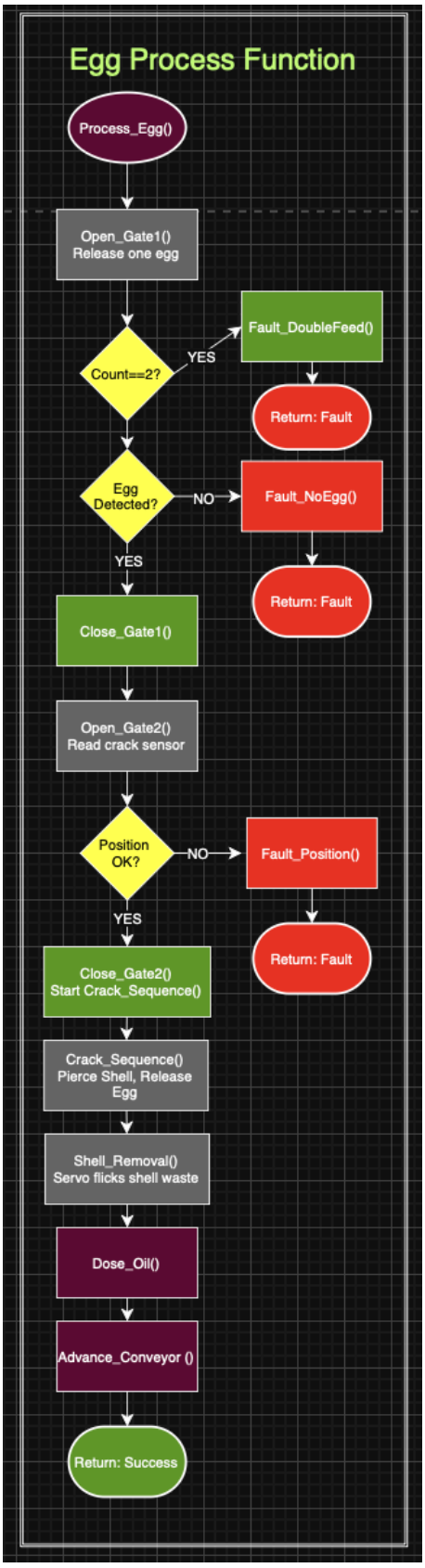
..

13.1.1. Conveyor Homing Function Flowchart



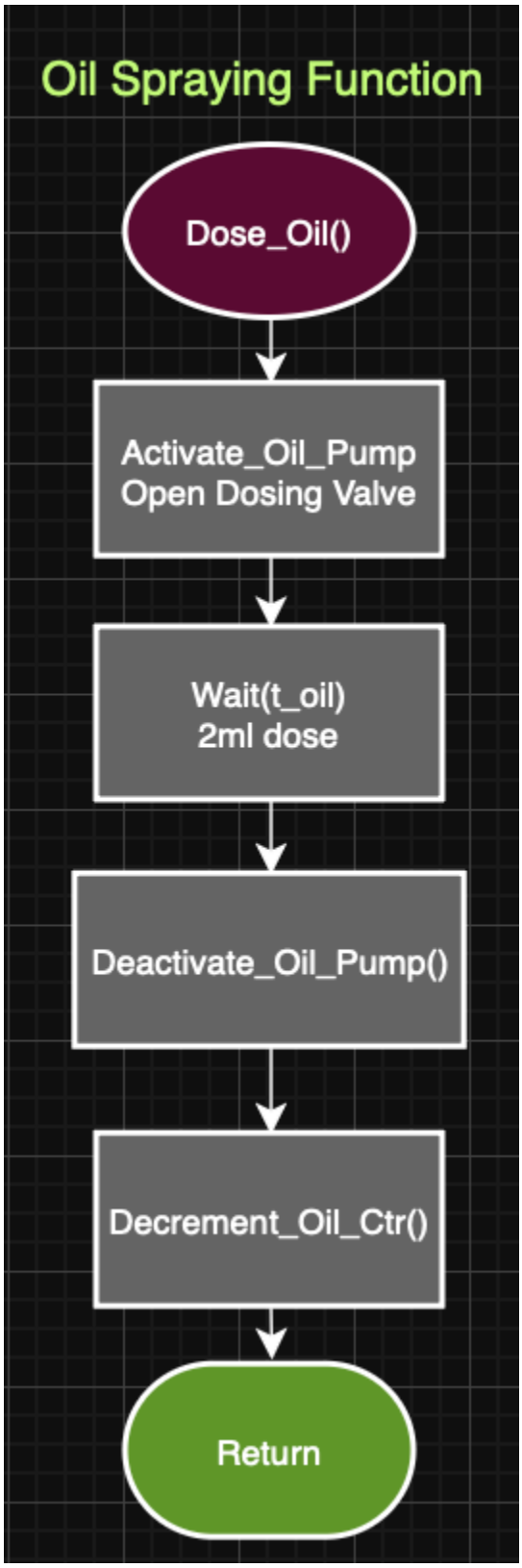
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13.1.2. Egg Process Function Flowchart



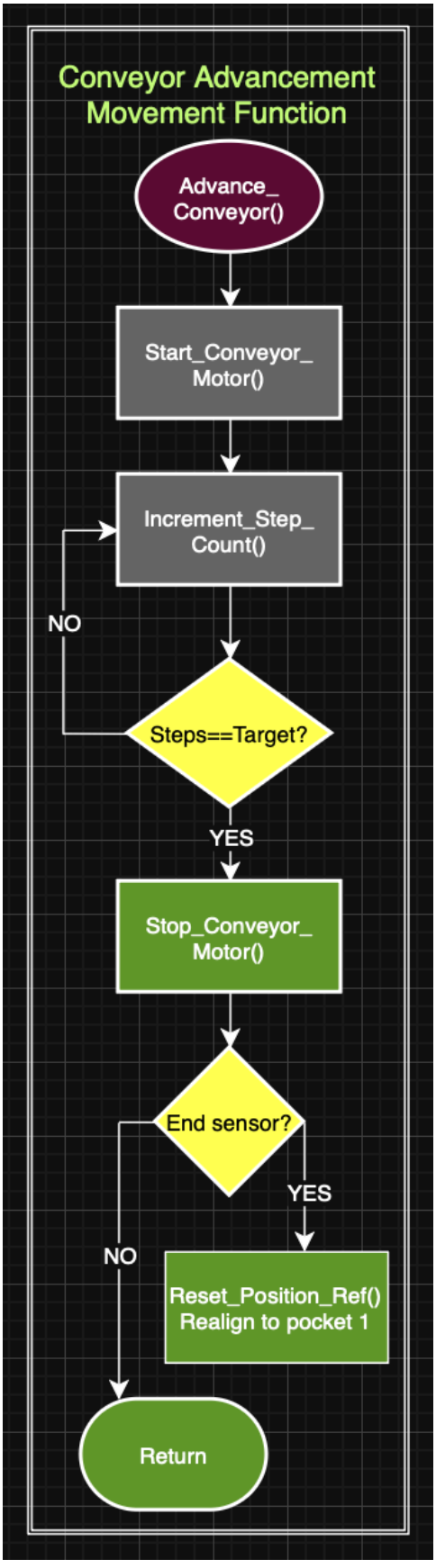
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13.1.3. Oil Spraying Function Flowchart



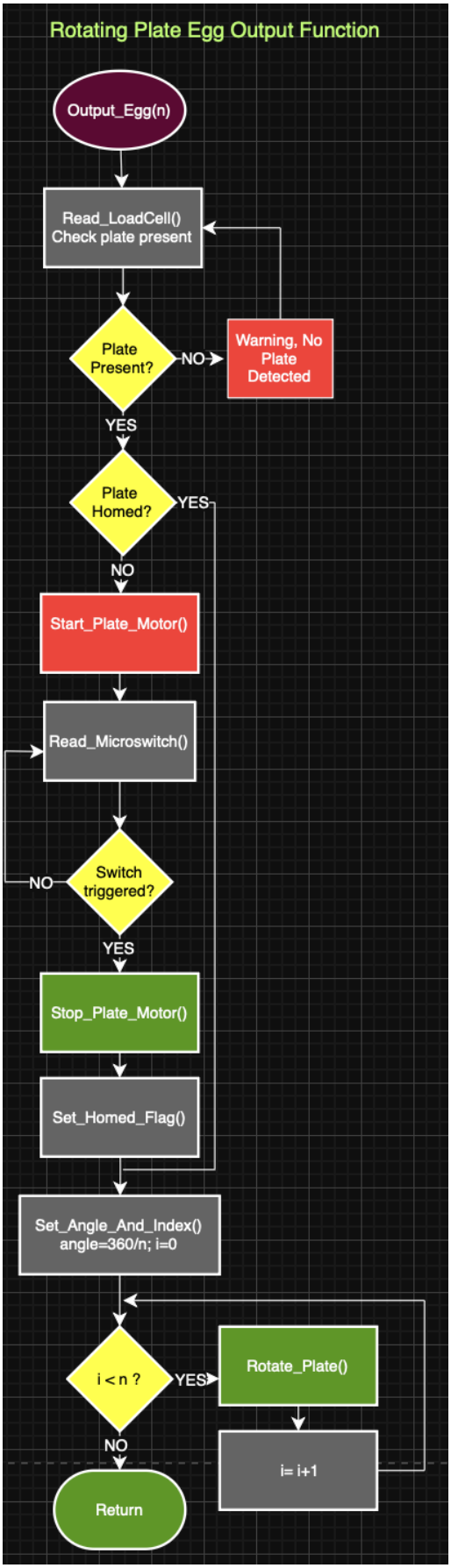
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13.1.4. Conveyor Advancement Movement Function Flowchart



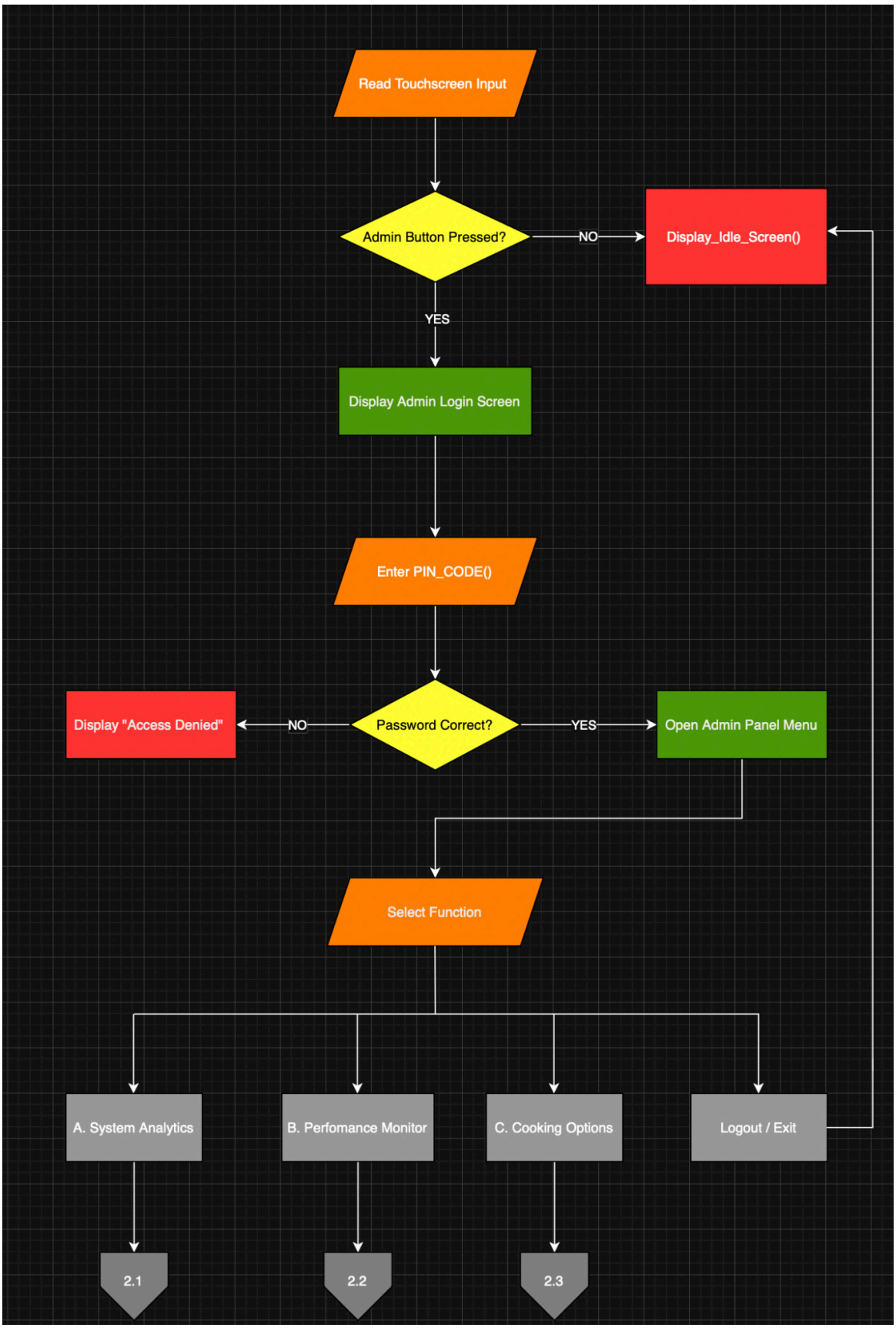
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13.1.5. Rotating Plate Output Function Flowchart



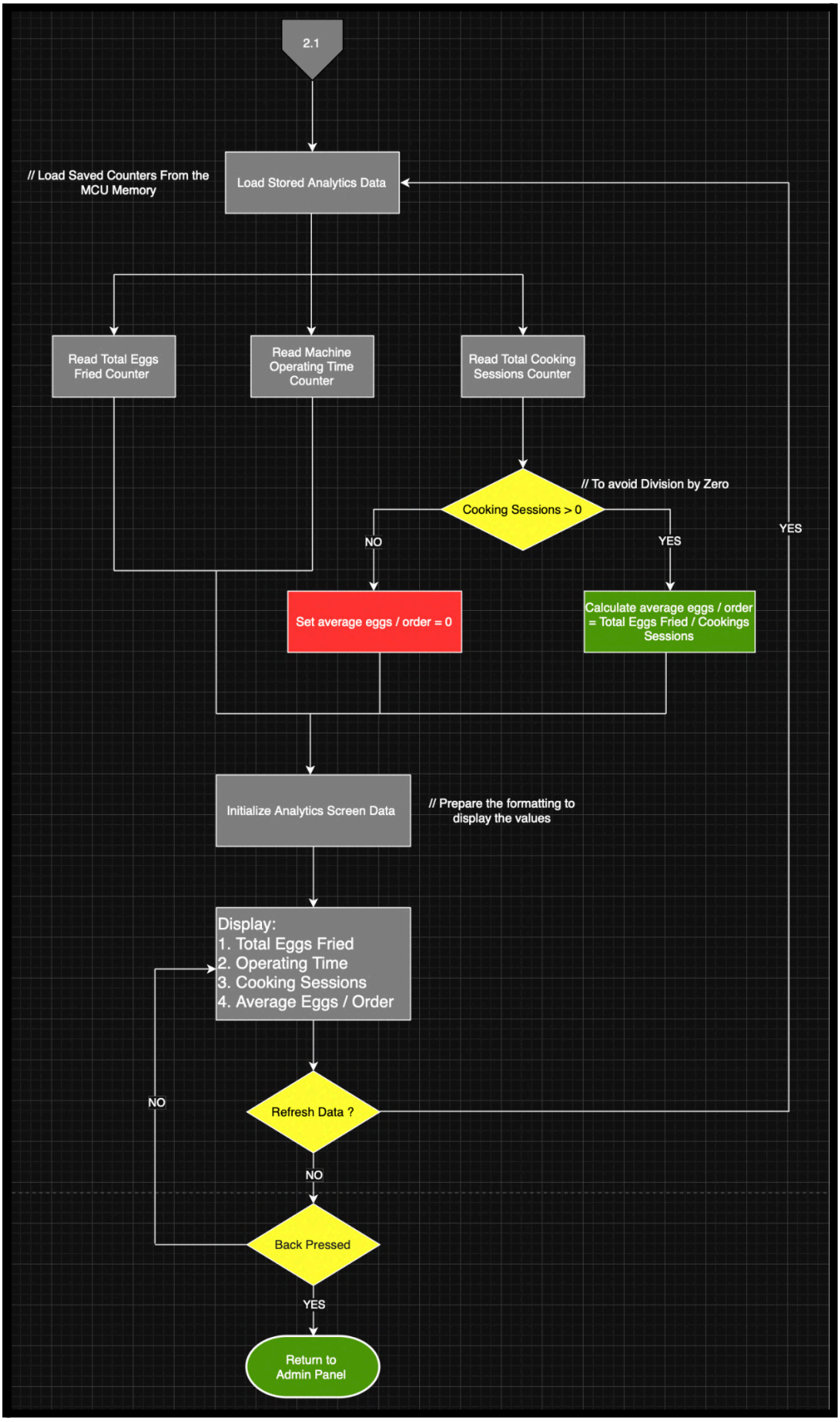
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13.2. Admin Panel Flowcharts



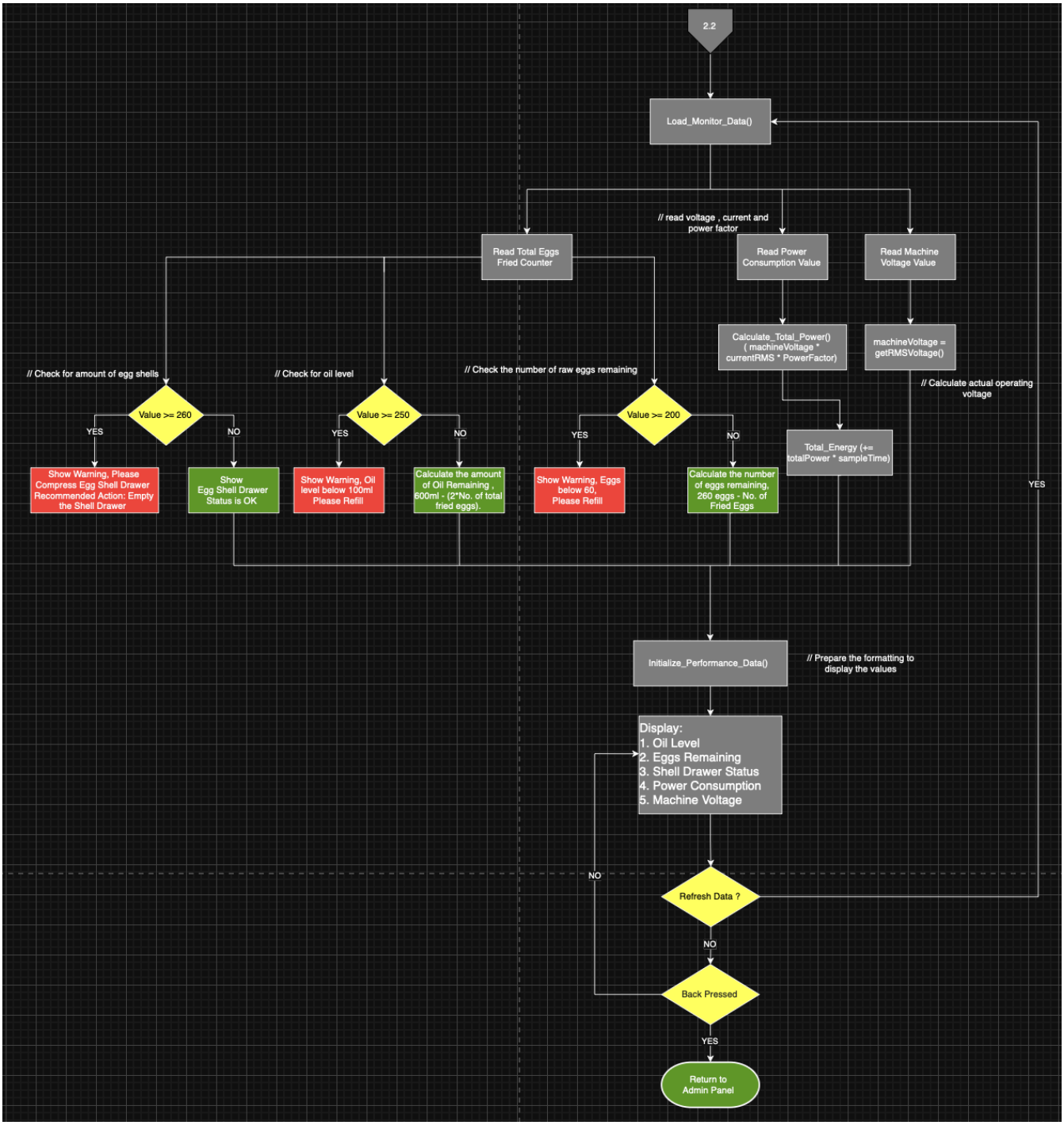
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13.2.1. Machine Analytical Data Flowchart



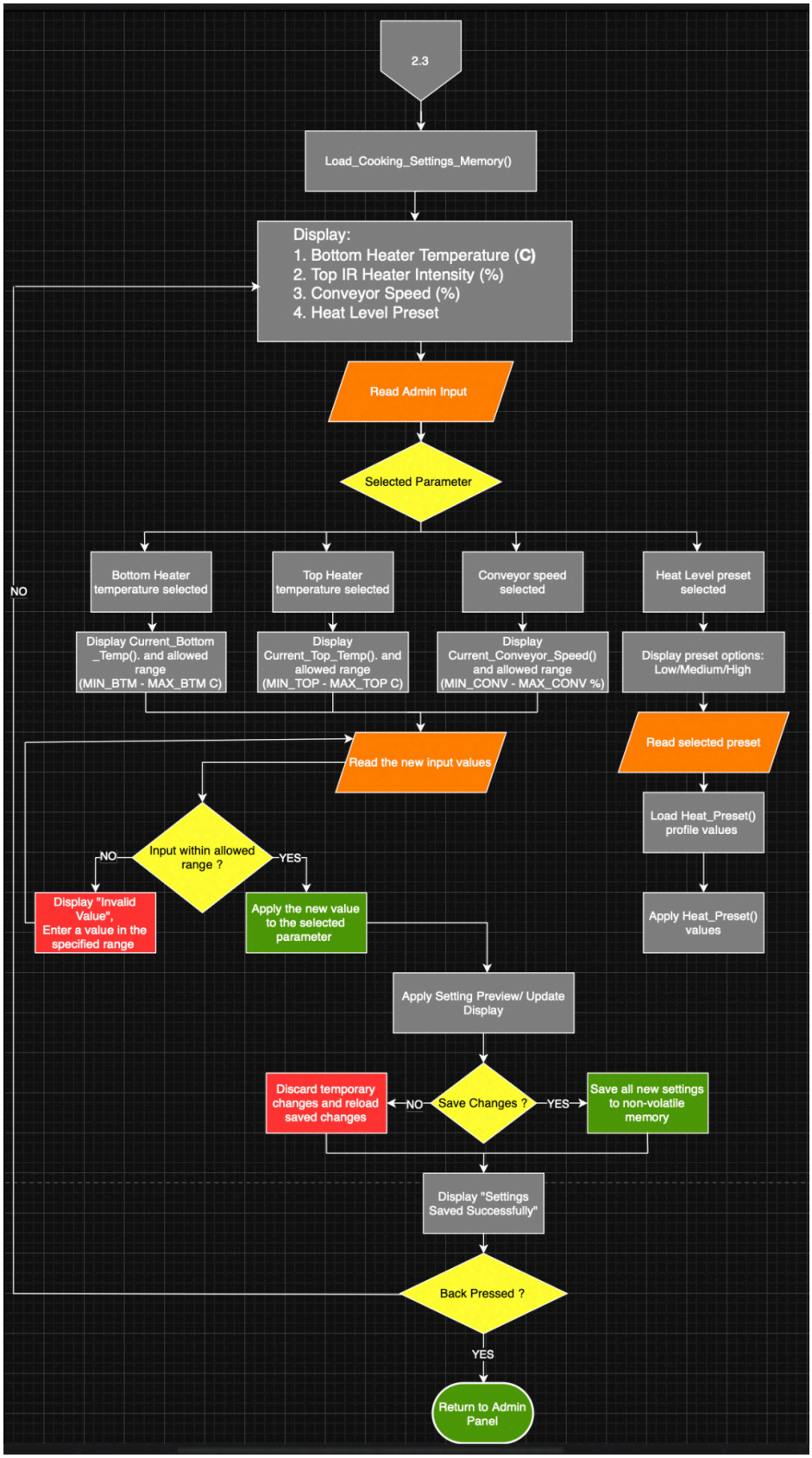
..

13.2.2. Machine Performance Data Flowchart



..

13.2.3. Machine Cooking Settings Flowchart



..

14. Bill of Materials (BOM)

Part NO.	Part Name (Make)	Material specification	Qty.	Technical Specification	Position in Technical assembly
1	Egg Storage Container	PP-C HI / Polypropylene Copolymer	1	Yield: ~22 MPa | Density: 0.90 g/cm³ | Temp: -20 to 100°C	1
2	Egg Tunnel	PP-C HI / Polypropylene Copolymer	1	Yield: ~22 MPa | Density: 0.90 g/cm³ | Temp: -20 to 100°C	2
3	Gates for Transportation Tunnel	Acetal Copolymer (POM-C)	2	Yield: ~62 MPa | Density: 1.41 g/cm³ | Temp: -40 to 100°C	4
4	Gate Shaft Holder for Tunnel	POM-C / Acetal Copolymer	2	Yield: ~62 MPa | Density: 1.41 g/cm³ | Temp: -40 to 100°C	5
5	Egg Tunnel Shaft	EN 1.4301 / AISI 304	2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	6
6	Egg Holding Shaft	EN AW-6082 T6	1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	7
7	Egg Holding Arm	EN AW-6082 T6	1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	8
8	Base of Spear	EN AW-6082 T6 (clear anodised)	1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	9

..

9	Spears (Left and Right)	EN 1.4116 / X50CrMoV15 (hardened)	2	Yield: ~800–1000 MPa | Density: 7.70 g/cm³ | Temp: -20 to 200°C	10
10	Solenoid Motor Mounting Bracket	EN 1.4301 / AISI 304	2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	11
11	Solenoid Motor Casing	EN AW-6082 T6 (clear anodised)	3	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	12
12	Main Drawer Box	EN 1.4301 / X5CrNi18-10	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	13
13	Drawer Front Cover	ABS-HI plastic	1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	14
14	Drawer Handle	ABS-HI plastic w/ brass inserts	1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	15
15	Inner Shaft (Drawer)	EN 1.4301 / X5CrNi18-10	2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	16
16	Manual Compression Plate	EN 1.4301 / X5CrNi18-10	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	17
17	Oil Storage Mount	EN AW-6082 T6	2	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	18
18	Pocket Plate	EN 1.4409 / GX2CrNiMo19-11-2 (cast)	4	Yield: ~185–205 MPa | Density: 7.90 g/cm³ | Temp: -196 to 400°C	19

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19	Sprocket for Conveyor Belt	EN 1.4404 / X2CrNiMo17-12-2	4	Yield: ~205 MPa | Density: 8.00 g/cm³ | Temp: -196 to 400°C	20
20	Driving Shaft / Idler Shaft	EN 1.4404 / X2CrNiMo17-12-2	2	Yield: ~205 MPa | Density: 8.00 g/cm³ | Temp: -196 to 400°C	21
21	Conveyor Side Frame	EN 1.4301 / X5CrNi18-10	2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	22
22	Conveyor Back Frame	EN 1.4301 / X5CrNi18-10	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	23
23	Conveyor Belt Hollow Support Frame	EN AW-6082 T6	1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	24
24	Nema 17/Gearbox/Encoder Motor Housing	EN AC-43500 / AlSi10MnMg (cast)	1	Yield: ~140–180 MPa | Density: 2.68 g/cm³ | Temp: -40 to 150°C	25
25	Nema Motor Mount	EN 1.4301 / AISI 304	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	26
26	TOP IR Shielding	EN 1.4301 / X5CrNi18-10	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C (shield, near IR zone)	27
27	Ceramic Block for Holding Quartz IR	96% Al_2O_3 ceramic	2	Flexural: ~300 MPa | Density: 3.70 g/cm³ | Temp: up to 1600°C	28
28	Side Lining for TOP IR	EN 1.4301 / X5CrNi18-10	2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	29

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29	Perforated Sheet	EN 1.4301 / X5CrNi18-10	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	30
30	TOP IR Back Frame	EN 1.4301 / X5CrNi18-10	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	31
31	Heating Block	EN AW-6082 T651	1	Yield: ~250 MPa | Density: 2.70 g/cm³ | Temp: -40 to 200°C	32
32	Side Frame (Bottom Heating)	EN 1.4301 / X5CrNi18-10	2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	33
33	Front Frame (Bottom Heating)	EN 1.4301 / X5CrNi18-10	2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	34
34	Bottom Frame (Bottom Heating)	EN 1.4301 / X5CrNi18-10	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	35
35	Mounting Brackets for Bottom Heating Block	EN 1.4301 / AISI 304	4	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	36
36	Aluminum Plate (Rotating Plate)	EN AW-6061 T6 (clear anodised)	1	Yield: ~275 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	37

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37	Top Load Cell Plate	EN AW-6061 T6 (clear anodised)	1	Yield: ~275 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	38
38	Bottom Load Cell Plate	EN AW-6082 T6 (clear anodised)	1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	39
39	Base Plate (Rotating Plate)	EN AW-6082 T6 (clear anodised)	1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	40
40	Stepper Motor Housing	EN AC-43500 / AlSi10MnMg (cast, clear anodised)	1	Yield: ~140–180 MPa | Density: 2.68 g/cm³ | Temp: -40 to 150°C	41
41	Bottom Base for Rotating Plate	EN 1.4301 / AISI 304	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	42
42	Electrical Box Housing	EN 1.4301 / AISI 304	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	43
43	Main Frame Housing	EN 1.4301 / AISI 304	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	44

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44	L Bracket for Servo Motor	EN 1.4301 / AISI 304	4	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	45
45	Plastic Back Plate (Display)	ABS-HI plastic w/ brass inserts	1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	46
46	Front Frame for Casing (Display)	ABS-HI plastic	1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	47
47	Main Case/Chassy Panels	ABS-HI plastic + purchased glass	1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	48

Part NO.	Part Name (buy)	Material specification	Link/Brand	Technical Specification	Position in Technical assembly
48	DIN Rail	zinc-plated steel	Hutschiene TS 35 x 7,5mm gelocht - L=326mm	35 x 7,5 x 326 mm	49
49	Cable Duct / Wire Raceway	PVC	Cable duct	25x25x1000mm	NA
50	Cable Glands / Strain Reliefs	Polyamide, V-2.	Kabelverschraubung, M 12 x 1,5, Ø 3,5 - 7 mm, schwarz, IP69K Kabelverschraubung, M 16 x 1,5, Ø 4,5 - 10 mm, schwarz, IP69K M 20 x 1,5, Ø 7 - 13 mm, schwarz, IP69K	M12x1.5 IP69k M16x1.5 IP69k M20x1.5 IP69K	NA
51	Cable Ties + Mounts	6/6 nylon	Cable mounts Cable ties	18.8mm x 18.8mm x 5.6mm	NA

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52	Axle Shaft	Aluminum	Aluminum shaft	Ø 10 mm x100 mm	50
53	Return Spring	Stainless steel - AISI 302	Spring	L0 - 20.57 Mm De - 7.62	NA
54	Thermal Insulation	LDPE	Thermal insulation	-70°C to105°C 500x330x20mm	NA
55	Inner Lining	Aluminum	Aluminum sheet	0.1x100x100mm	NA
56	Inline Check Valve	stainless steel	Duotight - 8mm Inline Check Valve	Ø8mm	NA
57	Silicone Tubing	Silicon	Silicon tubing	4mm x 1m	NA
58	Flat Fan Spray Nozzle	stainless steel	CDIYTOOL 8 Pieces 65 Degree High Pressure Spray Nozzle	0.4-5L/min 14mm hex width, 25mm long, 12.5 thread width	NA
59	Nozzle Mounting Bracket				NA
60	Roller Chain with Bent Attachments	Steel	IWIS - Rollenkette EuroChain	9,525 mm 5,72 mm 1505mm 06B-2	NA
61	2-Bolt Flange Bearing Block	Aluminum	Stehlagereinheiten / Flanschlagereinheiten 2- Loch KFL000	Ø 10mm	51
62	Flexible Clamp Motor Coupling	Aluminum	10 mm to 12 mm Flexible Shaft Coupling, Diameter 40 mm Length 66 mm Aluminium Motor Connector Flexible	bore diameter 10 mm to 12 mm total diameter 40 mm length 66mm	NA

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63	Planetary Gearbox for Motor	--	Planetengetriebe Schrittmotor 27:1	Planetarygear 27:1	52
64	Front Sealing Strip / Gap Cover	Silicon	EKI 1515 Silikonschaum	2mmx6mmx10m -60 °C bis +220 °C	NA
65	Thermal Compound	--	Wärmeleitpaste, Tube, 35gr, ungiftig	35g	NA
48	Stainless Fasteners Kit	A2 stainless steel	M2 M3 M4 M5 Schrauben Set – 1280 Stück	M2 M3 M4 M5	NA
Miscellanous					
67	Countersunk Raised Head Cross Recess Screw	A2 Stainless Steel	Senkkopfschrauben M10x25 Senkkopfschrauben M10x60	M10x25 M10x60	53
68	Pan Head Cross	A2 Stainless steel	Linsenschrauben M8x16 STM390400300S STM320300200E	DIN 7985 (ISO 7045) M8x16 M4x30 M3x20	54
69	Recess Screw	A2 stainless steel	Zylinderschraube Innensechskant DIN912 (ISO 4762) V2A - M6x20, 100 St. Zylinderschraube Innensechskant DIN912 (ISO 4762) V2A - M3x12, 100 St.	M6x20 M3x12	55
70	Socket Head Cap Screw	A2 stainless steel	DIN 912/ISO 4762 A2 – M3 x 25 mm 100 Stück DIN 912/ISO 4762 A2 – M3 x 10 mm 100 Stück	M3x25 M3 x 10 mm	56

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71	Pan Head Cross Recess Screw	Stainless steel	Flachkopfschraube ISO 7045 PZ M4x8-Z 4.8 galv.verz. 2000St.	ISO 7045 M4x8 M4 x 8mm	NA
72	Prevailing Torque Hex Nut	A2 Stainless steel	Sechskantmuttern M6 DIN 934 Edelstahl A2	M6	57
73	Plain Washer	A2 Stainless steel	Unterlegscheibe DIN125 (ISO 7089) V2A - 6,4 mm (M6), 1000 Stück	M6	58
74	Socket Set Screw, Cup Point	A2 Stainless Steel	Gewindestift DIN 916 / ISO 4029, M 5x20, Stahl 45 H	M5x20	59
75	Hex Thin Nut	A2 Stainless Steel	DIN 439/ISO 4035 Sechskantmutter Edelstahl A2 – M3 - 100 Stück DIN 439/ISO 4035 Sechskantmutter Edelstahl A2 – M4 - 100 Stück	M3 M4	60
76	Lazy Susan Bearing	Stainless Steel	Blickle 942239 DK-L-3 Stahlblech-Drehkranz 1 St.	140 x 110 mm	61
77	Rubber Drive Wheel	Polyoxymethylene	Freies Verschiffen 10PCS 8*32*16 flache typ PA nylon	8x32x16mm	62

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Electrical Parts

Part No.	Part name	Brand/link	Technical specifications	Location in technical assembly
78	IEC C20 Power Inlet	IEC 20 Power Inlet	16 A 250 VAC	63
79	MCB-Circuit Breaker	16A MCB- TYPE B Circuit Breaker	16A TYPE B	NA
80	RCCB / RCD-TYPE A	RCCB / RCD-TYPE A (30mA)	30mA TYPE A	NA
81	DIN Terminal Blocks	DIN Terminal Blocks (L/N/PE)	(L/N/PE)	NA
82	Protective Earth Terminal Block	Protective Earth Terminal Block	-	NA
83	AC Wire	LAPP 70261141 Strom Anschlusskabel Schwarz 3.00 m	3.00 m	NA
84	24V Industrial SMPS	24V Industrial SMPS 60W (2.5A at 24V)	60W (2.5A at 24V)	NA
85	24V → 12V LM2596T-12	LM2596T-12/LF03	4.5V to 40V 3A Low	NA
86	12V → 5V LM2596T-5	LM2596T-5.0/LF03	4.5V to 40V 3A Low	NA
87	24V → 6V LM2596S-ADJ (Adjusted to 6V)	LM2596S-ADJ/NOPB	24V DC to 6V DC, 2A continuous	NA
89	5V → 3.3V Regulator AMS1117-3.3	LM1117MPX-3.3/NOPB	5 mA 3.3 V -40 to 125 C	NA
90	DC Terminal Blocks	9-Way DC Terminal Block 5V 12V 24V for CCTV Power Supply	5V 12V 24V DC 9-way	NA
91	Inline Blade Fuse Holders	URAQT Fuse Holder Car Standard 16AWG	16AWG/14AWG/12AWG 30/20/10A	NA

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92	Blade Fuse Assortment	0287010.PXCN Littelfuse Inc. | Fuses | DigiKey	10A 32V	NA
93	Shielded Signal Cable	2M UL2547 28AWG 4 Cores PVC	81–600 V 4-core	NA
94	Ferrules and Crimp Terminals	1-966066-1 : Crimp Wire Pins, Tabs & Ferrules | TE Connectivity	15 AWG	NA
95	PCB Screw Terminal Connectors	20Paar 5.08mm Pitch Male Buchse PCB Screw Terminal Block | eBay.de	25 x 10 x 15mm AC 300V 15A	64
96	Axial fan 8414NGML	EBM Papst 8414NGML Axiallüfter 24 V/DC 42 m³/h (L x B x H) 80 x 80 x 25mm | » Freue dich auf voelkner	80x80x25mm 42 m^3/h 2pcs 24V DC	65
97	Main Microcontroller STM32H743ZITx	STM32H743ZIT6 STMICROELECTRONICS, ARM-MCU, STM32 Family	STM32 3.6V 550MHz	NA
98	Gate Actuators JF-0530B x3	JF-0530B DC12V 300mA 5N/10mm Pull-Push-Type Solenoid Electromagnet	DC 12V 300mA 5N/10mm 30x16x15mm	66
99	Flow Sensors IR sensor	IR Break Beam Sensors- 3mm LEDs : Adafruit Industries	3.3 - 5.5VDC 10mA @ 3.3V, 20mA @ 5V Response Time: <2 ms	NA
100	Flip Actuator MG996R	MG996R Digital Servo Motor mit Metall Getriebe	0.17 sec/60 deg at 4.8 V 4.8 to 7.2 V 40 x 19 x 43 mm	67
101	Platform Solenoid JF-0837B	JF-0837B DC 24V 1A 15N 10mm Push Pull Type Open Frame Linear Motion Solenoid Electromagnet | Harfington	DC 24V 1A 15N 37 x 20 x 26mm	68

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102	Solenoid Driver multi-channel MOSFET	Industrial-Grade MOSFET Driver Module – 10-100A Adjustable PWM Controller (12-60V DC)	12–60 V DC 10–100 A 12 x 13 x 1 cm	NA
103	Peristaltic Pump	DC 12V Dosing Pump Peristaltic Pump	DC12V 11ml/min Three roller	69
104	MOSFET Driver Module	Mosfet Mosfet 12V DC 8×6×2 Low Voltage Mosfet Module	12V DC 1-5MA	NA
105	Touchscreen Display Module — Nextion NX8048P070_011R	NEXTION 4.3/5.0/7.0" Intelligent LCD Touch Display	108x181x9.3mm	70
106	JST-XH 4-pin Connector	JST JST - Buchsengehäuse, 1x4-polig - XH	250V / 3A -25 to 80C	NA
107	Stepper Motor	NEMA 17	2.8 V 5 mm 2 phase	71
108	Stepper Motor Driver	Stepper Motor Driver	-40 to +85°C 12/24V 18 x 25 x 3 cm	NA
109	Control Wiring and Connectors	Control Wiring and Connectors	26AWG L-100 mm	NA
110	Hall Effect Sensor	A1104-class Hall effect switch sensor, 3-pin type	3.8 V to 24 V 25 mA -40 to 150 °C	NA
111	SmCo Magnet	SmCo	8 x 3 mm	NA
110	Incremental Rotary Encoder	AMT102-V	3,6V to 5,5V	72

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111	Cartridge Heaters	Cartridge Heaters	250W 230V (AC)	73
112	Temperature Sensor	PT100 temp. Sensor	-	74
113	Temperature Sensor Amplifier	MAX31865AAP+ Analog Devices DigiKey	3V to 3,6V -40°C to125°C	NA
114	Solid-state Relay	solid state relay 25A SSR-25AA , ZG3NC-325A Single phase	25A 90–480VAC	75
115	Heater Branch Fuse	0652C8000-11 Bel Fuse Inc. | Sicherungen | DigiKey	8 A 250 V	NA
116	High-temperature Wiring	Silikonkabel 18 AWG 0,5 mm² 200°C 300V flexibel verzinnt Meterware | akkuteile.de	18 AWG 200°C 600V	NA
117	Grounding Hardware	100m Erdkabel H07V-K 1x4mm² Grün/Gelb bei hd-digital kaufen,	AC Uo/U 450/750 V > 20 MΩ × km -30 °C to +80 °C	NA
118	IR Heater Element	IR HEATER	410 mm 250W 110V to 450V	76
119	IR Power Control / Relay	SSR Halbleiter Solid State Relais DC / AC, 230V AC 25 A	DC to AC 230V AC 25A	NA
120	Thermal Cutoff / Safety Sensor	THERMODISC Temperatursicherung, 16A, 184°C, kurz/lang | Temperatursicherungen günstig kaufen | reichelt elektronik	16A Max 184°C	NA
121	PT100 Sensor	PT100 temp. Sensor	-	77
122	SSR Heat Sink	Aluminium Heat Sink SSR	Aluminium	78

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		Dissipation For One Phase Solid State Relay 10a-40a For PID Thermostat Temperature Controller	Fit for: 10a-40a	
123	Ceramic Terminal Block	Elektrofachmarkt-online - Keramikklemme 2-polig mit Befestigungsloch	Ceramic clamp 2-pole max. 300°C , 450V, 32A	NA
124	Low-voltage Terminal Blocks	Phoenix Contact 1725672 MPT 0,5/ 4-2,54 Leiterpla	160 V Max 105 °C 4 cores	NA
125	Stepper Motor	FREI Standard-Schrittmotor, 5V	5V DC 4 Phase 34.3mN.m	NA
126	Stepper Driver Module	FREI Entwicklerboards - Motorsteuerung, ULN2003A	DC5V-12V 4-phase 5 line	NA
127	Microswitch	5Pcs Mini Micro Limit Switch NO NC 3 Pins PCB Terminals SPDT 5A 125V 250V 29mm Roller Arc lever Snap Action Push Microswitches	5A 125V/250V	NA
128	Load Cell and HX711 Amplifier Module	Stemedu 5 kg Load Cell Digital Weight Sensor for Portable Electronic with HX711 Ad Amplifier Module, 11 lbs Weighing Sensor Arduino Raspberry Pi	Four lines 5-10 V Dimensions: 12.7 mm, 12.7 mm, 80 mm. voltage range: 2.6-5.5 V	NA
129				
130	Wiring and Connectors	DKS-20MM-30 DigiKey Standard | Jumper Wire | DigiKey PVC Datenleitung LAPP UNITRONIC LiYY 3x0,25 - 0028303 | Automation24	-	NA
131	Emergency stop button	1NC 22mm Red Mushroom Emergency Stop Push Button Switch 600V 10A	22mm 600V 10A	NA

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15. Production Planning

15.1 Capacity Calculation

15.1.1 Production Target

From the first milestone, our projected sales for the first three years are :

First year (2027) : 500

Second Year (2028) : 900

Third Year (2029) : 1440

Fourth Year (2030) : 2016

Fifth year (2031) : 2419

Total number of working days in a year : 365 – weekends – public holidays

Total number of weekend days in Germany – 104 weekend days

Number of public holidays in Germany – 9 (only 6 falls on weekdays)

Total number of working days in a year : 255 working days.

Since the machine is sold on more of an MTO (make to order) basis, we will make as many machines that are being sold in that given year. Any surplus machines would be sent to companies for live demos and showcasing, or for other purposes.

15.1.2 Customer Takt time

Takt time – the maximum amount of time a company has to produce a product to meet customer demand.

Shift Specifics :

Number of shifts per day = 2

Duration per shift = 8 hours

Effective time per shift = 7.5 hours

Weekly Production time

Total time per week = 2 shifts/day x 7.5 hours/shift x 5 days/week = 75 hours

For planning, it’s practical to round to whole-machine daily targets:

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Year 1: 2 machines per day
Year 2: 3 machines per day.
Year 3: 6 machines per day.
Year 4: 8 machines per day.
Year 5: 9 machines per day.

Year	Target units/year	Takt time (days per machine)	Takt time (hours per machine)
1	500	255/500 = 0.51	12.2
2	900	255/900 = 0.28	6.8
3	1440	255/1440 = 0.18	4.3
4	2016	255/2016 = 0.13	3.0
5	2419	255/2419 = 0.11	2.5

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15.2. Machine Selection

Machine used	Specifications	Price per unit
CNC machining centre Haas UMC-750 (5-axis) 	• Travels (X/Y/Z) : 762 × 508 × 508 mm • Trunnion table : 630 × 500 mm, 300 kg capacity • Spindle speed : 8,100 rpm • Spindle torque/power : 122 Nm at 2,000 rpm, 22.4 kW • Tool changer : 40+1 side-mount (CT/BT40)	€70,000–85,000
CNC turning centre Haas ST-20Y 	• Y-axis travel : ±2" (50.8 mm) • Spindle speed : 35–4,000 rpm • Bar capacity : 63.5 mm (2.5") • Chuck size : 210 mm (8.3") • Turret : 12-station BMT-65, live tooling with C-axis	€70,500
CNC fibre laser Trumpf TruLaser 1030 fiber 	• Work area : 3,000 × 1,500 mm • Laser power : 2,000 W (TruDisk) • Max cutting speed : 40 m/min • Positioning accuracy : ±0.03 mm avg • Max sheet thickness : 6 mm • Power connection : 400 V, 26 kVA	2014–2015 (2–3 kW): €40,000–50,000

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Press brake — Amada HFE 	• Press beam length : 4,280 mm • Stroke : 200 mm (or 350 mm option) • Bending speed : 10–20 mm/s • Open height : 470–620 mm	€16,000 – €30,000
TIG welder — Miller Dynasty 210 	• Process : AC/DC TIG (GTAW), Pulsed TIG • Amperage range : 1–210 A • Duty cycle : 210 A @ 18.4 V, 60% • Max material thickness : 1/4 in (6.35 mm) steel/aluminium • Input powerAuto-Line 120–480 V, 1 or 3-phase	€3,000 – €10,000
Injection moulding machine Arburg Allrounder 270 A 	• Clamping force : 350–600 kN (25–60 ton class) • Shot weight : Up to ~60–90 g (screw diameter dependent) • Platen size : Compact (~296×296 mm typical) • Cycle time : Typically 10–30 sec for small parts	€10,000 – €20,000

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Ultrasonic welder — Branson (Emerson) GSX-E1 	• Frequency : 20 kHz or 30/40 kHz • Weld control : Digital closed-loop amplitude/force control • Cycle time : Typically 0.5–2 sec weld time • Actuator : Electric or pneumatic press	€35,000 – €55,000
CNC router — MultiCam 3000 Series 	• Working area : 2,032 × 3,099 mm (or 50"×100" variant) • Z-axis travel : 304 mm (or 6" clearance variant) • Spindle speed : Up to 32,000 rpm • Max cutting speed : 35.5 m/min • Tool changer : Linear, 6 positions	€20,000–35,000

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15.2.2. Production time per piece

For each manufacturing process for each part, the total time is estimated in 3 steps based on part characteristics (number of holes, size, complexity, etc.)

For CNC Milling, Turning, Thread Turning and Milling: The loading, processing and unloading time are estimated (machine-based scheduling).

For Injection Molding: The heating, injection and cooling (and ejection) times are estimated.

Part	**Process**	**Step 1 time (min)**	**Step 2 time (min)**	**Step 3 time (min)**	**Total time per part process (min)**
Egg Storage					
Egg storage container	Injection molding	0.13	0.37	0.08	0.58
	CNC drilling	0.50	1.00	0.50	2.00
Egg transportation					
Egg tunnel	Injection molding (upper half)	0.15	0.58	0.11	0.84
	Injection molding (lower half)	0.15	0.58	0.11	0.84
	Ultrasonic Welding	0.33	0.05	0.17	0.55
Gate for tunnel	Injection molding	0.10	0.29	0.06	0.45
Egg Breaking Mechanism					
Egg holding Shaft	CNC turning & facing	0.75	2.50	0.50	3.75
	Pocket milling/tapping	0.75	3.00	0.75	4.50

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Egg holding arm	Billet sawing & setup	1.00	0.50	0.25	1.75
	5- axis milling	0.25	6.50	0.25	7.00
	Hub milling, drilling & reaming	0.25	2.00	0.25	2.50
Base of Spear	Billet Sawing & setup	0.75	0.75	0.25	1.75
	Top side milling	0.25	4.50	0.25	5.00
	Side Machining	0.50	2.50	0.50	3.50
Spear - left	Billet Sawing & setup	0.75	0.75	0.25	1.75
	Mounting block milling	0.25	4.00	0.25	4.50
	Shaft hole & counterbore	0.50	3.00	0.50	4.00
	Taper & underside milling	0.50	3.50	0.50	4.50
Spear - right	(same as left)				15.25
Main drawer box	Laser cutting	0.50	0.80	0.30	1.60
	Bending & hemming	0.40	1.20	0.40	2.00
	TIG Welding & finishing	0.50	2.50	0.70	3.70
Drawer Front Cover	Injection molding	0.12	0.42	0.08	0.62
Drawer Handle	Injection molding	0.12	0.42	0.08	0.62
	Insert Pressing	0.20	0.40	0.20	0.80
Inner Shaft	CNC turning	0.50	2.00	0.50	3.00
Oil System					

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Oil System housing	Block sawing & setup	0.75	0.75	0.25	1.75
	Milling	0.50	8.00	0.50	9.00
	Drilling & countersinking	0.50	2.00	0.50	3.00
Conveyor System					
Pocket Plate	Gate removal & blasting	0.75	9.00	0.75	10.50
	5 axis CNC milling	0.75	9.00	0.75	10.50
	Drilling & Polishing	0.50	2.00	1.00	3.50
Sprocket	Turning & boring	0.50	3.50	0.50	4.50
	Hobbing & tooth deburring	0.50	4.00	1.00	5.50
	Locking hole Drilling & tapping	0.50	2.00	0.50	3.00
Driving shaft	CNC turning	0.50	4.50	0.50	5.50
	Centreless Grinding	0.50	3.00	0.50	4.00
Idler Shaft	CNC turning	0.50	3.00	0.50	4.00
	Centreless grinding	0.50	2.50	0.50	3.50
Side Frame	CNC milling	0.75	7.00	0.75	8.50
	Drilling & Boring	0.50	3.00	0.50	4.00
	Laser Cutting & finishing	0.50	1.20	0.40	2.10
	Hole finishing	0.40	1.50	0.40	2.30
Back frame	Laser Cutting	0.50	1.20	0.40	2.10
	Hole finishing	0.40	1.50	0.40	2.30

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Hollow Support Frame	Rough/finish milling	0.75	7.50	0.75	9.00
	Drilling & threading	0.50	2.00	0.50	3.00
Nema Gearbox Housing	CNC face milling & boring	0.75	6.00	0.75	7.50
	Drilling & thread forming	0.50	3.00	0.50	4.00
Top IR System					
Top IR Shielding	Laser Cutting	0.50	1.20	0.40	2.10
	Press Brake Bending	0,50	2.00	0.50	3.00
Ceramic Block	Pressing & green machining	0.75	3.50	0.75	5.00
	Diamond Grinding & hole finishing	0.50	4.00	0.50	2.00
Side Lining	Laser Cutting & finishing	0.40	0.60	0.30	1.30
	TIG Welding into shield	0.40	1.30	0.40	2.10
Perforated Sheet	Punching & laser cutting	0.75	2.50	0.75	4.00
Top IR back frame	Laser Cutting	0.50	0.80	0.40	1.70
	TIG welding to shield	0.40	1.50	0.50	2.40
Bottom Heating System					
Heating Block	Milling	0.50	5.00	0.50	6.00
	Deep hole drilling	0.50	3.00	0.50	4.00
Side frame strip	Laser cutting and finishing	0.40	0.60	0.30	1.30

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Front frame strip	Laser cutting and finishing	0.40	0.60	0.30	1.30
Bottom frame strip	Laser cut & finish	0.40	0.60	0.30	1.30
	TIG assembly	0.60	2.00	0.80	3.40
Rotating Plate					
Aluminum rotating plate	Turning	0.50	3.00	0.50	4.00
	Milling underside features	0.50	4.50	0.50	5.50
	Drilling & Countersinking	0.40	1.50	0.40	2.30
Top load cell plate	Milling & hole machining	0.50	4.00	0.50	5.00
Bottom load cell plate	Milling & hole machining	0.50	4.00	0.50	5.00
Base plate	Milling	0.75	6.00	0.75	7.50
	Drilling & threading	0.50	2.50	0.50	3.50
Stepper housing	CNC face & bore	0.75	6.00	0.75	7.50
	Drilling & thread forming	0.50	3.00	0.50	4.00
	Deburring & coating handling	0.50	0.75	0.50	1.75
Bottom base	Laser cutting & bending	0.60	1.20	0.40	2.20
	Welding & threading	0.50	2.00	0.50	3.00
Electrical box	Laser cutting	0.60	0.80	0.40	1.80
	Bending & assembly	0.60	1.40	0.50	2.50
	TIG welding & finishing	0.50	2.00	0.75	3.25

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Main Housing					
L bracket for servo motor	Milling & forming	0.50	1.80	0.40	2.70
	Drilling & slotting	0.40	1.20	0.40	2.00
Main frame housing	Laser cutting	1.00	1.50	0.50	3.00
	Bending flanges & stiffeners	0.75	3.00	0.75	4.50
	Welding & final finishing	1.00	4.00	1.50	6.50
Pump holding blocks	Sawing	0.40	0.80	0.30	1.50
	Milling	0.50	2.00	0.50	3.00
	Drilling & tapping	0.40	1.20	0.40	2.00
Side panel with glass	CNC routing	0.50	2.50	0.50	3.50
	Drilling	0.30	0.80	0.30	1.40
	Edge deburring & finishing	0.40	0.80	0.30	1.50
Main frame outer cover	CNC routing	0.50	2.00	0.50	3.00
	Drilling	0.30	0.80	0.30	1.40
	Edge finishing	0.40	0.60	0.30	1.30

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15.2.3. Lot Size

Part	Quantity per unit	Lot size	Total time per part process (min)	Total time per unit	Total time per day (min)
Egg Storage					
Egg storage container	1	1	0.58 2.00	2.58	5.16
Egg transportation					
Egg tunnel	1	1	0.84 0.84 0.55	2.23	4.46
Gate for tunnel	1	1	0.45	0.45	0.9
Egg Breaking Mechanism					
Egg holding Shaft	1	1	3.75 4.50	8.25	16.5
Egg holding arm	1	1	1.75 7.00 2.50	11.25	22.5
Base of Spear	1	1	1.75 5.00 3.50	10.25	20.5
Spears	2	2	1.75	29.50	59

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			4.50		
			4.00		
			4.50		
Main drawer box	1	1	1.60	7.30	14.6
			2.00		
			3.70		
Drawer Front Cover	1	1	0.62	0.62	1.24
Drawer Handle	1	1	0.62	1.42	2.84
			0.80		
Inner Shaft	1	1	3.00	3.00	6
Oil System					
Oil System housing	1	1	1.75	13.75	27.5
			9.00		
			3.00		
Conveyor System					
Pocket Plate	4	4	10.50	98.00	196
			10.50		
			3.50		
Sprocket	2	2	4.50	26.00	52
			5.50		
			3.00		
Driving/idler shaft	2	2	5.50	19.00	38
			4.00		
Side Frame	1	1	8.50	16.90	33.8

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			4.00 2.10 2.30		
Back frame	1	1	2.10 2.30	4.40	8.8
Hollow Support Frame	1	1	9.00 3.00	12.00	24
Nema Gearbox Housing	1	1	7.50 4.00	11.50	23
Top IR System					
Top IR Shielding	1	1	2.10 3.00	5.10	10.2
Ceramic Block	2	2	5.00 2.00	20.00	40
Side Lining	1	1	1.30 2.10	3.40	6.8
Perforated Sheet	1	1	4.00	4.00	8
Top IR back frame	1	1	1.70 2.40	4.10	8.2
Bottom Heating System					
Heating Block	1	1	6.00 4.00	10.00	20
Side frame strip	1	1	1.30	10.00	20
Front frame	1	1	1.30	1.30	2.6

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strip					
Bottom frame strip	1	1	1.30 3.40	1.30	2.6
Rotating Plate					
Aluminum rotating plate	1	1	4.00 5.50 2.30	11.80	23.6
Top load cell plate	1	1	5.00	5.00	10
Bottom load cell plate	1	1	5.00	5.00	10
Base plate	1	1	7.50 3.50	11.00	22
Stepper motor housing	2	2	7.50 4.00 1.75	26.50	53
Bottom base	1	1	2.20 3.00	5.20	10.4
Electrical box	1	1	1.80 2.50 3.25	7.55	15.1
Main Housing					
L bracket for servo motor	1	1	2.70 2.00	18.80	37.6

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Main frame housing	1	1	3.00 4.50 6.50	14.00	28
Pump holding blocks	1	1	1.50 3.00 2.00	6.50	13
Side panel with glass	1	1	3.50 1.40 1.50	6.40	12.8
Main frame outer cover	1	1	3.00 1.40 1.30	5.70	11.4

15.2.4. Machine Allocation

Machine / process type	**Required time per day (min)**	**Daily setup time (min)**	**Total time incl. setup (min)**	**Available time per day (min)**	**Utilisation incl. setup (%)**
Injection molding (egg storage, tunnel halves, gates, drawer cover, handle)	7.9	90 (high mold-change effort)	98	810	12.1%
Ultrasonic welding (egg tunnel)	1.1	15 (short daily setup for horn/tooling)	16.1	810	2.0%
CNC fibre laser	51.6	20 (daily	71.6	810	8.8%

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cutting (drawer box, frames, strips, shields, perforated sheet, bottom base, electrical box, main frame blanks)		program/tool change and sheet setup)			
Press-brake bending (drawer box, TOP-IR shielding, electrical box, main frame housing)	24.0	20 (tooling setup and angle checks)	44.0	810	5.4%
Band-saw cutting (egg holder arm billet, base of spear billet, spears billets, oil-housing billet, pump blocks)	20.5	15 (blade check and guide setup)	35.5	810	4.4%
CNC turning (egg holding shaft, inner shaft, sprockets, driving/idler shafts, rotating plate pre-turning)	99.5	15	114.5	810	14.1%
CNC milling / drilling / boring (all billets, pocket plates, frames, housings, heating block, plates, base plate, stepper housings, etc.)	531.4	20 (for tool and program changes)	551.4	810	68.1%
TIG / stitch welding (drawer box, side lining, TOP-IR back frame, bottom-heating frame assembly, rotating-plate base, electrical box housing, main frame housing)	41.9	20 (one daily setup for torch, gas, fixtures)	61.9	810	7.6%

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15.2.5.OEE

OEE stands for Overall Equipment Effectiveness. It is the gold-standard manufacturing metric used to identify the percentage of production time that is truly productive. An OEE score of 100% means perfect production: producing only good parts, at maximum speed, with zero downtime.

The OEE for each machine can be calculated using the formulas below;

OEE = Availability x Performance x Quality

Availability = Operating time / Planned Production Time

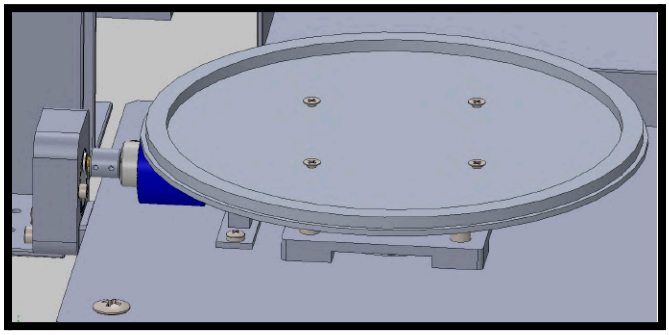
Performance = (Ideal Cycle Time x Total Output) / Operating Time

Quality = Good parts / Total parts produced

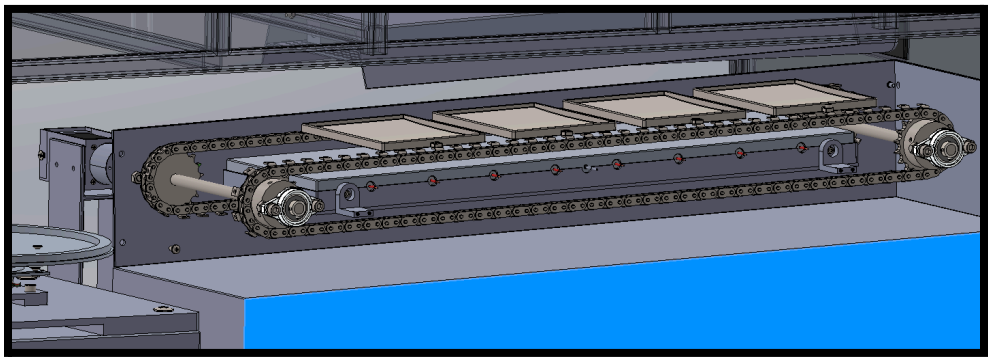
An Overall OEE of 90% is assumed. This is a realistic target considering the machine selection and volume of production. The new available working hours based on this OEE calculation is 810 mins/day.

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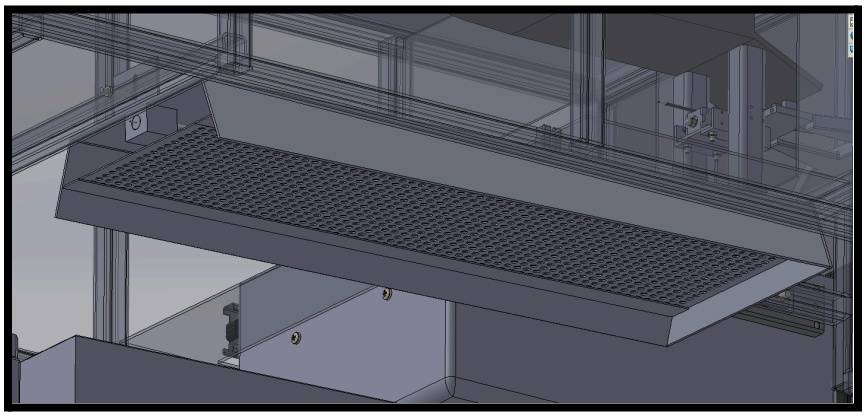
15.3. Assembly Sequence

Step	Assembly activity	Time per step (min)
	Rotating Plate	
1	First, the hollow bottom base is positioned inside the main housing and fixed with socket head cap screws. The common base plate is then placed on top of the hollow bottom base and screwed down, creating the main reference surface for the rotating plate subsystem. Next, the load cell bottom plate is mounted to the base plate,load-cell and the top load-cell plate, which are screwed to the load cell according to the required load direction. After that, the lazy Susan bearing is attached to the top load-cell plate, and the rotating steel/aluminium plate is fixed to the bearing using countersunk screws so that the screw heads sit flush with the rotating surface. The microswitch housing is then screwed onto the base plate, and the microswitch sensor is mounted inside its housing. Finally, the motor subassembly is prepared by fitting the motor shaft, coupling/rubber wheel, and related drive parts together. The motor is screwed into its motor housing, and the complete motor housing is then mounted to the base plate with screws so that the drive element aligns with the rotating plate mechanism. 	30 min

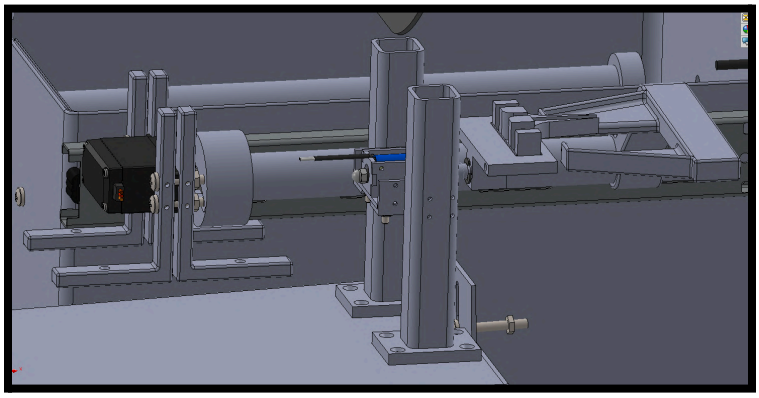
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	Conveyor system	
2	First, the hollow conveyor support frame is screwed to the base of the main frame. The conveyor back frame is then welded between the side walls so that the side frames stay aligned and square. After the welded frame is checked, the NEMA motor casing mount is fixed to the conveyor frame, and the NEMA motor casing is screwed onto this mount. Next, the stepper motor, planetary gearbox, and encoder are assembled inside the NEMA motor casing and fixed with screws. The coupler is attached to the gearbox output shaft, and the drive shaft is connected through the coupler. The drive shaft is then inserted through the bearing blocks and conveyor side frames. The same procedure is repeated on the opposite side with the idler shaft, but without the motor and coupler. The sprockets are fitted onto both ends of the drive shaft and idler shaft. Their set screws are tightened onto the shafts after the sprockets are positioned correctly. Both left and right sprocket pairs are aligned so the two conveyor chains run parallel and stay synchronized. The roller chains are then assembled from chain links, connecting pins, and attachment links. The K1 attachment links are positioned at the required spacing so the egg pocket plates can be mounted evenly. After the chains are placed onto the sprocket teeth, the chain tension is adjusted and both sides are checked to move together without phase shift. The egg pocket plates are then fixed to the attachment links using nuts, bolts, and washers. The aluminium heating block is installed inside the conveyor system with its mounting brackets and adjusted so the clearance to the moving pocket plates is correct. Finally, a thin layer of high-temperature, long-life food-area chain lubricant is applied to the chains, sprockets, and moving contact points, while keeping lubricant away from the egg-contact cooking surfaces. 	75 min

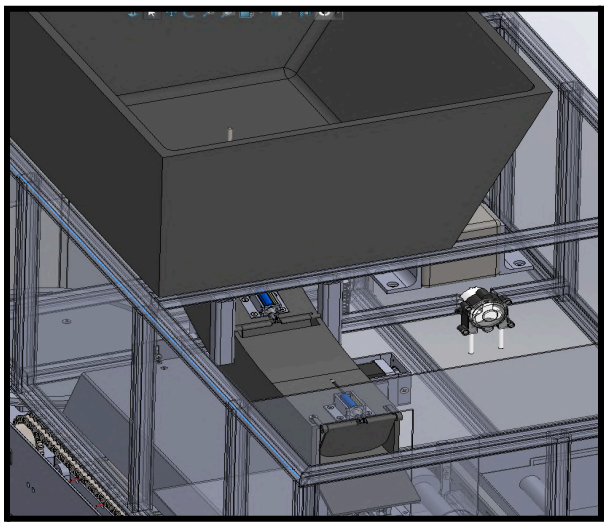
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	Top IR system	
3	First, the Top IR shield is positioned above the conveyor belt and fixed to the main frame through the bent mounting tabs. The slotted holes in the tabs are used to adjust the shield height according to the assembly drawing before the screws are fully tightened. Next, the two side lining rails are welded to the inner side walls of the IR shield. These rails form the support ledges for the perforated protection sheet. The back frame is then welded to the rear opening of the IR shield so the hood becomes stiff and closed at the back. After the shield body is prepared, the ceramic holder blocks are mounted to the inside/top area of the shield using screws through the ceramic block inserts. High-temperature ceramic adhesive is used only where required to secure the inserts or prevent loosening under heat cycling. The quartz IR heating element is then placed between the two ceramic blocks so the tube is supported at both ends and remains electrically insulated from the metal shield. Finally, the perforated stainless sheet is slid into the welded side lining rails below the IR tube. It is kept removable for cleaning and replacement, while still protecting the user and preventing large particles from reaching the heating element. 	25

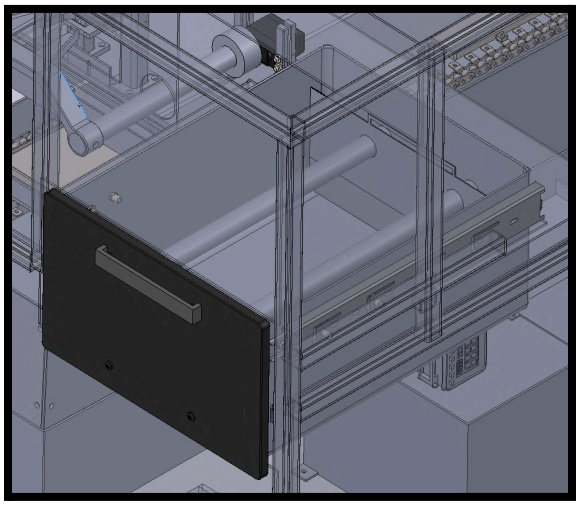
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Egg Breaking System		
4	First, the support brackets and welded frame parts for the egg-breaking mechanism are screwed to the main frame. The L-brackets are also fixed to the main frame and used as the mounting points for the shaft/motor housing and the servo-driven holding arm. Next, the servo motor for the holding arm is screwed onto its L-bracket. The arm shaft is connected to the servo output and fixed with screws so that the shaft rotates together with the servo. The egg holding arm is then mounted onto the shaft and screwed in place, making the arm rotate with the servo during operation. After that, the solenoid motor and its casing are mounted to the breaking-mechanism bracket. The solenoid output shaft is aligned with the shaft hole of the breaking mechanism and fixed using the designed shaft connection or retaining adhesive where required. The upper guide/support blocks for the spear mechanism are welded or fixed onto the breaking-mechanism base so that the spears stay aligned during motion. Finally, the left and right spears are installed into the guide blocks and connected to the solenoid-driven mechanism. The return springs are mounted between the spears in tension, so after the solenoid opens or drives the spears, the springs pull them back to their original position. 	40

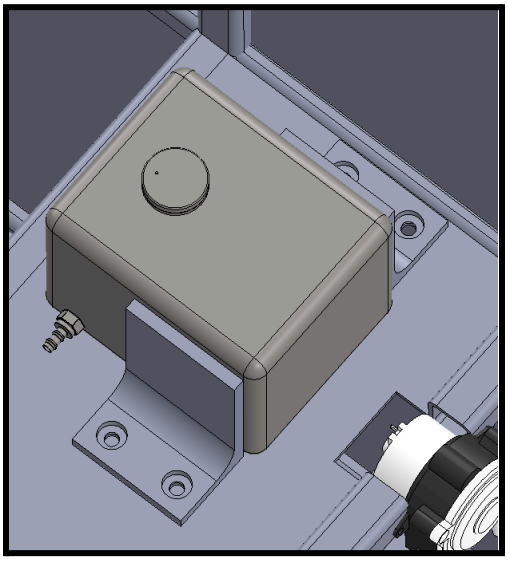
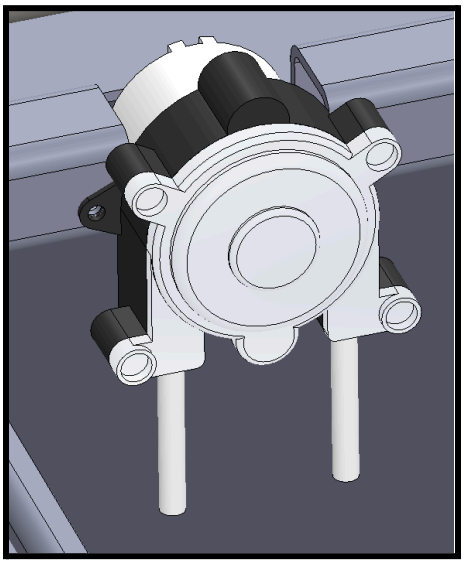
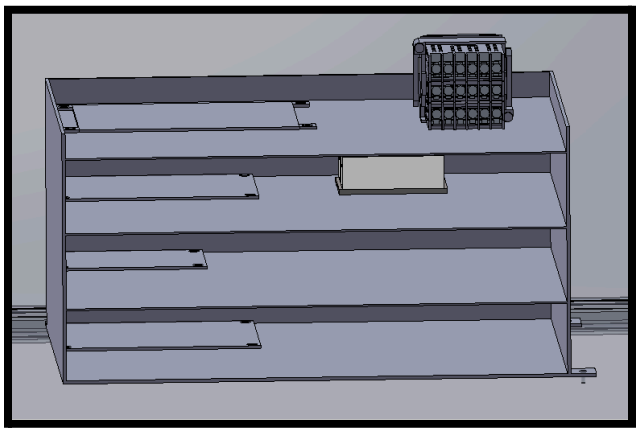
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Egg Storage and Transportation Unit		
5	First, the solenoid motor is assembled into its motor casing, and the complete solenoid motor casing is screwed onto the egg storage tunnel. After that, the solenoid output shaft is connected to the breaking/gate mechanism shaft so the motor can drive the motion correctly. Next, the side support mounts next to the motor are fixed to the tunnel. These mounts form the supports for the gate. The gate is then inserted into position between the mounts, and the hinge shaft/pins are slid through the mounts and the gate hub so the gate is held in place and can rotate freely. After the gate mechanism is completed, the tube weldments that hold Motor 1 are welded to the top side of the main frame. The egg storage tunnel is then brought into position and aligned with the shared mounting holes of the storage section. Finally, the egg storage tunnel assembly and the motor support structure are screwed to the chassis/main frame, making sure the tunnel, motor casing, and gate mechanism are all aligned correctly before final tightening. 	25

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Egg Shell Drawer		
6	First, the purchased drawer sliding rails are mounted to the left and right sides of the main frame using the prepared hole pattern. The rails are aligned parallel to each other so the drawer can slide in and out without jamming. Next, the sheet-metal drawer box is prepared outside the frame. The internal shield/liner is placed inside the drawer first, and the compression shaft is then installed and screwed into position. The compression shaft must move freely and stay aligned with the drawer guide direction. After that, the front plastic cover is positioned on the front face of the drawer box and fixed with screws. The front handle is then placed onto the plastic cover and screwed in place so the drawer can be pulled out for emptying and cleaning. Finally, the moving sections of the sliding rails are pulled outward. The drawer box is aligned with the hole pattern on the moving rail sections and screwed to the rails. After tightening, the drawer is pushed in and pulled out several times to check smooth motion, correct alignment, and full closing inside the main frame. 	15

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Oil System		
7	The oil tank is screwed into the main frame firmly using the big L brackets. The peristaltic pump is then screwed onto the main frame as well and the cap is screwed into its tank.  	15
Electrical Box		
8	The electrical box is screwed down to the base of the machine. The SSR is screwed down inside the box. The PID temperature controller is screwed on top of the electrical housing and the 4 Custom PCBs are screwed into each layer. 	20

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Electrical Assembly and Wiring		
9	Install oil-system tubing and pump electrics: Cut and route silicone tubing from stainless oil tank to peristaltic pump and flat-fan spray nozzle; install inline check valve, hose adapter, nozzle mounting bracket, and pump vibration mounts; connect MOSFET driver module and pump wiring into the control harness.	20
10	Install tunnel sensors and wiring: Mount flow/egg-detection sensors in the egg storage/tunnel area, position brackets, run shielded signal cables along the tunnel, terminate to the sensor/actuator interface PCB using ferrules and PCB screw terminals	25
11	Install conveyor home sensor, Hall sensor, smco magnet, and encoder wiring: Mount the home/reference sensor on the conveyor, install the incremental rotary encoder on the conveyor shaft or gearbox, run mixed-gauge control wiring from sensors and encoder to the main PCB and stepper driver, secure with cable ties and chain guards	25
12	Wire bottom heating subsystem: Insert cartridge heaters into the aluminium heating block, mount temperature sensor(s), thermal fuse/safety cutoff, and SSR; connect high-temperature wiring from IEC C20 inlet/MCB/RCCB through the SSR to the heaters, add grounding hardware and thermal sensor leads back to the temperature controller and MCU .	30
13	Wire top IR heating subsystem: Mount IR heater element, reflector shield, ventilated guard cover, and small ventilation fan; connect IR heater through its relay/SSR and high-temperature wiring, add thermal cutoff/safety sensor wiring, and route the fan supply wiring to the appropriate DC rail	25
14	Install rotating plate electrics: Mount geared stepper motor, ULN2003 driver module, lever microswitch for home position, HX711 amplifier, and associated wiring; connect motor leads, microswitch, and load-cell harness to the motion/load-cell PCB, fix cable strain reliefs near the rotating plate assembly.	30
15	Install touchscreen / screen system wiring: Fit the Nextion touchscreen into the front panel using M3 screws and standoffs; install JST-XH connector and FFC/FPC cables; connect 5 V supply and UART TX/RX between touchscreen and main STM microcontroller; secure the display harness and check clearance.	25
16	Power-supply and main harness routing: In the electrical box, mount remaining DIN rail components (terminal blocks, inline blade fuse holders, DC terminal blocks, buck converters), connect IEC inlet to MCB/RCCB and PE terminal, distribute DC rails (24/12/6/5/3.3 V) via terminal blocks, route shielded signal cables and mixed-gauge wiring through cable ducts, install	35

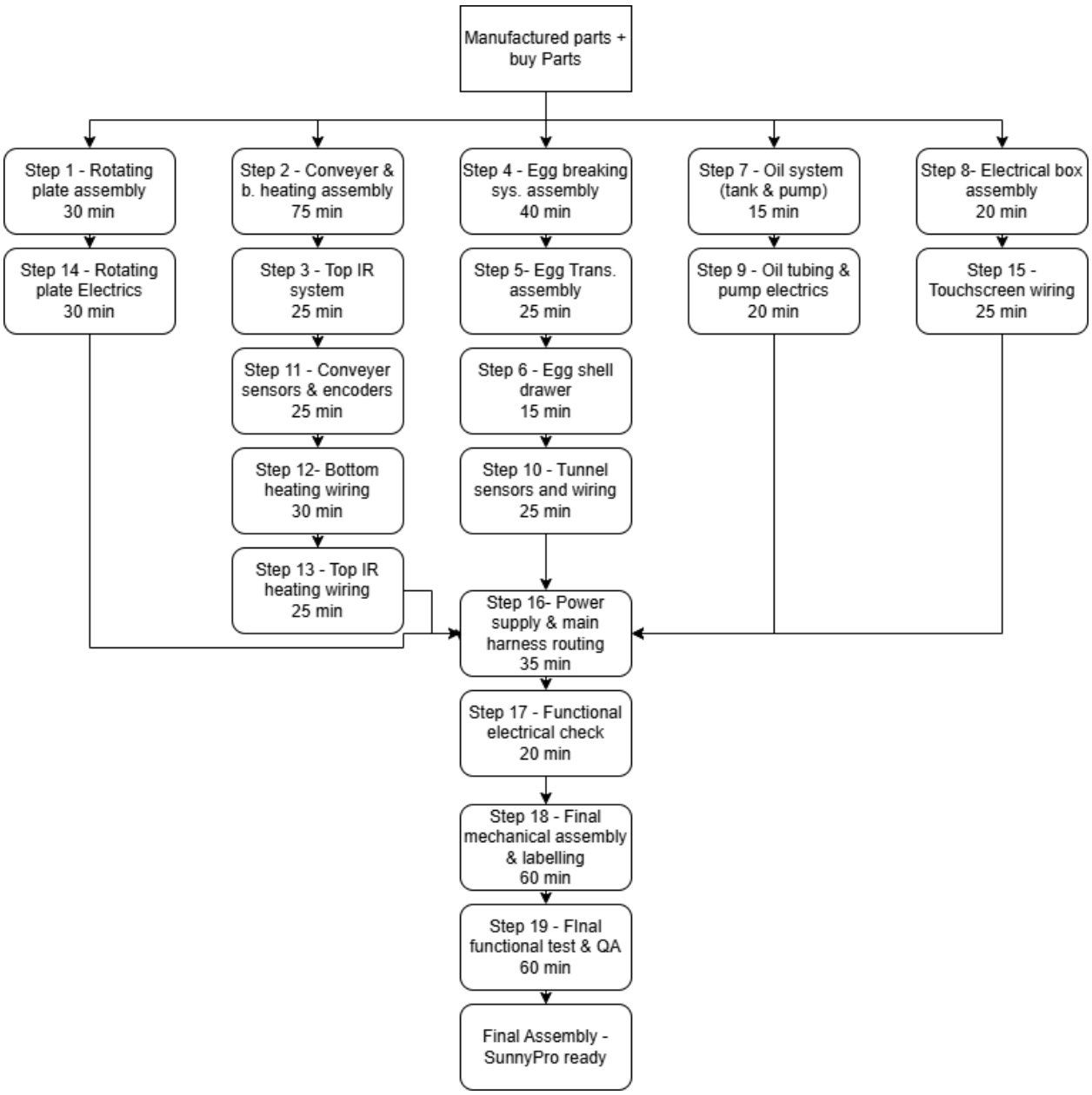
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	cable glands/strain reliefs, and finalize labeling and cable ties.	
17	Functional electrical check: Quick continuity check on critical circuits (heaters, pumps, motors, sensors), visual inspection of polarity and earth connections, and power-on smoke test with low-risk subsystems (no heating load), confirming that all tunnel sensors, conveyor sensors, touchscreen, and rotating plate signals are seen by the MCU.	20
Final Assembly		
18	Fit all outer covers, front bezel and touchscreen trim; align and fasten with specified screws and gaskets. Route and secure remaining external cable runs and tubing; fit cable glands, strain reliefs, hose clamps and protective wraps. Apply serial/CE plate, safety labels (hot surface, mains), and terminal/wire ID labels. Quick torque check on critical fasteners (heating block, conveyor bearings, base plate), fans are screwed on side, electrical inlet is installed	60
19	Power MCU, check touchscreen, run conveyor indexing, verify sensors and encoder signals, exercise solenoids/servos and pumps; log results. Controlled heat verification: with empty plates, run a short heating cycle to confirm SSR switching, thermocouple readings, oil dosing timing and safety cutoffs. Clean food-contact surfaces, fit removable pocket plates, verify seals and liners. Complete final QA checklist, note any defects, and record serial/test log; if passes, prepare machine for packing.	60

Total assembly time : 245 minutes + 235 minutes + 120 = 10 h per machine.

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15.3.1. Process flow diagram



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15.4. Workforce Calculation

Workforce per year

- Two shifts per day, 7.5 h effective per shift → 15 h/day of plant time, but per worker we assume one 7.5 h shift/day.
- Worker available time per day

Time of Worker = 7.5 h × 60 = 450 min/day
Efficiency factor (breaks, minor delays, etc.) : E = 0.90,
Effective worker time:
Effective time = 450 × 0.90 = 405 min/day

Planned output (machines/day):

- Year 1: 2 machines per day
- Year 2: 3 machines per day
- Year 3: 6 machines per day
- Year 4: 8 machines per day
- Year 5: 9 machines per day

Approximate total machine-processing time per SunnyPro (all machining/fabrication):
Machining time ≈ 900 min

Operator time ≈ 50% of machine-time (one operator can attend more than one semi-automatic machine)

→ operator-time per machine: T ≈ 450 min

Year	Machines per day	Assembly time per day	Required assembly workers	Machine operator minutes/ day	Required machine-process operators
1	2	2×600=1200 min	1200/405 ≈ 3 assembly workers	2×450= 900 min	900/405 = 2 Machine operators
2	3	3×600=1800 min	1800/405 ≈ 4 assembly workers	3×450= 1350 min	1350/405 = 3 machine-process operators

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3	6	6×600=3600 min	3600/405 ≈ 9 assembly workers	6×450= 2700 min	2700/405 = 6 machine-process operators
4	8	8×600=4800 min	4800/405≈12 assembly workers	8×450= 3600 min	3600/405 = 9 machine-process operators
5	9	9×600=5400 min	5400/405≈14 assembly workers	9×450= 4050 min	4050/405 = 10 machine-process operators

Since there is a variety of manufacturing machines, one machine operator would have to handle all of the manufacturing for the many make parts of our product, so we will have +1 machine operators for every year, to distribute the workload and include more experience in the team.

So for each new year we will have **2, 3, 6, 9 and 10** machine operators respectively.

Workload distribution:

Process	**Worker 1**	**Worker 2**	**Worker 3**	**Worker 4**	**Worker 5**
CNC Milling	-	-		150 min	100 min
CNC Turning	-	-		54 min	36 min
Laser/Router	-	-		20 min	20 min
Press-Brake	-	-		18 min	18 min
TIG welding	-	-		18 min	18 min
Injection Moulding	-	-		28 min	28 min
Ultrasonic welding	-	-		5 min	5 min
Assembly Line	400 min	400 min	400 min	-	-
Total time per worker	400 min	400 min	400 min	293 min	225 min

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16. Cost Calculation

In order to calculate full cost per unit, surcharge calculation structure will be applied. This ensures the final offer price can be calculated with various costs included while ensuring the product remains profitable.

16.1. Material Costs

16.1.1. Production Material Costs - Buy Parts

Summarized from below:

	Total Cost (in €) per Unit
From Mechanical BOM	
- Mechanical parts	311.09
- Screws, clips, nuts, etc…	43.18
From Electrical BOM	457.45
Total cost for one unit (buy parts)	**811.72**

Cost Breakdown:

From Mechanical BOM:

Part	**Supplier Link**	**Quantity**	**Cost in €/ Unit**	**Total cost in €/Unit**
DIN Rail	Hutschiene TS 35 x 7,5mm _ gelocht - L=326mm	1	3.49	3.49
Cable Duct / Wire Raceway	Cable duct	1	2.50	2.50
Cable Glands / Strain Reliefs	-Kabelverschraubung, M 12 _ x 1,5, Ø 3,5 - 7 mm, schwarz, IP69K	4	2.23	8.92
Cable Mounts	Cable mounts	1	0.26	0.26
Cable Ties	Cable ties	1	0.19	0.19
Axle Shaft	Aluminum shaft	1	6.03	6.03

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Return Spring	Spring	2	3.54	7.08
Thermal Insulation	Thermal insulation	1	12.30	12.30
Inner Lining	Aluminum sheet	1	1.25	1.25
Inline Check Valve	Duotight - 8mm Inline Check Valve	1	3.46	3.46
Silicone Tubing	Silicon tubing	1	0.69	0.69
Flat Fan Spray Nozzle	CDIYTOOL 8 Pieces 65 Degree High Pressure Spray Nozzle	1	14.99	14.99
Roller Chain with Bent Attachments	Z24	1	148.54	148.54
2-Bolt Flange Bearing Block	Stehlagereinheiten / Flanschlagereinheiten 2- Loch KFL000	4	5.44	21.76
Flexible Clamp Motor Coupling	10 mm to 12 mm Flexible Shaft Coupling, Diameter 40 mm Length 66 mm Aluminium Motor Connector Flexible	1	10.39	10.39
Planetary Gearbox for Motor	Planetengetriebe Schrittmotor 27:1	1	29.26	29.26
Thermal Compound	Wärmeleitpaste, Tube, 35gr, ungiftig	1	4.99	4.99
Stainless Fasteners Kit	M2 M3 M4 M5 Schrauben Set – 1280 Stück	1	34.99	34.99
Screws, nuts etc.				
Countersunk Raised Head Cross Recess Screw	For M4x16: Haus&Dach	4	0.09	0.36

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Pan Head Cross	-Linsenschrauben M8x16 -STM390400300S -STM320300200E	• 2 • 2 • 2	0.35 0.20 0.25	1.60
Recess Screw	-Zylinderschraube Innensechskant DIN912 (ISO 4762) V2A - M6x20, 100 St. -Zylinderschraube Innensechskant DIN912 (ISO 4762) V2A - M3x12, 100 St.	• 2 • 2	0.05 0.01	0.12
Socket Head Cap Screw	-DIN 912/ISO 4762 A2 – M3 x 25 mm 100 Stück -DIN 912/ISO 4762 A2 – M3 x 10 mm 100 Stück	• 9 • 4	0.07 0.04	0.79
Pan Head Cross Recess Screw	Flachkopfschraube ISO 7045 PZ M4x8-Z 4.8 galv.verz. 2000St.	2	0.04	0.08
Prevailing Torque Hex Nut	Sechskantmuttern M6 DIN 934 Edelstahl A2	4	0.17	0.68
Plain Washer	Unterlegscheibe DIN125 (ISO 7089) V2A - 6,4 mm (M6), 1000 Stück	4	0.01	0.04
Socket Set Screw, Cup Point	Gewindestift DIN 916 / ISO 4029, M 5x20, Stahl 45 H	2	0.08	0.16
Hex Thin Nut	-for M3: Der Schraubenladen -for M4: “ -for M6: “ -for M2.5: “	• 8 • 10 • 4 • 2	0.04 0.04 0.05 0.04	1.00
Lazy Susan Bearing	Stahlblech-Drehkranz 1 St.	1	24.36	24.36
Rubber Drive Wheel	Freies Verschiffen 10PCS 8*32*16 flache typ PA nylon	1	13.99	13.99

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From Electrical BOM:

Part	Supplier	Quantity	Cost in €/ Unit	Total cost in €/Unit
IEC C20 Power Inlet	IEC 20 Power Inlet	1	3.13	3.13
MCB-Circuit Breaker	16A MCB- TYPE B Circuit Breaker	1	20.95	3.13
RCCB / RCD-TYPE A	RCCB / RCD-TYPE A (30mA)	1	45.80	45.80
DIN Terminal Blocks	DIN Terminal Blocks (L/N/PE)	1	2.83	2.83
Protective Earth Terminal Block	Protective Earth Terminal Block	1	2.88	2.88
AC Wire	BACHMANN	1	11.95	11.95
24V Industrial SMPS	24V Industrial SMPS 60W (2.5A at 24V)	1	21.50	21.50
24V → 12V LM2596T-12	LM2596T-12/LF03	1	2.82	2.82
12V → 5V LM2596T-5	LM2596T-5.0/LF03	1	2.82	2.82
24V → 6V LM2596S-ADJ	LM2596S-ADJ/NOPB	1	4.41	4.41
5V → 3.3V Regulator AMS1117-3.3	LM1117MPX-3.3/NOPB	1	0.61	0.61
DC Terminal Blocks	9-Way DC Terminal Block 5V 12V 24V for CCTV Power Supply	1	11.89	11.89
Inline Blade Fuse Holders	URAQT Fuse Holder Car Standard 16AWG	4	1.05	4.20
Blade Fuse Assortment	0287010.PXCN Littelfuse Inc. | Fuses | DigiKey	1	0.22	0.22

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Shielded Signal Cable	2M UL2547 28AWG 4 Cores PVC	1	9.98	9.98
Ferrules and Crimp Terminals	1-966066-1 : Crimp Wire Pins, Tabs & Ferrules | TE Connectivity	1	0.17	0.17
PCB Screw Terminal Connectors	20Paar 5.08mm Pitch Male Buchse PCB Screw Terminal Block | eBay.de	1	18.20	18.20
Axial fan	EBM Papst 8414NGML Axiallüfter 24 V/DC 42 m³/h (L x B x H) 80 x 80 x 25mm | » Freue dich auf voelkner	2	27.26	54.52
Main Micro-controller STM32H743ZITx	STM32H743ZIT6 STMICROELECTRONICS, ARM-MCU, STM32 Family	1	8.57	8.57
Gate Actuators	JF-0530B DC12V 300mA 5N/10mm Pull-Push-Type Solenoid Electromagnet	2	3.49	6.98
Flow Sensors IR sensor	IR Break Beam Sensors- 3mm LEDs : Adafruit Industries	3	2.85	8.55
Flip Actuator	MG996R Digital Servo Motor mit Metall Getriebe	1	6.30	6.30
Platform Solenoid JF-0837B	JF-0837B DC 24V 1A 15N 10mm Push Pull Type Open Frame Linear Motion Solenoid Electromagnet | Harfington	1	12.47	12.47
Solenoid Driver multi-channel MOSFET	Industrial-Grade MOSFET Driver Module – 10-100A Adjustable PWM Controller (12-60V DC)	1	4.39	4.39
Peristaltic Pump	DC 12V Dosing Pump Peristaltic Pump	1	7.29	7.29
MOSFET Driver Module	Mosfet Mosfet 12V DC 8×6×2 Low Voltage Mosfet Module	1	9.29	9.29

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Touchscreen Display Module –Nextion NX8048P070_011R	NEXTION 4.3/5.0/7.0" Intelligent LCD Touch Display	1	45.39	45.39
JST-XH 4-pin Connector	JST JST - Buchsengehäuse, 1x4-polig - XH	1	0.11	0.11
Stepper Motor	NEMA 17	1	12.40	12.40
Stepper Motor Driver	Stepper Motor Driver	1	5.89	5.89
Control Wiring and Connectors	Control Wiring and Connectors	1	0.74	0.74
Hall Effect Sensor	A1104-class Hall effect switch sensor, 3-pin type	1	0.89	0.89
SmCo Magnet	SmCo	1	0.33	0.33
Incremental Rotary Encoder	AMT102-V	1	18.30	18.30
Cartridge Heaters	Cartridge Heaters	1	15.43	15.43
Temperature Sensor	PT100 temp. Sensor	1	6.59	6.59
Temperature Sensor Amplifier	MAX31865AAP+ Analog Devices DigiKey	1	5.86	5.86
Solid-state Relay	solid state relay 25A SSR-25AA , ZG3NC-325A Single phase	1	11.59	11.59
Heater Branch Fuse	0652C8000-11 Bel Fuse Inc. | Sicherungen | DigiKey	1	0.38	0.38
High-temperature Wiring	Silikonkabel 18 AWG 0,5 mm² 200°C 300V flexibel verzinnt Meterware | akkuteile.de	1	0.86	0.86
Grounding Hardware	100m Erdkabel H07V-K 1x4mm² Grün/Gelb bei hd-digital kaufen.	1	1.62	1.62

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IR Heater Element	IR HEATER	1	15.00	15.00
IR Power Control / Relay	SSR Halbleiter Solid State Relais DC / AC, 230V AC 25 A	1	15.14	15.14
Thermal Cutoff / Safety Sensor	THERMODISC Temperatursicherung, 16A, 184°C, kurz/lang | Temperatursicherungen günstig kaufen | reichelt elektronik	1	2.30	2.30
PT100 Sensor	PT100 temp. Sensor	1	1.25	1.25
SSR Heat Sink	Aluminium Heat Sink SSR Dissipation For One Phase Solid State Relay 10a-40a For PID Thermostat Temperature Controller	1	8.59	8.59
Ceramic Terminal Block	Elektrofachmarkt-online - Keramikklemme 2-polig mit Befestigungsloch	1	1.04	1.04
Low-voltage Terminal Blocks	Phoenix Contact 1725672 MPT 0,5/ 4-2,54 Leiterpla	1	1.17	1.17
Stepper Motor	FREI Standard-Schrittmotor, 5V	1	1.50	1.50
Stepper Driver Module	FREI Entwicklerboards - Motorsteuerung, ULN2003A	1	1.35	1.35
Microswitch	5Pcs Mini Micro Limit Switch NO NC 3 Pins PCB Terminals SPDT 5A 125V 250V 29mm Roller Arc lever Snap Action Push Microswitches	1	0.78	0.78

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Load Cell and HX711 Amplifier Module	Stemedu 5 kg Load Cell Digital Weight Sensor for Portable Electronic with HX711 Ad Amplifier Module, 11 lbs Weighing Sensor Arduino Raspberry Pi	1	10.98	10.98
Wiring and Connectors	DKS-20MM-30 DigiKey Standard | Jumper Wire | DigiKey PVC Datenleitung LAPP UNITRONIC LiYY 3x0,25 - 0028303 | Automation24	1	0.98	0.98
Emergency stop button	1NC 22mm Red Mushroom Emergency Stop Push Button Switch 600V 10A	1	6.09	6.09

16.1.2. Production Raw Materials Costs - Make Parts

The approximate price basis was determined with online market research for typical German/European supplier prices. For each material, a price range was identified and the average value was used as the approximate price basis. The material quantities are based on the SolidWorks mass properties of the CAD models.

**Cost per unit = Quantity in kg/Unit * Approximate price basis in €/kg*

Material name	**Quantity (in kg)/Unit**	**Appr. Price basis (in €/kg)**	***Cost/Unit (in €)**
Polypropylene Copolymer	4.3	1.80	7.74
POM-C / Acetal copolymer	0.221	5.00	1.11
ABS-HI plastic	11.495	2.50	28.74
AISI 304 Stainless Steel Sheet, 1.0 mm	5.039	2.55	12.85
AISI 304 Stainless Steel Sheet, 2.0 mm	86.586	2.35	203.48
AISI 304 Stainless Steel	1.147	2.25	2.58

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Sheet, 3.0 mm			
AISI 304 Stainless Steel Round Bar	1.513	5.5	8.32
AISI 304 Stainless Steel Plate	2.586	1.15	2.97
Martensitic Stainless Steel EN 1.4116 Billet	0.022	2.35	0.05
EN AW-6082 T6 Aluminum Plate	4.751	4.00	19.00
EN AW-6082 T6 Aluminum Billet	6.462	3.50	22.62
Clear Anodised EN AW-6082 T6 Aluminium Plate	1.157	5.00	5.79
Clear Anodised EN AW-6082 T6 Aluminium Billet	0.044	4.50	0.20
Clear Anodised EN AW-6061 T6 Aluminium Plate	0.815	5.20	4.24
EN 1.4409 / GX2CrNiMo19-11-2 for stainless steel precision casting	1.536	4.50	6.91
EN AC-43500 / EN AC-AlSi10MnMg Cast Aluminium Alloy	0.285	3.90	1.11
96% Aluminium Oxide Ceramic, Al_2O_3 C799	0.042	15.00	0.63
Brass Threaded Inserts	8	0.18	1.44
Total cost for one unit (raw material)			**329.77**

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16.1.3. Production Material Costs

Production Material Costs/unit = Total cost (buy parts) + Total cost (raw material)
= 811.72€ + 329.77€= ***1 141.49€***

16.1.4. Total Material Costs per unit

*Material overhead cost = Total Production Material cost * 15% = 200.07€*

Total material cost/unit = 1 141.49€ + 171.22€ = ***1 312.71€***

16.2 Manufacturing Cost

16.2.1. Production Wages

Position	Job Details	Number of Employee	Hourly rate (in €) per worker	*Salary per year (in €)	Total Salary per year (in €)
Assembly worker	Assemble the machine	3	16	30 600	91 800
Plant Operator	Operates machine	2	14	26 775	53 550
Total		4	-	-	145 350

**Salary per year = Hourly rate * Effective time per shift * Total number of working days = Hourly rate * 7.5 * 255*

Total production wages *= 145 350€ / 500 units =* ***290.70 €/unit***

16.2.2. Machine and Depreciation Cost

Since the planned production volume in the first year is relatively low, we decided to purchase second-hand machines. This reduced our initial investment costs while still providing sufficient manufacturing capacity. Suitable used machines can be bought from industrial machine platforms such as Machineseeker and Machinio.

A depreciation rate of 10% per year is being used with an assumed machine life of 10 years. This matched the standard rate for cnc equipment used in the manufacturing industry.

In our first year we are planning a production and sales volume of 500 Sunny Pro units.

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Name of Machine	Cost per Machine (in €)	* Depreciation Cost per Year (in €)	* Depreciation Cost /Unit (in €)
CNC machining centre - Haas UMC-750 (5-axis)	85 000	8 500	17.00
CNC turning centre - Haas ST-20Y	66 415	6 641.5	13.28
CNC fibre laser - Trumpf TruLaser 1030 fiber	56 000	5 600	11.20
Press brake - Amada HFE	26 000	2 600	5.20
TIG welder - Miller Dynasty 210	3 000	300	0.60
Injection moulding machine - Arburg Allrounder 270 A	8 650	865	1.73
Ultrasonic welder - Branson (Emerson) GSX-E1	8 700.21	870.02	1.74
CNC router - MultiCam 3000 Series	19 673.84	1 967.38	3.94
Total	281 439.05 €		54.69 €

** Depreciation cost per year = Machine purchase price * 10%*

**Depreciation cost per unit = Depreciation cost per year / 500 units*

16.2.3. Maintenance and Energy Cost

Machine	*Maintenance Cost per Year (in €)	Power consumption (in kW)	*Electricity cost per year (in €)
CNC milling	2 550.00	22.4	8 064
CNC turning	1 992.45	14.9	5 364

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CNC fibre laser cutting	1 680.00	10.0	3 600
Press-brake bending	780.00	16.5	5 940
TIG welding equipment	90.00	3.1	1 116
Injection moulding	259.50	15.0	5 400
Ultrasonic welding	261.01	5.5	1 980
CNC router	590.22	9.0	3 240

According to OXMAINT, a maintenance cost rate of 3% is within a healthy range for equipment maintenance.[22]
*Maintenance cost per year= Machine price * 3%

**Electricity cost per year* = *Power consumption in kW * Operating hours/year * Electricity price €/kWh*

Available machine hours/year: 2,000 h/year (calculated above in production planning)
According to the official website of the Bundesnetzagentur, the industrial electricity price in January 2025 was approximately 0.18€/kWh, so this value is used for Electricity price in €/kWh.[23]

16.2.4. Rental cost

Considering all the factors in terms of employee numbers, work stations for assembly, machine sizes and storage space, we would need a place of at least 400m². According to the REALOGIS Market Report, the average industrial rent in Düsseldorf is about 7.00€/m². [24]

Rental cost calculation with 400 m²:

Rental cost	= 400 m² * 7 €/m²/month	= 2 800 €/month
Annual rental cost	= 2 800 €/month * 12 months	= 33 600 €

[22] Johnson. Maintenance budgeting & cost analysis for manufacturing. *Oxmaint*. Published March 31, 2026. Accessed July 6, 2026. https://oxmaint.com/industries/manufacturing-plant/maintenance-budgeting-cost-analysis-manufacturing
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[24] REALOGIS. Market report North Rhine Westphalia full year 2025. Accessed July 6, 2026. https://www.realogis.com/market-report/logistics/leasing/north-rhine-westphalia-full-year/2025

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Rental cost/unit = 33 600€ / 500 units = ***67.20 €/unit***

16.2.5. Hourly Machine Cost Table

Formula:

Machine hour rate = (Depreciation + Maintenance + Energy + Rent) / Available machine hours per year

Machine	**Depreciation Cost per Year (in €)**	***Maintenance Cost per Year (in €)**	***Electricity cost per year (in €)**	**Rental cost /year for one unit (in €)**	**Available machine hours per year**	**Hourly Machine Rate (in €/h)**
CNC milling	8 500.00	2 550.00	8 064	4200	2000	11.66
CNC turning	6 641.50	1 992.45	5 364	4200	2000	9.10
CNC fibre laser cutting	5 600.00	1 680.00	3 600	4200	2000	7.54
Press-brake bending	2 600.00	780.00	5 940	4200	2000	6.76
TIG welding equipment	300.00	90.00	1 116	4200	2000	2.85
Injection moulding	865.00	259.50	5 400	4200	2000	5.36
Ultrasonic welding	870.02	261.01	1 980	4200	2000	3.66
CNC router	1 967.38	590.22	3 240	4200	2000	5.00

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16.2.6. Production Overhead Costs

Machine	Total time incl. setup (in min) per unit	Total time incl. setup (in h) per unit	Hourly Machine Rate (in €/h)	Cost per unit (in €)
CNC milling	551.4	9.19	11.66	107.16
CNC turning	114.5	1.91	9.10	17.38
CNC fibre laser cutting	71.6	1.20	7.54	9.05
Press-brake bending	44.0	0.74	6.76	5.07
TIG welding equipment	61.9	1.03	2.85	2.94
Injection moulding	98.0	1.64	5.36	8.79
Ultrasonic welding	16.1	0.27	3.66	0.99
CNC router	35.5	0.59	5.00	2.95
Total production overhead cost per unit				**154.33**

16.2.7. Manufacturing Costs

Manufacturing costs/unit = Production wages/unit + Production overhead costs/unit
= 290.70 €/unit + 154.33€/unit
= 445.03 €/unit

16.3. Cost of Goods Sold

Production cost/unit = Total material cost/unit + Total manufacturing cost/unit
= 1 312.71€/unit + 445.03 €/unit
= 1 757.74 €/unit

Administrative overhead (10% of Production cost) = 1 757.74€/unit * 10%
= 175.77 €/unit

Marketing and Sales overhead (10% of Production cost) = 1 645.57 €/unit * 10%
= 175.77 €/unit

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Cost of Goods Sold= Production cost + Administrative overhead + Marketing and sales overhead
= 1 757.74 €/unit + 175.77 €/unit + 175.77 €/unit
= 2 109.28 €/unit

16.4. Selling Price

We decided that our profit margin should be 32%, because our SunnyPro is designed for the Restaurant and Dining industry, so that our business generates sustainable growth.[25]

Profit = Cost of Goods Sold * Profit Margin
= 2 109.28 €/unit * 32%
= 674.97 €/unit

Cash Selling Price = Cost of good sold + Profit
= 2 109.28 €/unit + 674.97 €/unit
= 2 784.26 €/unit

Cash Discount = Cash selling price * 4%
= 2 784.26 €/unit * 4%
= 111.37 €/unit
(This percentage was selected in order to lure customers for early payment.)

Commission = Cash selling price * 5 %
= 2 784.26 €/unit * 5 %
= 139.21 €/unit
(We decided to go with a commission rate of 5%, because our product is already expensive and even a lower commission will still yield a good sum for the salesperson.)[26]

Target selling price= Cash selling price + Cash discount + Commission
= 2 784.26 €/unit + 111.37 €/unit + 139.21 €/unit
= 3 034.84 €/unit

Discount = Target selling price * 3%

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[26] Pal D. Manufacturers sales rep commission rates: how to calculate? *Salesmate*. Published June 2023. Accessed July 6, 2026. https://www.salesmate.io/blog/manufacturers-sales-rep-commission-rates/

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= 3 034.84 €/unit * 3%
= 91.05 €/unit

Offer Price = Target selling price + Discount
= 3 034.84 €/unit + 91.05 €/unit
= 3 125.89 €/unit

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